

Finite Screen Spacetime

Entropy-Response Geometry from Causal-Screen Ledgers

FDS-G1 Complete Series v1.2

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

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Abstract. FDS-G1 proposes a finite causal-screen entropy-response geometry in which gravitational stiffness arises from screen-ledger capacity, not from a metric/field primitive. Its leading late-time dark-sector branch ($M_{3/4}$) is projection-locked to

$$s < 3, \quad \mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a), \quad \hat{R}_H(1) = 1,$$

a sparse background-Weyl residual rather than a flexible dark-energy or modified-growth fit. A completed medium-prior 8-seed production-refined nested-evidence audit over seven models selects the hierarchy

$$M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE-2} > \text{G1DE-1} > \text{CPL} > \Lambda\text{CDM},$$

with margins $\Delta \log Z = 0.856, 1.775, 6.523, 7.402, 9.603, 11.884$ against the six controls (all models share medium-prior configuration, eight seeds, $d \log Z = 0.1$). KiDS-1000 shear-only diagnostics independently support the Weyl-channel signal under nuisance profiling, with free- κ relaxation returning $\kappa \approx 0.746$; deterministic mock injection passes. **This is a production-refined diagnostic evidence result, not a final cosmological discovery.** The likelihood tests the output-normalized shape $\hat{R}_H(a)$; it does not identify a unique microscopic source-kernel factorization $K_H * J_H$. Open limitations: real (P_{nn})/clustering data missing; full 3×2 pt lensing likelihoods pending; D0 calibration lemmas remain open; CMB/BBN/nonlinear compatibility not yet tested.

v1.2 integrated release. Version 1.2 integrates the finalized KiDS BandPower (EE+nE) diagnostic bridge (unchanged from v1.2-dev) with an audited theory-hardening architectural layer—finite-ledger, causal-measurement, retarded-kernel, modular-response, entropy-inequality, and μ/Σ production-interface modules—across the Core, Companion, and D-series papers. All new theoretical material is marked by claim type: definition, admissibility rule, conditional target, open lemma, or empirical protocol. All empirical claims and boundaries remain unchanged from the v1.2-dev BandPower-only release. The *nn*/clustering channel remains unavailable locally.

Archive document. Recommended entry points: Core, G1DE- $M_{3/4}$, D0, D1, and the D8–D10 bootstrap overview.

Single Main Claim of FDS–G1. Finite causal-screen entropy response generates a metricized residual geometry whose leading late-time dark-sector branch ($M_{3/4}$) is locked to

$$s < 3, \quad \mu = 1, \quad \Sigma - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a), \quad \hat{R}_H(1) = 1,$$

and is selected by a completed production-refined 7-model 8-seed homogeneous nested-evidence audit with KiDS-1000 shear-only Weyl-channel diagnostic support.

What this does NOT claim:

- Not a completed first-principles derivation of General Relativity.
- Not a unique microscopic source–kernel factorization for $\hat{R}_H(a)$ (D6/D7 open).
- Not a validated production cosmological constraint (full 3×2 pt, CMB, nonlinear pending).
- Not a proof of the D0 calibration lemmas (finite-index density, boost modular covariance — open).
- Not a replacement for GR+ Λ CDM in strong-field, nonlinear, or early-universe regimes.

Evidential status: production-refined medium-prior evidence-selected (pilot grade); diagnostic KiDS-1000 Weyl-channel support; not yet production-confirmed.

Status of this paper. This is a bridge program, not a completed theory of gravity.

This work presents FDS–G1 as a finite-screen entropy-response bridge program, not as a completed first-principles derivation of General Relativity or quantum gravity. The current completed claims are: a finite-screen response architecture, a typed residual and demotion discipline, a projection-locked late-time output branch, and a production-refined medium-prior 8-seed nested-evidence audit for the $M_{3/4}$ branch.

What this paper does NOT claim:

- Not a completed first-principles derivation of General Relativity.
- Not a unique microscopic source–kernel factorization for $R_b^H(a)$ (D6/D7 open).
- Not a production cosmological constraint (full 3×2 pt, CMB, nonlinear pending).
- Not a proof of the D0 calibration lemmas (finite-index density, boost modular covariance — open).
- Not a replacement for GR+ Λ CDM in strong-field, nonlinear, or early-universe regimes.

The following are not yet completed: a microscopic derivation of the finite-screen ledger; the D0 finite-index density lemma; the D0 boost modular covariance lemma; a unique source–kernel factorization for $R_b^H(a)$; full 3×2 pt production validation; noisy mock ensembles; independent pipeline replication; and CMB/BBN/nonlinear-structure compatibility. Existing v1.2 empirical results and theory-hardening modules are integrated as separately marked claim types; integration does not imply that open lemmas are solved or that production cosmological validation is complete.

v1.2 integrated update. Version 1.2 integrates two independent, strictly separated developments:

1. **Empirical track:** It records the completed KiDS BandPower $EE + nE$ diagnostic bridge and its full validation layers, exactly as in the v1.2-dev BandPower-only release. All empirical results and boundary constraints remain unchanged.
2. **Theory-hardening architectural track:** It upgrades the finite-screen theoretical closure framework by adding integrated finite-ledger, causal-measurement, retarded-kernel, modular-response, entropy-inequality, and production-interface modules.

The theory-hardening modules are formal, body-level components of v1.2, with their individual proof status explicitly audited and itemized in the theory-hardening claim ladder. Integration into the body does not imply that open lemmas are solved, that a complete microscopic derivation is finished, or that production cosmological validation has been completed.

Existing empirical results and legacy paragraphs are not rewritten. The v1.2 theory-hardening material is integrated into the relevant body sections as explicitly marked claim-type audited modules.

Series Map

Paper	Current role	Closure role	Main theorem target	Failure consequence
Core	finite-screen entropy response	defines closure spine	screen entropy $\rightarrow$ response $\rightarrow$ metricization	bridge demotion, not empirical failure
A	screen statistics	area-law substrate	area-law universality	no screen-statistical basis
B	screen dynamics	capacity-flow normal form	Raychaudhuri-limit recovery	normal-form demotion
C	scalar-capacity EFT	covariant capacity mode	no-free scalar residual	scalar branch demotion
D	likelihood tests	empirical penalty layer	no-free-fit protocol	branch demotion
E	horizon occupancy	physical residual branch	conserved occupancy residual	G1DE demotion
F	holography/horizons	dictionary consistency	response-density dictionary	dictionary demotion
G1DE- $M_{3/4}$	evidence + KiDS shear-only stress interface	output-level evidence and Weyl-channel diagnostic test	$M_{3/4}$ evidence lock	empirical branch demotion, or shape-lock demotion under full $3 \times 2\text{pt}$ / binned- Σ stress
D0	calibration	strong derivation route	finite-index + boost modular lemmas	strong derivation demotion
D1	sheaf geometry	gluing obstruction	response obstruction class	residual demotion
D2	3/4 law	projection uniqueness	optical MaxEnt uniqueness	3/4 demotion
D3	non-eq. entropy	entropy ledger	optical-stress residual accounting	thermodynamic demotion
D4	influence functional	source-kernel structure	quotient-controlled influence	realization demotion
D5	Ward identifiability	admissibility rules	no-free- $A(a, k)$ theorem	generic dark-stress demotion
D6	memory quotient	output quotient	horizontal lift	source-kernel uniqueness forbidden
D7	Markov realization	existence model	finite-state passive lift	uniqueness not claimed
D8	reconstruction	universal metric envelope	causal access + entropy density $\rightarrow$ metric envelope	nonmetric / Finsler / no-continuum demotion
D9	obstruction classification	residual exact sequence	no-free-dark-sector classification	residual rejected or demoted
D10	G1DE bootstrap	minimal Weyl branch tested by G1fit-real and KiDS shear-only diagnostics	$M_{3/4}$ uniqueness / prediction compression	branch demotion to M_κ , const- Σ , binned- Σ , G1DE-2, or rejection

D8–D10 do not replace D0–D7. They form a bootstrap extension: D8 reconstructs

the metric envelope from finite causal access and entropy-density data, D9 classifies admissible obstruction residuals through an exact sequence, and D10 proves the minimal Weyl branch structure behind $M_{3/4}$.

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Closure Contract and Non-Absorption Rule

G1 is closed only if every arrow is typed, every residual is admissible, and every failure is non-absorbable.

Every major statement must be classified as one of:

1. Definition
2. Formal theorem
3. Conditional theorem
4. Bridge assumption
5. Normal-form approximation
6. Open lemma
7. Empirical output normal form
8. Falsification / demotion rule

Closure = Typed arrows + Admissible residuals +
Non-absorbable failures + Identifiable outputs.

Non-absorption:

- Failed calibration $\nrightarrow$ free residual survival.
- Failed empirical branch $\nrightarrow$ failure of screen entropy primitive.
- Successful $M_{3/4}$ evidence $\nrightarrow$ unique microscopic derivation.

No failed bridge, residual, or empirical branch may be reabsorbed by redefining $R_{\mu\nu}^{\text{scr}}$, $R_H(a)$, $A(a, k)$, or the source–kernel factorization.

Scope and Claim Status

This archive presents FDS–G1 as a bridge program, not as a completed first-principles derivation of General Relativity. Each component states its assumptions, identifies which standard physical bridges it uses, and specifies the conditions under which its claims would be weakened, demoted, or falsified.

The core G1 bridge identifies finite causal-screen entropy response as the substrate of gravitational stiffness. D0 strengthens this bridge by reducing the Bekenstein–Hawking coefficient and Einstein-calibration problem to sharper targets: finite-index screen-algebra density and boost modular covariance lemmas. These lemmas remain open; their role is to make the calibration problem precise rather than to claim that it has already been fully

solved.

Memory-output discipline. The current G1DE likelihood tests an output normal form for the horizon response $\mathcal{R}_H(a)$. It does not by itself identify a unique source–kernel factorization $K_H * J_H$. D6 formulates this distinction as a memory quotient and supplies a maximum-caliber horizontal-lift principle for selecting canonical passive realizations.

Output-normal-form discipline. The likelihood tests the output-normalized horizon-response shape $\hat{R}_H(a) \equiv X_b(a)$. It does not identify a unique microscopic factorization $K_H * J_H$. A successful $M_{3/4}$ evidence result supports the output branch, not a unique finite-screen microdynamics. The output shape $\hat{R}_H(a)$ is a production-normalized diagnostic quantity; its redshift dependence is not a uniquely derived microphysical prediction.

Microscopic-realization discipline. D7 supplies a finite-state detailed-balance Markov-screen prototype realizing the D6 passive lift assumptions. It is an existence/realization model, not yet a unique derivation of the physical finite-screen state family or the occupancy functional C_{used} .

The complete series should therefore be read in three layers: (i) the finite-screen entropy-response architecture; (ii) the G1DE dark-sector closure and prediction lock; (iii) the production-refined medium-prior evidence pipeline (8 seeds, $d \log Z = 0.1$), now supplemented by an independent KiDS-1000 shear-only diagnostic stress test with deterministic mock-injection false-positive audit, and with full $3 \times 2\text{pt}$ validation still pending (medium-prior production refinement completed).

Bridge-closure modules integrated in v1.2. The remaining G1 closure work separates into four bridge modules rather than new FDS-Core theorems. First, a finite screen-ledger representation module must specify the operational microstructure of the screen ledger. Second, a causal measurement module must define screen observables, probe couplings, finite record channels, and survey-observable maps. Third, a retarded finite-response kernel module must express screen perturbations through causal, finite-description, Ward/Bianchi-compatible response kernels. Fourth, an early-universe cosmology module must state radiation-era residual bounds, BBN and CMB compatibility conditions, transfer assumptions, and kill tests. Failure of any such module demotes the corresponding G1 bridge claim, not the formal FDS-Core layer.

Demotion paths for the G1 dark-sector branch

Every empirical branch of the framework carries pre-specified consequences of failure. The main demotion paths for the $M_{3/4}$ branch are:

Failure mode	Consequence	Severity
Free- κ decisively beats $\kappa = 3/4$	Isotropic 3/4 lock demoted	High
Constant- Σ decisively beats $R_b^H(a)$ shape	Shape lock demoted; amplitude may survive	Medium
$ \mu - 1 \sim \Sigma - 1 $ required	Ricci-leakage branch fails	High
Independent $A(a, k)$ required	G1DE becomes generic dark-stress model	Critical
Λ CDM or CPL wins production evidence	G1DE observational branch demoted	Critical
Full 3×2 pt does not support Weyl channel	Dark-sector interpretation fails	Critical
Noisy mocks show systematic false positives	Current evidence interpretation demoted	High
CMB/BBN requires unacceptable residual	Early-universe branch demoted	Medium
D0 calibration lemmas provably false	Strong derivation route demoted; bridge program continues	High

A detailed kill-test table with expanded failure modes appears in Section “Suggested Evaluation Protocol” (Table 5).

Series map

Paper	Role
Core	Finite screen entropy response geometry — primitive object, response chain, normal-form dynamics, integrable closure
Companions A–F	Supporting technical modules — screen statistics (A), transition dynamics (B), scalar-capacity EFT (C), likelihood interface (D), horizon occupancy residuals (E), holographic dictionaries (F)
G1DE- $M_{3/4}$	Matched exact likelihoods and homogeneous medium-prior nested evidence for seven late-time cosmology models
D0–D10	G1DE dark-sector closure sequence — modular anomaly inflow and Einstein calibration (D0), finite-access sheaf foundation (D1), algebraic projection (D2), non-equilibrium entropy (D3), unified influence functional (D4), production identifiability rules (D5), memory quotient and maximum-caliber horizontal lift (D6), finite-state Markov-screen realization prototype (D7), metric reconstruction and universal envelope (D8), obstruction exact sequence and no-free-dark-sector classification (D9), G1DE bootstrap and minimal Weyl branch uniqueness (D10)

Current empirical status

FDS–G1 has advanced from a finite-screen response bridge program to a theory-structured, evidence-selected and KiDS shear-only stress-tested production-refined dark-sector branch. This is not final cosmological confirmation; it is a coherent, low-parameter, falsifiable residual branch selected by a completed medium-prior 8-seed production-refined nested-evidence audit and independently stress-tested through KiDS-1000 shear-only Weyl-channel diagnostics.

The leading empirical dark-sector branch is the projection-locked $M_{3/4}$ model,

$$s < 3, \quad \mu(a, k) = 1 \ (\mu_0 = 0), \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\widehat{R}_H(a), \quad \widehat{R}_H(1) = 1,$$

(equivalently $\Sigma_0 = -\frac{3}{4}(3 - s)$ under this normalization), a sparse projection-locked background–Weyl amplitude law rather than a generic modified-growth or flexible dark-energy fit. G1DE-2 serves as a free-response adversarial envelope; G1DE-1 is the growth-only negative control supporting the interpretation that the primary signal is $H + \Sigma$ with $\mu \simeq 1$. G1DE-1 is included as a full member of the eight-seed production-refined comparison. The earlier mixed-provenance raw audit is retained in the provenance appendix and has been superseded for model ranking.

Pending items:

1. Full prior-family stress testing beyond the existing top-control wide-prior sensitivity.

2. Full 3×2 pt lensing likelihoods including galaxy-galaxy lensing and clustering for production shape discrimination. KiDS shear-only diagnostic completed; does not replace 3×2 pt.
3. Noisy-ensemble mock injection and independent pipeline replication.
4. CMB/BBN/nonlinear-structure compatibility where relevant.

Bridge-closure Module Formal Roadmap (v1.2 hardening orientation)

This section provides a high-level orientation to the four theory-hardening modules integrated into the v1.2 body. Full formal definitions appear in the relevant Core, Companion, and D-series sections and are summarized in the unnumbered Theory-Hardening Architecture section below. This integration preserves existing empirical results while upgrading the theoretical body text through explicitly marked claim-type audited modules.

Module orientation. The four modules are:

1. **P4 Finite screen ledger:** Formal operational definition of the finite causal-screen ledger object, integrated in Core §5.
2. **P1 Admissible finite response kernel:** Admissibility rule for retarded, finite-description screen response kernels, integrated in Core §11 and D4.
3. **P2 Causal measurement protocol:** Formal operational definition of screen measurement channels, integrated in Companion B and D10.
4. **P3 Early-universe compatibility:** Kill-test interface for early-universe residual constraints, integrated in Companion E.

Roadmap alignment. The modules are aligned to the existing G1 closure spine, with explicit claim status documented in the Theory-Hardening claim ladder below. No modules are presented as completed derivations or theorem results.

Dependency-ordered open problems (v1.2 update). This updated list extends the living v1.2 roadmap, ordered from microscopic to observational:

1. **Finite screen ledger microstructure:** define full correlation geometry, screen algebra, and Markov realization chart for the finite ledger object
2. **Causal measurement protocol:** finalize operational definitions of screen observables, probe coupling, record stability, and survey mapping
3. **Retarded response kernel:** derive complete finite-description/passive/Ward-compatible admissible kernel class

4. **Early-universe compatibility:** formalize CMB/BBN/radiation-era transfer and kill-test interfaces
5. **Coefficient problem:** resolve D0 finite-index density lemma and boost modular covariance lemma
6. **Production empirical validation:** complete full 3×2 pt analysis, noisy mock ensembles, independent pipeline replication, and CMB/BBN/nonlinear compatibility testing

Theory-Hardening Architecture

Prior-art anchors. The following hardening modules adapt mathematical and methodological tools from adjacent literatures: relative-entropy gravity and entropic action principles (Bianconi 2024; Nusbaumer 2026; Ahmad and Klinger 2025), local causal-diamond thermodynamics (Jacobson 1995; Barzi et al. 2026), finite-capacity horizon thermodynamics (Antiba 2026), finite causal-diamond phase spaces and modular/soft edge modes (Ciambelli, He, and Zurek 2026), QNEC-type entropy inequalities (Bousso 1999), and mainstream modified-gravity μ/Σ production parameterizations (Bellini and Sawicki 2014; DESI Collaboration 2025). These references supply hardening templates, not completed G1 derivations. The G1-specific primitive remains the finite causal-screen ledger and its entropy-response geometry.

H1. Relative-Entropy Variational Closure

Claim type: Conditional derivation route

The G1 finite-screen calibration problem can be framed as a constrained relative-entropy variational principle that unifies the D0 finite-index density, boost modular covariance, and D8 metric envelope constraints:

$$\mathcal{I}_{G1} = D_{\text{rel}}[\rho_{\text{scr}} \parallel \rho_{\text{env}}] + \lambda_{\text{cap}} C_{\text{cap}} + \lambda_{\text{Ward}} C_{\text{Ward}} + \lambda_{\text{q}} C_{\text{quotient}}$$

where C_{cap} enforces finite capacity constraints, C_{Ward} enforces Ward/Bianchi consistency, and C_{quotient} enforces gauge/quotient invariance. This variational target provides a unifying formal framework for future resolution of the D0 open calibration lemmas (Bianconi 2024; Nusbaumer 2026; Ahmad and Klinger 2025).

Boundary: This is a conditional derivation route, not a completed Einstein gravity derivation.

—

H2. Local Causal-Diamond First-Law Protocol

Claim type: Conditional theorem target

The D0 finite-screen calibration route can be formalized using a local causal-diamond first-law template, with thermodynamic objects:

$$\{\delta S_{\text{scr}}, \delta A, \delta K_{\text{mod}}, \delta Q_{\text{matter}}\}$$

The finite causal-diamond phase-space literature now provides an additional structural anchor for treating causal-diamond area/radial fluctuations and their conjugate variables as genuine finite-region gravitational phase-space data (Ciambelli, He, and Zurek 2026).

The equilibrium limit of this identity provides a conditional derivation route to the Einstein/Weyl-transverse sector of general relativity (Jacobson 1995; Barzi et al. 2026):

$$\delta S_{\text{scr}} = \frac{\delta Q}{T} \implies \text{Einstein/Weyl-transverse equilibrium sector.}$$

Boundary: This is a proof template, not a completed full general relativity recovery derivation.

H3. Modular Screen Phase Space and Optical Susceptibility

Claim type: Conditional theorem target / open hardening lemma

The D2 3/4 isotropic optical projection result can be framed as a modular susceptibility ratio for the screen optical phase space $\{\Gamma_{\Sigma}, \Omega_{\Sigma}, H_{\Sigma}, K_{\Sigma}\}$. The optical algebra decomposes as

$$T_I GL(2) = A^{(1)} \oplus S^{(2)} \oplus \Omega^{(1)},$$

where the symmetric shear sector S carries multiplicity two. The normalized unimodular response coefficient is therefore

$$\kappa = \frac{2\chi_S + \chi_{\Omega}}{\chi_A + 2\chi_S + \chi_{\Omega}} \stackrel{\text{isotropic}}{=} \frac{3}{4},$$

where $\chi_A, \chi_S, \chi_{\Omega}$ are the sector susceptibilities. At the isotropic entropy-Hessian fixed point $\chi_A = \chi_S = \chi_{\Omega}$, giving $\kappa = (2 + 1)/(1 + 2 + 1) = 3/4$.

This phase-space target is aligned with recent work identifying the soft sector of asymptotically flat gravity with the phase space of a finite causal diamond, where the causal-diamond radial fluctuation and its symplectic partner play the role of finite-region edge degrees of freedom (Ciambelli, He, and Zurek 2026).

Boundary: This is a conditional theorem target, not a completed modular derivation of the 3/4 projection coefficient.

H4. Modular Observable Origin of the Horizon-Response Shape

Claim type: Realization target

The standard output-normal-form definition of the horizon response $R_H(a) = R_0 R_b^H(a)$ (used for all likelihood and diagnostic tests) can be connected to a microscopic modular fluctuation target:

$$R_H(a) \sim \frac{\delta \langle K_{\Sigma(a)} \rangle}{\delta \langle K_{\Sigma(1)} \rangle} \quad \text{or} \quad R_H(a) \sim \frac{\text{Var}(K_{\Sigma(a)})}{\text{Var}(K_{\Sigma(1)})}$$

where K_{Σ} is the screen modular Hamiltonian. This provides a formal realization route for the output shape without resolving the source-kernel factorization ambiguity (Song 2025).

Boundary: This does not replace the existing output-normal-form likelihood definition, and does not claim to identify a unique source-kernel factorization.

H5. Screen-QNEC-Type Residual Admissibility

Claim type: Conjectural admissibility filter

Candidate G1 screen residuals in the hardening class are required to pass a proposed entropy-convexity and null-positivity filter, defining an admissible class:

$$R_{\mu\nu}^{\text{scr}} k^{\mu} k^{\nu} \in \mathcal{A}_{\text{Ward}} \cap \mathcal{A}_{\text{quotient}} \cap \mathcal{A}_{\text{entropy convexity}}$$

Admissible residuals must satisfy a null entropy-variation inequality analogous to the Quantum Null Energy Condition (QNEC), acting as a filter for valid residual constructions (Bousso 1999).

Boundary: This is a conjectural filter for residual construction, not a proved inequality satisfied by all G1 residuals.

H6. Mainstream Modified-Gravity Production Interface

Claim type: Empirical production protocol

The G1 $M_{3/4}$ residual is mapped to the standard modified-gravity μ/Σ coordinate system for direct comparison to mainstream literature and future survey constraints:

$$\mu(a, k) = 1 + \epsilon_{\mu}(a, k), \quad \Sigma(a, k) = 1 - \frac{3}{4}(3 - s)R_b^H(a) + \epsilon_{\Sigma}(a, k)$$

where leading suppressed Ricci leakage is:

$$\epsilon_{\mu} = O(R_H^2) + O(R_H H^2 / k^2).$$

Production comparisons must evaluate $M_{3/4}$ against $\mu\Sigma$ -parametric MG, $w_0 w_a$, Horndeski/EFT, CPL, Λ CDM, free- κ , constant- Σ , binned- Σ , and G1DE-2 controls (Bellini and Sawicki 2014; DESI Collaboration 2025).

Boundary: This is an empirical comparison protocol, not production evidence.

v1.2 Body-Integrated Theory-Hardening Claim Ladder

Hardening item	Claim type	v1.2 status	Strengthens	Forbidden reading
Finite ledger object	Definition	integrated	Core §5 / D1 / D7	unique microscopic ontology
Measurement protocol	Definition/protocol	integrated	B / D / D10	completed survey theorem
Retarded kernel	admissibility rule	integrated	Core §11 / D4 / D6	derived fundamental kernel
Relative entropy action	derivation route	open	D0/D8	completed Einstein derivation
Causal diamond first law	theorem target	open	D0	full GR derivation
3/4 modular susceptibility	theorem target	open	D2/D3	modular theorem target
$R_b^H(a)$ modular observable	realization target	open	D6/D7	unique source-kernel factorization
Screen-QNEC filter	conjectural admissibility	open	D9	proved QNEC theorem
μ/Σ interface	empirical protocol	pending	D10/G1fit	production evidence
Finite causal-diamond soft/edge phase-space anchor	prior-art alignment	external support	H2/H3/D1/D3 finite-region phase-space bridge	proof of FDS-G1; proof of M3/4; completed microscopic derivation

Table 1: Audited theory-hardening claim boundary table. All items are explicitly categorized to avoid misinterpretation.

Existing Evidence vs Future Prediction Lock

The current evidence status and the future prediction lock are deliberately separated. The existing-data result is now a completed medium-prior 8-seed production-refined nested-evidence audit for the stage-2d exact likelihood: the canonical hierarchy is

$$M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE-2} > \text{G1DE-1} > \text{CPL} > \Lambda\text{CDM},$$

with all models sharing the same medium-prior configuration and eight independent seeds. It supports the claim that $M_{3/4}$ is the leading tested G1 dark-sector branch among the current control set, with $\Delta \log Z(M_{3/4}, M_\kappa) = 0.856$, $\Delta \log Z(M_{3/4}, \text{const-}\Sigma) = 1.775$, $\Delta \log Z(M_{3/4}, \text{G1DE-1}) = 7.402$, and $\Delta \log Z(M_{3/4}, \Lambda\text{CDM}) = 11.884$. It is not yet a final cosmological discovery, a production survey constraint, or a proof of the microscopic finite-screen realization. Medium-prior production refinement with $d \log Z = 0.1$ and eight seeds is complete. Wider prior-family stress, independent replication, and full $3 \times 2\text{pt}$ validation remain pending.

The future prediction lock is sharper: future weak-lensing, E_G , RSD/full-shape, and $3 \times 2\text{pt}$ analyses must select the specific G1 channel pattern

$$s < 3, \quad \mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)R_{bH}(a), \quad R_{bH}(1) = 1,$$

without allowing a free $A(a, k)$, a retuned κ , or independent redshift kernels.

G1DE-1 Negative-Control Audit

The growth-only negative-control model G1DE-1 has been run as a nested-evidence control under the same medium-prior configuration as the other six models. In the production-

Layer	Current status	What it does not prove	Next locked test
Existing-data production-refined audit	$M_{3/4}$ leads the production-refined medium-prior eight-seed audit hierarchy and survives top-control wide-prior sensitivity	final cosmological discovery; finite-index screen algebra; boost modular covariance; unique source-kernel realization	full wide/physical prior-family stress; independent replication; full 3×2 pt validation
Future prediction lock	channel pattern fixed: $s < 3$, $\mu = 1$, $\Sigma - 1 = -\frac{3}{4}(3 - s)R_{bH}$	no post-hoc retuning of κ , $A(a, k)$, or independent μ, Σ shapes	KiDS shear-only diagnostic passed; next locked tests: full 3×2 pt validation, wide/physical prior-family stress, RSD/full-shape Ricci-leakage tests, and independent replication

Table 2: Separation between the current production-refined evidence result and the future ex-ante prediction lock.

refined configuration (eight seeds, $d \log Z = 0.1$) it gives

$$\log Z_{\text{G1DE1}} = -901.751 \pm 0.180, \quad \Delta \log Z(M_{3/4}, \text{G1DE1}) = 7.402, \quad B \simeq 1.65 \times 10^3.$$

Thus the growth-only negative control is decisively below the projection-locked background-Weyl branch in the medium-prior production refinement. This supports the interpretation that the present signal is not adequately explained by a hidden growth-only residual. The medium-prior 8-seed production-refined audit is complete.

Suggested Evaluation Protocol

The following protocol is suggested for evaluating the framework on its stated logical and empirical terms. It does not immunize any claim from mathematical or empirical refutation. Its purpose is to separate technical assessment from sociological dismissal, to identify which assumptions or bridges are actually being tested, and to state the failure conditions under which specific branches should be demoted.

Before listing the evaluation principles, we fix two reference objects: the current production-refined evidence baseline and the no-hidden-fit checklist for the locked branch.

Table 3: Reference production-refined evidence baseline for the evaluation protocol: seven-model medium-prior nested-evidence audit using $d \log Z = 0.1$ and eight independent seeds per model. All models share the same medium-prior configuration. $\Delta \log Z_i = \log Z_{M_{3/4}}^{\text{mean}} - \log Z_i^{\text{mean}}$. The seed scatter column reports the standard deviation across eight independent runs. Full prior-family stress, independent replication, and full $3 \times 2\text{pt}$ validation remain pending.

Model	ndim	seeds	$\log Z_{\text{mean}}$	seed scatter	$\Delta \log Z$	$B_{M_{3/4},i}$
G1DE- $M_{3/4}$	4	8	-894.349	0.130	0.000	1
G1DE- M_κ	5	8	-895.205	0.129	0.856	2.35
G1DE-const- Σ	5	8	-896.124	0.069	1.775	5.90
G1DE-2	6	8	-900.872	0.191	6.523	6.8×10^2
G1DE-1	5	8	-901.751	0.180	7.402	1.65×10^3
CPL	5	8	-903.952	0.105	9.603	1.5×10^4
ΛCDM	3	8	-906.233	0.100	11.884	1.45×10^5

Five of the seven completed production blocks ($M_{3/4}$, M_κ , const- Σ , CPL, and ΛCDM) satisfy the strict seed-scatter target $\sigma_{\log Z} < 0.15$. G1DE-1 and G1DE-2 show slightly larger scatter, 0.180 and 0.191, respectively, but their evidence gaps from $M_{3/4}$ are $\Delta \log Z = 7.402$ and 6.523, so the production ranking is unaffected. This is a production-refined medium-prior pass, not a final cosmological discovery, not a full survey-grade constraint, and not a completed full $3 \times 2\text{pt}$ validation.

No Hidden Fit / No Circularity Checklist

The projection-locked branch is not defined by a hidden free response function. The following quantities are fixed or constrained before evidence comparison:

$$R_H(a) = R_0 R_{bH}(a), \quad R_0 = 3 - s, \quad R_{bH}(1) = 1,$$

$$\kappa = \frac{3}{4}, \quad \mu(a, k) = 1, \quad A(a, k) \notin \Theta_{\text{G1DE}}.$$

The model does not fit a separate optical amplitude $A(a, k)$, does not fit an independent source J_H and kernel K_H , and does not allow independent redshift functions for $\mu(a, k)$ and $\Sigma(a, k)$. The adversarial controls are M_κ , constant- Σ , binned- $\Sigma(z)$ (Weyl-shape diagnostic, not a G1DE branch), and G1DE-2. If these controls decisively outperform $M_{3/4}$, the relevant lock is demoted.

The production-refined evidence update strengthens the statistical audit of this checklist but does not alter the locked model definition: $R_H(a)$, $\kappa = 3/4$, $\mu = 1$, and the no-free- $A(a, k)$ rule are unchanged.

Quantity	Status in $M_{3/4}$	Failure consequence
$R_0 = 3 - s$	amplitude tied to background filling slope	free-amplitude branch required
$R_{bH}(1) = 1$	normalization fixed	identifiability fails
$\kappa = 3/4$	fixed isotropic optical projection	demote to M_κ if free- κ wins
$\mu(a, k) = 1$	leading-order Ward-suppressed Ricci leakage	demote if $ \mu - 1 \sim \Sigma - 1 $
$A(a, k)$	not a fitted parameter	generic dark-stress model if required
$K_H * J_H$	not uniquely identified by likelihood	D6/D7 realization problem, not evidence claim

Table 4: Reference no-hidden-fit checklist for the evaluation protocol and for the M3/4 branch.

Non-absorption rule for $M_{3/4}$. If the projection-locked branch fails, it may not be rescued by freely redefining $R_b^H(a)$, introducing an independent $A(a, k)$, retuning κ after the fact, or separating the μ and Σ kernels. Such changes define demoted successor branches, not the original $M_{3/4}$ model. Conversely, success of $M_{3/4}$ would support the output normal form; it would not by itself prove a unique microscopic finite-screen source-kernel factorization. Nor would it validate the D0 calibration lemmas, the full D8 metric reconstruction, or the uniqueness of the D10 bootstrap.

Amplitude Lock vs Shape Lock

The current empirical support separates into two levels. The stronger level is the background–Weyl amplitude lock: $s < 3$, $\mu \simeq 1$, and $\Sigma_0 = -\frac{3}{4}(3 - s)$. This amplitude-level lock is supported by the production-refined evidence hierarchy and independently by KiDS shear-only free- κ relaxation ($\kappa \approx 0.746$). The more demanding level is the redshift-shape lock $\Sigma(a) - 1 \propto R_b^H(a)$. The KiDS shear-only stress test and free- κ relaxation primarily support the amplitude-level projection lock. Full discrimination of the redshift-shape law requires full 3×2 pt lensing likelihoods, binned- $\Sigma(z)$ controls, noisy mock ensembles, and independent replication. The amplitude lock can survive even if the redshift-kernel shape is unresolved.

Lock level	Fixed quantities	Current support	Failure condition
Amplitude lock	$s < 3$, $\mu \simeq 1$, $\Sigma_0 = -\frac{3}{4}(3 - s)$	Production evidence hierarchy; KiDS free- κ relaxation ($\kappa \approx 0.746$)	Free- κ decisively beats 3/4 in production
Shape (redshift-kernel) lock	$\Sigma(a) - 1 \propto \widehat{R}_H(a)$	Preliminary, nuisance-entangled	Constant- Σ or binned- $\Sigma(z)$ robustly beats $\widehat{R}_H(a)$ shape

This distinction is essential: the amplitude lock is empirically stronger and can survive even if the redshift-kernel shape is later demoted.

Paradigm-level candidate status. FDS-G1 is not presented as an incremental parameterization within an already fixed model space. It proposes a shift of primitive objects: from metric/field-first descriptions toward finite causal-screen ledgers, entropy response, integrable geometry, and residual gluing defects. In this structural sense, it should be evaluated as a paradigm-level candidate: by logical closure, explanatory compression, empirical risk, falsifiability, and prediction lock, rather than by immediate familiarity or consensus alone. This classification is methodological, not evidential; it does not imply that the framework is correct, only that the appropriate evaluation standard is not the same as for a small phenomenological extension.

Historical cases such as heliocentrism, Darwinian evolution, and Wegener’s continental-drift hypothesis illustrate a narrow methodological point: theories that change primitives can be socially unfamiliar before they are technically resolved. The analogy is methodological, not evidential. Historical resistance does not validate a new theory, but it does show why consensus cannot substitute for technical evaluation.

1. Consensus is not a premise. The validity of a derivation within this framework depends on its internal logical rigor, its bridge assumptions, and its empirical predictions, not on its current level of social acceptance within any academic community. Established empirical constraints and proven mathematical theorems remain binding. If an FDS-G1 claim contradicts a well-established experimental result or a proven theorem under the same assumptions, that contradiction counts as evidence against the claim. The methodological point is only that consensus is not itself a premise.

2. Objections should identify the failed premise. Where this framework derives a limit, prediction, or closure condition from finite distinguishability, causal-screen access, entropy response, Ward/Bianchi closure, Weyl normalization, finite-access modular calibration, or a specified likelihood implementation, an objection should identify which premise, bridge, normalization, or data step fails. A generic objection of the form “a future model may explain this differently” is not a direct rebuttal unless it specifies the alternative mechanism and the premise it replaces.

3. Ex-ante prediction locks are not post-hoc fits. If a model identity, prediction equation, control hierarchy, and falsification ladder were publicly timestamped before the relevant future data, the result should not be dismissed merely as post-hoc curve fitting. The correct evaluation is whether the prediction was actually locked, whether the likelihood implementation matches the locked model, whether adversarial controls were included, and whether the result survives prior and pipeline stress tests. This does not imply that the derivation is correct; it only means that the post-hoc-fit objection is not the appropriate criticism.

4. Flexible fitting is not explanatory compression. A model is not scientifically strengthened merely by fitting data with additional freedom. Evaluation should distinguish explanatory compression from flexible response fitting. For FDS-G1, the relevant question

is whether the projection-locked branch

$$\mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a)$$

outperforms or remains competitive against free- κ , constant- Σ , free-response G1DE-2, G1DE-1, CPL, and Λ CDM controls under matched likelihood and prior stress.

5. Scientific claims require failure conditions. Every empirical branch of the framework must carry a pre-specified consequence of failure. For the G1 dark-sector branch, the main demotion paths are:

1. if free- κ decisively beats $M_{3/4}$, the exact 3/4 isotropic-compliance lock is demoted;
2. if constant- Σ decisively beats the output-response-shape model, the $\hat{R}_H(a)$ shape is demoted while the Weyl-amplitude branch may survive;
3. if data require $|\mu - 1| \sim |\Sigma - 1|$, the Ward-suppressed Ricci-leakage branch fails;
4. if free $A(a, k)$ is required, the model becomes a generic dark-stress source rather than G1DE;
5. if production evidence loses to CPL or Λ CDM, the G1DE observational branch is demoted;
6. if full 3×2 pt lensing data do not support the Weyl-channel signal, the current dark-sector interpretation fails or must be downgraded;
7. if deterministic or noisy mock injection shows that Λ CDM, constant- Σ , or binned- Σ truths are systematically misclassified as $M_{3/4}$, the KiDS shear-only interpretation is demoted.

A claim with no possible demotion, observational penalty, or failure condition is not an empirical scientific claim in this framework. It may retain heuristic, metaphysical, or programmatic value, but not confirmed scientific status.

6. Evidence is layer-specific. Existing-data evidence for $M_{3/4}$ supports the tested output normal form, not the full microscopic derivation. In particular, evidence for

$$R_H(a) = R_0 R_{bH}(a)$$

does not prove a unique source–kernel factorization $K_H * J_H$, finite-index screen algebras, or boost modular covariance. Conversely, failure of the data-facing residual branch does not by itself falsify the screen-entropy representation theorem or the finite-screen response architecture.

Table 5: What would falsify or demote $M_{3/4}$ and the G1 dark-sector branch. Severity: Critical = branch-terminating; High = major demotion; Medium = partial demotion.

Failure mode	Consequence	Severity
Free- κ decisively beats $\kappa = 3/4$	Exact isotropic 3/4 lock demoted	High
Constant- Σ decisively beats $R_b^H(a)$ shape	Shape lock demoted; amplitude branch may survive	Medium
Binned- $\Sigma(z)$ robustly beats locked shape	Redshift-kernel lock demoted	Medium
$ \mu - 1 \sim \Sigma - 1 $ required by RSD/full-shape	Ward-suppressed Ricci-leakage branch fails	High
Independent $A(a, k)$ required	G1DE becomes generic dark-stress model	Critical
Λ CDM or CPL decisively wins production evidence	G1DE observational branch demoted	Critical
Full 3×2 pt does not support Weyl channel	Current dark-sector interpretation fails or downgrades	Critical
Noisy mocks classify Λ CDM/const- Σ /binned- Σ truths as $M_{3/4}$	Current evidence interpretation demoted	High
CMB/BBN requires unacceptable high-redshift residual occupancy	Early-universe branch demoted	Medium
Independent replication fails	Production evidence status withdrawn	Critical
D0 calibration lemmas provably false	Strong derivation route demoted; bridge program continues at reduced calibration claim level	High

Scope Clarifications and Anticipated Objections

- 1. Is FDS–G1 a replacement for GR, or a residual extension of GR?** The current framework is a finite-screen response framework with GR-like equilibrium limits. G1DE is a late-time residual branch, not a full replacement for numerical relativity or strong-field GR production systems.
- 2. If GR is recovered as an equilibrium branch, does G1 automatically inherit all GR successes?** No. The framework has a standard-limit structure, but the microscopic derivation and production-level implementation are not yet complete. Inheriting a limit is not equivalent to completing a global proof.
- 3. Is Λ CDM contained as a limit, or merely fitted as a benchmark?** Λ CDM appears as the $s = 3$, vanishing-residual consistency limit of the G1DE hierarchy. This does not imply that all CMB, BBN, and full-shape constraints have been verified at production level.
- 4. Is the evidence result driven by prior volume?** Top-control wide-prior sensitivity has been completed and the locked ranking is stable. Full wide-prior baselines for CPL, G1DE-2, and Λ CDM remain recommended as robustness checks.
- 5. Is the signal really $H + \Sigma$, or is it hidden growth modification?** The current interpretation is a primary signal in the background and Weyl channel with $\mu \simeq 1$. The G1DE-1 growth-only negative control and the $M_{3/4}$ locked branch are designed precisely to distinguish $H + \Sigma$ from growth-only alternatives.
- 6. Why is $\mu(a, k) = 1$ not a weakness?** Because in the locked branch $\mu = 1$ is not a hidden fit assumption; it is the leading-order Ward-suppressed Ricci-leakage limit of a Weyl-normalized optical residual. D5 gives

$$\Sigma(a, k) - 1 = -\kappa R_H(a) + O(R_H^2),$$
 while

$$\mu(a, k) - 1 = O(R_H^2) + O\left(R_H \frac{H^2}{k^2}\right).$$
 On linear subhorizon survey scales, the leading-order branch is therefore

$$\mu(a, k) = 1.$$
 If full-shape, RSD, or 3×2 pt data require $|\mu - 1| \sim |\Sigma - 1|$, the Ward-suppressed Ricci-leakage branch is demoted.

7. Why does 3/4 matter? It is not an arbitrary numerical fit parameter, but it is also not claimed as an unconditional geometric constant. It is the isotropic full-optical-algebra branch of the projection closure:

$$T_1GL(2) = A^{(1)} \oplus S^{(2)} \oplus \Omega^{(1)}, \quad \kappa = \frac{\dim(S \oplus \Omega)}{\dim(A \oplus S \oplus \Omega)} = \frac{3}{4}.$$

If optical compliance is anisotropic, the weighted branch is

$$\kappa = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega}.$$

If free- κ decisively beats $M_{3/4}$, the exact isotropic lock is demoted. If a reduced optical branch removes twist/holonomy, a 2/3-type branch becomes a control rather than a contradiction of the full-algebra result.

8. Does the compressed E_G dataset make the lensing result too weak? Yes for the SN+BAO+RSD+ E_G evidence block: compressed E_G alone is not a production weak-lensing constraint. However, v1.1 adds an independent KiDS-1000 shear-only CLASS/covariance stress test. This supports the Weyl-channel signal and the 3/4 projection lock under $m_i + \Delta z_i + A_{IA}$ nuisance profiling. It remains diagnostic: full 3×2 pt data are still required for production shape discrimination.

9. What if constant- Σ or binned- $\Sigma(z)$ beats the output-response-shape model in future data? That would demote the $\hat{R}_H(a)$ output-shape functionality but not necessarily kill the Weyl-amplitude branch. In KiDS shear-only, binned- $\Sigma(z)$ finds lower raw- χ^2 basins, but these are nuisance-entangled and boundary-sensitive, so they stress but do not cleanly demote the shape lock. The amplitude lock can survive even if the redshift-kernel shape is unresolved.

9b. What is the difference between amplitude lock and shape lock? The current evidence is strongest at the amplitude level: a suppressed Weyl response with $\Sigma_0 < 0$ and an optical projection coefficient near $\kappa = 3/4$. The redshift-shape lock

$$\Sigma(a) - 1 \propto R_{bH}(a)$$

is more demanding. The KiDS shear-only stress test strengthens the amplitude-level lock: free- κ relaxation returns $\kappa \approx 0.746$, and the Weyl signal survives nuisance profiling. The shape lock remains under controlled stress: binned- $\Sigma(z)$ controls find redshift-shape information that is present but nuisance-entangled. If constant- Σ or binned- $\Sigma(z)$ robustly wins in future full 3×2 pt analyses, the $R_{bH}(a)$ shape lock is demoted, but the Weyl-amplitude branch may survive.

10. What if a free $A(a, k)$ function is required? The model ceases to be G1DE and becomes a generic dark-stress model. The Weyl-normalized production convention explicitly forbids sampling $A(a, k)$ as an independent function.

11. Does G1 explain the cosmological constant problem? No. It reframes late-time residuals as finite-screen modular/gluing defects, but the microscopic vacuum-energy problem and the full calibration of the zero-point remain separate open tasks.

12. What about CMB and BBN? The current branch must remain compatible with early-universe constraints. If high-redshift residual occupancy is too large, the branch is demoted. Full CMB/BBN production validation is not yet included.

13. What about nonlinear structure formation and baryonic feedback? The current G1DE test is a linear, optical, output-level branch. Nonlinear and baryonic production pipelines are future work and are not claimed to be replaced at this stage.

14. What about black holes and gravitational waves? General Relativity remains the production theory for current strong-field and wave-form-level tests. FDS-G1 provides a finite-screen interpretation and possible residual/memory directions at the conceptual level, not yet waveform-level predictions.

15. Is the framework overbuilt? The number of technical notes should not be judged by volume alone. Prediction compression is the relevant metric. The key empirical claim ultimately reduces to $s < 3$, $\mu = 1$, and $\Sigma - 1 = -\frac{3}{4}(3 - s)\hat{R}_H$.

16. Is this just thermodynamic gravity / Jacobson / Verlinde under a new name? No. It uses those bodies of work as prior bridges, but adds the finite-screen entropy-response chain, the G1DE background-Weyl residual, the 3/4 projection lock, the D0 calibration route, and the evidence/failure hierarchy.

17. Is D0 a completed derivation of Einstein gravity? No. D0 is a calibration-reduction route: it compresses the Einstein-calibration problem into two sharper lemmas (finite-index screen-algebra density and boost modular covariance). Those lemmas remain open.

18. Does “paradigm-level candidate” mean “confirmed paradigm”? No. It is a structural classification, not a triumphal declaration of victory. Candidate status arises from primitive shift, prediction lock, falsifiability, and evidence interface. Confirmed status requires future data, replication, and production closure.

19. What would make G1 stronger than Λ CDM in the relevant domain? Production-level evidence, full 3×2 pt confirmation of the Weyl-channel signal, retained $\kappa \simeq 3/4$, low false-positive rate in noisy mock ensembles, and independent pipeline replication.

20. What would kill the dark-sector branch? A combination of: CPL or Λ CDM decisively beating the G1DE branch in production evidence; full 3×2 pt data not supporting the Weyl-channel signal; free-response G1DE-2 or free $A(a, k)$ models being required; the μ channel strongly deviating from one; full wide-prior sensitivity collapsing the ranking; or systematic $M_{3/4}$ false positives in mock-injection ensembles.

Does production breadth decide the theory’s status? No. Production breadth is an implementation-readiness dimension, not a truth criterion. GR+ Λ CDM currently dominates strong-field waveforms, nonlinear simulations, CMB/BBN pipelines, and community infrastructure. FDS–G1 does not yet replace that production apparatus. The theoretical comparison concerns different dimensions: primitive depth, bridge assumptions, prediction lock, explanatory compression, and falsifiability.

Limitations

The current framework has seven major open limitations.

1. **Microscopic ledger:** The finite causal-screen ledger is not uniquely derived. D7 provides a finite-state Markov-screen existence prototype, but the physical microdynamics of the screen ledger remain unspecified at the microscopic level.
2. **Strong calibration:** The D0 finite-index screen-algebra density lemma and boost modular covariance lemma remain open. The Einstein-calibration problem is therefore reduced to sharper targets but not completed.
3. **3/4 coefficient:** The 3/4 projection law is a conditional isotropic optical closure result and modular-response target. It is not yet a fully derived microscopic universality theorem, and anisotropic demotion paths (2/3 scalar-shear branch) remain logically open.
4. **Source–kernel factorization:** The likelihood tests the output shape $R_b^H(a)$, not a unique microscopic factorization $K_H * J_H$. A successful evidence result supports the output branch but does not identify the unique source–kernel lift.
5. **Survey-grade validation:** Current KiDS support is diagnostic (shear-only and Band-Power EE+nE). Full 3×2 pt lensing likelihoods, noisy mock ensembles, independent pipeline replication, CMB/BBN compatibility, and nonlinear-structure tests remain required for production confirmation.
6. **Production survey interface:** The current μ/Σ mapping is a protocol definition, not a validated production pipeline. Full integration with DESI/Euclid/LSST pipelines and mock surveys remains future work.
7. **Prior sensitivity:** The medium-prior configuration is the current baseline. Full wide/physical prior-family stress testing for all seven models at production grade is not yet complete.

Scope summary: FDS–G1 is strongest as a finite-screen entropy-response architecture with a locked falsifiable residual branch. It is weakest at the microscopic ledger level (D0 lemmas open), the source–kernel uniqueness level (D6/D7 realizations not unique), and the production validation level (full $3 \times 2\text{pt}$, CMB, nonlinear pending).

Near-Term Falsifiable Prediction Lock

FDS–G1 is not only a structural framework; its current leading empirical branch makes a short-horizon falsifiable prediction. DESI DR2 cosmology products are already available, and Euclid DR1 is expected in October 2026. The projection-locked $M_{3/4}$ branch therefore faces near-term tests in the background, Weyl/lensing, and growth channels.

The locked prediction is

$$s < 3, \quad \mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a), \quad \hat{R}_H(1) = 1.$$

The key near-term test is not whether future data prefer “some” evolving dark energy, but whether they select the specific G1 pattern: background deviation plus suppressed Weyl response, with near-GR growth and $\kappa = 3/4$.

The relevant tests are channel-specific. They do not merely ask whether the model can improve a fit; they ask whether the improvement appears in the predicted channels and fails in the predicted ways.

Minimal Kill Tests for the $M_{3/4}$ Branch

The shortest decisive tests of the current G1DE branch are not broad philosophical tests. They are three concrete empirical kill tests:

Near-term prediction channels and demotion criteria for the G1 $M_{3/4}$ branch.

Prediction channel	Support for $M_{3/4}$	Demotion / failure condition
Background expansion	DESI/BAO/SN continue favoring $s < 3$ over $s = 3$.	Production evidence returns to Λ CDM or CPL dominance.
Weyl / lensing response	KiDS shear-only supports $M_{3/4}$ vs Λ CDM under nuisance. Full 3×2 pt needed for production.	3×2 pt does not support Weyl-channel signal.
Growth response	$\mu(a, k) \simeq 1$, maintaining near-GR matter growth.	Data require $ \mu - 1 \sim \Sigma - 1 $, indicating unsuppressed Ricci leakage.
Projection lock	Free- κ KiDS returns $\kappa \approx 0.746$; supports $3/4$ lock.	Free- κ decisively beats the $M_{3/4}$ exact lock in production.
Output-response shape	KiDS binned- $\Sigma(z)$ finds shape info but nuisance-entangled.	Binned- Σ or constant- Σ robustly beats $R_{bH}(a)$ in full 3×2 pt.
False-positive risk	Deterministic mock injection passed.	Noisy ensemble shows systematic $M_{3/4}$ false positives.
Model identity	No free $A(a, k)$ amplitude is required.	Free $A(a, k)$ is required, reducing G1DE to a generic dark-stress model.

Kill test	Required analysis	Demotion consequence
Nested-evidence kill test	medium-prior production-refined nested sampling completed; remaining kill-test extensions are wide/physical-prior families, independent reruns, and full 3×2 pt shape discrimination	G1DE observational branch demoted if CPL/ Λ CDM or free-response controls decisively win
Weyl-shape kill test	full 3×2 pt comparison of $R_{bH}(a)$ against constant- Σ and binned- $\Sigma(z)$	shape lock demoted if constant- Σ decisively wins; amplitude branch may survive
Ricci-leakage kill test	RSD/full-shape/growth tests of $\mu(a, k) = 1$ against leading-order $\mu \neq 1$ alternatives	Ward-suppressed Ricci-leakage branch fails if $ \mu - 1 \sim \Sigma - 1 $ is required
False-positive kill test	deterministic + noisy mock injection under Λ CDM, $M_{3/4}$, const- Σ , binned- Σ truths	branch interpretation demoted if pipeline systematically misclassifies non- $M_{3/4}$ truths as $M_{3/4}$

Table 6: Minimal empirical kill tests for the current $M_{3/4}$ dark-sector branch.

Comparative Structural Positioning

The purpose of this matrix is not to assign a total score. It separates dimensions that are often conflated: primitive layer, bridge assumptions, prediction lock, flexibility, empirical maturity, and known stress. All frameworks require primitives, calibration assumptions, and observational interfaces; the relevant question is whether those bridges are established, open, flexible, or prediction-locking.

Framework	Primitive layer	Bridge / calibration assumptions	Prediction lock	Flexibility risk	Current maturity / stress
GR+ Λ CDM	Metric geometry, stress-energy, constant Λ , cold dark matter	Equivalence principle, smooth manifold, Einstein-Hilbert action, conservation, initial conditions, astrophysical/baryonic modeling, unexplained Λ	High as production baseline; low for the origin of Λ	Low baseline freedom, but unexplained dark sectors	Established full-stack baseline; stressed by Λ problem and late-time tensions
CPL / $w_0 w_a$	Effective dark-energy fluid	Smooth equation-of-state expansion plus perturbation/closure priors	Low–medium	Moderate phenomenological flexibility	Useful benchmark; weak mechanism
Horndeski / EFT of DE	Scalar-tensor / EFT operator basis	Field content, stability, screening, local tests, GW constraints, EFT truncation	Usually low unless restricted	High without strong priors	Mature formalism; often flexibility-limited
$f(R)$, $f(Q)$, related MG	Modified geometric action or connection sector	Chosen function, screening, local tests, growth consistency, GW propagation	Medium if function fixed	Medium–high	Strong literature; constrained by local/growth/GW tests
Thermodynamic / emergent gravity	Entropy, horizon thermodynamics, information principles	Horizon entropy, local first law, Raychaudhuri bridge, metric/Ward closure	Often low unless residual law added	Model-dependent	Conceptually adjacent; usually lacks locked data-facing residual
Quantum-spacetime programs	String/AdS-CFT/LQG/causal-set primitives	Compactification, holographic dictionary, semiclassical/continuum recovery	Low near-term dark-sector lock	Varies; often model-multiplicity or long bridge	High ambition; long empirical path for late-time cosmology
FDS-G1 / $M_{3/4}$	Finite causal-screen ledger, entropy response, integrable geometry, gluing defects	Finite distinguishability, causal-screen localization, entropy ledger, stiffness bridge, D0 modular calibration lemmas, Ward/Bianchi closure, Weyl normalization, optical projection, likelihood implementation	High: $\mu = 1$, $\Sigma - 1 = -\frac{3}{4}(3 - s)\widehat{R}_H$	Low in locked branch; controlled by free- κ , const- Σ , binned- $\Sigma(z)$, G1DE-2	Production-refined evidence-selected + KiDS shear-only diagnostic; top-control wide-prior stable; pending full 3×2 pt, production-scale noisy mocks, CMB/BBN/nonlinear compatibility, and independent replication

A closing expanded version of this matrix, with additional columns for dark-sector specificity and falsification / demotion clarity, appears at the end of the complete series.

Evidence Summary: Medium-Prior Production-Refined Audit

The completed medium-prior 8-seed production-refined nested-evidence audit (see Table on p. 17) gives the canonical hierarchy

$$M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE-2} > \text{G1DE-1} > \text{CPL} > \Lambda\text{CDM}.$$

All seven models share the same medium-prior configuration and were run with $d \log Z = 0.1$ and eight independent seeds. The locked $M_{3/4}$ branch exceeds the free- κ control by $\Delta \log Z = 0.856$ ($B \simeq 2.35$), constant- Σ by $\Delta \log Z = 1.775$ ($B \simeq 5.90$), G1DE-2 by $\Delta \log Z = 6.523$ ($B \simeq 6.8 \times 10^2$), G1DE-1 by $\Delta \log Z = 7.402$ ($B \simeq 1.65 \times 10^3$), CPL by $\Delta \log Z = 9.603$ ($B \simeq 1.5 \times 10^4$), and ΛCDM by $\Delta \log Z = 11.884$ ($B \simeq 1.45 \times 10^5$).

AIC/BIC matched comparison: $M_{3/4}$ also leads by AIC ($\Delta\text{AIC} = 0$) and BIC ($\Delta\text{BIC} = 0$), matching the G1DE-2 χ^2 to numerical precision with two fewer response parameters.

Interpretation: The medium-prior production refinement confirms that the earlier three-seed hierarchy was not a seed artifact. The data favor a sparse projection-locked background–Weyl residual over phenomenological dark-energy flexibility and unconstrained response freedom. The projection-locked model $M_{3/4}$ is selected by production-refined medium-prior evidence; the amplitude lock $\kappa = 3/4$ is favored (survives top-control wide-prior sensitivity) and remains a conditional projection-locking result. The output-response shape $X_b(a) = \hat{R}_H(a)$ has preliminary support and requires expanded lensing data. G1DE-2 serves as a free-response adversarial envelope; G1DE-1 is the growth-only negative control supporting the interpretation that the primary signal is $H + \Sigma$ with $\mu \simeq 1$. G1DE-1 is included as a full member of the eight-seed production-refined comparison. Within the tested late-time residual sector, $M_{3/4}$ is presently the most constrained and best-supported G1 branch: it combines a locked projection coefficient, parameter compression, matched production-refined evidence, and explicit demotion paths. G1DE-1 and G1DE-2 show slightly larger seed scatter than the strict 0.15 stability target, but their large evidence gaps from $M_{3/4}$ ensure the production ranking is unaffected.

D7 gives a finite-state Markov-screen realization prototype for an admissible passive lift, but the likelihood remains output-level: it samples $R_H(a) = R_0 \hat{R}_H(a)$, not the microscopic generator $\mathcal{L}_N$.

Note on supersession. The master v1.2 production-refined evidence table (p. 17) supersedes earlier embedded component-level statements that described the $d \log Z = 0.1$, eight-seed medium-prior run as pending. The standalone included components are retained as synchronized body material unless explicitly regenerated.

Wide-prior sensitivity (top-control set): A wide-prior pass over the three closest models confirms that the evidence preference for $M_{3/4}$ is not a medium-prior-volume artifact. The projection-locked ranking survives prior broadening (Table 7).

Table 7: Wide-prior nested-evidence comparison for the three closest models. The projection-locked ranking survives prior broadening.

Model	$\log Z$ (medium)	$\log Z$ (wide)	$\Delta \log Z$ (wide)	$B_{M_{3/4},i}$ (wide)
$M_{3/4}$	-894.34	-896.81	0.00	1
M_κ	-895.06	-897.66	0.85	2.3
const- Σ	-896.13	-898.76	1.96	7.1

Appendix: Closure Theorems, Bridge Principles, and Open Lemmas

The following seven items form the mathematical closure spine of the G1 program. Items 1, 2, 5 are formal theorems. Item 3 is a classification rule. Item 4 comprises open lemmas (strong-route targets). Item 6 is a conditional admissibility rule. Item 7 is a conditional theorem.

1. **Screen Ledger Representation Theorem (Core, Theorem 1).**

$$S_{\text{scr}} = k_{\text{B}} \log N_{\text{scr}}.$$

2. **Quotient-Invariance Theorem (Core, Theorem 2).**

$$\mathcal{X}_{\text{phys}} = \mathcal{X}_{\text{raw}} / \mathcal{G}_{\text{scr}}.$$

3. **Response Obstruction Theorem (Core, Theorem 3).**

$d\omega \neq 0 \Rightarrow \{\text{incomplete quotient, missing variable, memory, non-identifiability, physical residual}\}.$

Only the final case enters $R_{\mu\nu}^{\text{scr}}$.

4. **Calibration Lemma Stack (D0, Lemmas D0.1–D0.5).** Weak bridge: BH entropy.
Strong route: finite-index density + boost modular covariance + anomaly cancellation.

5. **Null Response Metricization Theorem (Core, Theorem 4).**

$$\mathcal{F}_p(k) = E_{\mu\nu}(p)k^\mu k^\nu$$

up to trace.

6. **No-Free-Residual Theorem (Core, §11).**

$$R_{\mu\nu}^{\text{scr}} \in \ker \nabla^\mu \cap \text{Obs}(\mathcal{C}_{\text{resp}}) \cap \text{Quot}(\mathcal{G}_{\text{scr}}) \cap \text{Penalty}(\mathcal{L}).$$

7. **3/4 Projection Uniqueness Theorem (D2, Theorem D2.1).**

$$\kappa = \frac{\dim(S \oplus \Omega)}{\dim(A \oplus S \oplus \Omega)} = \frac{3}{4}$$

under isotropic optical MaxEnt closure.

These theorems are conditional on their stated bridge assumptions. The open lemmas (D0 finite-index density, boost modular calibration, and all-null Ward/Bianchi completion) remain explicit mathematical targets rather than claimed results.

Bootstrap Extension Theorems and Classification Results

The following items extend the D0–D7 closure spine with the D8–D10 reconstruction/obstruction/bootstrap layer:

8. **Metric Reconstruction / Universal Envelope Principle (D8).** Causal access reconstructs conformal cone data; universal quadratic principal symbol gives metricity; entropy density fixes scale only after calibration.
9. **Obstruction Filtration / No-Free-Dark-Sector Classification (D9).** Residuals are quotient / obstruction / Ward-admissible classes, not free dark functions.
10. **Minimal Weyl Branch Uniqueness Theorem (D10).** Under finite-description, no-free- $A(a, k)$, Ward-suppressed Ricci leakage, passive memory shape, and isotropic optical compliance, the unique minimal branch is

$$\mu = 1, \quad \Sigma - 1 = -\frac{3}{4}(3 - s)R_{bH}(a).$$

Provenance Appendix: Superseded Raw Audit

The earlier mixed-provenance seven-model evidence table is retained here for provenance only. It has been superseded for model ranking by the production-refined table (p. 17). The raw audit contains heterogeneous prior configurations: $M_{3/4}$, M_κ , and constant- Σ inherited from wide-prior legacy exact-pilot runs, CPL inherited from recovered chains, and G1DE-1, G1DE-2, and Λ CDM from medium-prior exact-pilot runs. The raw audit served to standardize metadata, expose provenance lineage, and complete the G1DE-1 negative-control audit before the homogeneous reruns were run. It should not be used for model ranking now that the homogeneous table is available.

Table 8: Raw seven-model provenance-transparent audit table (superseded). Prior labels are mixed across models. Retained for provenance documentation only; the canonical medium-prior 8-seed production-refined audit supersedes this table.

Model	ndim	seeds	$\log Z$	$\Delta \log Z$	Prior	Run family
G1DE- $M_{3/4}$	4	3	-896.81	0.00	wide	legacy exact-pilot
G1DE- M_κ	5	3	-897.66	0.85	wide	legacy exact-pilot
G1DE-const- Σ	5	3	-898.76	1.95	wide	legacy exact-pilot
G1DE-2	6	3	-900.95	4.14	medium	legacy exact-pilot
G1DE-1	5	3	-901.75	4.94	medium	legacy exact-pilot
CPL	5	3	-903.69	6.88	recovered	legacy recovered
Λ CDM	3	3	-906.22	9.41	medium	legacy exact-pilot

v1.2 Theory-Hardening Integration Map

This section records where the v1.2 hardening modules are integrated in the body:

- Finite ledger representation $\mathcal{L}_{\text{scr}}$: Core §5
- Retarded finite response kernel rule: Core §11
- Causal measurement protocol: Companion B §7, D10 §3
- Early-universe kill tests: Companion E §6
- Relative-entropy action principle: D0 §4
- Modular susceptibility 3/4 law target: D2 §5
- Modular $R_b^H(a)$ target: D6 §4
- Screen-QNEC admissibility filter: D9 §5
- μ/Σ mainstream interface: Companion D §8, D10 §4, G1DE-M3/4 §4

The formal definitions are no longer overlays; they appear in the relevant Core, Companion, and D-series sections. Legacy empirical results are unchanged; theoretical body text has been updated through explicitly marked v1.2 hardening modules.

Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers

From Finite Screen Capacity to Local Coupling, Capacity-Flow Normal Forms, and Integrable Geometry Closure

v1.2 Body-Integrated Release

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

Finite Distinction Systems — Gravity 1 (FDS-G1) Core consolidates the finite screen-capacity gravity program into a single flagship paper. This flagship paper, together with six technical companions (Companions A–F covering screen statistics, transition dynamics, scalar-capacity EFT, likelihood tests, horizon occupancy residuals, and holographic dictionaries), forms the complete FDS-G1 program. The primitive object is a finite causal-screen entropy functional

$$S_{\text{scr}} : \mathcal{X}_{\text{phys}} \rightarrow \mathbb{R},$$

defined on a physical quotient state space of screen variables. Its response one-form

$$\omega = \eta_i(X) dX^i, \quad \eta_i = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta X^i}$$

contains the local area response

$$\eta_{\text{loc}}(x) = \omega(\delta_A(x)) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}.$$

Under the standard horizon-entropy and local-screen thermodynamic bridges, this area response determines the local inverse coupling, or gravitational stiffness:

$$G_{\text{loc}}^{-1}(x) = \frac{4\hbar}{c^3} \eta_{\text{loc}}(x), \quad G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}.$$

The paper organizes the G1 program around the chain

$$\text{finite screen statistics} \rightarrow S_{\text{scr}}[X] \rightarrow \omega \rightarrow \eta_{\text{loc}} \rightarrow G_{\text{loc}} \rightarrow \theta\text{-flow} \rightarrow \text{integrable geometry} \rightarrow \mathcal{R}_{\mu\nu}.$$

Finite screen statistics supplies the area-extensive substrate; interacting screens renormalize the response density; stochastic entropy-gradient screen dynamics gives Raychaudhuri-like capacity-flow normal forms; integrable all-null response data generate Einstein-like metric closure; non-integrable response data appear as residual sectors. Scalar-capacity EFT, holographic dictionaries, horizon thermodynamics, and likelihood-ready observational tests are treated as companion interfaces or manuscripts. The paper is a bridge program: the derivation is conditional on explicitly stated assumptions. The framework is

weakened if finite screen statistics fails to produce an area-extensive response substrate, if local response data cannot be metric-integrated, or if residual sectors fail conservation and likelihood penalties. It aims to provide a precise statistical and response-geometric substrate for the area-entropy coefficient that controls gravity-like coupling. The choice of S_{scr} is motivated by a primitive FDS argument: finite maintainable distinctions require causal boundary accounting, and the coordinate-free ledger of finite screen state-count is entropy. A stronger calibration route is developed in D0, where the usual horizon-entropy bridge is replaced by a conditional modular anomaly-inflow argument: the primitive coefficient is a finite-access capacity density η_0 ; under boost modular calibration and all-null Ward/Bianchi closure one obtains $G = c^3/(4\hbar\eta_0)$ and $S_H/k_B = \eta_0 A_H = A_H/(4\ell_P^2)$.

v1.2 additions. This version adds body-level hardening modules: finite ledger representation, admissible retarded finite response kernels, and updated dependency-ordered open problems.

1 Scope, Claim Status, and Bridge Assumptions

This paper presents the core formulation of the FDS-G1 finite screen-capacity program. The framework is organized around the response-geometric chain

$$\text{finite screen statistics} \rightarrow S_{\text{scr}}[X] \rightarrow \omega \rightarrow \eta_{\text{loc}} \rightarrow G_{\text{loc}} \rightarrow \theta\text{-flow} \rightarrow \text{integrable metric closure} \rightarrow \mathcal{R}_{\mu\nu}.$$

Table 1: v1.2 Core Claim Status Update

Claim	Type	Status
finite screen ledger tuple $\mathcal{L}_{\text{scr}}$	Definition	integrated in v1.2
retarded finite response kernel	admissibility rule	integrated in v1.2
relative-entropy / causal-diamond calibration	conditional target	open

FDS-G1 is not introduced as a phenomenological modification of the Friedmann equation or of the Einstein equation. It is a finite-screen response program in which local inverse coupling, capacity-flow normal forms, metric-integrability conditions, and residual sectors arise from a common entropy-response architecture. The Core defines the finite-screen entropy-response spine. D8 upgrades the geometric side into a reconstruction/envelope problem; D9 upgrades residual admissibility into an obstruction classification; D10 upgrades the data-facing $M_{3/4}$ branch into a minimal Weyl-branch bootstrap theorem.

1.1 Closure Contract and Non-Absorption Rule

A G1 claim is closed only when its assumptions, proof type, failure mode, and demotion consequence are explicitly stated. No failed bridge, residual, or empirical branch may be reabsorbed into the theory by redefining $R_{\mu\nu}^{\text{scr}}$, $R_H(a)$, $A(a, k)$, or the microscopic source–kernel factorization.

The governing standard is: every major statement must be classified as one of

1. Definition
2. Formal theorem
3. Conditional theorem
4. Bridge assumption
5. Normal-form approximation
6. Open lemma

7. Empirical output normal form
8. Falsification / demotion rule

The master closure rule for the series is

Closure = Typed arrows + Admissible residuals + Non-absorbable failures + Identifiable outputs.

The non-absorption rule:

- Failed calibration $\not\Rightarrow$ free residual survival.
- Failed empirical branch $\not\Rightarrow$ failure of screen entropy primitive.
- Successful $M_{3/4}$ $\not\Rightarrow$ unique microscopic derivation.

1.2 Why finite screen entropy is the primitive object

The choice of a finite causal-screen entropy functional is not an ornamental assumption added after the fact. It is the minimal gravitational primitive obtained by applying the FDS ontology to a causal spacetime setting. The argument is programmatic rather than a derivation of the Einstein equation:

1. Physical distinctions must be finite and maintainable.
2. Maintainable distinctions require a boundary.
3. In gravity, the relevant boundary is causal: an operational screen or horizon.
4. A finite causal screen has a finite state-count ledger.
5. The coordinate-free ledger of finite state-count is entropy.
6. Therefore the primitive gravitational object is a screen entropy functional

$$S_{\text{scr}} : \mathcal{X}_{\text{phys}} \rightarrow \mathbb{R}.$$

7. Geometry is the integrable response envelope of this ledger.

Theorem 1 (Screen Ledger Representation Theorem). *Let Σ be a finite causal screen with accessible state-class set $\mathcal{C}_\Sigma$, $N_\Sigma = |\mathcal{C}_\Sigma|$. Suppose a ledger L satisfies*

$$L(\Sigma_1 \sqcup \Sigma_2) = L(\Sigma_1) + L(\Sigma_2), \quad N_{\Sigma_1 \sqcup \Sigma_2} = N_{\Sigma_1} N_{\Sigma_2},$$

coarse-graining monotonicity

$$\pi : \mathcal{C}_\Sigma \rightarrow \mathcal{C}'_\Sigma \quad \Rightarrow \quad L(\mathcal{C}'_\Sigma) \leq L(\mathcal{C}_\Sigma),$$

and standard regularity/extensivity assumptions (continuity, or monotonicity over positive integers, extensivity to rational/real capacities). Then, up to scale, the unique physically admissible ledger is

$L(\Sigma) = \alpha \log N_\Sigma.$

With $\alpha = k_B$,

$S_{\text{scr}} = k_B \log N_{\text{scr}}.$

Proof. Additivity of L on disjoint unions and multiplicativity of state-count under independent composition, together with coarse-graining monotonicity and standard regularity/extensivity assumptions, force L to be the unique logarithmic form. The scale α is fixed by identifying L with thermodynamic entropy under the standard temperature-identification bridge. $\square$

This theorem proves that screen entropy is not merely an analogy. It is the unique additive coordinate of finite screen capacity.

Thus, within FDS, finite screen entropy is not a decorative analogy. It is what finite, boundary-maintained distinguishability becomes when the physical access relation is gravitational and causal. Coupling, capacity flow, metric closure, and dark residual sectors are then read as response, integrability, and failure modes of this entropy ledger.

Equivalently, to reject $S_{\text{scr}}[\mathcal{X}_{\text{phys}}]$ as the primitive starting point for the G1 branch, one must reject at least one of the following FDS commitments: finite distinguishability, boundary-maintained access, causal-screen localization, entropy as a coordinate-free capacity ledger, or geometry-as-response.

The central claim is conditional but sharply defined. Given the standard horizon-entropy normalization and local-screen thermodynamic bridge, FDS-G1 identifies the local inverse gravitational coupling, or stiffness, with the local entropy-area response of a finite causal screen:

$$G_{\text{loc}}^{-1}(x) = \frac{4\hbar}{c^3} \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}, \quad G_{\text{loc}}(x) = \frac{c^3}{4\hbar} \left[\frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)} \right]^{-1}.$$

The purpose of this paper is not to replace the Bekenstein-Hawking entropy formula, Jacobson-type local thermodynamic arguments, Raychaudhuri focusing, Lovelock uniqueness, spin-2 consistency, scalar-tensor EFT, or holographic duality. Rather, it supplies a finite screen-capacity substrate and response-geometric language for the area-entropy coefficient that appears in these standard structures.

Claims in this paper are classified as derived toy results, response principles, bridge assumptions, EFT embeddings, normal-form results, or falsification interfaces. This classification distinguishes what follows internally from finite-screen assumptions, what is imported from standard horizon or holographic physics, and what remains an open closure condition.

Central slogan. A useful but nonliteral slogan is:

Gravity-like geometry is the integrable hydrodynamic response of finite causal-screen entropy.

The technical reading is that local screen entropy response defines inverse coupling (stiffness); stochastic screen dynamics defines capacity flow; and metric geometry is the integrable envelope of compatible local response laws.

What would count as a stronger derivation? To upgrade the program from bridge to derivation, one would need to derive internally, from FDS primitives alone, the Bekenstein-Hawking coefficient $1/(4\ell_P^2)$, the local screen first law, Raychaudhuri focusing, all-null tensor closure, and the observed gravitational constants. G1 does not yet do this. It instead provides a sequence of toy theorems, response principles, EFT embeddings, and falsification interfaces. D0 supplies a stronger conditional calibration route: the Bekenstein-Hawking coefficient problem is reduced to two sharper lemmas—construct a causal finite-index screen-algebra net with universal local capacity density η_0 , and prove boost modular calibration $K_S = (2\pi/\hbar)Q_\chi$ for local causal screens; once these hold, $G = c^3/(4\hbar\eta_0)$ and $S_H/k_B = A_H/(4\ell_P^2)$ follow.

The comparative claim is structural. A competing macro-spacetime residual theory is not equivalent merely because it can fit a distance curve or introduce effective functions. It must provide a comparable primitive layer, a comparable response chain, and a comparable falsifiable residual fingerprint.

1.3 Claim-Type Table

Table 2: Claim-type audit for FDS-G1 Core. Every major statement is classified by proof type, dependency, and failure consequence.

Claim	Type	Depends on	Failure consequence
finite screen entropy	representation theorem target	finite accessible screen states	primitive weakened
primitive $G^{-1} \sim \eta_{\text{loc}}$	bridge / D0 lemma target	BH bridge or D0 calibration	stiffness bridge demoted
θ -flow	normal form	local stochastic screen dynamics	normal-form demotion
all-null metricization	theorem target	quadratic null response	metric closure fails
$R_{\mu\nu}^{\text{scr}}$	admissible residual	obstruction + conservation	residual rejected
$M_{3/4}$	empirical output normal form	projection lock + likelihood	branch demotion

1.4 Strong vs Weak Closure

Two levels of closure are distinguished.

Weak closure uses standard horizon entropy, local first law, Raychaudhuri, and Ward/Bianchi closure as bridges. It is the current active bridge level.

Strong closure derives the calibration through D0 finite-index density and boost modular covariance. D0 reduces the BH coefficient and Einstein calibration problem to finite-index screen-algebra density and boost modular covariance lemmas, which remain open. The strong route is a conditional target, not a completed derivation.

If D0 finite-index density fails, strong calibration fails but the weak horizon bridge may survive. The distinction prevents empirical branch success from being misread as strong-route closure.

2 Prior-art boundary

The paper uses established results as bridges. Area entropy and black-hole thermodynamics trace to Bekenstein and Hawking [1, 2]. The local thermodynamic route from horizon first law to Einstein equations follows Jacobson-type reasoning [4]. Raychaudhuri focusing is standard congruence geometry [3]. Lovelock uniqueness constrains low-derivative metric closure [5]; spin-2 consistency routes are associated with Weinberg and Deser [6, 7]. Scalar-tensor and Horndeski theories provide the conservative EFT home for variable-coupling sectors [8, 9]. AdS/CFT and holographic entropy are represented by Maldacena, Ryu-Takayanagi, and FLM [10, 11, 12]. GW170817 motivates luminal tensor-sector choices in late-time modified gravity [13].

FDS-G1’s contribution is orthogonal to these results: it supplies the finite screen-capacity interpretation and the entropy-response formalism that organizes them into a single bridge program.

Antiba’s finite-capacity thermodynamics of causal horizons studies horizon capacity as a limitation on local classical representability [16]. G1 is adjacent but distinct: it treats finite causal-screen entropy response as the source of local inverse coupling and as input data for metric-integrability and residual-closure structure. Bousso’s covariant entropy conjecture links light-sheet area to entropy content [17] and provides a key prior constraint for any causal-screen entropy program. The holographic dictionary section draws on the Brown–Henneaux central-charge relation $c_{\text{CFT}} = 3L/(2G_3)$ as a concrete example of screen capacity density controlling asymptotic coupling [18]. The horizon thermodynamics and first-law discussion complements the Bekenstein–Hawking and Jacobson references with the Iyer–Wald Noether-charge formalism for diffeomorphism-invariant theories [19]. The local thermodynamic route to gravitational

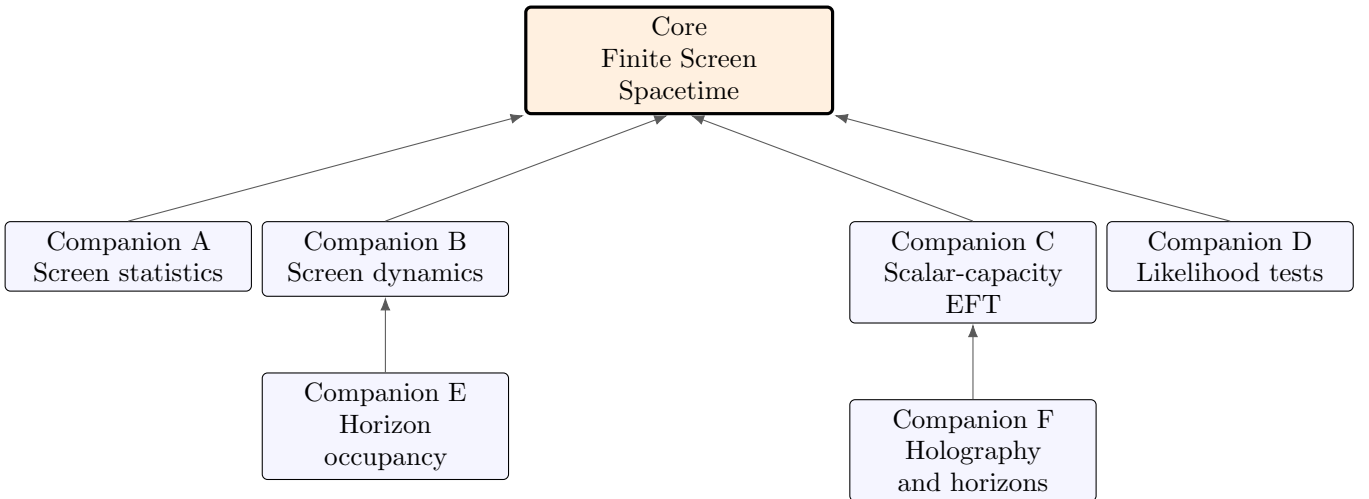
equations is further supported by Jacobson’s entanglement-equilibrium approach to the Einstein equation [20].

Recent adjacent work has also moved toward finite-capacity, finite-resolution, and information-ledger descriptions of gravitational screens. Barzi et al. study finite heat-capacity corrections to Unruh/Jacobson thermodynamics and the associated Rényi-entropy structure [21]. Nusbaumer formulates semiclassical gravity on causal diamonds with finite-resolution boundary algebras and a relative-entropy variational principle [22]. Song develops a local entropy-balance ledger on causal horizons, balancing area entropy, modular energy, and irreversible record updates [23]. Ahmad and Klinger develop an algebraic quantum-probability route to emergent area terms and generalized entropy [24]. Bianconi formulates an entropic action from quantum relative entropy between spacetime and matter-induced metrics [25]. These approaches are adjacent to G1 but differ in emphasis: G1 identifies the local inverse coupling with the marginal entropy-area response of a finite causal screen and then follows that response through capacity-flow, metric-integrability, residual closure, and likelihood-ready tests.

Relation to the FDS series. FDS-G1 is the gravitational branch of the broader Finite Distinction Systems program. The formal FDS core defines active finite distinction systems as finite-capacity boundary-maintaining systems and introduces the proof-status discipline used here [28]. FDS-T1 develops observer-relative distinguishability budgets [29]; FDS-T2 formulates effective geometry as horizon boundary accounting [30]; FDS-T3 studies capacity overflow, effective stochasticity, and Markov closure under finite projection [31]; and FDS-X1 gives the predecessor horizon-ledger dark-energy bridge [32]. FDS-X3 decomposes the four known interactions into functional distinction-operation classes, identifying gravity as global causal-geometric / stress-energy accounting [34]. FDS-G1 develops this gravitational row by modeling global boundary accounting as finite causal-screen entropy response. FDS-X2 supplies the weak-sector orientation-capacity module used by X3, but is not a direct input to the gravitational response bridge developed here [33]. The present paper differs from those predecessors by promoting the screen entropy functional $S_{\text{scr}}[X]$, its response one-form ω , and the local area response η_{loc} into the central gravity-specific architecture.

3 Program architecture and dependency graph

The G1 program is organized as one flagship core paper together with six technical companions:



Internal theoretical chain:

$$\text{finite screen statistics} \rightarrow S_{\text{scr}}[X] \rightarrow \omega \rightarrow \eta_{\text{loc}} \rightarrow G_{\text{loc}}^{-1}/G_{\text{loc}} \rightarrow \theta\text{-flow} \rightarrow \text{integrable metric closure} \rightarrow \mathcal{R}_{\mu\nu}.$$

Table 3: Claim-status audit for FDS-G1 Core.

Claim	Meaning	Status
Finite cell area law	$S_A/k_B = (A/a_0) \ln m$	toy theorem
Interacting area law	$S_A/k_B = \eta_{\text{eff}} A + o(A)$	statistical bridge
Screen quotient state space	$\mathcal{X}_{\text{phys}} = \mathcal{X}_{\text{raw}}/\mathcal{G}_{\text{scr}}$	definition/framework
FDS primitive screen ledger	finite maintainable distinctions require causal boundary entropy ledger	programmatic / ontological bridge
Response one-form	$\omega = \eta_i dX^i$	definition
Local coupling bridge	$G_{\text{loc}} = (c^3/4\hbar)[\eta_{\text{loc}}]^{-1}$	conditional bridge
Capacity-flow normal form	$\dot{\theta} = -a\theta^2 - b\sigma^2 - c\Phi + \Xi$	stochastic normal-form result
Metric all-null closure	null response reconstructs tensor up to trace	standard tensor bridge
Ward/Bianchi closure	response conservation	gauge-to-Ward bridge
Scalar-capacity EFT	$\varphi = \ln(\eta/\eta_0)$	conservative scalar-tensor EFT
Residual taxonomy	non-integrable response classes	framework
Likelihood interface	SN/BAO/growth/local- G blocks	medium-prior evidence executed; top-control wide-prior sensitivity completed; medium-prior production refinement (8 seeds, $d \log Z = 0.1$) completed

4 Finite screen statistical mechanics

Definition 1 (Finite distinction screen). *A finite distinction screen Σ_A is a causal boundary patch with operational area A and finite screen-local distinguishability units. In the simplest model,*

$$N_A = \frac{A}{a_0}, \quad (1)$$

where a_0 is an effective area per independent cell.

If each cell carries m distinguishable states and the cells are independent,

$$\Omega_A = m^{N_A}, \quad S_A = k_B \ln \Omega_A = k_B \frac{A}{a_0} \ln m. \quad (2)$$

Therefore

$$\frac{S_A}{k_B} = \eta_{\text{nat}} A, \quad \eta_{\text{nat}} = \frac{\ln m}{a_0}. \quad (3)$$

Matching the standard horizon bridge gives

$$\eta_{\text{BH}} = \frac{1}{4\ell_P^2} = \frac{c^3}{4G\hbar}, \quad G = \frac{c^3}{4\hbar} [\eta_{\text{BH}}]^{-1}. \quad (4)$$

This identifies the screen-capacity density that must be supplied to reproduce the standard horizon coefficient; the coefficient $1/4$ itself remains an external input.

Definition 2 (Screen area-law universality class). $\mathcal{U}_{\text{scr}}$ is the class of finite screen systems whose degrees of freedom are screen-local, finite-state, finite-range or sufficiently short-range, finite-correlation in the regime considered, stable under screen-local coarse-graining, and not dominated by hidden bulk-volume degrees of freedom or global nonlocal constraints.

For systems in $\mathcal{U}_{\text{scr}}$ one expects the thermodynamic form

$$\frac{S_A}{k_B} = \eta_{\text{eff}} A + S_{\text{sub}}(A), \quad \frac{S_{\text{sub}}(A)}{A} \rightarrow 0 \quad (A \rightarrow \infty). \quad (5)$$

Finite interactions renormalize η_{eff} rather than destroying the leading area law. Criticality, long-range constraints, infinite local Hilbert spaces, hidden bulk volume degrees of freedom, or dominant nonlocal constraints are controlled failure modes.

5 Screen state space and quotient structure

A screen entropy functional must be defined on physical screen states, not on arbitrary coordinate labels. Write raw variables as

$$X_{\text{raw}} = \{A(x), q_{AB}(x), \chi(x), \Phi(x), M(x), \dots\}, \quad (6)$$

where A is area or cell density, q_{AB} is screen shape, χ is occupancy or residual capacity, Φ is flux load, and M denotes memory or hidden screen modes. The physical state is the quotient

$$\boxed{\mathcal{X}_{\text{phys}} = \mathcal{X}_{\text{raw}} / \mathcal{G}_{\text{scr}}}, \quad (7)$$

where $\mathcal{G}_{\text{scr}}$ includes screen coordinate reparametrizations, null-generator reparametrizations, cell relabelings, and internal labels that do not change causal access or entropy response.

Definition 3 (Screen entropy functional). *The entropy functional is*

$$S_{\text{scr}} : \mathcal{X}_{\text{phys}} \rightarrow \mathbb{R}. \quad (8)$$

Functional derivatives are understood on the quotient tangent space, or after gauge fixing and projection away from pure gauge directions.

Definition 4 (Finite screen ledger representation). *The finite screen operational representation is defined as the tuple*

$$\mathcal{L}_{\text{scr}} = (V_{\text{scr}}, E_{\text{corr}}, C, \pi_{\text{cap}}, S_{\text{scr}}, \mathcal{G}_{\text{scr}}, \tau)$$

where:

- V_{scr} is a finite set of screen record carriers
- E_{corr} is a finite correlation/adjacency structure on V_{scr}
- C stores capacity-truncated correlation classes or weights
- π_{cap} is the finite capacity projection operator
- S_{scr} is the screen entropy ledger
- $\mathcal{G}_{\text{scr}}$ is the quotient/gauge groupoid of screen automorphisms
- τ is the screen update window

This is an operational finite representation, not a unique microscopic ontology.

Example 1 (Cell relabeling). If cells $i = 1, \dots, N$ have states σ_i , a permutation $i \mapsto \pi(i)$ changes labels but not area, entropy, or response. Thus $\{\sigma_i\} \sim \{\sigma_{\pi(i)}\}$.

Example 2 (Generator reparametrization). A monotonic change $\lambda \mapsto f(\lambda)$ changes the coordinate expression for $\theta = d \ln A / d\lambda$. It does not change the cross-sectional entropy-area response $\eta_{\text{loc}} = k_B^{-1} \delta S / \delta A$.

Theorem 2 (Quotient-Invariance Requirement). A response quantity is physically admissible only if it descends to $\mathcal{X}_{\text{phys}}$. Require

$$S_{\text{scr}}(g \cdot x) = S_{\text{scr}}(x), \quad \omega_{g \cdot x}(Dg v) = \omega_x(v).$$

Any residual computed before quotient projection is not physical.

Failure rule. If $d\omega \neq 0$ disappears after quotient correction, the apparent residual is a gauge artifact, not $R_{\mu\nu}^{\text{scr}}$.

6 Response one-form and local coupling

Let $s_{\text{scr}} = S_{\text{scr}}/k_B$. The response one-form is

$$\omega = \eta_i(X) dX^i, \quad \eta_i(X) = \frac{\delta s_{\text{scr}}}{\delta X^i}. \quad (9)$$

For field variables this denotes

$$\omega = \int_{\Sigma} d^{d-2}x \eta_i(x) \delta X^i(x). \quad (10)$$

Let $\delta_A(x)$ be the physical tangent vector that changes the local area element $A(x)$ while holding the other physical screen variables fixed. Then

$$\boxed{\eta_{\text{loc}}(x) = \omega(\delta_A(x)) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}}. \quad (11)$$

This makes η_{loc} the area component of the full response one-form.

Principle 1 (Local response coupling). Under the standard horizon-entropy and local-screen thermodynamic bridges, the local area response determines the local inverse coupling, or gravitational stiffness:

$$\boxed{G_{\text{loc}}^{-1}(x) = \frac{4\hbar}{c^3} \eta_{\text{loc}}(x), \quad G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}}. \quad (12)$$

For homogeneous screens,

$$\eta_{\text{loc}} \rightarrow \eta_{\delta}(A) = \frac{d(S/k_B)}{dA}, \quad G_{\delta}(A) = \frac{c^3}{4\hbar} [\eta_{\delta}(A)]^{-1}, \quad G_{\delta}^{-1}(A) = \frac{4\hbar}{c^3} \eta_{\delta}(A). \quad (13)$$

For finite-area corrections

$$\frac{S(A)}{k_B} = \eta_0 A + \alpha \ln(A/A_0) + \gamma + b(A/A_0)^{-p} + \dots, \quad (14)$$

one obtains

$$\eta_{\delta}(A) = \eta_0 + \frac{\alpha}{A} - \frac{pb}{A_0} (A/A_0)^{-p-1} + \dots. \quad (15)$$

Thus macroscopic finite-area corrections are naturally suppressed by inverse area unless a separate residual or scalar-capacity mode is activated.

7 Response Integrability and Obstruction Classes

If the response one-form derives from a scalar entropy potential s_{scr} , then

$$\omega = ds_{\text{scr}}, \quad \boxed{d\omega = 0}. \quad (16)$$

In coordinates,

$$d\omega = \frac{1}{2}(\partial_i \eta_j - \partial_j \eta_i) dX^i \wedge dX^j. \quad (17)$$

A nonzero response curl indicates missing variables, memory, nonlocal constraints, stochastic projection, or physical residual modes.

Theorem 3 (Response Exactness and Obstruction). *Define*

$$s_{\text{scr}} = S_{\text{scr}}/k_B, \quad \omega = \eta_i(X) dX^i, \quad \mathcal{C}_{\text{resp}} = d\omega.$$

Closed response corresponds to $\omega = ds_{\text{scr}}$, $d\omega = 0$. The response obstruction is classified as

$$d\omega \neq 0 \Rightarrow \begin{cases} \text{incomplete quotient,} \\ \text{missing coarse variable,} \\ \text{projected memory mode,} \\ \text{source–kernel non-identifiability,} \\ \text{true physical residual.} \end{cases}$$

Only the final case may enter $R_{\mu\nu}^{\text{scr}}$. All others must be resolved by quotient correction, variable enlargement, or memory-lift before a physical residual is declared.

Let $[\mathcal{C}_{\text{resp}}] \in H_{\text{resp}}^2(\mathcal{X}_{\text{phys}})$ denote equivalence classes of response non-closure after quotient, missing-variable, and memory-lift attempts have been exhausted. H_{resp}^2 is a bookkeeping device, not a full cohomology theory; it identifies which obstructions survive all structural corrections before being admitted as physical residuals. D1 supplies the mathematical home for this obstruction via finite-access sheaf geometry (see D1, §4).

Example 3 (Two-variable exact response). *Let*

$$s(A, \chi) = \eta_0 A + \alpha \ln A + \beta A \chi - \frac{\lambda}{2} \chi^2. \quad (18)$$

Then

$$\eta_A = \frac{\partial s}{\partial A} = \eta_0 + \frac{\alpha}{A} + \beta \chi, \quad \eta_\chi = \frac{\partial s}{\partial \chi} = \beta A - \lambda \chi. \quad (19)$$

The response one-form $\omega = \eta_A dA + \eta_\chi d\chi$ is exact, since

$$\partial_\chi \eta_A = \beta = \partial_A \eta_\chi, \quad d\omega = 0. \quad (20)$$

Example 4 (Apparent curl from hidden memory). *Suppose the true entropy depends on a hidden memory mode M ,*

$$s(A, \chi, M) = \eta_0 A + \alpha \ln A + \beta A \chi + \gamma A M - \frac{\lambda}{2} \chi^2 - \frac{\mu}{2} M^2. \quad (21)$$

On the full space (A, χ, M) , $\omega = ds$ is closed. If M is projected out but depends on history, the reduced apparent response $\tilde{\omega} = \tilde{\eta}_A dA + \tilde{\eta}_\chi d\chi$ can satisfy $d\tilde{\omega} \neq 0$. This is an epistemic or memory residual, not necessarily new fundamental physics.

8 Response Hessian, fluctuations, and scalar-capacity modes

The entropy-response Hessian is

$$\mathcal{H}_{ij} = -\frac{\delta^2 s_{\text{scr}}}{\delta X^i \delta X^j}. \quad (22)$$

Near equilibrium, $\mathcal{H}$ controls stability, fluctuations, and slow hydrodynamic response modes. In a Gaussian approximation,

$$\langle \delta X^i \delta X^j \rangle \sim (\mathcal{H}^{-1})^{ij}. \quad (23)$$

The scalar-capacity variable

$$\varphi(x) = \ln \frac{\eta(x)}{\eta_0} \quad (24)$$

can be interpreted as a long-wavelength mode of the area-response density. Its EFT potential is a coarse-grained free-energy cost for moving away from the equilibrium response density. Thus the scalar-capacity EFT is not introduced as a generic dark scalar; it is the hydrodynamic EFT of a slow screen-response mode.

9 Stochastic screen dynamics and capacity-flow normal forms

Let screen macrostate variables $X^i(\lambda)$ evolve by an entropy-gradient stochastic process. A generic Onsager form is

$$\dot{X}^i = L^{ij} \frac{\delta s_{\text{scr}}}{\delta X^j} + \zeta^i, \quad (25)$$

with positive semidefinite response matrix L^{ij} and noise ζ^i . A large-deviation or Onsager-Machlup path weight may be written schematically as

$$P[X(\lambda)] \sim \exp[-N_A \mathcal{I}_{\text{scr}}[X]], \quad (26)$$

$$\mathcal{I}_{\text{scr}}[X] = \frac{1}{2} \int d\lambda (\dot{X}^i - F^i[X]) D_{ij}^{-1} (\dot{X}^j - F^j[X]), \quad (27)$$

where $F^i = L^{ij} \delta s / \delta X^j$ in local equilibrium.

Define

$$N(\lambda) = \frac{A(\lambda)}{a_0}, \quad x(\lambda) = \ln N(\lambda), \quad \theta = \dot{x} = \frac{d \ln A}{d\lambda}. \quad (28)$$

Then

$$\dot{\theta} = \frac{\ddot{A}}{A} - \left(\frac{\dot{A}}{A} \right)^2. \quad (29)$$

The negative quadratic term is therefore kinematic: it follows from logarithmic capacity expansion. Shape modes contribute the positive scalar $\sigma^2 = \sigma_{AB} \sigma^{AB}$; positive flux load focuses the screen. The low-order affine equilibrium normal form is

$$\boxed{\dot{\theta} = -a\theta^2 - b\sigma^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + \zeta_{\theta} + \mathcal{O}(3)}. \quad (30)$$

Equation (30) is a normal-form bridge: it shows that Raychaudhuri-like terms are stable low-order structures in stochastic finite screen dynamics. The identification of Φ_{flux} with $R_{\mu\nu} k^\mu k^\nu$ or $T_{\mu\nu} k^\mu k^\nu$ is the standard local-screen geometry bridge.

Proposition 1 (Raychaudhuri-Limit Recovery). *In the metric-integrable equilibrium limit, the capacity-flow normal form must reduce to the standard Raychaudhuri equation:*

$$\frac{d\theta}{d\lambda} = -\frac{1}{D-2} \theta^2 - \sigma_{\mu\nu} \sigma^{\mu\nu} + \omega_{\mu\nu} \omega^{\mu\nu} - R_{\mu\nu} k^\mu k^\nu + \mathcal{R}_{\text{scr}}^\theta.$$

For hypersurface-orthogonal null screens, $\omega_{\mu\nu} = 0$. Therefore

$$a = \frac{1}{D-2}, \quad b = 1$$

in the standard geometric limit. The remaining scalar focusing residual $\mathcal{R}_{\text{scr}}^\theta = \Xi_{\text{res}} + \zeta_\theta + \mathcal{O}(3)$ is the finite-screen correction. If these constants do not match, the model remains a finite-screen normal form but is not a GR-equilibrium recovery.

10 Metric integrability and all-null closure

Entropy integrability is not yet metric integrability. For each event p and null direction k^μ , local screen response data may define a null focusing response $\mathcal{F}_p(k)$.

Theorem 4 (Null Response Metricization). *Let screen response data define a null focusing function $\mathcal{F}_p(k)$ at each event p and null direction k^μ . Assume:*

1. *locality: $\mathcal{F}_p(k)$ depends only on data at p ;*
2. *smoothness in p ;*
3. *homogeneous quadratic dependence on k : $\mathcal{F}_p(\lambda k) = \lambda^2 \mathcal{F}_p(k)$;*
4. *compatibility with a Lorentzian null cone;*
5. *nondegenerate screen response in at least two independent null directions.*

Then there exists a symmetric tensor $E_{\mu\nu}(p)$ such that

$$\mathcal{F}_p(k) = E_{\mu\nu}(p) k^\mu k^\nu \tag{31}$$

up to a trace term $\Phi g_{\mu\nu}$. Because $g_{\mu\nu} k^\mu k^\nu = 0$, all-null data cannot fix the trace. If

$$E_{\mu\nu} k^\mu k^\nu = \kappa T_{\mu\nu} k^\mu k^\nu \quad \forall k,$$

then $E_{\mu\nu} - \kappa T_{\mu\nu} = \Phi g_{\mu\nu}$, and Ward/Bianchi closure determines whether Φ becomes a cosmological integration sector or an admissible residual.

All-null data reconstructs $E_{\mu\nu}$ up to a trace term, since $g_{\mu\nu} k^\mu k^\nu = 0$. Thus if

$$E_{\mu\nu} k^\mu k^\nu = \kappa T_{\mu\nu} k^\mu k^\nu \quad \forall k^\mu \text{ null}, \tag{32}$$

then

$$E_{\mu\nu} - \kappa T_{\mu\nu} = \Phi g_{\mu\nu}. \tag{33}$$

Ward/Bianchi closure fixes the trace freedom into the usual cosmological integration sector. In the Einstein-like equilibrium case,

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} + \mathcal{R}_{\mu\nu}. \tag{34}$$

Here $\mathcal{R}_{\mu\nu}$ is not an arbitrary dark term; it represents non-integrable or unmodeled screen response data after attempting metric closure.

D8 extends this theorem in two directions: causal-access reconstruction of the conformal cone and principal-symbol metricity.

11 Ward/Bianchi dynamical closure

If the screen response action or its hydrodynamic envelope is invariant under screen reparametrizations that become spacetime diffeomorphism redundancy in the metric envelope, then the effective geometric tensor obeys a Ward identity

$$\nabla^\mu E_{\mu\nu} = 0. \quad (35)$$

Therefore residual sectors must satisfy

$$\boxed{\nabla^\mu (T_{\mu\nu} + \mathcal{R}_{\mu\nu}) = 0.} \quad (36)$$

This is the conservation-closure condition that prevents variable capacity from becoming an unconstrained varying- G ansatz.

If memory is present, local response may become nonlocal:

$$\eta_i(\lambda) = \int^\lambda d\lambda' K_{ij}(\lambda, \lambda') X^j(\lambda'), \quad (37)$$

or covariantly,

$$\mathcal{R}_{\mu\nu}^{\text{mem}}(x) = \int d^4x' \sqrt{-g(x')} K_{\mu\nu}{}^{\alpha\beta}(x, x') T_{\alpha\beta}(x'). \quad (38)$$

Such terms must still obey the total conservation law (36).

Theorem 5 (No-Free-Residual Rule). *Define the effective geometric equation*

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} + R_{\mu\nu}^{\text{scr}}.$$

Admissibility requires $\nabla^\mu (T_{\mu\nu} + R_{\mu\nu}^{\text{scr}}) = 0$. If matter is separately conserved,

$$\nabla^\mu R_{\mu\nu}^{\text{scr}} = 0.$$

Then the residual must satisfy the intersection condition

$$\boxed{R_{\mu\nu}^{\text{scr}} \in \ker \nabla^\mu \cap \text{Obs}(\mathcal{C}_{\text{resp}}) \cap \text{Quot}(\mathcal{G}_{\text{scr}}) \cap \text{Penalty}(\mathcal{L}).}$$

A residual that fails any of these conditions is rejected: it is either a gauge artifact, a missing-variable projection, a non-conserved effective stress, or a function that carries no likelihood penalty.

D9 reformulates this admissibility condition as an exact sequence from raw response defects to conserved residual classes.

Admissibility rule: Retarded finite response kernels. All G1 screen residual kernels must satisfy the form

$$R_{\mu\nu}^{\text{scr}}(x) = \Pi_{\text{cap}} \int_{\mathcal{J}^-(x)} K_{\mu\nu}^{\text{scr}, \text{AB}}(x, y; \tau) \delta \mathcal{L}_A(y) dV_y + \epsilon_{\mu\nu}^{\text{cap}}(x)$$

and constraints:

- Strictly retarded (support limited to the past causal cone $\mathcal{J}^-(x)$)
- Finite-description or finite-rank after capacity projection Π_{cap}
- Quotient invariant under $\mathcal{G}_{\text{scr}}$
- Ward/Bianchi consistent
- No free $A(a, k)$ or unpenalized functional degrees of freedom

Kernels failing these constraints are rejected from the G1DE admissible class.

Table 4: Residual admissibility hierarchy.

Residual type	Must first pass	If fails
epistemic	missing-variable lift	not physical residual
memory	D6 quotient / horizontal lift	realization ambiguity
stochastic	T3 projection/lumpability test	modeling noise
capacity	scalar-capacity EFT consistency	scalar branch only
horizon occupancy	conservation + D5 identifiability	G1DE demotion
geometric	null metricization + Ward closure	nonmetric response theory

12 Residual taxonomy

The residual sector is classified by admissibility, not used as a catch-all.

This classification makes explicit what survives when a residual model fails. For example, if a horizon occupancy model is ruled out, the finite screen statistics and entropy-response framework may survive. If metric closure fails, the program moves beyond standard metric geometry and must state so directly.

13 Scalar-capacity EFT interface

The local response density may have a long-wavelength mode

$$\varphi = \ln \frac{\eta}{\eta_0}, \quad \frac{G_{\text{eff}}}{G_0} = e^{-\varphi}. \quad (39)$$

A conservative scalar-capacity EFT is

$$S = \int d^4x \sqrt{-g} \left[\frac{1}{2} F(\varphi) R - \frac{1}{2} Z(\varphi) (\nabla \varphi)^2 - V(\varphi) + \mathcal{L}_m + \mathcal{L}_{\text{res}} \right]. \quad (40)$$

With $F(\varphi) = M_{*,0}^2 e^\varphi$, this is a scalar-tensor/Horndeski-compatible sector, not a new field-equation form. Its novelty is the screen-capacity interpretation and prior structure. A minimal Horndeski embedding is

$$G_2 = Z(\varphi) X - V(\varphi), \quad G_3 = 0, \quad G_4 = \frac{1}{2} F(\varphi), \quad G_5 = 0. \quad (41)$$

In EFT-of-dark-energy language,

$$M_*^2 = F, \quad \alpha_M = \frac{d \ln F}{d \ln a}, \quad \alpha_T = 0, \quad \alpha_B = -\alpha_M \quad (42)$$

for the minimal nonminimally coupled model. Stability requires $F > 0$, positive kinetic normalization, absence of gradient instabilities, and compatibility with local-gravity and gravitational-wave propagation constraints.

14 Observational signatures and likelihood interface

G1 predicts no generic freedom to drift G macroscopically. A visible drift obeys

$$\frac{\dot{G}}{G} \simeq -\frac{\dot{\eta}}{\eta}, \quad (43)$$

and must be carried by a conservation-closed residual or scalar-capacity sector. The three refined signature tests are:

1. **No free macroscopic G -drift.** Any variation of G is constrained by local-gravity, PPN, and propagation priors.
2. **Background–Weyl residual pattern.** Horizon occupancy residuals should produce a structured fingerprint $s < 3$, $\Sigma_0 < 0$, $\mu_0 \simeq 0$: background filling deviation and suppressed Weyl response through a common ledger variable, with the growth response near GR.
3. **Finite-area suppression.** Log or finite-area corrections give $\Delta G/G \sim -\alpha/(\eta_0 A)$ and are negligible on macroscopic screens unless an additional residual sector is activated.

A compressed likelihood interface has the schematic form

$$\chi_{\text{tot}}^2 = \chi_{\text{SN}}^2 + \chi_{\text{BAO}}^2 + \chi_{\text{RSD}}^2 + \chi_{E_G}^2 + \chi_{\text{priors}}^2. \quad (44)$$

Companion likelihood manuscripts compare Λ CDM, CPL, G1DE, and scalar-capacity variants. G1fit demonstrations have now been upgraded to a matched medium-prior production-refined nested-evidence comparison (G1fit-real), using Pantheon+ full covariance, DESI DR2 BAO covariance, curated RSD, and compressed E_G . The completed medium-prior 8-seed production-refined audit gives the canonical hierarchy G1DE- $M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE2} > \text{G1DE1} > \text{CPL} > \Lambda\text{CDM}$, with the projection-locked model ranking first. Top-control wide-prior sensitivity has been completed; full survey-grade validation (including prior-family stress, independent replication, and full $3 \times 2\text{pt}$) remains pending.

A G1DE-1 nested-evidence negative-control audit under the same medium-prior production-refined configuration gives $\log Z_{\text{G1DE1}} = -901.751 \pm 0.180$ (eight seeds, $d \log Z = 0.1$), with $\Delta \log Z(M_{3/4}, \text{G1DE1}) = 7.402$ ($B \simeq 1.65 \times 10^3$), strongly supporting the conclusion that the growth-only negative control is decisively worse than the projection-locked branch in the production-refined comparison. D10 supplies the theoretical bootstrap behind the $M_{3/4}$ branch; G1fit-real remains the empirical test.

This evidence result is layer-specific. It supports the projection-locked output normal form within the tested late-time residual sector. It does not prove the D0 finite-index calibration route, does not identify a unique microscopic source–kernel factorization, and does not replace production-level CMB/BBN, nonlinear-structure, or waveform pipelines. Its role is to select a sparse data-facing residual branch with explicit demotion paths.

14.1 Empirical status of the residual sector

The present status of the G1 residual sector is no longer purely schematic. A matched exact-likelihood and medium-prior nested-evidence comparison has been completed. The G1DE hierarchy uses an output-level memory response

$$\mathcal{R}_H(a) = R_0 \hat{\mathcal{R}}_H(a), \quad \hat{\mathcal{R}}_H(1) = 1, \quad R_0 = 3 - s,$$

implemented through a Weyl-normalized transfer operator

$$\mathcal{W}_{\text{G1}} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}}.$$

This is an output normalization; it should not be confused with a unique microscopic source–kernel factorization $K_H * J_H$, which is treated in D6 as a deeper horizontal-lift layer.

$$\Sigma(a, k) - 1 = -\kappa R_0 \hat{\mathcal{R}}_H(a) + O(R_0^2), \quad \mu(a, k) - 1 = O(R_0^2) + O(R_0 \hat{\mathcal{R}}_H \mathcal{H}^2 / k^2).$$

The projection-locked branch $\kappa = 3/4$, $\hat{\mathcal{R}}_H = X_b$ (with $X_b(1) = 1$ as the production-normalized output-response shape) gives $\Sigma(a) = 1 - \frac{3}{4}(3 - s)X_b(a)$, $\mu \simeq 1$. In the completed medium-prior 8-seed production-refined audit, G1DE- $M_{3/4}$ ranks first among the tested seven-model control set. The detailed results and falsification ladder are in Companions D and E; the data paper is G1fit-real.

The refined empirical claim-status is therefore:

Table 5: Updated empirical status for the G1 residual sector.

Claim	Status
Finite screen entropy response	theoretical bridge
G1DE-0 background test	real-data full-covariance implemented
$s = 3$ Λ CDM-like limit	numerically validated
G1DE-2 production-refined control	included in the completed medium-prior 8-seed audit
$\Sigma_0 < 0$ Weyl response	pilot-level indication
$\mu_0 < 0$ growth response	not supported; $\mu_0 \simeq 0$
Medium-prior evidence completed (8-seed homogeneous); top-control wide-prior sensitivity completed; full survey-grade validation pending	

A post-pilot projection-locked variant, G1DE-M_{3/4}, imposes $\mu(a, k) = 1$ (equivalently $\mu_0 = 0$) and $\Sigma(a) = 1 - \frac{3}{4}(3 - s)X_b(a)$. In the completed medium-prior 8-seed production-refined audit (G1fit-real), this four-parameter locked model ranks first among the tested control hierarchy; see Companions D, E for status. This is an empirical refinement of the residual sector, not a core bridge assumption.

The factorized production form is

$$\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1, \quad R_0 = 3 - s,$$

implemented through a Weyl-normalized transfer operator $\mathcal{W}_{G1} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{GR}$. The active-filling shape $X_{\text{fill}}(a)$ (peak-normalized) and the production-normalized memory shape $\hat{R}_H(a) = X_{\text{fill}}(a)/X_{\text{fill}}(1)$ are related by a constant factor; the peak-normalized form is used in the internal model definition, while $\hat{R}_H(1) = 1$ ensures identifiability of the projection coefficient κ and the residual amplitude R_0 .

15 Horizon occupancy residuals

A minimal horizon occupancy residual model uses

$$\rho_H(a) = \chi_H(a) \rho_c(a), \quad \rho_c(a) = \frac{3H^2(a)}{8\pi G}. \quad (45)$$

A lowest-order filling-closure law is

$$\frac{d\chi_H}{d \ln a} = s \chi_H \left(1 - \frac{\chi_H}{\chi_\infty} \right), \quad (46)$$

with solution

$$\chi_H(a) = \frac{\chi_\infty}{1 + Ba^{-s}}. \quad (47)$$

The effective equation of state is

$$w_H(a) = -1 - \frac{1}{3} \frac{d \ln[\chi_H(a) H^2(a)]}{d \ln a}. \quad (48)$$

The model is not intended as an arbitrary background fit. Its distinctiveness is a structured background–Weyl residual pattern

$$\chi_H(a) \Rightarrow H(a), \quad \Sigma(a), \quad \mu(a) \simeq 1, \quad \text{capacity closure}, \quad (49)$$

with the growth response near GR ($\mu_0 \simeq 0$) as a consistency condition rather than a required nonzero signal. This is more restrictive than a generic dark-energy or modified-gravity parametrization because the background and Weyl response share the same active-filling shape $\hat{X}(a)$.

16 Holography and horizons

The holographic dictionary is naturally written in response-density language:

$$\frac{1}{4G_N} = \eta_{\text{screen}}. \quad (50)$$

For $\text{AdS}_3/\text{CFT}_2$,

$$c = \frac{3L}{2G_3}, \quad \eta_{\text{AdS}_3} = \frac{1}{4G_3} = \frac{c}{6L}. \quad (51)$$

Thus central charge is interpreted as a boundary version of bulk screen capacity density. The Ryu-Takayanagi and FLM formulas are read as realizations of the screen-response principle:

$$S_R = \frac{A(\gamma_R)}{4G_N} + S_{\text{bulk}} + \cdots. \quad (52)$$

Similarly, black-hole and de Sitter horizons are exterior-accessible distinction ledgers with

$$S_H/k_B = \eta_{\text{BH}} A_H. \quad (53)$$

For Kerr-Newman horizons the first law

$$dM = T_H dS_H + \Omega_H dJ + \Phi_H dQ \quad (54)$$

is read as a multi-channel capacity update law.

17 Non-Absorption Rules

The following rules enforce the closure contract: no failed bridge, residual, or empirical branch may be silently reabsorbed by redefining $R_{\mu\nu}^{\text{scr}}$, $R_H(a)$, $A(a, k)$, or the microscopic source–kernel factorization.

A residual branch is not admissible if it survives only by adding untyped functional freedom. In particular, the G1DE branch forbids an independently sampled optical amplitude $A(a, k)$. The output response is normalized as

$$R_H(a) = R_0 R_{bH}(a), \quad R_{bH}(1) = 1,$$

and the data-facing branch tests $R_H(a)$, not a separately fitted microscopic source J_H and memory kernel K_H . If a free $A(a, k)$ or independent source–kernel factorization is required by the data, the model exits the G1DE class.

Table 6: Non-Absorption rules for FDS-G1 Core.

Failure condition	What fails	What may survive
No screen area extensivity	statistical core of G1	other FDS ideas outside gravity
No finite-index screen algebra net	D0 strong calibration	Core response framework as bridge
No boost modular calibration	$(2\pi/\hbar)$ modular route	standard horizon bridge (weak closure)
No all-null metricization	Einstein-like closure	nonmetric response theory; residual cannot absorb failure
No metricization of anomaly-free sector	D0 Einstein derivation route	finite-access / residual framework
Unconstrained residuals	predictive value	core response formalism if residual taxonomy tightened

Failure condition	What fails	What may survive
Scalar-capacity EFT ruled out	variable-capacity field model	microscopic/horizon-only corrections
Normal form not robust	G1micro hardening	response coupling and EFT sectors
No passive horizon remnant	G1DE anomaly-remnant interpretation	phenomenological Weyl-transfer branch
Free $A(a, k)$ required	Weyl normalization	generic dark-stress model, not G1DE
G1DE loses to constant- Σ	output shape $X_b(a)$	Weyl amplitude branch
G1DE loses to free- κ	exact 3/4 isotropic compliance	weighted compliance branch
Data require $ \mu - 1 \sim \Sigma - 1 $	Ward-suppressed Ricci leakage	finite-screen residual with density component
Production evidence loses to Λ CDM/CPL	G1DE observational branch	statistical and response core; screen entropy primitive unchanged
Likelihood pipeline fails with real data	observational branch	theoretical bridge remains provisional
$M_{3/4}$ decisive evidence success	—	does not prove finite-index screen algebras, boost modular covariance, or unique source-kernel factorization

18 Companion manuscripts and archive structure

The recommended release structure is one flagship Core paper, six technical companions (A–F), one data-confrontation paper (G1fit-real), and the D0–D10 dark-sector closure sequence:

1. **Companion A: Finite Screen Statistical Mechanics.** Independent cells, interacting screens, and differential capacity.
2. **Companion B: Stochastic Screen Transitions.** Microtransition and Raychaudhuri-like capacity flow.
3. **Companion C: Scalar-Capacity EFT.** Horndeski and EFT-of-dark-energy embedding.
4. **Companion D: Likelihood and Falsification.** Predictions, observational interface, and compressed-data testing.
5. **Companion E: Horizon Occupancy Residuals.** Dark-sector ledger dynamics and model comparison.
6. **Companion F: Holography and Horizons.** Black-hole, de Sitter, and AdS/CFT dictionaries.
7. **G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual.** Matched exact likelihoods and homogeneous medium-prior nested evidence for seven late-time cosmology models.
8. **D0: Screen Modular Anomaly Inflow.** Einstein calibration of finite-access entropy density.
9. **D1: Finite-Access Sheaf Geometry.** Finite causal ledgers as a sheaf; gluing defects as residuals.
10. **D2: The 3/4 Projection Law.** Optical-stress closure and the projection coefficient.

11. **D3: Non-Equilibrium Screen Entropy.** Thermodynamic closure of the G1 residual.
12. **D4: Screen Influence Functional.** Unified influence framework for the dark-sector closure.
13. **D5: Weyl-Normalized Ward Completion.** Production likelihood interface and identifiability rules.
14. **D6: Maximum-Caliber Horizontal Lift.** Memory quotient, source–kernel identifiability, and maximum-caliber passive horizontal lift.
15. **D7: Finite Markov-Screen Realization.** A finite-state detailed-balance realization prototype for the D6 passive lift assumptions, showing how optical symmetry, a slow horizon mode, and Ward-stiff Ricci leakage realize the G1DE- $M_{3/4}$ normal form.
16. **D8: Metric Reconstruction and the Universal Envelope.** Conformal reconstruction and principal-symbol metricity from finite causal-screen response data.
17. **D9: Obstruction Exact Sequence and No-Free-Dark-Sector.** Sheaf-gluing theorem and obstruction classification that prevents dark-sector over-flexibility.
18. **D10: G1DE Bootstrap and Minimal Weyl Branch Uniqueness.** Prediction compression theorem and minimal Weyl residual uniqueness.

A code/data archive should include reproducible notebooks, demo compressed data, generated CSV files, figures, claim-status notes, prior-art boundaries, notation tables, and a failure registry.

19 Conclusion

FDS-G1 Core recasts the G1 program as entropy-response geometry. The central object is no longer a bare capacity density but a screen entropy functional $S_{\text{scr}}[X]$ on a quotient state space. Its response one-form defines local area response; the area component controls local inverse coupling (stiffness) through Eq. (12), while G_{loc} is the corresponding coupling. The response Hessian controls stability and fluctuations; Onsager-type stochastic screen dynamics produces Raychaudhuri-like capacity-flow normal forms; all-null metric integrability and Ward/Bianchi closure generate Einstein-like equilibrium sectors; non-integrable response data appear as residuals.

The most compact statement is

$$\boxed{\text{Gravity-like geometry is the integrable hydrodynamic response of finite causal-screen entropy.}} \quad (55)$$

The reason finite screen entropy appears as the central object is that, in the FDS ontology, finite maintainable distinctions become physically accessible only through boundary ledgers; in gravity those ledgers are causal screens. The paper remains a bridge program. Its strength is not the claim to derive all of gravity from scratch, but the construction of a finite screen-capacity substrate, a local response principle, a normal-form dynamics, an EFT embedding, and a falsification interface that can be compared with existing gravitational, holographic, and cosmological frameworks. In the completed medium-prior 8-seed production-refined audit reported by G1fit-real, the projection-locked residual model G1DE- $M_{3/4}$ is preferred over free- κ , constant- Σ , G1DE-2, G1DE-1, CPL, and Λ CDM controls. This is a completed medium-prior production-refined audit; full survey-grade validation remains necessary.

A Notation and response hierarchy

This appendix collects the main symbols. The purpose is to keep the response hierarchy explicit:

$$S_{\text{scr}}[X] \longrightarrow \omega = \eta_i \, dX^i \longrightarrow \eta_{\text{loc}} = \omega(\delta_A) \longrightarrow G_{\text{loc}} = \frac{c^3}{4\hbar} [\eta_{\text{loc}}]^{-1}.$$

The average density $\bar{\eta}(A)$, the thermodynamic density η_{eff} , and the local response density η_{loc} are related but not interchangeable.

Table 7: Notation table for FDS-G1 Core.

Symbol	Meaning	Role
$S_{\text{scr}}[X]$	screen entropy functional on quotient state space	primitive object
$\mathcal{X}_{\text{raw}}$	raw screen variables including gauge labels	pre-quotient bookkeeping
$\mathcal{X}_{\text{phys}}$	physical quotient state space $\mathcal{X}_{\text{raw}}/\mathcal{G}_{\text{scr}}$	domain of S_{scr}
$A(x)$	local area element or cell-count density	area direction
q_{AB}	screen shape metric	shape/shear sector
χ	occupancy or horizon residual fraction	dark/residual variable
Φ	flux load crossing screen	focusing source proxy
M	memory or hidden response mode	non-Markovian sector
ω	response one-form $\eta_i dX^i$	differential response object
η_{loc}	area component of ω	local inverse-coupling response density
$\eta_{\delta}(A)$	homogeneous differential density	finite-area response
η_{eff}	thermodynamic area-law density	interacting screen limit
G_{loc}	local effective coupling under horizon bridge; $G_{\text{loc}}^{-1} \propto \eta_{\text{loc}}$	inverse response stiffness
$\mathcal{H}_{ij}$	entropy response Hessian	stability/fluctuation metric
θ	logarithmic screen expansion	capacity-flow variable
σ_{AB}	traceless shape flow	shear-like dissipation
Ξ_{res}	residual forcing in capacity flow	nonintegrable/stochastic sector
$\mathcal{R}_{\mu\nu}$	covariant residual representative	metric closure defect
φ	scalar-capacity field $\ln(\eta/\eta_0)$	EFT slow mode

B Detailed claim map: derived, bridged, and modeled

The word “derivation” is local to a specified model. A screen-cell entropy formula may be derived in a toy model, while an Einstein-like equation remains a conditional bridge unless all geometric inputs are derived internally.

Table 8: Detailed claim map.

Statement	Status	Reason
$S_A/k_B = (A/a_0) \ln m$	toy derivation	follows from independent finite cells
$S_A/k_B = \eta_{\text{eff}} A + o(A)$	statistical universality claim	requires screen-local finite-state noncritical assumptions
$\eta_{\text{loc}} = k_B^{-1} \delta S / \delta A$	definition	area component of response one-form
$G_{\text{loc}} = (c^3/4\hbar)[\eta_{\text{loc}}]^{-1}$	conditional bridge	requires horizon entropy normalization
$\dot{\theta} = -a\theta^2 - b\sigma^2 - c\Phi + \Xi$	normal-form derivation	follows from low-order stochastic transition assumptions
$\Phi \leftrightarrow R_{\mu\nu} k^\mu k^\nu$	geometry bridge	standard local-screen/Raychaudhuri identification

Statement	Status	Reason
Einstein-like closure	conditional bridge	needs all-null integrability and Ward/Bianchi conservation
$\varphi = \ln(\eta/\eta_0)$ scalar EFT	model	conservative scalar-tensor realization of capacity mode
G1DE occupancy residual	phenomenological model	tests correlated back-ground/growth/lensing deviations
Holographic dictionary	compatibility bridge	interprets $1/(4G_N)$ and central charge as response density

C Expanded entropy-response toy model

The simple two-variable model illustrates how finite-area corrections, occupancy residuals, Hessian stability, and apparent nonintegrability fit into one framework. Let

$$s(A, \chi) = \eta_0 A + \alpha \ln(A/A_0) + \beta A \chi - \frac{\lambda}{2} \chi^2. \quad (56)$$

The area and occupancy responses are

$$\eta_A = \eta_0 + \frac{\alpha}{A} + \beta \chi, \quad \eta_\chi = \beta A - \lambda \chi. \quad (57)$$

The homogeneous local coupling is

$$G_A(A, \chi) = \frac{c^3}{4\hbar} \left[\eta_0 + \frac{\alpha}{A} + \beta \chi \right]^{-1}. \quad (58)$$

Thus finite-area corrections and occupancy residuals enter the same response coefficient.

The Hessian is

$$\mathcal{H} = - \begin{pmatrix} \partial_A^2 s & \partial_A \partial_\chi s \\ \partial_\chi \partial_A s & \partial_\chi^2 s \end{pmatrix} = \begin{pmatrix} \alpha/A^2 & -\beta \\ -\beta & \lambda \end{pmatrix}. \quad (59)$$

Local stability in this sector requires

$$\alpha/A^2 > 0, \quad \frac{\alpha\lambda}{A^2} - \beta^2 > 0. \quad (60)$$

If the area-occupancy coupling is too strong, the response Hessian loses positivity. In a physical model this signals instability, a missing stabilizing variable, or the need for a different coarse-grained description.

Now add a hidden memory variable

$$s(A, \chi, M) = \eta_0 A + \alpha \ln(A/A_0) + \beta A \chi + \gamma A M - \frac{\lambda}{2} \chi^2 - \frac{\mu}{2} M^2. \quad (61)$$

On (A, χ, M) the response one-form is exact. If the observed coarse variables are only (A, χ) but M follows a history-dependent relation $M = M[A, \chi; \text{path}]$, the reduced response is generally not exact. The apparent curl is then a diagnostic of projection, memory, or genuine residual physics. This is the simplest mathematical model for the epistemic/physical residual distinction.

D Three levels of integrability

The term “integrability” appears in three distinct senses.

Level I: entropy-potential integrability. This is the condition

$$\omega = ds_{\text{scr}}, \quad d\omega = 0. \quad (62)$$

It asks whether the chosen screen response variables are sufficient to define an entropy potential. Failure at this level often indicates missing coarse variables or memory.

Level II: metric all-null integrability. This asks whether local null response data can be represented by a symmetric tensor field,

$$\mathcal{F}_p(k) = E_{\mu\nu}(p)k^\mu k^\nu. \quad (63)$$

All-null data determines only the trace-free part, because $g_{\mu\nu}k^\mu k^\nu = 0$. The undetermined trace becomes an integration freedom, represented in Einstein gravity by the cosmological constant sector.

Level III: Ward/Bianchi dynamical integrability. This asks whether the tensor envelope satisfies a covariant conservation identity,

$$\nabla^\mu E_{\mu\nu} = 0. \quad (64)$$

Residual sectors are admissible only if the total effective source is conserved,

$$\nabla^\mu (T_{\mu\nu} + \mathcal{R}_{\mu\nu}) = 0. \quad (65)$$

These three notions should not be conflated. A closed entropy response one-form does not by itself prove metric geometry; metric all-null reconstruction does not by itself prove dynamics; and Bianchi closure is a further gauge/Ward condition.

E Open problems

FDS-G1 Core leaves eight major scientific tasks ordered by dependency:

1. **Finite ledger microstructure.** Construct consistent microscopic realizations of the $\mathcal{L}_{\text{scr}}$ tuple that reproduce area-extensive entropy and local response.
2. **Causal measurement protocol.** Formalize the complete empirical access map from screen ledger transitions to observational records.
3. **Retarded response kernel derivation.** Derive the constrained kernel form from first-principles ledger update rules.
4. **Modular susceptibility and 3/4 law.** Prove the modular-susceptibility origin of the 3/4 projection coefficient.
5. **Modular origin of $R_b^H(a)$.** Derive the horizon-occupancy output shape from modular fluctuation spectra.
6. **Early-universe compatibility.** Verify that residual kernels satisfy BBN, recombination, and CMB constraints without fine tuning.
7. **D0 finite-index / boost modular calibration.** Complete the conditional derivation of the Bekenstein-Hawking normalization from modular anomaly inflow.
8. **Production empirical validation.** A matched medium-prior production-refined nested-evidence comparison (G1fit-real) has been executed, and a KiDS-1000 shear-only diagnostic stress test has been completed. Remaining tasks are full prior-family stress testing, full 3×2 pt lensing including galaxy-galaxy lensing and clustering, noisy mock ensembles, CMB/BBN/nonlinear compatibility, and independent pipeline replication.

These are not defects in the stated bridge program. They are the natural next steps if the program is to be promoted from finite-capacity bridge to a candidate microscopic account of gravity-like geometry.

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* 7, 2333 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* 43, 199 (1975).
- [3] A. Raychaudhuri, “Relativistic cosmology. I,” *Phys. Rev.* 98, 1123 (1955).
- [4] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* 75, 1260 (1995).
- [5] D. Lovelock, “The Einstein tensor and its generalizations,” *J. Math. Phys.* 12, 498 (1971).
- [6] S. Weinberg, “Photons and gravitons in S-matrix theory,” *Phys. Rev.* 135, B1049 (1964).
- [7] S. Deser, “Self-interaction and gauge invariance,” *Gen. Rel. Grav.* 1, 9 (1970).
- [8] G. W. Horndeski, “Second-order scalar-tensor field equations in a four-dimensional space,” *Int. J. Theor. Phys.* 10, 363 (1974).
- [9] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *JCAP* 07, 050 (2014).
- [10] J. M. Maldacena, “The large N limit of superconformal field theories and supergravity,” *Adv. Theor. Math. Phys.* 2, 231 (1998).
- [11] S. Ryu and T. Takayanagi, “Holographic derivation of entanglement entropy from AdS/CFT,” *Phys. Rev. Lett.* 96, 181602 (2006).
- [12] T. Faulkner, A. Lewkowycz, and J. Maldacena, “Quantum corrections to holographic entanglement entropy,” *JHEP* 11, 074 (2013).
- [13] J. M. Ezquiaga and M. Zumalacarregui, “Dark energy after GW170817: dead ends and the road ahead,” *Phys. Rev. Lett.* 119, 251304 (2017).
- [14] E. Verlinde, “On the origin of gravity and the laws of Newton,” *JHEP* 04, 029 (2011).
- [15] T. Padmanabhan, “Thermodynamical aspects of gravity: new insights,” *Rep. Prog. Phys.* 73, 046901 (2010).
- [16] A. Antiba, “Finite-capacity thermodynamics of causal horizons,” *Entropy* 28, 557 (2026).
- [17] R. Bousso, “A covariant entropy conjecture,” *JHEP* 07, 004 (1999) [hep-th/9905177].
- [18] J. D. Brown and M. Henneaux, “Central charges in the canonical realization of asymptotic symmetries: an example from three-dimensional gravity,” *Commun. Math. Phys.* 104, 207 (1986).
- [19] V. Iyer and R. M. Wald, “Some properties of the Noether charge and a proposal for dynamical black hole entropy,” *Phys. Rev. D* 50, 846 (1994).
- [20] T. Jacobson, “Entanglement equilibrium and the Einstein equation,” *Phys. Rev. Lett.* 116, 201101 (2016) [arXiv:1505.04753].
- [21] F. Barzi, H. El Moumni, and K. Masmar, “Modified Unruh thermodynamics in emergent gravity: finite heat capacity and Rényi entropy,” *Phys. Rev. Research* (2026), arXiv:2509.03470.
- [22] P. Nusbaumer, “Relative-entropy variational principle for semiclassical gravity with finite-resolution boundaries,” *Preprints* (2026), doi:10.20944/preprints202601.1205.v4.

- [23] K. Song, “Spacetime dynamics and local entropy balance on causal horizons,” arXiv:2512.17961 (2025).
- [24] A. Ahmad and D. Klinger, “Emergent geometry from quantum probability,” Phys. Rev. D 110, 126004 (2024), arXiv:2411.07288.
- [25] G. Bianconi, “Gravity from entropy,” arXiv:2408.14391 (2024).
- [26] M. Abdul Karim et al. (DESI Collaboration), “DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations,” arXiv:2503.14738 (2025).
- [27] A. G. Adame et al. (DESI Collaboration), “DESI 2024 III: Baryon Acoustic Oscillations from Galaxies and Quasars,” JCAP 02, 021 (2025), arXiv:2404.03000.
- [28] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), <https://doi.org/10.5281/zenodo.20382013>.
- [29] Y. Wu, “FDS-T1: Finite Distinguishability Budgets and Maintenance Bounds for Physical Observers,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20234249>.
- [30] Y. Wu, “FDS-T2: Effective Geometry as Horizon Boundary Accounting,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20284911>.
- [31] Y. Wu, “FDS-T3: Capacity Overflow, Effective Stochasticity, and Phase-B Invariants,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20250367>.
- [32] Y. Wu, “FDS-X1: Horizon-Maintenance Dark Energy and Horizon Ledger Occupancy (Pre-Euclid Pre-registration),” Zenodo (2026), <https://doi.org/10.5281/zenodo.20290215>.
- [33] Y. Wu, “FDS-X2 v2: Minimal Nonzero CP/T Orientation and Full Weak-Sector Consistency,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20289955>.
- [34] Y. Wu, “FDS-X3: Functional Decomposition of the Four Fundamental Interactions,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20274519>.
- [35] Y. Wu, “D0: Screen Modular Anomaly Inflow and the Einstein Calibration of Finite Access Entropy,” FDS-G1 technical note (2026).

Companion A: Finite Screen Statistical Mechanics

Independent Cells, Interacting Screens, Differential Capacity, and the Statistical Substrate of Screen Stiffness

v1.2 Body-Integrated Release

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This companion develops the statistical-mechanical substrate used by the FDS-G1 entropy-response geometry program. Its purpose is to show how finite screen-local distinguishability produces area-extensive entropy, how interactions renormalize the capacity density, and why the differential response

$$\eta_\delta(A) = \frac{d(S/k_B)}{dA}$$

rather than the average density alone controls local response. We begin with independent finite cells, for which

$$S_A/k_B = (A/a_0) \ln m.$$

We then formulate a screen area-law universality class: finite-state, screen-local systems with finite-range interactions and finite correlation length have leading entropy proportional to screen area, with subleading boundary, topological, logarithmic, or finite-size corrections. We give fixed-occupancy thermodynamics, an Ising-like interacting screen example, and a finite-size correction model that separates the average density

$$\bar{\eta}(A) = \frac{S(A)/k_B}{A}$$

from the marginal density

$$\eta_\delta(A) = \frac{d(S/k_B)}{dA}.$$

Under the standard horizon-entropy bridge, the latter determines the effective local coupling $G_\delta \propto \eta_\delta^{-1}$, while $G_\delta^{-1} \propto \eta_\delta$ is the inverse coupling or screen stiffness,

$$G_\delta(A) = \frac{c^3}{4\hbar} [\eta_\delta(A)]^{-1}, \quad G_\delta^{-1}(A) = \frac{4\hbar}{c^3} \eta_\delta(A).$$

The results supply the technical statistical foundation for the flagship paper *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*.

1 v1.2 Ledger-Geometry Alignment

Companion A supplies statistical realizations of the finite ledger object $\mathcal{L}_{\text{scr}}$. The independent-cell basis is an operational statistical chart, not a unique microscopic atomistic ontology.

2 Statistical Scope and Claim Status

This companion supplies the statistical-mechanical substrate for the FDS-G1 entropy-response framework. Its purpose is to establish when finite screen-local systems support an area-extensive entropy and why the marginal capacity density

$$\eta_\delta(A) = \frac{d(S/k_B)}{dA}$$

is the response variable relevant for local screen stiffness under the standard horizon bridge. The companion derives area-extensive entropy in explicit toy models, formulates a finite-correlation area-law universality class, distinguishes average from differential capacity, and identifies the conditions under which the statistical substrate feeds the full G1 response chain. The finite-capacity language follows the broader FDS formal core [12] and the observer-relative distinguishability budgets developed in FDS-T1 [13], but the present companion specializes those ideas to screen-local statistical mechanics.

The G1 program does not require that every finite-capacity screen system produces gravity-like coupling. It only requires that gravitational screens (horizons, holographic boundaries, causal diamonds) belong to a finite-capacity, area-extensive universality class under the standard horizon/local-screen bridge.

The role of this companion is summarized by the chain

$$\text{finite screen cells} \Rightarrow S_{\text{scr}}[A] \sim k_B \eta A \Rightarrow \eta_\delta(A) = \frac{d(S/k_B)}{dA} \Rightarrow G_\delta(A) = \frac{c^3}{4\hbar} [\eta_\delta(A)]^{-1}, \quad (1)$$

$$G_\delta^{-1}(A) = \frac{4\hbar}{c^3} \eta_\delta(A), \quad (\text{inverse coupling / stiffness}) \quad \text{under the standard horizon bridge.} \quad (2)$$

In the full G1 program, the statistical results here feed into the entropy-response geometry chain

$$S_{\text{scr}}[X] \rightarrow \omega = \eta_i(X) dX^i \rightarrow \eta_{\text{loc}}(x) = \omega(\delta_A(x)) \rightarrow G_{\text{loc}}(x) \rightarrow \theta\text{-flow} \rightarrow \text{metric closure / residuals.} \quad (3)$$

3 Notation and claim status

4 Finite screen state space

Definition 1 (finite screen). *A finite distinction screen is an effective codimension-two screen patch Σ_A with area A and a finite set of screen-local degrees of freedom. In the simplest discretized model,*

$$N_A = \left\lfloor \frac{A}{a_0} \right\rfloor \quad (4)$$

counts the number of effective screen cells.

Table 1: Basic notation.

Symbol	Meaning	Role
A	screen area	extensive screen measure
a_0	effective area per cell	microscopic or coarse-grained scale
N_A	number of screen cells	$N_A \simeq A/a_0$
m	states per independent cell	finite distinguishability
S_A	screen entropy	$S_A = k_B \ln \Omega_A$
η_{nat}	nats per area	$S/k_B = \eta A$
η_{bit}	bits per area	$\eta_{\text{nat}} = \eta_{\text{bit}} \ln 2$
$\bar{\eta}(A)$	average density	$(S/k_B)/A$
$\eta_\delta(A)$	differential density	$d(S/k_B)/dA$
$G_\delta(A)$	response coupling; $G_\delta^{-1} \propto \eta_\delta$	$c^3/(4\hbar)[\eta_\delta]^{-1}$ under bridge

Table 2: Claim status in this companion.

Claim	Status	Section
Independent cells give $S \propto A$	toy theorem	5
Fixed occupancy gives $dE = TdS$	thermodynamic identity in toy model	6
Finite-range interacting screens have leading area-extensive entropy	statistical bridge / universality claim	7
η_δ controls local response	G1 response principle	8
$G_\delta = c^3/(4\hbar)[\eta_\delta]^{-1}$ and $G_\delta^{-1} \propto \eta_\delta$	conditional bridge using standard horizon normalization	9
Log/finite-size corrections generate $G_\delta(A)$ corrections	effective model	10

Definition 2 (screen-locality). *Screen-locality means that the active degrees of freedom counted in S_{scr} live on the screen or on an effective boundary/code subspace associated with the screen, not in an uncontrolled bulk volume. Screen-locality is the assumption that makes an area law possible.*

The cell model should be read as an effective statistical representation. It does not yet identify the cells as fundamental atoms of spacetime. They may represent horizon microstates, entanglement-cut channels, boundary-code subspace units, or coarse-grained causal-screen distinguishability cells.

Remark 1 (Relation to the Screen Ledger Representation Theorem). *The state-count formalism ties directly to the Core’s Screen Ledger Representation Theorem (Core, Theorem 1):*

$$N_A = |\mathcal{C}_{\Sigma_A}|, \quad S_A = k_B \log N_A.$$

The independent-cell model gives $N_A = m^{A/a_0}$ as a simple realization of the finite-state-count ledger. The logarithmic form is not an empirical fit—it is the unique additive coordinate of finite screen capacity under the theorem’s additivity, multiplicativity, and coarse-graining monotonicity assumptions. The independent-cell model is a toy derivation; the representation theorem explains why the logarithm is the correct ledger coordinate.

Theorem 1 (Area-Law Universality Class). *For screen-local finite-state systems with finite interaction range, finite correlation length, no dominant hidden bulk-volume degrees of freedom, and stable screen-local coarse-graining,*

$$S_A/k_B = \eta_{\text{eff}} A + S_{\text{sub}}(A), \quad S_{\text{sub}}(A)/A \rightarrow 0.$$

Failure cases. *The area law is demoted under: critical long-range correlations; infinite local Hilbert spaces; hidden bulk volume modes; dominant nonlocal constraints; failure of screen-local coarse-graining. These are not small corrections—they are demotion cases for the entire statistical substrate.*

5 Independent finite cells

The minimal model has N_A independent cells, each with m distinguishable states. The number of microstates is

$$\Omega_A = m^{N_A}. \quad (5)$$

For $A \gg a_0$, ignoring the floor correction,

$$N_A \simeq \frac{A}{a_0}. \quad (6)$$

Proposition 1 (independent cells give an area law). *For independent m -state screen cells,*

$$\frac{S_A}{k_B} = \ln \Omega_A = \frac{A}{a_0} \ln m. \quad (7)$$

Thus

$$\eta_{\text{nat}} = \frac{\ln m}{a_0}, \quad \eta_{\text{bit}} = \frac{\log_2 m}{a_0}. \quad (8)$$

Proof. Since $\Omega_A = m^{A/a_0}$, one has

$$S_A = k_B \ln \Omega_A = k_B \frac{A}{a_0} \ln m. \quad (9)$$

□

Figure 1 shows the linear dependence of S_A/k_B on A/a_0 for several m .

5.1 Matching the standard horizon coefficient

The standard Bekenstein-Hawking entropy is

$$\frac{S_H}{k_B} = \frac{A}{4\ell_P^2} = \frac{c^3 A}{4G\hbar}. \quad (10)$$

If the independent-cell density is matched to this coefficient, then

$$\frac{\ln m}{a_0} = \frac{1}{4\ell_P^2}, \quad (11)$$

and hence

$$a_0 = 4\ell_P^2 \ln m. \quad (12)$$

For binary cells, $m = 2$, the effective area per bit is

$$a_{\text{bit}} = 4\ell_P^2 \ln 2. \quad (13)$$

This is a matching relation, not a first-principles derivation of the Bekenstein-Hawking coefficient.

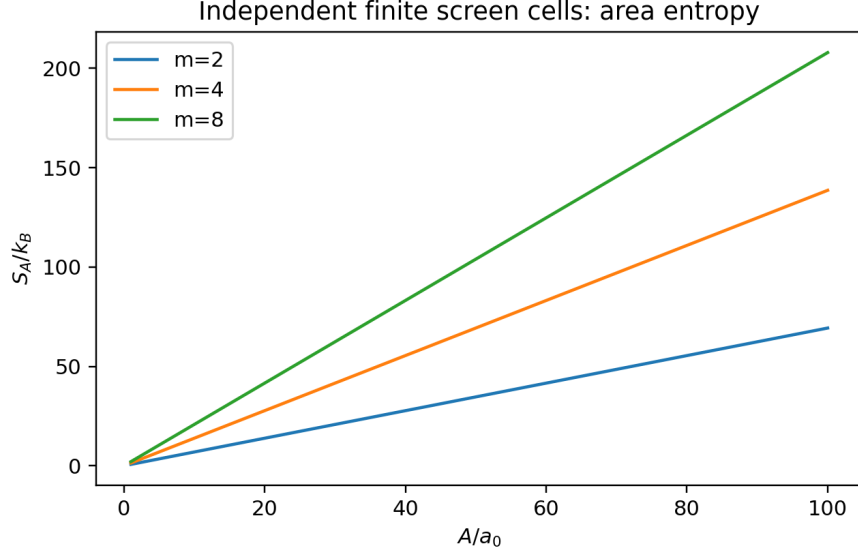


Figure 1: Independent finite screen cells produce exactly area-extensive entropy.

6 Fixed-occupancy screen thermodynamics

The independent m -state model explains area-extensive entropy but does not yet include energetic occupancy. A minimal thermodynamic screen has binary cells

$$\sigma_i \in \{0, 1\}, \quad (14)$$

with n active cells out of N . The number of configurations is

$$\Omega(N, n) = \binom{N}{n}. \quad (15)$$

Let

$$p = \frac{n}{N}. \quad (16)$$

Using Stirling's approximation,

$$\frac{S}{k_B N} \circ H_2(p) = -p \ln p - (1-p) \ln(1-p). \quad (17)$$

If each active distinction carries energy ϵ , then

$$E = \epsilon n. \quad (18)$$

The microcanonical temperature is defined by

$$\frac{1}{T} = \frac{\partial S}{\partial E} = \frac{1}{\epsilon} \frac{\partial S}{\partial n}. \quad (19)$$

For $0 < p < 1/2$, this yields the positive-temperature branch

$$T(p) = \frac{\epsilon}{k_B \ln[(1-p)/p]}. \quad (20)$$

Proposition 2 (screen first-law identity). *For the fixed-occupancy model with $E = \epsilon n$ and $S = k_B \ln \Omega(N, n)$, the thermodynamic definition of T implies*

$$dE = T dS \quad (21)$$

for quasi-static occupancy updates.

Figures 2 and 3 show the entropy density, positive-temperature branch, and finite- N convergence.

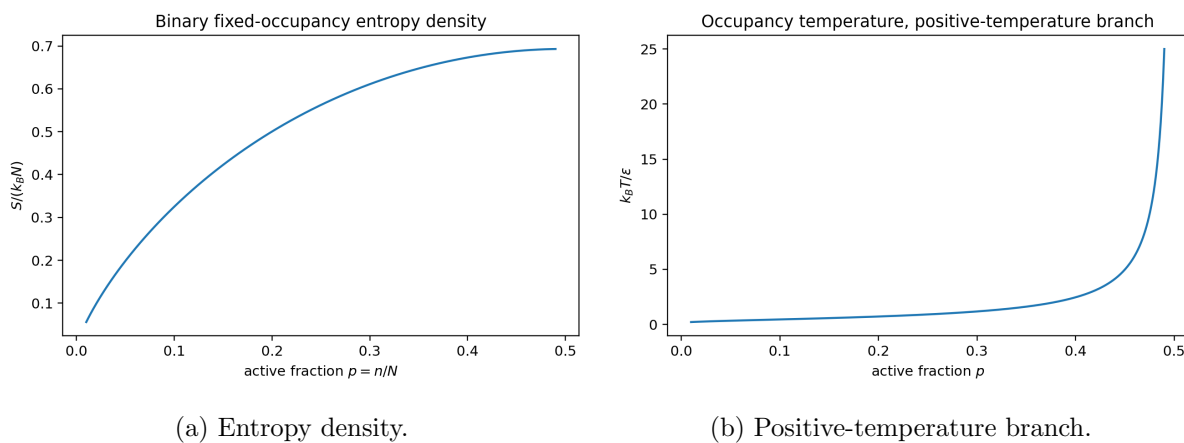


Figure 2: Binary fixed-occupancy thermodynamics.

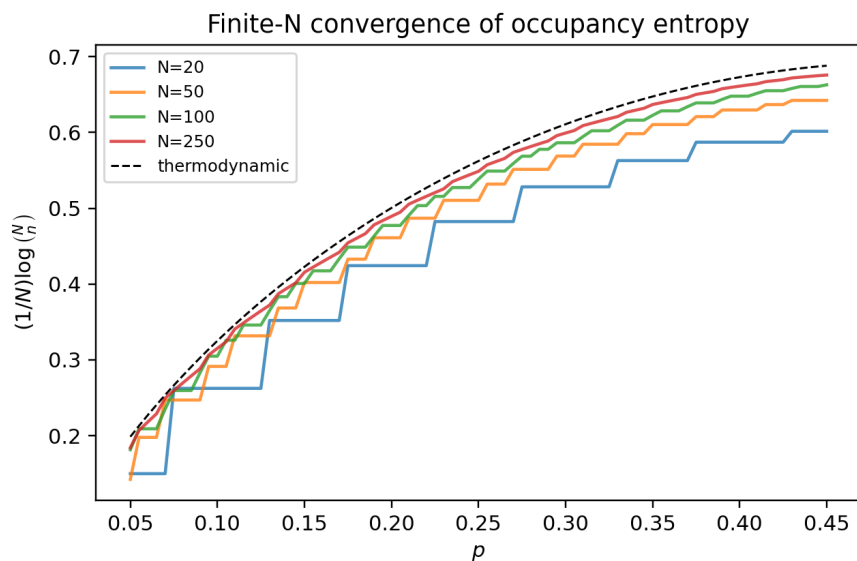


Figure 3: Finite- N convergence of $(1/N) \log \binom{N}{n}$ to the binary entropy function.

7 Interacting screen cells and area-law universality

The independent model is intentionally simple. A more robust framework must allow local interactions and coarse-graining. Let a screen have cell variables

$$\sigma_i \in \mathcal{S}, \quad |\mathcal{S}| = m, \quad (22)$$

with local Hamiltonian

$$H_{\text{scr}}[\sigma] = \sum_i h_i(\sigma_i) + \sum_{\langle ij \rangle} J_{ij}(\sigma_i, \sigma_j) + \cdots, \quad (23)$$

where interactions are finite range on the screen. The partition function is

$$Z_A(\beta) = \sum_{\{\sigma_i\}} e^{-\beta H_{\text{scr}}[\sigma]}. \quad (24)$$

The entropy is

$$S_A = k_B (\ln Z_A + \beta \langle H_{\text{scr}} \rangle). \quad (25)$$

Definition 3 (screen area-law universality class). *A finite screen model belongs to the G1 screen area-law universality class if:*

1. *degrees of freedom are screen-local;*
2. *each local cell has finite state space;*
3. *interactions are finite range or sufficiently fast decaying on the screen;*
4. *the state is non-critical, with finite correlation length;*
5. *no hidden bulk-volume degrees of freedom dominate the entropy;*
6. *coarse-graining preserves screen-locality.*

Claim 1 (Area extensivity under finite correlation length). *For a screen-local finite-state system in the above universality class,*

$$\frac{S_A}{k_B} = \eta_{\text{eff}} A + S_{\text{sub}}(A), \quad (26)$$

with

$$\eta_{\text{eff}} = \lim_{A \rightarrow \infty} \frac{S_A/k_B}{A} \quad (27)$$

finite, and $S_{\text{sub}}(A) = o(A)$ for non-critical short-range phases. This is a statistical bridge statement, not a proven theorem from first principles. It relies on the standard large-volume thermodynamic-limit existence conditions for finite-state, finite-range systems, adapted to the screen-local setting.

Motivation. Finite state space bounds the entropy per cell. Screen-local finite-range interactions imply that a thermodynamic entropy density exists in the large-screen limit under standard subadditivity and finite-correlation assumptions. Since the number of screen cells scales as $N_A \sim A/a_0$, the leading entropy is extensive in A , not in a bulk volume. Correlations and constraints renormalize the entropy density and generate subleading terms. $\square$

7.1 Boundary, critical, and topological corrections

The leading area term may be corrected by

$$\frac{S_A}{k_B} = \eta_{\text{eff}} A + \alpha \ln(A/a_0) + \gamma + C_{\partial} \sqrt{A/a_0} + O(A^{-p}). \quad (28)$$

The interpretation of these terms depends on the model:

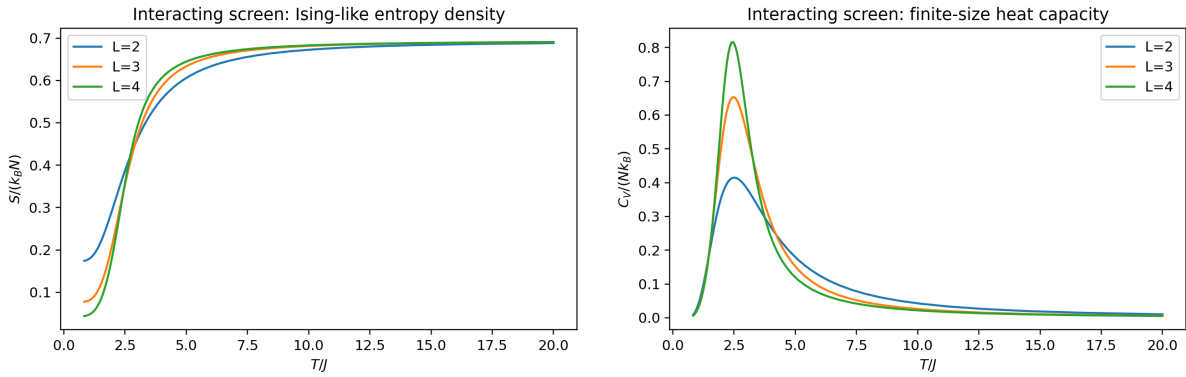
- $C_{\partial} \sqrt{A/a_0}$ can arise from a boundary of the screen patch.
- $\alpha \ln A$ can arise near criticality, from constraints, or from finite-size corrections.
- γ may encode topological or global constraint information.

7.2 Ising-like screen example

An illustrative interacting model has

$$\sigma_i = \pm 1, \quad H = -J \sum_{\langle ij \rangle} \sigma_i \sigma_j. \quad (29)$$

Exact enumeration on small periodic $L \times L$ screens shows a finite entropy density $S/(k_B N)$, with finite-size heat-capacity structure. This example is not proposed as a horizon microstate model; it illustrates how interactions renormalize the capacity density without destroying area extensivity.



(a) Entropy density.

(b) Finite-size heat capacity.

Figure 4: Exact enumeration of small Ising-like interacting screens.

8 Average and differential capacity

The G1 core theory distinguishes two capacities.

Definition 4 (average capacity density). *The average density is*

$$\bar{\eta}(A) = \frac{S(A)/k_B}{A}. \quad (30)$$

It measures the total stored screen capacity per unit area.

Definition 5 (differential capacity density). *The differential density is*

$$\eta_\delta(A) = \frac{d(S/k_B)}{dA}. \quad (31)$$

It measures the marginal entropy-area response of a quasi-static screen update.

Principle 1 (Which capacity feeds local stiffness). *Only the differential capacity $\eta_\delta(A)$, not the average capacity $\bar{\eta}(A)$, enters local screen stiffness:*

$$G_\delta(A) = \frac{c^3}{4\hbar} [\eta_\delta(A)]^{-1}.$$

The average capacity controls the total entropy budget; the differential capacity controls the local gravitational response. This distinction is essential for understanding why finite-area corrections appear in total entropy but are suppressed in local coupling (Core, §6).

For a strictly area-extensive entropy

$$S(A)/k_B = \eta_0 A, \quad (32)$$

one has

$$\bar{\eta}(A) = \eta_\delta(A) = \eta_0. \quad (33)$$

With subleading terms, the two differ.

Principle 2 (local response principle). *In a quasi-static local screen update, the relevant inverse coupling (stiffness) variable is the differential response,*

$$\delta S = k_B \eta_\delta(A) \delta A, \quad (34)$$

not only the average stored capacity.

This is the homogeneous version of the local response formula used in the G1 core paper,

$$\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}. \quad (35)$$

For a uniform screen, $\eta_{\text{loc}} \rightarrow \eta_\delta(A)$.

9 Effective coupling under the horizon bridge

Under the standard horizon-entropy bridge,

$$\frac{S_H}{k_B} = \frac{c^3 A}{4G\hbar}, \quad (36)$$

the area response can be read as

$$\eta_{\text{BH}} = \frac{c^3}{4G\hbar}. \quad (37)$$

Therefore, for a screen with marginal capacity density $\eta_\delta(A)$, the corresponding response coupling and its inverse (stiffness) are

$$G_\delta(A) = \frac{c^3}{4\hbar} [\eta_\delta(A)]^{-1}, \quad G_\delta^{-1}(A) = \frac{4\hbar}{c^3} \eta_\delta(A). \quad (38)$$

This is a conditional bridge, not a first-principles derivation of G . It states that if the standard horizon/local-screen normalization is used, then the inverse gravitational stiffness (G^{-1}) is read as marginal screen distinguishability density, and the coupling G is inversely proportional to it.

10 Finite-size corrections and running response

Consider the effective entropy model

$$\frac{S(A)}{k_B} = \eta_0 A + \alpha \ln(A/a_0) + \gamma + b(A/a_0)^{-p}. \quad (39)$$

Then

$$\bar{\eta}(A) = \eta_0 + \frac{\alpha \ln(A/a_0) + \gamma + b(A/a_0)^{-p}}{A}, \quad (40)$$

whereas

$$\eta_\delta(A) = \eta_0 + \frac{\alpha}{A} - \frac{bp}{a_0}(A/a_0)^{-p-1}. \quad (41)$$

In units $a_0 = 1$, this becomes

$$\eta_\delta(A) = \eta_0 + \frac{\alpha}{A} - bpA^{-p-1}. \quad (42)$$

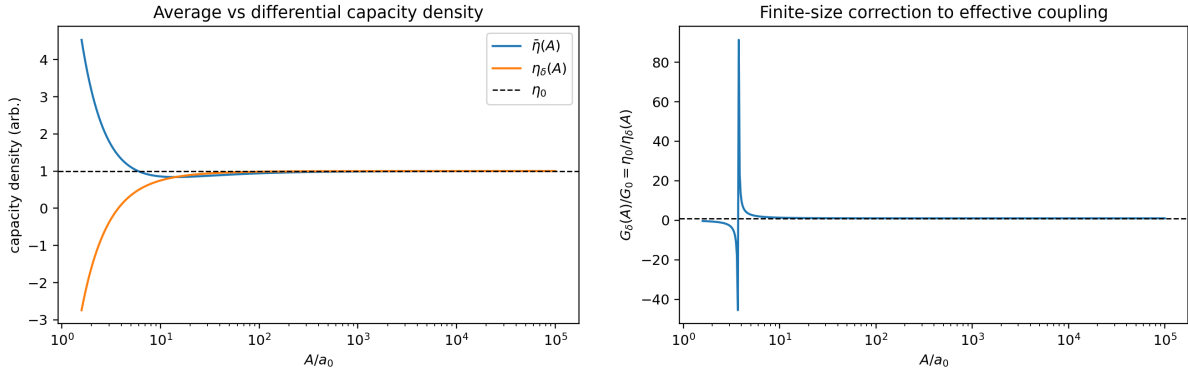
The response coupling is

$$\frac{G_\delta(A)}{G_0} = \frac{\eta_0}{\eta_\delta(A)}. \quad (43)$$

For large A ,

$$\frac{\Delta G}{G_0} \simeq -\frac{\alpha}{\eta_0 A} + O(A^{-p-1}, A^{-2}). \quad (44)$$

Thus macroscopic finite-area corrections are suppressed unless the screen is near cutoff scale, early-universe scale, or coupled to a genuine residual/capacity field.



(a) Average vs differential density.

(b) Response coupling correction.

Figure 5: Finite-size corrections separate stored capacity from marginal response.

11 Failure modes

The screen statistical foundation can fail in several controlled ways.

This failure registry is important: the G1 program does not require every screen-like system to be gravitational. It only requires that gravitational screens belong to a finite-capacity, area-extensive universality class under the standard horizon/local-screen bridge.

Table 3: Failure registry for finite screen statistics.

Failure condition	Consequence	What survives
Dominant bulk-volume degrees of freedom	leading entropy becomes volume-like	finite cell models may remain useful elsewhere
Infinite local state space without cutoff	η may diverge	need UV completion or constraint
Long-range critical correlations	log or anomalous scaling can dominate subleading terms	response framework survives with corrections
Strong nonlocal global constraints	simple density may fail	residual / topology sector needed
No standard horizon bridge	$G^{-1} \sim \eta$ interpretation fails	screen statistics remains independent result
Macroscopic G variation without closure	violates Bianchi/local gravity constraints	scalar-capacity or residual sector needed

12 Interfaces to other G1 companions

The results of this paper feed into the other G1 companion papers as follows.

Table 4: Interfaces with the rest of the G1 program.

Companion	Input from screen statistical mechanics
FDS-G1 Core / Finite Screen Spacetime	$S_{\text{scr}}[X], \eta_{\text{loc}} = (1/k_B)\delta S/\delta A$
Microtransition and Capacity Flow	$N = A/a_0, \theta = d \ln N/d\lambda$
Scalar-Capacity EFT	long-wavelength mode $\varphi = \ln(\eta/\eta_0)$
Likelihood and Falsification	finite-size $G_\delta(A)$ corrections and no-free- G drift
Horizon Occupancy Residuals	capacity fraction and occupancy variables χ_H
Holography and Horizons	$1/(4G_N)$ as screen capacity density

Remark 2 (Scope boundary). *Companion A supplies the finite-screen statistical substrate: S_{scr} , η_{eff} , and η_δ . By itself it does not imply null focusing, metric closure, Einstein-like equations, or cosmological residuals. Those bridges are treated in Companions B–F and in FDS-G1 Core. Companion A supplies statistical routes to an area-extensive finite screen density. D0 addresses the distinct calibration question: how such a density η_0 is related to boost modular energy and hence to gravitational stiffness G^{-1} . This separation prevents any misinterpretation that finite screen entropy alone derives gravity-like geometry. The area/differential capacity substrate provided here is extended to optical deformation sectors $A(1) \oplus S(2) \oplus \Omega(1)$ in companion notes on optical projection and screen influence, where the relative-entropy Hessian $K_{IJ} = \partial_I \partial_J S_{\text{rel}}(\rho_S(q) \parallel \rho_S(0))|_{q=0}$ controls the compliance projection coefficient κ_{BW} . The matched medium-prior production-refined nested-evidence comparison (G1fit-real) now selects the projection-locked G1DE- $M_{3/4}$ model, which has further been stress-tested via independent KiDS-1000 shear-only data; Companion A does not predict these signs from the statistical-mechanical substrate alone.*

13 Screen Statistics and Distinguishability RG

The differential capacity density $\eta_\delta(A) = d(S/k_B)/dA$ varies with scale $\ell \sim \sqrt{A}$. One may define a distinguishability RG flow

$$\eta(\ell) = \frac{d(S_\ell/k_B)}{dA_\ell}, \quad \beta_\eta = \frac{d\eta}{d \ln \ell},$$

whose fixed points and relevant perturbations connect to the D8 metric-envelope scale calibration and the D9 obstruction classification. The full RG-scale treatment belongs to D8/D9 rather than the present companion.

14 Conclusion

This companion establishes the statistical-mechanical part of the G1 program. The central results are:

1. independent finite screen cells give $S_A/k_B = (A/a_0) \ln m$;
2. fixed occupancy gives an explicit screen thermodynamics with $dE = TdS$;
3. finite-state, finite-range, finite-correlation interacting screens have leading entropy extensive in screen area;
4. interactions and constraints renormalize the effective density η_{eff} ;
5. finite-size terms separate average density $\bar{\eta}$ from differential response η_δ ;
6. under the standard horizon bridge, $G_\delta(A) = c^3/(4\hbar)[\eta_\delta(A)]^{-1}$ and $G_\delta^{-1}(A) = 4\hbar\eta_\delta(A)/c^3$, so η_δ controls the inverse coupling (stiffness).

The paper therefore supplies the statistical substrate for the G1 core claim:

$$\text{gravity-like inverse coupling (stiffness) is marginal screen entropy-area response.} \quad (45)$$

A Numerical artifacts

The accompanying script `fds_g1A_screen_statistics.py` generates all figures and CSV files used in this paper:

- independent area entropy;
- binary occupancy entropy and temperature;
- finite- N occupancy convergence;
- exact enumeration of small Ising-like screens;
- finite-size differential-capacity corrections.

B Relation to standard references

This paper uses standard statistical mechanics and horizon-entropy conventions. The following references are relevant background, not claims of novelty.

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Physical Review D* 7, 2333 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Communications in Mathematical Physics* 43, 199 (1975).
- [3] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Physical Review Letters* 75, 1260 (1995).
- [4] W. Rindler, “Kruskal space and the uniformly accelerated frame,” *American Journal of Physics* 34, 1174 (1966).
- [5] L. Onsager, “Reciprocal relations in irreversible processes,” *Physical Review* 37, 405 (1931).
- [6] D. Ruelle, *Statistical Mechanics: Rigorous Results*, W. A. Benjamin (1969).
- [7] R. Bousso, “A covariant entropy conjecture,” *Journal of High Energy Physics* 07, 004 (1999).
- [8] S. Ryu and T. Takayanagi, “Holographic derivation of entanglement entropy from AdS/CFT,” *Physical Review Letters* 96, 181602 (2006).
- [9] T. Faulkner, A. Lewkowycz, and J. Maldacena, “Quantum corrections to holographic entanglement entropy,” *Journal of High Energy Physics* 11, 074 (2013).
- [10] V. Iyer and R. M. Wald, “Some properties of the Noether charge and a proposal for dynamical black hole entropy,” *Physical Review D* 50, 846 (1994).
- [11] A. Antiba, “Finite-capacity thermodynamics of causal horizons,” *Entropy* 28, 557 (2026).
- [12] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), <https://doi.org/10.5281/zenodo.20382013>.
- [13] Y. Wu, “FDS-T1: Finite Distinguishability Budgets and Maintenance Bounds for Physical Observers,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20234249>.

Companion B: Screen Transition and Response Dynamics

Stochastic Screen Transitions, Entropy-Gradient Response, and Raychaudhuri-like Capacity Flow
v1.2 Body-Integrated Release

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This companion paper develops the dynamical bridge used by the FDS-G1 Core framework. Companion A establishes that finite screen-local statistical systems can produce area-extensive entropy and a marginal capacity density. Here we ask how a finite screen state updates along a null or causal-screen parameter and why its hydrodynamic continuum limit naturally contains Raychaudhuri-like focusing terms. The primitive dynamical variable is the screen cell count $N(\lambda) = A(\lambda)/a_0$ and the logarithmic expansion $\theta = d \ln N / d\lambda$. We model screen evolution by a Markov or near-Markov transition process, derive a Kramers-Moyal/Fokker-Planck continuum limit, and show that, under locality, screen extensivity, affine parametrization, rotational invariance of shape modes, and positive flux focusing, the lowest-order deterministic normal form is

$$\frac{d\theta}{d\lambda} = -a\theta^2 - b\sigma_{\text{scr}}^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + \zeta_{\theta}.$$

The term $-\theta^2$ follows kinematically from logarithmic capacity expansion; the $-\sigma^2$ term is the lowest positive-definite shear dissipation scalar; and $-\Phi_{\text{flux}}$ represents positive flux load reducing expansion. This is a normal-form bridge: it shows why Raychaudhuri-like capacity-flow equations are stable hydrodynamic forms for stochastic finite-screen update dynamics, without deriving the geometric Raychaudhuri equation or Einstein gravity from microscopic cells. The standard geometric identification of Φ_{flux} with null focusing data remains an external local-screen bridge.

1 Dynamical Scope and Normal-Form Claim

This companion hardens the transition from finite screen update dynamics to a Raychaudhuri-like capacity-flow normal form. For a broad class of local finite-screen stochastic transition processes, the hydrodynamic evolution of logarithmic screen expansion has a lowest-order normal form with negative quadratic expansion, shear dissipation, positive flux focusing, and residual/noise terms.

The finite-capacity formalism and the treatment of effective stochasticity under capacity overflow follow the FDS formal core [7] and FDS-T3 [8].

The relation to standard gravity enters later through a bridge:

$$\Phi_{\text{flux}} \longleftrightarrow R_{\mu\nu} k^\mu k^\nu \longleftrightarrow \kappa T_{\mu\nu} k^\mu k^\nu,$$

which is established separately through the standard local-screen geometry bridge. This paper explains why the left-hand side has a stable capacity-flow structure; the geometric identification belongs to local screen thermodynamics and the broader FDS-G1 Core response-geometry framework.

1.1 Position within the G1 architecture

Caveat 1 (What this companion does and does not derive) *Companion B does not derive the geometric Raychaudhuri equation from microscopic cells. It derives the stable capacity-flow normal form from finite screen transition statistics. The geometric Raychaudhuri equation is recovered only after the Core's metricization and local-screen bridge assumptions are imposed. The constants a, b, c are normal-form parameters; they become $1/(D-2), 1, \kappa$ only in the metric-integrable equilibrium limit.*

The dependency chain is

$$\text{finite screen statistics} \longrightarrow S_{\text{scr}}[X] \longrightarrow \eta_{\text{loc}} \longrightarrow G_{\text{loc}} \longrightarrow \theta\text{-flow} \longrightarrow \text{metric/Ward closure}.$$

Companion A supplies the statistical substrate. This companion supplies the transition-to-flow bridge. Companion C supplies scalar-capacity effective field theory. Companion D supplies likelihood/falsification interfaces. Companion E treats horizon-occupancy residuals. Companion F treats horizon and holographic dictionaries.

2 Finite screen state and expansion

Let a local causal screen patch have area $A(\lambda)$ and effective cell area a_0 . Define

$$N(\lambda) = \frac{A(\lambda)}{a_0}. \quad (1)$$

The logarithmic screen expansion is

$$\theta(\lambda) = \frac{d}{d\lambda} \ln N(\lambda) = \frac{1}{A} \frac{dA}{d\lambda}. \quad (2)$$

This matches the usual expansion variable when the screen patch is identified with a cross-sectional area of a null congruence. In the present paper, however, θ is defined operationally as the relative rate of change of maintained screen cells.

A minimal macroscopic screen state is

$$X(\lambda) = (x, \theta, \sigma_{\text{scr}}, \chi, \Phi, M, \dots), \quad x = \ln N. \quad (3)$$

Here σ_{scr} denotes a coarse shear/shape-distortion amplitude, χ an occupancy variable, Φ a flux/load variable, and M possible memory or hidden screen modes.

Table 1: Claim status in this companion.

Claim	Meaning	Status
$N = A/a_0$ and $\theta = d \ln N / d \lambda$	Cell-count definition of screen expansion	Definition
Kramers-Moyal limit	Markov transition process yields drift/noise continuum variables	Standard stochastic bridge
$-\theta^2$ term	Kinematic term from logarithmic expansion	Toy theorem
$-\sigma^2$ term	Lowest positive-definite shear dissipation scalar	Normal-form argument
$-\Phi_{\text{flux}}$ term	Positive load focuses screen expansion	Physical sign assumption
Identification $R_{\mu\nu} k^\mu k^\nu$ with	Geometry bridge	Standard bridge / not derived
Residuals	Nonclosure, memory, noise, coarse-graining defects	Model-building sector

2.1 Gauge and parametrization cautions

The parameter λ may represent an affine null parameter in a geometric bridge, but at the screen-transition level it is simply an update parameter. Reparametrization changes the numerical value of θ unless an affine or local-equilibrium gauge is fixed. Therefore the normal form below is stated in an affine/local-equilibrium update gauge; non-affine parametrization introduces a linear term $\kappa_\lambda \theta$.

3 Stochastic transition process

Assume a Markov or near-Markov transition kernel

$$P(X_{\lambda+\Delta\lambda} | X_\lambda). \quad (4)$$

For the pair (x, θ) a minimal update is

$$\Delta x = \theta \Delta \lambda + O(\Delta \lambda^{3/2}), \quad (5)$$

$$\Delta \theta = A_\theta(X) \Delta \lambda + \sqrt{B_\theta(X) \Delta \lambda} \xi, \quad (6)$$

where ξ is a unit-variance random variable. The Kramers-Moyal expansion gives the Fokker-Planck equation

$$\partial_\lambda P = -\partial_i [A^i(X) P] + \frac{1}{2} \partial_i \partial_j [B^{ij}(X) P] + \dots \quad (7)$$

Equivalently, the Langevin form is

$$\dot{X}^i = A^i(X) + \zeta^i, \quad \langle \zeta^i(\lambda) \zeta^j(\lambda') \rangle = B^{ij}(X) \delta(\lambda - \lambda'). \quad (8)$$

For large screen cell number $N \gg 1$, path probabilities often admit a large-deviation form

$$P[X(\lambda)] \asymp \exp[-N \mathcal{I}_{\text{scr}}[X]], \quad (9)$$

where $\mathcal{I}_{\text{scr}}$ is the screen update action. Near local equilibrium one expects an Onsager-Machlup form

$$\mathcal{I}_{\text{scr}}[X] = \frac{1}{2} \int d\lambda (\dot{X}^i - F^i) D_{ij}^{-1} (\dot{X}^j - F^j). \quad (10)$$

If the drift is entropy-gradient driven,

$$F^i = L^{ij} \frac{\delta S_{\text{scr}}}{\delta X^j}, \quad (11)$$

then stochastic screen dynamics obeys a fluctuation-dissipation pattern, schematically $D^{ij} \propto k_B L^{ij}$ near equilibrium. This relation will later constrain residual noise.

4 Normal-form derivation of capacity flow

4.1 Defence: why this is not a copy of Raychaudhuri

The normal form

$$\dot{\theta} = -a\theta^2 - b\sigma_{\text{scr}}^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + \zeta_{\theta}$$

is not obtained by copying the geometric Raychaudhuri equation. It is obtained by asking what the lowest-order scalar drift of logarithmic screen-cell expansion can contain under the following assumptions:

- finite screen updates;
- Markov or near-Markov transition kernel;
- screen-extensive hydrodynamic limit ($N \gg 1$);
- affine or local-equilibrium update gauge;
- screen rotational invariance;
- positive flux focusing;
- finite- N noise scaling.

Each term has a distinct operational origin. The $-\theta^2$ term is kinematic (logarithmic expansion derivative, §4.2). The $-\sigma_{\text{scr}}^2$ term is the lowest positive-definite scalar dissipation term compatible with cross-section rotations. The $-\Phi_{\text{flux}}$ term is a sign assumption about positive load. The geometric Raychaudhuri equation is recovered only after an additional bridge that identifies Φ_{flux} with null focusing and fixes conventional coefficients.

Theorem 1 (Screen transition normal form) *Let a finite causal screen have cell count $N(\lambda) = A(\lambda)/a_0$ and logarithmic expansion $\theta = d \ln N / d\lambda$. Assume the screen update is governed by a Markov or near-Markov transition with finite cell noise, an affine or local-equilibrium gauge, screen rotational invariance, and positive flux focusing. Then, in the hydrodynamic limit $N \gg 1$, the lowest-order deterministic drift of θ is*

$$\boxed{\dot{\theta}_{\text{det}} = -a\theta^2 - b\sigma_{\text{scr}}^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + O(3)}, \quad a, b, c > 0. \quad (12)$$

The full Langevin equation including finite-cell stochastic noise is

$$\dot{\theta} = \dot{\theta}_{\text{det}} + \zeta_{\theta}, \quad \zeta_{\theta} = O(N^{-1/2}). \quad (13)$$

Here a is a positive coefficient that depends on dimension, expansion convention, and the affine bridge. It is not fixed by screen kinematics alone; calibration requires matching the transverse dimension $n = D - 2$ and the geometric convention for θ . The normal-form theorem fixes the allowed structure and signs, not the exact Raychaudhuri coefficients.

4.2 The universal kinematic term

The first robust term follows from logarithmic expansion. Since $\theta = \dot{x}$ and $x = \ln N$,

$$\dot{\theta} = \ddot{x} = \frac{\ddot{N}}{N} - \left(\frac{\dot{N}}{N} \right)^2. \quad (14)$$

If the cell-count acceleration has a regular local expansion

$$\frac{\ddot{N}}{N} = F_0(X), \quad (15)$$

then

$$\dot{\theta} = F_0(X) - \theta^2. \quad (16)$$

Thus a negative quadratic expansion term is forced by the use of logarithmic area/cell-count expansion. It is not inserted to imitate Raychaudhuri; it is a kinematic term.

In a more general convention, the coefficient is written $a > 0$:

$$\dot{\theta} = -a\theta^2 + \dots. \quad (17)$$

The precise coefficient depends on dimension, normalization of θ , and bridge conventions. In the null Raychaudhuri equation for a two-dimensional transverse screen ($n = D - 2 = 2$) in four spacetime dimensions the corresponding coefficient is $1/2$. The theorem leaves a as a free positive parameter because $-\theta^2$ is forced by kinematics ($\ddot{x} = \ddot{N}/N - \dot{x}^2$) but its exact coefficient relative to the geometric Raychaudhuri convention is a bridge calibration, not a screen-internal result.

4.3 Shape modes and shear dissipation

Let q_{AB} denote a coarse metric or shape tensor on the screen cross-section of dimension $n = D - 2$. Decompose its update into trace and traceless pieces:

$$\theta = \frac{1}{2} q^{AB} \dot{q}_{AB}, \quad \sigma_{AB} = \frac{1}{2} \dot{q}_{AB} - \frac{1}{n} \theta q_{AB}, \quad q^{AB} \sigma_{AB} = 0. \quad (18)$$

This companion treats the twist-free screen-flow sector; nonzero twist is an extension or failure channel. At lowest order, locality and rotational invariance allow the positive scalar

$$\sigma_{\text{scr}}^2 = \sigma_{AB}\sigma^{AB}. \quad (19)$$

If shear converts coherent expansion into irreversible shape distortion or unresolved mixing, it enters with negative sign:

$$\dot{\theta} \supset -b\sigma_{\text{scr}}^2, \quad b > 0. \quad (20)$$

This is a normal-form statement: it is the lowest positive-definite scalar dissipation term compatible with screen rotations.

4.4 Flux focusing

Let Φ_{flux} be the positive scalar load carried by energy, information, or distinguishability demand crossing the screen along the chosen generator. Its role is defined at three layers:

Screen-update level: Φ_{flux} is the rate of external load crossing or consuming screen distinguishability capacity.

Thermodynamic bridge: $\Phi_{\text{flux}} \sim \frac{\delta Q}{T \delta \lambda \delta A}$, relating flux to heat exchange per screen area per update step.

Geometric bridge: $\Phi_{\text{flux}} \leftrightarrow R_{\mu\nu}k^\mu k^\nu \leftrightarrow \kappa T_{\mu\nu}k^\mu k^\nu$, the standard local-screen identification with null focusing.

The dimensions are $[\Phi_{\text{flux}}] = [\lambda^{-2}]$. The focusing sign assumption is

$$\Phi_{\text{flux}} \geq 0 \implies \dot{\theta} \leq 0. \quad (21)$$

Thus

$$\dot{\theta} \supset -c\Phi_{\text{flux}}, \quad c > 0. \quad (22)$$

Under the standard local-screen bridge one may identify Φ_{flux} with either $T_{\mu\nu}k^\mu k^\nu$ or with the geometric focusing scalar $R_{\mu\nu}k^\mu k^\nu$ after using screen first-law relations. This identification is not part of the stochastic derivation; it is a physical bridge.

4.5 Residuals and noise

Unresolved memory, nonlocal constraints, stochastic finite-cell fluctuations, and coarse-graining defects enter as

$$\Xi_{\text{res}} + \zeta_\theta. \quad (23)$$

If the process is near equilibrium and Markovian, the noise covariance is local. For finite cell count $N \gg 1$, the noise amplitude scales as

$$\langle \zeta_\theta(\lambda) \zeta_\theta(\lambda') \rangle = \frac{2D_\theta}{N} \delta(\lambda - \lambda'), \quad \zeta_\theta = O(N^{-1/2}), \quad (24)$$

so that deterministic macroscopic capacity flow is recovered as $N \rightarrow \infty$ and finite-cell stochasticity appears as a controlled $O(N^{-1/2})$ residual. If memory is present, a nonlocal kernel may appear:

$$\Xi_{\text{mem}}(\lambda) = \int^\lambda K(\lambda, \lambda') \Phi(\lambda') d\lambda'. \quad (25)$$

This case generally cannot be represented by a local screen potential without enlarging the state space.

4.6 Capacity-flow normal form

Collecting the allowed lowest-order terms gives

$$\boxed{\frac{d\theta}{d\lambda} = \kappa_\lambda \theta - a\theta^2 - b\sigma_{\text{scr}}^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + \zeta_\theta + \mathcal{O}(\theta^3, \theta\sigma^2, \nabla\Phi).} \quad (26)$$

Here $\kappa_\lambda \theta$ is a non-affine/reparametrization term. In the affine local-equilibrium gauge,

$$\kappa_\lambda = 0, \quad (27)$$

so the normal form becomes

$$\boxed{\dot{\theta} = -a\theta^2 - b\sigma_{\text{scr}}^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + \zeta_\theta.} \quad (28)$$

5 Relation to the geometric Raychaudhuri equation

For a hypersurface-orthogonal null congruence, the geometric Raychaudhuri equation is

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - R_{\mu\nu}k^\mu k^\nu, \quad (29)$$

up to twist terms and convention-dependent factors. The capacity-flow normal form matches its structure under the bridge

$$a \rightarrow \frac{1}{2}, \quad b\sigma_{\text{scr}}^2 \rightarrow \sigma_{\mu\nu}\sigma^{\mu\nu}, \quad c\Phi_{\text{flux}} \rightarrow R_{\mu\nu}k^\mu k^\nu. \quad (30)$$

This paper shows how stochastic finite-screen dynamics leads to a robust focusing-like normal form. The geometric bridge identifies this normal form with null congruence focusing when the response data are integrable into a Lorentzian metric envelope.

5.1 Raychaudhuri-Limit Matching

In the metric-integrable equilibrium limit, the normal-form coefficients must reduce to the geometric Raychaudhuri values:

$$a \rightarrow \frac{1}{D-2}, \quad b \rightarrow 1$$

in the standard geometric limit ($a \rightarrow 1/2$ for the convention-dependent factor in the geometric equation above). The remaining term

$$\mathcal{R}_{\text{scr}}^\theta = \Xi_{\text{res}} + \zeta_\theta + \mathcal{O}(3)$$

is the finite-screen correction. If these constants do not match, the model remains a finite-screen normal form but is not a GR-equilibrium recovery. The geometric Raychaudhuri equation is thus the metric-integrable fixed point of the broader capacity-flow normal form.

6 Integrability into response geometry

In FDS-G1 Core, the local area response is

$$\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}, \quad (31)$$

with local coupling

$$G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}, \quad G_{\text{loc}}^{-1}(x) = \frac{4\hbar}{c^3} \eta_{\text{loc}}(x). \quad (32)$$

The present companion adds transition dynamics. For the screen response data to define a metric geometry, the all-null focusing data must be representable as a quadratic form

$$\mathcal{F}_p(k) = E_{\mu\nu}(p) k^\mu k^\nu. \quad (33)$$

Since $g_{\mu\nu} k^\mu k^\nu = 0$ for null k , all-null data reconstructs $E_{\mu\nu}$ only up to trace terms. This is why a cosmological-constant-like integration freedom naturally appears in all-null closure arguments.

The dynamical consistency condition is the Ward/Bianchi closure

$$\nabla^\mu (T_{\mu\nu} + \mathcal{R}_{\mu\nu}) = 0, \quad (34)$$

where $\mathcal{R}_{\mu\nu}$ is the covariant representative of response nonintegrability, memory, stochasticity, or omitted screen modes.

6.1 Update Time, Modular Time, and Boost Time

The capacity-flow parameter λ in B is a finite-screen update-flow affine parameter. Its alignment with modular/Brown-York time and boost time is studied in D8, where the all-null response envelope requires a specific time-alignment calibration. B supplies the normal-form dynamics; D8 supplies the geometric reconstruction.

7 Residual taxonomy

The residual term in the capacity-flow normal form has multiple origins. Table 2 separates them so that Ξ_{res} does not become an unconstrained catch-all.

If the residual is physical rather than epistemic, it must either be included as an explicit state variable, represented by a conserved residual stress, or modeled by a nonlocal kernel compatible with Ward/Bianchi closure.

Classification of Ξ_{res} . The capacity-flow residual Ξ_{res} in the normal form (Core, eq. 30) must be classified before being admitted as a geometric residual. The five admissible categories are:

1. **Quotient artifact.** Apparent residual removed after screen gauge fixing.
2. **Missing variable.** Apparent residual resolved by enlarging the screen state space.
3. **Memory term.** Non-Markovian projection removable by horizontal lift (D6).
4. **Stochastic projection.** Finite-cell noise from the transition process (T3 lumpability test).
5. **Physical residual.** Survives all corrections—the only category that may feed Core $R_{\mu\nu}^{\text{scr}}$.

Table 2: Residuals in screen transition dynamics: B origin and FDS-G1 Core alignment.

Residual type	Screen origin	B label	FDS-G1 label	Core label
Epistemic	Missing coarse variables; apparent non-closure from projection	epistemic	epistemic	
Stochastic	Finite-cell noise in transition process	stochastic	stochastic	
Memory	Non-Markovian update kernels	memory	memory	
Capacity	Slow variation of η_{loc}	capacity	scalar-capacity	
Flux	Unresolved crossing load or exchange	flux	EFT	
Geometric	No single metric envelope integrates response data	geometric	flux	
			geometric	

8 v1.2 Causal Measurement Protocol

Claim type: Formal operational protocol A screen transition acquires empirical meaning only after specification of the finite record channel and update map:

$$Y_{\text{scr}} : \mathcal{X}_{\text{phys}} \rightarrow Z_{\text{obs}}, \quad U_{\text{scr}} : (M_t, Y_t, A_t) \rightarrow M_{t+1}$$

where:

- Y_{scr} maps physical screen states to accessible, stable record equivalence classes
- U_{scr} defines the ledger update under probe intervention A_t

The expectation value of the ledger update under intervention is:

$$\mathbb{E}[\ell_{t+k} \mid do(U_{\text{scr}})] \quad \text{vs} \quad \mathbb{E}[\ell_{t+k} \mid do(U_{\emptyset})]$$

Boundary note: This protocol defines empirical access to screen transitions. It does not by itself prove survey likelihood success.

9 Numerical toy demonstrations

This section summarizes simple numerical examples included with the companion package. They are not observational fits; they illustrate qualitative behavior of the normal form.

9.1 Deterministic focusing cases

Figure 1 compares three deterministic cases: pure kinematic contraction, flux focusing, and flux plus shear. Positive flux and shear accelerate the decrease of θ , as expected.

9.2 Stochastic ensemble

Figure 2 shows an ensemble with finite-cell noise. The mean trajectory follows the deterministic focusing trend while individual paths fluctuate. Such fluctuations become candidates for stochastic residuals in the geometric envelope.

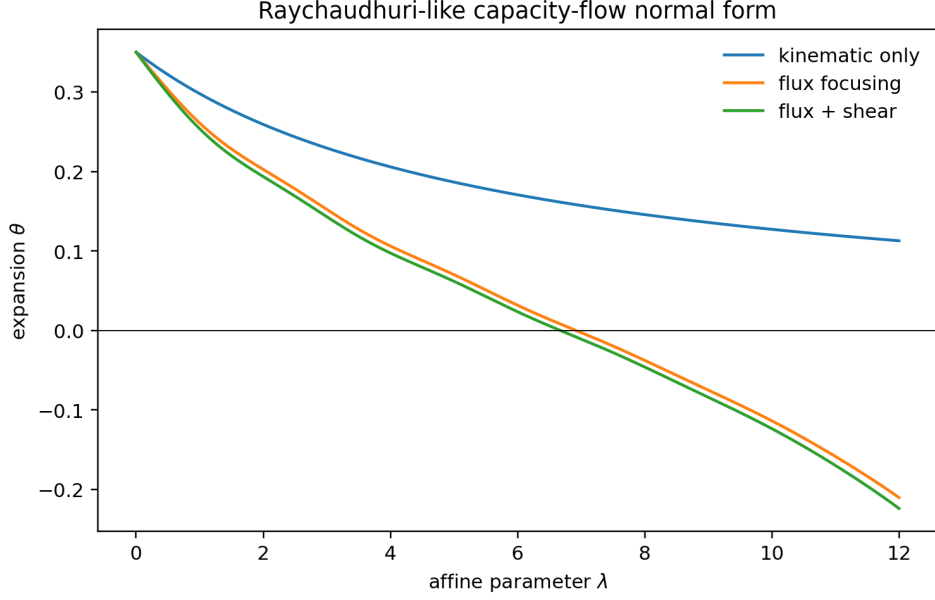


Figure 1: Deterministic capacity-flow examples. The flux and shear terms drive stronger focusing than the kinematic $-\theta^2$ term alone.

9.3 Memory residual

Figure 3 illustrates a nonlocal memory residual generated by a kernel convolution of prior flux. A memory term changes the response even when instantaneous flux is the same, demonstrating why a local Markov state may be insufficient.

9.4 Drift decomposition

Figure 4 decomposes the deterministic drift into normal-form contributions. This is useful for diagnosing whether focusing is dominated by expansion kinematics, shear, flux, or residual terms. All numerical parameters (a, b, c, D_θ, Φ) are illustrative; no observational fit is claimed.

Table 3: Simulation parameters for toy demonstrations.

Parameter	Value
a	1.0
b	0.5
c	1.0
D_θ	0.1
$\Phi_{\text{flux}}(\lambda)$	step or Gaussian pulse
σ_{scr}	constant or decaying
N	10^4

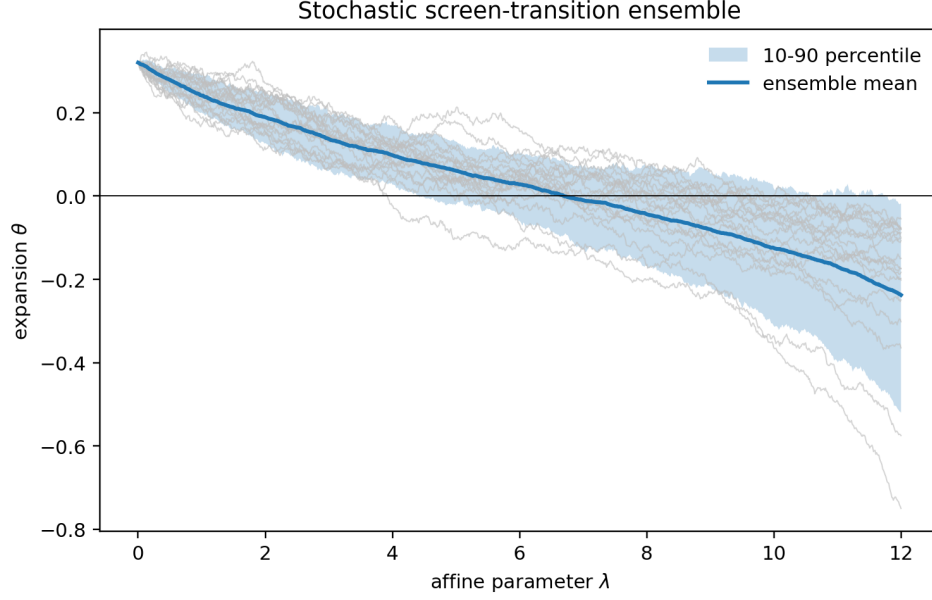


Figure 2: Stochastic screen-transition ensemble. Grey curves show representative sample paths; the band gives the 10–90 percentile range.

10 Failure conditions

This companion fails or must be modified if any of the following occurs.

1. **No screen-local state variable.** If a screen cannot be represented by a local extensive cell count or area variable, $\theta = d \ln N / d\lambda$ loses its operational basis.
2. **Dominant nonlocality.** If transition rates depend on global screen history at leading order, a local normal form is insufficient.
3. **Infinite local state capacity.** If finite cell capacity fails, stochastic hydrodynamic scaling may break.
4. **No positive flux focusing.** If positive flux does not reduce expansion, the sign of the focusing term changes and the gravity bridge fails.
5. **No metric integrability.** If all-null response data cannot be represented by a tensor quadratic form, a single Lorentzian metric envelope is not available.

These failure modes are useful: they identify exactly where a modified or nonmetric residual sector must enter.

11 Interface with companion papers

- **Companion A** supplies $S_{\text{scr}}[X]$ and η_δ from finite screen statistical mechanics.
- **This companion** supplies stochastic screen transition dynamics and θ -flow normal forms.

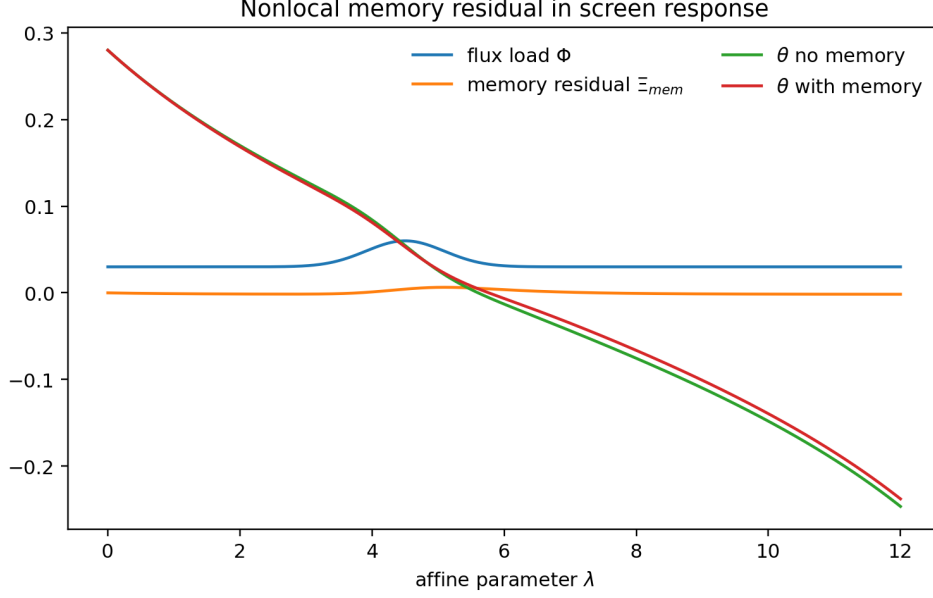


Figure 3: A memory residual produced by a nonlocal kernel. Memory modifies θ compared with the local no-memory case.

- **Companion C** promotes slow capacity response modes to scalar-capacity EFT.
- **Companion D** tests signatures such as no-free macroscopic G drift and the refined background-Weyl residual pattern ($s < 3$, $\Sigma_0 < 0$, $\mu_0 \simeq 0$).
- **Companion E** applies residual dynamics to horizon occupancy cosmology.
- **Companion F** interprets horizon and holographic entropy formulas as screen-capacity dictionaries.

The matched medium-prior production-refined nested-evidence comparison (G1fit-real) now selects G1DE-M_{3/4} over free-response, constant- Σ , CPL, and Λ CDM controls, a result strengthened by independent KiDS-1000 shear-only stress testing. The leading residual is implemented as a Weyl-normalized transfer correction,

$$\mathcal{W}_{\text{G1}} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}},$$

rather than an independent dark stress source. The accompanying Ward-suppressed leakage hierarchy gives

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O(\mathcal{R}_H \mathcal{H}^2 / k^2).$$

This is consistent with a screen residual affecting the Weyl optical channel more strongly than local matter focusing, as predicted by the normal-form bridge between screen transition dynamics and cosmological response.

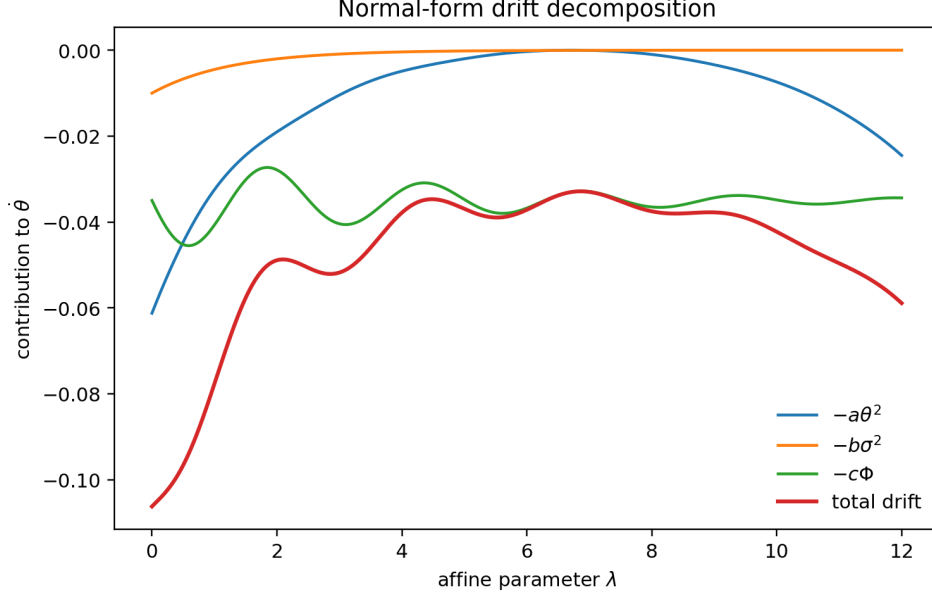


Figure 4: Contribution of each normal-form term to $\dot{\theta}$.

12 Conclusion

The central result is not a microscopic derivation of spacetime curvature. It is a dynamical normal-form theorem for finite screen transitions:

$$\dot{\theta} = -a\theta^2 - b\sigma_{\text{scr}}^2 - c\Phi_{\text{flux}} + \Xi_{\text{res}} + \zeta_{\theta} + \dots \quad (35)$$

The negative quadratic term is kinematic, the shear term is the lowest dissipative scalar, and the flux term implements positive-load focusing. When this response data is integrable into a Lorentzian metric envelope and combined with the standard local-screen bridge, it aligns with Raychaudhuri-type focusing. Nonintegrable, noisy, or memory-dependent components become residual sectors.

Thus Companion B supports the FDS-G1 Core claim:

Gravity-like geometry is the integrable hydrodynamic response of finite causal-screen entropy.

It does so by explaining why the hydrodynamic response of finite screen updates naturally has a Raychaudhuri-like capacity-flow form.

References

- [1] A. Raychaudhuri, “Relativistic cosmology. I,” *Phys. Rev.* 98, 1123 (1955).
- [2] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* 75, 1260 (1995).
- [3] L. Onsager, “Reciprocal relations in irreversible processes,” *Phys. Rev.* 37, 405 (1931).
- [4] L. Onsager and S. Machlup, “Fluctuations and irreversible processes,” *Phys. Rev.* 91, 1505 (1953).

- [5] C. Gardiner, *Handbook of Stochastic Methods*, Springer.
- [6] H. Risken, *The Fokker-Planck Equation*, Springer.
- [7] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), <https://doi.org/10.5281/zenodo.20382013>.
- [8] Y. Wu, “FDS-T3: Capacity Overflow, Effective Stochasticity, and Phase-B Invariants,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20250367>.

Companion C: Scalar-Capacity Effective Field Theory

Horndeski Embedding of Variable Screen Response
v1.2 Body-Integrated Release

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This companion paper gives the effective-field-theory closure of the FDS-G1 entropy-response program. The core paper interprets local gravitational coupling as the inverse marginal entropy-area response; stiffness G_{loc}^{-1} is proportional to the response itself.

$$G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}, \quad \eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}. \quad (1)$$

If the response density becomes a long-wavelength field, its minimal covariant description is scalar-tensor-like. We define the scalar-capacity mode

$$\varphi(x) = \ln \frac{\eta(x)}{\eta_0}, \quad \frac{G_{\text{eff}}}{G_0} = e^{-\varphi}, \quad (2)$$

and embed the resulting conservative EFT into the luminal Horndeski / EFT-of-dark-energy language. The central point is not that the mathematical form is a new scalar-tensor theory; it is a screened, capacity-interpreted subclass with a restricted prior structure: positive capacity, conservation closure, large-area suppression unless a genuine capacity mode is excited, and a luminal tensor sector. This paper consolidates the earlier minimal scalar-capacity EFT and Horndeski notes into the official Companion C to *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*.

v1.2 relation to the μ/Σ interface. The scalar-capacity branch remains distinct from the M3/4 Weyl-optical residual; the μ/Σ production interface tests the locked Weyl branch, not an unconstrained scalar-capacity drift.

1 EFT Scope and Capacity Interpretation

This companion embeds variable screen response into the standard language of scalar-tensor gravity. The core claim of the FDS-G1 program is the entropy-response bridge. The Scalar-Capacity EFT is a conservative local-field embedding of capacity variation; it is not the unique data-facing G1DE residual. The Weyl-transfer branch ($\mathcal{W}_{\text{G1}} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}}$, $\Sigma - 1 = -\kappa R_0 \hat{R}_H$, $\mu \simeq 1$) is more constrained than generic scalar EFT and is currently selected by medium-prior nested evidence. The FDS formal core [13] provides the finite-capacity substrate,

and FDS-T2 [14] develops effective geometry as horizon boundary accounting, on which the scalar-capacity EFT builds.

$$\boxed{\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}, \quad G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}} \quad (3)$$

under the standard horizon/local-screen thermodynamic bridge. Companion A supplies the screen statistical mechanics substrate; Companion B supplies stochastic screen transition dynamics and Raychaudhuri-like capacity-flow normal forms. This paper asks what happens if the local response density in Eq. (3) becomes a long-wavelength covariant field.

The answer is deliberately conservative: it becomes a scalar-tensor-like EFT, interpreted as a capacity interpretation and prior restriction on a known field-theory class, not as a replacement for Horndeski theory or the EFT of dark energy.

Mathematically, minimal scalar-capacity dynamics is a luminal scalar-tensor / Horndeski subclass. Physically, the scalar is not an arbitrary dark field; it is the hydrodynamic mode of marginal screen entropy-area response.

1.1 Claim-status table

Statement	Status	Meaning
$\varphi = \ln(\eta/\eta_0)$	Definition	Long-wavelength capacity-response mode.
$G_{\text{eff}}/G_0 = e^{-\varphi}$	Bridge relation	Follows from $G^{-1} \propto \eta$ under the horizon-response bridge.
Minimal action with $F(\varphi)R$	EFT model	Conservative covariant completion of variable response.
Horndeski embedding	Standard mapping	Places G_{1eff} in the usual scalar-tensor language.
$\alpha_T = 0$	Model choice	Luminal tensor sector; avoids non-luminal GW propagation in the minimal model.
Capacity priors	G1 contribution	Positivity, closure, suppression, and residual interpretation of the scalar.

No Double Counting rule. The scalar-capacity mode $\phi = \ln(\eta/\eta_0)$ may not duplicate the G1DE Weyl-transfer residual. If a residual is assigned to ϕ , it must not also appear independently as $R_H(a)$ or $A(a, k)$. Double-counting demotes the scalar branch to generic scalar-tensor phenomenology.

In the D9 classification, the scalar-capacity EFT is the conserved scalar branch of response-density flow, distinct from the Weyl-optical residual branch tested by $M_{3/4}$.

Remark 1 (Which G ?). *This paper uses several related but distinct quantities. Table 1 clarifies their roles and should be consulted before interpreting any constraint.*

These quantities coincide only in restricted limits. In scalar-tensor theories, scalar perturbations, mass, screening, and frame convention can make local and cosmological effective couplings differ.

2 From response density to scalar-capacity field

In the core response formulation, the local area component of the response one-form is

$$\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}. \quad (4)$$

Table 1: Effective gravitational couplings used in this paper and the G1 programme.

Symbol	Meaning	Domain
G_{loc}	screen-response bridge coupling; $G_{\text{loc}} \propto [\eta_{\text{loc}}]^{-1}$	G1 entropy-response
$G_{\text{eff}} \sim [8\pi F(\varphi)]^{-1}$	Jordan-frame Planck-mass coupling	EFT / action level
$G_{\text{Cav}} (G_{\text{N,local}})$	Newton constant measured in laboratory or solar system	local gravity
$\mu(a, k)$	perturbation growth effective coupling	cosmological linear perturbations
$\Sigma(a, k)$	lensing effective coupling	Weyl / lensing sector

When the response is homogeneous on a screen patch, this reduces to a scalar density. If the density varies slowly over spacetime, define the dimensionless scalar-capacity field

$$\varphi(x) = \ln \frac{\eta(x)}{\eta_0}. \quad (5)$$

The local effective coupling is then

$$\frac{G_{\text{eff}}(x)}{G_0} = e^{-\varphi(x)}. \quad (6)$$

Equivalently, the effective Planck-mass-squared function is

$$F(\varphi) = \frac{1}{8\pi G_{\text{eff}}} = M_{*,0}^2 e^\varphi, \quad (7)$$

where conventions differ by factors of 8π ; throughout this paper F is the coefficient multiplying $R/2$ in the gravitational action.

Definition 1 (Scalar-capacity mode). *The scalar-capacity mode is the long-wavelength field $\varphi = \ln(\eta/\eta_0)$ encoding deviations of the local screen entropy-area response from its equilibrium value. It is a response variable, not a fundamental microscopic cell label.*

3 Frame convention

The action (11) is written in the **Jordan frame**: matter is minimally coupled to $g_{\mu\nu}$, so $\nabla^\mu T_{\mu\nu}^{(m)} = 0$ and test-particle geodesics follow the Jordan-frame metric. The Planck-mass function is

$$F(\varphi) = M_{*,0}^2 e^\varphi, \quad G_{\text{eff}} = \frac{1}{8\pi F}. \quad (8)$$

An Einstein-frame transformation,

$$\tilde{g}_{\mu\nu} = \frac{F(\varphi)}{M_*^2} g_{\mu\nu}, \quad M_*^2 = \frac{1}{8\pi G_0}, \quad (9)$$

diagonalizes the gravitational sector but introduces matter couplings to φ . The kinetic no-ghost proxy in the Einstein frame is

$$\mathcal{K}_E = \frac{Z}{F} + \frac{3}{2} \left(\frac{F_{,\varphi}}{F} \right)^2 > 0, \quad (10)$$

which for $F \propto e^\varphi$ becomes $\mathcal{K}_E = Z/F + 3/2$. All stability conditions below are stated in a frame-invariant manner.

4 Minimal scalar-capacity EFT

The minimal covariant scalar-capacity EFT is

$$S_{\text{cap}} = \int d^4x \sqrt{-g} \left[\frac{1}{2} F(\varphi) R - \frac{1}{2} Z(\varphi) g^{\mu\nu} \nabla_\mu \varphi \nabla_\nu \varphi - V(\varphi) + \mathcal{L}_m + \mathcal{L}_{\text{res}} \right]. \quad (11)$$

Here $\mathcal{L}_{\text{res}}$ denotes residual sectors required when variable capacity is not fully captured by the scalar mode: nonlocal memory, stochastic screen noise, horizon occupancy residuals, or unresolved coarse variables.

4.1 Field equations

Varying the metric gives

$$F G_{\mu\nu} = T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\varphi)} + T_{\mu\nu}^{(\text{res})} + \nabla_\mu \nabla_\nu F - g_{\mu\nu} \square F, \quad (12)$$

where

$$T_{\mu\nu}^{(\varphi)} = Z \nabla_\mu \varphi \nabla_\nu \varphi - g_{\mu\nu} \left[\frac{1}{2} Z (\nabla \varphi)^2 + V(\varphi) \right]. \quad (13)$$

Varying φ yields

$$Z \square \varphi + \frac{1}{2} Z_{,\varphi} (\nabla \varphi)^2 - V_{,\varphi} + \frac{1}{2} F_{,\varphi} R + \frac{\delta \mathcal{L}_{\text{res}}}{\delta \varphi} = 0. \quad (14)$$

4.2 Conservation closure

The Bianchi identity requires

$$\nabla^\mu \left(T_{\mu\nu}^{(m)} + T_{\mu\nu}^{(\varphi)} + T_{\mu\nu}^{(\text{res})} \right) + \text{terms exchanged with } F(\varphi) R = 0. \quad (15)$$

If matter is minimally coupled, $\nabla^\mu T_{\mu\nu}^{(m)} = 0$ holds in the Jordan frame. The scalar and residual sectors must then close the remaining response current. This is the field-theory version of the G1 rule:

variable screen capacity cannot appear as a free $G(x)$ drift without a conserved closure sector.

(16)

5 Horndeski embedding

The general Horndeski class contains the most general four-dimensional single-scalar tensor theory with second-order field equations in the usual formulation. The minimal G1 scalar-capacity EFT is embedded as

$$G_2(\varphi, X) = Z(\varphi)X - V(\varphi), \quad G_3 = 0, \quad (17)$$

$$G_4(\varphi) = \frac{1}{2} F(\varphi), \quad G_5 = 0, \quad (18)$$

where

$$X = -\frac{1}{2} g^{\mu\nu} \nabla_\mu \varphi \nabla_\nu \varphi. \quad (19)$$

This is a conformally coupled scalar-tensor subclass, not a full general Horndeski model.

Theorem 1 (Scalar-Capacity Admissibility). *The minimal scalar-capacity EFT in Eq. (11) is a luminal scalar-tensor/Horndeski subclass with $G_3 = G_5 = 0$ and $G_4 = F(\varphi)/2$. Its distinctiveness is not the mathematical Lagrangian form but the interpretation of φ as a coarse-grained screen-capacity response mode. A scalar-capacity branch is admissible only if it: (1) descends from η_{loc} ; (2) respects Ward/Bianchi closure; (3) avoids free $A(a, k)$; (4) does not double-count the G1DE Weyl residual; (5) carries independent likelihood penalty. Failure demotes it to generic scalar-tensor phenomenology.*

6 EFT-of-dark-energy alpha functions

On a spatially flat FLRW background, the Bellini-Sawicki alpha-function language describes many linear scalar-tensor deviations from GR by a time-dependent effective Planck mass and functions $(\alpha_M, \alpha_K, \alpha_B, \alpha_T)$.

For the minimal capacity subclass,

$$M_*^2 = F(\varphi) = M_{*,0}^2 e^\varphi, \quad (20)$$

so

$$\alpha_M = \frac{d \ln M_*^2}{d \ln a} = \frac{d \varphi}{d \ln a}. \quad (21)$$

Under the Bellini-Sawicki sign convention adopted here, the conformally coupled minimal subclass has

$$\alpha_T = 0, \quad \alpha_B = -\alpha_M. \quad (22)$$

The kineticity is schematically

$$\alpha_K \sim \frac{Z(\varphi) \dot{\varphi}^2}{H^2 F} + \text{normalization terms}, \quad (23)$$

with detailed normalization convention-dependent.

Remark 2 (Why $\alpha_T = 0$ is chosen). *The minimal G1eff sector intentionally sets $\alpha_T = 0$. This is a conservative tensor sector compatible with the strong constraints on the gravitational-wave propagation speed following GW170817/GRB170817A. Non-luminal extensions would require a separate, tightly constrained companion analysis.*

7 Stability and viability conditions

The minimal scalar-capacity EFT is viable only in the region satisfying standard scalar-tensor stability and observational requirements.

7.1 Positivity of capacity and gravitational stiffness

Because $\eta > 0$,

$$F(\varphi) > 0, \quad G_{\text{eff}} > 0. \quad (24)$$

This is not optional; negative capacity would destroy the FDS interpretation.

7.2 No-ghost condition

In the Einstein-frame kinetic normalization, a simple positivity proxy is

$$\mathcal{K}_E(\varphi) = \frac{Z(\varphi)}{F(\varphi)} + \frac{3}{2} \left(\frac{F_{,\varphi}}{F} \right)^2 > 0. \quad (25)$$

For $F \propto e^\varphi$ this becomes

$$\mathcal{K}_E = \frac{Z}{F} + \frac{3}{2} > 0. \quad (26)$$

A sufficient conservative choice is $Z > 0$.

7.3 Gradient stability and mass scale

The scalar sound speed must be positive,

$$c_s^2 > 0, \quad (27)$$

and the effective mass

$$m_{\text{eff}}^2 \sim \frac{d^2 V_{\text{eff}}}{d\varphi^2} \quad (28)$$

should not induce an uncontrolled tachyonic instability on scales where the EFT is applied.

7.4 Local-gravity constraints

Since

$$\frac{\dot{G}_{\text{eff}}}{G_{\text{eff}}} = -\dot{\varphi} = -H\alpha_M, \quad (29)$$

local bounds on $\dot{G}/G$, PPN parameters, and screening mechanisms constrain the allowed motion of φ . A macroscopic G drift without closure or screening is not a viable G1 model.

8 Local gravity, screening, and decoupling

A light scalar coupled to matter mediates a fifth force and modifies local gravity. The minimal scalar-capacity EFT requires at least one of the following mechanisms to remain viable:

1. **Small running.** $|\alpha_M| \ll 1$ ensures $|\dot{G}/G|$, PPN, and growth constraints are satisfied without screening.
2. **Heavy field.** $m_{\text{eff}}^2 \gg H_0^2$ suppresses the scalar force below local experimental scales; the field is cosmologically relevant only through its residual or coupling effects.
3. **Environmental screening / residual confinement.** The capacity mode may be active at cosmological or horizon-residual scales but frozen or screened in the solar system and Cavendish experiments through a screening mechanism (chameleon, symmetron, Vainshtein) or through nonlocal residual confinement.

Without one of these mechanisms, a macroscopic scalar-capacity mode is generically ruled out by local fifth-force and PPN constraints.

9 Capacity priors: how G1 differs from generic scalar-tensor EFT

The same Lagrangian form can describe many scalar-tensor models. G1 adds a restricted prior interpretation.

Capacity prior	Meaning
$\eta > 0$	Screen distinction density is positive; $G_{\text{eff}} > 0$.
Large-area suppression	Finite-area corrections scale as $\Delta G/G \sim -\alpha/(\eta_0 A)$ unless a genuine long-wavelength capacity mode is excited.
No free G drift	$\dot{G}/G$ requires a Bianchi-closed scalar or residual sector.
Luminal tensor sector	Minimal model sets $\alpha_T = 0$.
Residual closure	Unmodeled capacity changes appear as conserved $T_{\mu\nu}^{(\text{res})}$ or nonlocal memory kernels.
Occupancy linkage	If coupled to horizon occupancy, expansion, growth, and lensing deviations should be correlated rather than arbitrary.

10 Toy alpha-function profiles

For an illustrative capacity field relaxing to equilibrium today, take

$$\varphi(a) = \epsilon(1 - a^s). \quad (30)$$

Then

$$\frac{G_{\text{eff}}}{G_0} = e^{-\epsilon(1-a^s)}, \quad \alpha_M = \frac{d\varphi}{d \ln a} = -\epsilon s a^s. \quad (31)$$

This toy profile is not a fit; it is a diagnostic used to show how scalar-capacity running maps to the EFT variables.

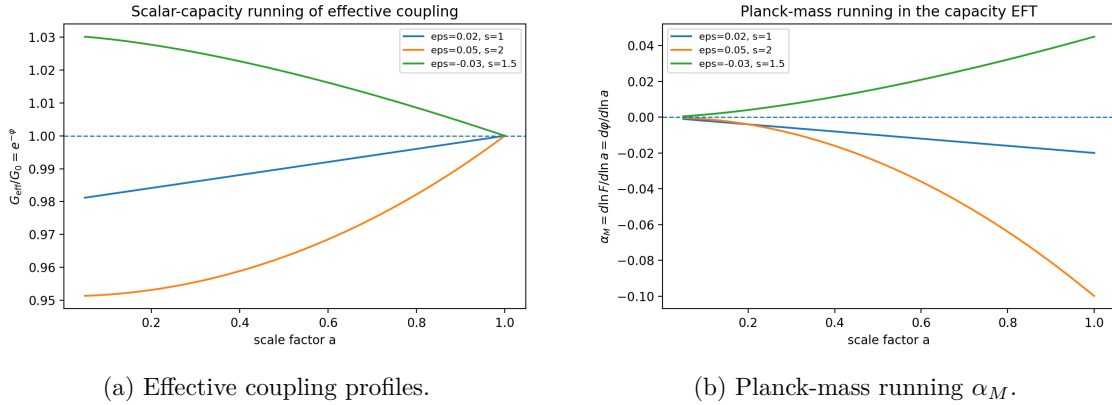


Figure 1: Illustrative scalar-capacity EFT diagnostics for $\varphi(a) = \epsilon(1 - a^s)$, satisfying $\varphi(1) = 0$ and $G_{\text{eff}}(1) = G_0$. $\epsilon > 0$ gives G_{eff} lower in the past (high early η); $\epsilon < 0$ gives G_{eff} higher in the past (low early η). s controls relaxation sharpness. These plots are diagnostic maps, not observationally allowed contours.

11 EFT interpretation of the production-refined likelihood audit

The completed medium-prior 8-seed production-refined audit (G1fit-real) selects the projection-locked G1DE-M_{3/4} model across a seven-model hierarchy, a result independently stress-tested via KiDS-1000 shear-only data, with

$$\mu_0 \simeq 0, \quad \Sigma_0 < 0.$$

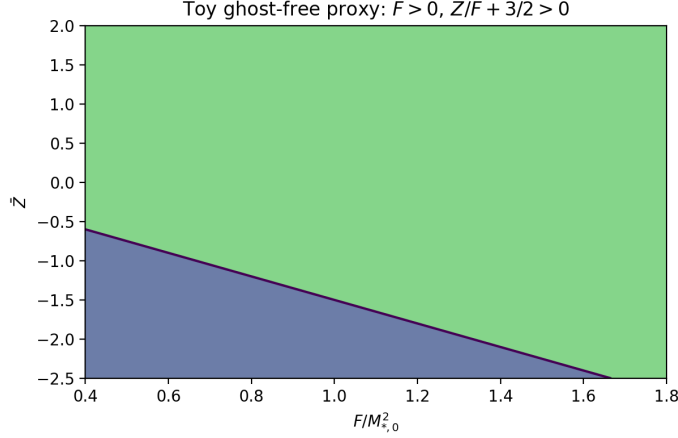


Figure 2: A simple no-ghost proxy region for $F > 0$ and $\mathcal{K}_E = Z/F + 3/2 > 0$. Positive Z and positive F are safely inside the conservative region.

In EFT-of-dark-energy language, this suggests that the leading phenomenological opening is not an effective Newton coupling shift in the matter-growth equation ($\mu(a, k) \simeq 1$), but a Weyl-response or gravitational-slip channel ($\Sigma(a, k) < 1$). Therefore the G1 scalar-capacity EFT should be constrained to the luminal, weak-growth-deviation regime:

$$\alpha_T = 0, \quad \mu(a) \simeq 1, \quad \Sigma(a) < 1$$

over the active filling window. This is a phenomenological location statement, not a derivation from Horndeski. It is compatible with, but not predicted by, a wide class of luminal scalar-tensor theories; the G1-specific claim is the shared ledger shape (a) for both background and Weyl response.

12 Connection to companions

Companion	Relation to scalar-capacity EFT
FDS-G1 Core	Defines η_{loc} and G_{loc} as entropy-response variables.
A: Screen statistics	Supplies $\eta_\delta(A)$ and finite-area corrections that motivate scalar-capacity modes.
B: Transition dynamics	Supplies response dynamics and residual noise/memory channels.
D: Likelihood/falsification	Constrains α_M , $\dot{G}/G$, growth, and lensing signatures.
E: Horizon occupancy	Provides a homogeneous residual sector that may couple to φ .
F: Holography/horizons	Interprets $1/(4G_N)$ as screen density in horizon and AdS/CFT settings.

13 Failure registry

Failure condition	Consequence
$F \leq 0$	Negative capacity or negative effective stiffness; model fails.
Ghost or gradient instability	Scalar-capacity EFT invalid on the scale considered.
Large unscreened $\dot{G}/G$	Excluded by local gravity unless residual closure or screening is present.
Non-luminal tensor sector	Not part of the minimal model; strongly constrained.
Arbitrary φ potential with no screen prior	Reduces to generic scalar-tensor phenomenology; loses G1 distinctiveness.
No connection to η_{loc}	Breaks the capacity interpretation.
No viable screening or mass suppression	Ruled out by local fifth-force and PPN constraints.
φ independent of screen-response priors	Collapses to generic scalar-tensor model without G1 interpretation.

14 Conclusion

Companion C positions variable screen response within the standard language of scalar-tensor gravity. The minimal G1 scalar-capacity EFT is not advertised as a new field-equation form. Its contribution is the interpretation and restriction of a conservative luminal scalar-tensor subclass by finite screen-capacity response:

$$\boxed{\eta > 0, \quad G_{\text{eff}}^{-1} \propto \eta, \quad \alpha_T = 0, \quad \alpha_B = -\alpha_M,} \quad (32)$$

with the dictionary

$$\boxed{\varphi = \ln \frac{\eta}{\eta_0}, \quad \frac{G_{\text{eff}}}{G_0} = e^{-\varphi}, \quad \alpha_M = \frac{d\varphi}{d \ln a}.} \quad (33)$$

A further restriction comes from conservation closure: variable screen capacity cannot appear as a free $G(x)$ drift without a conserved residual sector or screening mechanism. The decoupling limit is

$$m_{\text{eff}}^2 \gg H_0^2, \quad |\alpha_M| \ll 1, \quad (34)$$

recovering local gravity and Λ CDM-like cosmology when the capacity mode is heavy or frozen.

The scalar-capacity G_{eff} is an action-level inverse Planck stiffness; observable quantities (G_{Cav} , growth rate $\mu(a, k)$, lensing $\Sigma(a, k)$) require perturbation theory, screening, and frame specifications beyond the minimal φ -only dictionary. This prevents conflation of G_{eff} with local Newton’s constant in survey constraints.

References

- [1] G. W. Horndeski, “Second-order scalar-tensor field equations in a four-dimensional space,” *International Journal of Theoretical Physics* **10**, 363–384 (1974).
- [2] C. Deffayet, X. Gao, D. A. Steer, and G. Zahariade, “From k-essence to generalised Galileons,” *Physical Review D* **84**, 064039 (2011).

- [3] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *Journal of Cosmology and Astroparticle Physics* **07**, 050 (2014).
- [4] J. Gleyzes, D. Langlois, F. Piazza, and F. Vernizzi, “Essential building blocks of dark energy,” *Journal of Cosmology and Astroparticle Physics* **08**, 025 (2013/2014 literature stream).
- [5] T. Baker, E. Bellini, P. G. Ferreira, M. Lagos, J. Noller, and I. Sawicki, “Strong constraints on cosmological gravity from GW170817 and GRB 170817A,” *Physical Review Letters* **119**, 251301 (2017).
- [6] J. M. Ezquiaga and M. Zumalacarregui, “Dark energy after GW170817: dead ends and the road ahead,” *Physical Review Letters* **119**, 251304 (2017).
- [7] C. Brans and R. H. Dicke, “Mach’s principle and a relativistic theory of gravitation,” *Physical Review* **124**, 925 (1961).
- [8] T. Clifton, P. G. Ferreira, A. Padilla, and C. Skordis, “Modified gravity and cosmology,” *Physics Reports* **513**, 1–189 (2012).
- [9] T. Kobayashi, “Horndeski theory and beyond: a review,” *Reports on Progress in Physics* **82**, 086901 (2019).
- [10] A. Joyce, B. Jain, J. Khoury, and M. Trodden, “Beyond the cosmological standard model,” *Physics Reports* **568**, 1–98 (2015).
- [11] B. P. Abbott et al. (LIGO/Virgo Collaboration), “GW170817: observation of gravitational waves from a binary neutron star inspiral,” *Phys. Rev. Lett.* **119**, 161101 (2017).
- [12] P. Vaillant and D. Scott, “A complex tension origin for dilaton gravity,” *Entropy* **28**, 544 (2026).
- [13] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), <https://doi.org/10.5281/zenodo.20382013>.
- [14] Y. Wu, “FDS-T2: Effective Geometry as Horizon Boundary Accounting,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20284911>.

Companion D: Likelihood-Ready Tests and Falsification

Predictions, Compressed-Data Interfaces, and Model Comparison for Finite Screen-Capacity

Gravity

v1.2 Body-Integrated Release

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This companion paper collects the falsification and likelihood interface for the FDS-G1 finite screen-capacity program. The core G1 claim is not that finite distinction systems derive gravity from first principles, but that finite causal-screen statistics provide an entropy-response substrate for the area coefficient that controls gravity-like coupling under the standard horizon and local-screen bridges. Companion D translates this claim into observational diagnostics through four layers. Layer 1 is the original demonstration interface (transparent demo compressed data), retained for reproducibility. Layer 2 is the matched full-likelihood pilot, now recorded through G1fit-real[12], which uses Pantheon+ full covariance, DESI DR2 BAO, curated RSD, and compressed E_G . Layer 3 is the medium-prior production-refined nested-evidence comparison, completed as of G1fit-real v1.2, with the projection-locked G1DE-M_{3/4} ranking first in the canonical hierarchy (eight seeds, $d \log Z = 0.1$). Layer 4 is an independent KiDS-1000 shear-only diagnostic stress test of the Weyl channel, including scale cuts, CLASS $P(k, z)$, shear calibration, photo- z , minimal IA nuisance, free- κ , constant- Σ , binned- Σ , and deterministic mock-injection false-positive audits (G1fit-real v1.1). We define modular likelihoods for supernova distances, BAO distances, growth-rate data, E_G data, and shear correlation functions. We compare baseline Λ CDM, CPL dark energy, and the G1DE residual models. We emphasize three falsification principles: no free macroscopic drift of the effective gravitational coupling without conservation closure; a background-Weyl residual fingerprint sharing a common screen ledger variable; and inverse-area suppression of ordinary finite-area corrections. A central purpose of this companion is to prevent residual sectors from functioning as arbitrary dark components: every residual must either close a conservation law, correlate observables through a common ledger variable, or be demoted by information criteria. The purpose is not to claim final cosmological confirmation, but to make the framework empirically penalizable and to report the current production-refined evidence status under explicit control models.

1 Likelihood Scope and Falsification Status

This companion turns the screen-response framework into a likelihood-ready testing protocol, following the FDS formal core [10] and the pre-registration and demotion conditions registered in FDS-X1 [11]. It defines a modular compressed likelihood for supernova distances, BAO distances,

growth-rate data, local gravity priors, and gravitational-wave propagation flags. The companion uses demonstration compressed data generated from a fiducial Λ CDM model with small seeded perturbations. These data are included only to verify the code path and the information-criterion bookkeeping. Production inference must use real data vectors and covariance matrices, for example Pantheon+ supernovae, DESI BAO measurements, growth compilations, CMB distance priors, Lunar Laser Ranging bounds, Cassini PPN bounds, and gravitational-wave speed constraints.

Status. This is a likelihood interface and falsification protocol. A matched full-likelihood and medium-prior production-refined nested-evidence comparison (G1fit-real[12]) has now been executed with Pantheon+ full covariance, DESI DR2 BAO compressed distances, a curated RSD $f\sigma_8$ subset, and compressed E_G data. A completed medium-prior 8-seed production-refined audit gives the canonical hierarchy

$$\text{G1DE-M}_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE2} > \text{G1DE1} > \text{CPL} > \Lambda\text{CDM},$$

with the four-parameter projection-locked model ranking first. G1DE-1 is included as a full member of the production-refined comparison, not as a separate appendix audit. Medium-prior production refinement (eight seeds, $d \log Z = 0.1$) is complete; full prior-family stress testing and full $3 \times 2\text{pt}$ production validation remain pending. KiDS shear-only diagnostic stress testing (G1fit-real v1.1) has been completed as a Weyl-channel check.

The model hierarchy $M_{3/4} \rightarrow M_\kappa \rightarrow \text{const-}\Sigma \rightarrow \text{G1DE-2}$ is interpreted in D10 as a local chart on the G1 residual branch moduli space.

Likelihood discipline. Likelihood success supports only the tested output normal form. It does not prove finite-index screen algebras, boost modular covariance, microscopic source–kernel uniqueness, or strong-field GR replacement.

2 Position in the G1 dependency structure

The FDS-G1 Core dependency chain is

$$\text{finite screen statistics} \rightarrow S_{\text{scr}}[X] \rightarrow \omega \rightarrow \eta_{\text{loc}} \rightarrow G_{\text{loc}} \rightarrow \theta\text{-flow} \rightarrow \text{metric/Ward closure} \rightarrow \text{residual sectors.} \quad (1)$$

Companion D acts at the final step. It takes residual or capacity-response sectors and asks whether they survive observational compression and parameter penalties.

The local response relation in the core theory is

$$\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}, \quad G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}. \quad (2)$$

If η_{loc} varies macroscopically, conservation closure is mandatory. In scalar-capacity language, one writes

$$\varphi = \ln \frac{\eta}{\eta_0}, \quad \frac{G_{\text{eff}}}{G_0} = e^{-\varphi}, \quad (3)$$

and the Horndeski/EFT-of-dark-energy embedding identifies

$$\alpha_M = \frac{d \ln F}{d \ln a}, \quad \alpha_T = 0 \quad (4)$$

for the conservative luminal-tensor sector discussed in Companion C.

3 Signature predictions as likelihood requirements

The G1 residual program is not defined by arbitrary $w(a)$ fitting. It is defined by constrained response patterns. Companion D uses the following three signatures as pass/fail conditions.

3.1 No free macroscopic G drift

A variable screen response implies

$$\frac{\dot{G}}{G} \simeq -\frac{\dot{\eta}}{\eta} = -H\alpha_M. \quad (5)$$

A model that requires a large unscreened drift of G is disfavored unless it supplies a conservation-closed residual, screening mechanism, or scalar-capacity field satisfying local-gravity priors. The key principle is

$$\nabla^\mu (T_{\mu\nu} + \mathcal{R}_{\mu\nu}) = 0. \quad (6)$$

Macroscopic capacity variation without this closure is not a valid G1 model.

3.2 Correlated expansion-growth-lensing residuals

A G1 horizon-occupancy residual uses the factorized memory response

$$\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1, \quad R_0 = 3 - s,$$

implemented through a Weyl-normalized transfer operator

$$\mathcal{W}_{G1} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{GR},$$

so that

$$\Sigma(a, k) - 1 = -\kappa R_0 \hat{R}_H(a) + O(R_0^2), \quad \mu(a, k) - 1 = O(R_0^2) + O(R_0 \hat{R}_H \mathcal{H}^2/k^2).$$

so that $0 \leq \hat{X}(a) \leq 1$ for $0 \leq \chi_H \leq \chi_\infty$. The growth and lensing response proxies are

$$\mu(a) = 1 + \mu_0 \hat{X}(a), \quad \Sigma(a) = 1 + \Sigma_0 \hat{X}(a). \quad (7)$$

Thus the falsifiable feature is not $w(a) \neq -1$ alone, but a structured residual fingerprint. The current matched medium-prior production-refined nested-evidence comparison (G1fit-real[12]) reports

$$s < 3, \quad \Sigma_0 < 0, \quad \mu_0 \simeq 0,$$

i.e. a background filling deviation and a suppressed Weyl response, with the growth response remaining consistent with GR. This pattern—expansion plus Weyl response sharing the same active-filling shape $\hat{X}(a)$, while matter growth stays near GR—is more restrictive than generic modified-gravity fits.

3.3 Finite-area corrections are suppressed

If a finite screen has

$$\frac{S(A)}{k_B} = \eta_0 A + \alpha \ln(A/a_0) + \gamma + \dots, \quad (8)$$

then the differential density is

$$\eta_\delta(A) = \frac{d(S/k_B)}{dA} = \eta_0 + \frac{\alpha}{A} + \dots, \quad (9)$$

and

$$\frac{\Delta G}{G} \simeq -\frac{\alpha}{\eta_0 A}. \quad (10)$$

Therefore ordinary finite-area or logarithmic corrections must be small on macroscopic screens. A large late-time cosmological signal must come from a residual/horizon-occupancy sector, a scalar-capacity mode, or another explicitly modeled nonlocal sector, not from a bare logarithmic area correction alone.

4 v1.2 Screen-to-Survey Observable Map

Claim type: Interface specification This map translates screen-residual quantities to standard cosmological observables used in mainstream MG pipelines: The projection-locked G1DE-M_{3/4}

Screen-side quantity	Survey-side observable	Failure consequence
$R_H(a)$ background ledger occupancy	$H(z), D_L, D_A, \text{BAO}, \text{SN}$	Background branch demoted
$\Sigma(a, k)$ Weyl optical response	weak lensing, E_G , BandPower (EE+nE)	Weyl branch demoted
$\mu(a, k)$ Ricci/growth leakage	RSD, $f\sigma_8$, full-shape	Ricci-leakage branch demoted
$R_b^H(a)$ memory shape	redshift-dependent lensing / $3 \times 2\text{pt}$	Shape lock demoted
Stochastic residual/noise kernel	covariance, mocks	Deterministic interpretation demoted

model has explicit interface form:

$$\mu(a, k) = 1 + \epsilon_\mu(a, k), \quad \Sigma(a, k) = 1 - \frac{3}{4}(3 - s)R_b^H(a) + \epsilon_\Sigma(a, k)$$

with $\epsilon_\mu, \epsilon_\Sigma = O(R_H^2) + O(R_H H^2/k^2)$, consistent with GW speed constraints.

Boundary note: This is an interface specification, not production evidence.

5 Compressed likelihood blocks

For the Layer-1 demonstration interface we use

$$\chi_{\text{demo}}^2 = \chi_{\text{SN}}^2 + \chi_{\text{BAO}}^2 + \chi_{\text{growth}}^2 + \chi_{\text{local}}^2. \quad (11)$$

For the matched exact pilot reported in G1fit-real[12], the implemented likelihood is

$$\chi_{\text{exact}}^2 = \chi_{\text{SN}}^2 + \chi_{\text{BAO}}^2 + \chi_{\text{RSD}}^2 + \chi_{E_G}^2. \quad (12)$$

Local/prior blocks ($\dot{G}/G$, PPN, GW speed) are retained for Layer-3 production; the pilot layer uses E_G as a compressed Weyl/lensing diagnostic. Each block is modular. The production form for a generic block is

$$\chi_{\text{block}}^2 = (\mathbf{d} - \mathbf{m})^T C^{-1} (\mathbf{d} - \mathbf{m}), \quad (13)$$

where $\mathbf{d}$ is the data vector, $\mathbf{m}$ the model prediction, and C the survey covariance matrix. The demo code uses a diagonal covariance,

$$C = \text{diag}(\sigma_i^2), \quad (14)$$

as a simplification. Production inference must replace these with survey covariance matrices. The demo is **Layer 1** (synthetic demonstration pipeline); production inference is **Layer 2** (survey-quality likelihood analysis). Every demo figure carries the caveat: synthetic demo data only; no cosmological constraint is claimed.

5.1 Supernova distances

For a model with Hubble function $E(z) = H(z)/H_0$, define

$$D_C(z) = \int_0^z \frac{dz'}{E(z')}, \quad D_L(z) = (1+z)D_C(z). \quad (15)$$

The relative distance modulus is

$$\mu(z) = 5 \log_{10} D_L(z) + \mathcal{M}, \quad (16)$$

where $\mathcal{M}$ is the SN absolute magnitude nuisance, absorbing the H_0 degeneracy. For a diagonal covariance, the analytic marginalization over $\mathcal{M}$ gives

$$\chi_{\text{SN,marg}}^2 = \Delta^T C^{-1} \Delta - \frac{(\mathbf{1}^T C^{-1} \Delta)^2}{\mathbf{1}^T C^{-1} \mathbf{1}}, \quad (17)$$

where $\Delta_i = \mu_{\text{obs},i} - 5 \log_{10} D_L(z_i)$ (setting $\mathcal{M} = 0$ for the residual) and $\mathbf{1}$ is the unit vector. This avoids fitting $\mathcal{M}$ as an explicit free parameter in the demo.

5.2 BAO compressed distances

The demonstration interface uses

$$\frac{D_M(z)}{r_d} = \frac{c/H_0}{r_d} D_C(z), \quad \frac{D_H(z)}{r_d} = \frac{c/H_0}{r_d} \frac{1}{E(z)}. \quad (18)$$

A real DESI analysis should import the official BAO data vector and covariance matrix.

5.3 Growth proxy

A full analysis should solve perturbation equations for a specified scalar-capacity or residual model. The production growth equation for the matter density contrast in the quasistatic limit is

$$\delta_m'' + \left(2 + \frac{H'}{H}\right) \delta_m' = \frac{3}{2} \Omega_m(a) \mu(a, k) \delta_m, \quad (19)$$

where $\mu(a, k)$ is the effective gravitational coupling for perturbations relative to GR and primes denote derivatives with respect to $\ln a$. The growth rate and redshift-space-distortion proxy are

$$f(a) = \frac{d \ln D}{d \ln a}, \quad f\sigma_8(a) = f(a) \sigma_{8,0} D(a). \quad (20)$$

The demo uses a growth-index proxy,

$$f(z) \simeq \Omega_m(z)^{0.55} \left[1 + \frac{1}{2}(\mu(z) - 1) \right], \quad (21)$$

and constructs $f\sigma_8(z)$ by integrating $f = d \ln D / d \ln a$. This is sufficient for a Layer 1 pipeline demonstration but not for Layer 2 precision inference.

5.4 Lensing proxy

Weak gravitational lensing responds to the Weyl potential $\Phi + \Psi$. In the quasistatic limit,

$$k^2(\Phi + \Psi) = -8\pi G a^2 \Sigma(a, k) \rho_m \Delta_m, \quad (22)$$

where $\Sigma(a, k)$ is the effective lensing coupling. The G1DE residual fingerprint correlates Σ with the growth coupling μ through the common ledger shape $\hat{X}(a)$,

$$\mu(a) = 1 + \mu_0 \hat{X}(a), \quad \Sigma(a) = 1 + \Sigma_0 \hat{X}(a), \quad \hat{X}(a) = \frac{4}{\chi_\infty} \chi_H(a) (1 - \chi_H(a)/\chi_\infty). \quad (23)$$

A G1DE model that requires independent $\mu(a)$ and $\Sigma(a)$ functions (beyond a common ledger amplitude) has lost its correlated residual claim.

5.5 Local and propagation priors

The local block may include priors on

$$\frac{\dot{G}}{G}, \quad \gamma_{\text{PPN}} - 1, \quad \alpha_T, \quad (24)$$

or model-specific stability flags. In the conservative scalar-capacity sector one imposes $\alpha_T = 0$ to remain in the luminal tensor-propagation class.

5.6 No Free Function rule

The Weyl amplitude $A(a, k)$ must not be sampled as an independent free function. If optimal likelihood or evidence requires free $A(a, k)$, the model is no longer G1DE—it becomes a generic dark-stress source. This rule is enforced at the inference level: any chain that samples $A(a, k)$ independently of the G1DE hierarchy is classified as a non-G1 dark-stress model.

6 Model classes

The demo compares four model classes. The same infrastructure can be extended to HDE, quintessence proxies, and scalar-tensor EFT models.

6.1 Λ CDM

$$E^2(z) = \Omega_{m0}(1+z)^3 + (1 - \Omega_{m0}). \quad (25)$$

This is the reference model.

6.2 CPL

$$w(a) = w_0 + w_a(1 - a), \quad (26)$$

$$E^2(a) = \Omega_{m0}a^{-3} + (1 - \Omega_{m0})a^{-3(1+w_0+w_a)}e^{-3w_a(1-a)}. \quad (27)$$

CPL is a flexible background-only model unless supplemented by perturbation prescriptions.

6.3 G1DE

The G1DE background is defined by a horizon occupancy residual $\chi_H(a)$. The filling-closure law is

$$\chi_H(a) = \frac{\chi_\infty}{1 + Ba^{-s}}, \quad B = \frac{\chi_\infty}{\chi_{H0}} - 1, \quad E^2(a) = \frac{\Omega_{m0}a^{-3}}{1 - \chi_H(a)}. \quad (28)$$

The Λ CDM-like limit occurs when $s = 3$ and $\chi_\infty = 1$. To see this, write

$$1 - \chi_H(a) = \frac{Ba^{-3}}{1 + Ba^{-3}}, \quad (29)$$

so that

$$E^2(a) = \frac{\Omega_{m0}a^{-3}(1 + Ba^{-3})}{Ba^{-3}} = \Omega_{m0}a^{-3} + \frac{\Omega_{m0}}{B}. \quad (30)$$

Matching $B = \Omega_{m0}/(1 - \Omega_{m0})$ gives $E^2(a) = \Omega_{m0}a^{-3} + (1 - \Omega_{m0})$, the flat- Λ CDM background. Thus G1DE reduces smoothly to Λ CDM when $s = 3$ and $\chi_\infty = 1$, and its departures are controlled by s and χ_∞ .

6.4 G1DE-1: background plus growth

The growth extension introduces

$$\mu(a) = 1 + \mu_0 \hat{X}(a), \quad \hat{X}(a) = \frac{4}{\chi_\infty} \chi_H(a) \left(1 - \frac{\chi_H(a)}{\chi_\infty} \right). \quad (31)$$

This class is penalized by information criteria unless the growth response improves the likelihood enough to justify the additional parameter.

7 Information criteria and falsification

For a model with k parameters and n effective data points,

$$\text{AIC} = \chi_{\min}^2 + 2k, \quad (32)$$

$$\text{BIC} = \chi_{\min}^2 + k \ln n. \quad (33)$$

These are not substitutes for a full Bayesian evidence, but they are useful guardrails against unconstrained residual sectors.

Table 1: Free parameters per model in the demo comparison. For correlated compressed data, BIC should be treated as an approximate guardrail, not a substitute for Bayesian evidence. Synthetic demo only — no cosmological constraint.

Model	Demo free parameters	k
Λ CDM	Ω_{m0}	1
CPL	Ω_{m0}, w_0, w_a	3
G1DE	Ω_{m0}, s	2
G1DE+ μ	Ω_{m0}, s, μ_0	3

Table 2: Demo compressed-data comparison. This table is generated from the included toy data and should not be read as a real cosmological constraint. Synthetic demo only — no cosmological constraint.

Model	$\chi^2_{\min}$	k	Δ AIC	Δ BIC
Λ CDM	3.21	1	0.00	0.00
CPL	2.43	3	3.22	6.27
G1DE	2.55	2	1.34	2.87
G1DE+ μ	2.54	3	3.33	6.38

The demonstration intentionally allows G1DE to lose to Λ CDM after parameter penalties. This is a feature, not a bug: a viable G1 residual sector must survive information criteria and not rely on arbitrary functional freedom.

Exact-likelihood reference. The G1fit-real[12] matched production-refined comparison gives the following chi-squared decomposition at the G1DE-2 best-fit point:

Table 3: Exact code-path validation for G1DE-2 (from G1fit-real). Matched production-refined values; nested-evidence comparison in G1fit-real.

block	χ^2
SN (Pantheon+ full covariance)	1749.609044
BAO (DESI DR2)	9.633696
RSD (curated $f\sigma_8$)	6.883906
E_G (compressed)	1.010038
total	1767.136683

8 Exact-likelihood pilot results (G1fit-real summary)

The matched exact-likelihood and medium-prior nested-evidence comparison for G1DE is reported in the companion data note G1fit-real[12]. We reproduce its core diagnostics here to support the falsification ladder below. For the full six-parameter posterior, corner plots, trace diagnostics, and data-block details, see G1fit-real.

The pilot chain setup is 3 seeds $\times$ 48 walkers $\times$ 5000 iterations, with combined $N_{\text{postburn}} = 360\,000$ post-burn samples. Cross-seed split- $\hat{R}$ is below 1.0013 for all parameters.

8.1 Exact code-path χ^2 decomposition

See Table 3 in the preceding section.

8.2 Posterior summary

Table 4: Combined three-seed exact posterior for G1DE-2 (from G1fit-real). Full table at G1fit-real Table 1.

parameter	mean	std	q_{50}	interpretation
s	2.5598	0.1175	2.5595	background filling below $s = 3$
μ_0	0.0449	0.2603	0.0232	growth response consistent with zero
Σ_0	-0.3303	0.1404	-0.3375	negative Weyl response

8.3 Sign probabilities

Table 5: Sign-probability diagnostics for G1DE-2 (from G1fit-real).

diagnostic	probability
$P(s < 3)$	0.999922
$P(\mu_0 < 0)$	0.464564
$P(\Sigma_0 < 0)$	0.984525
$P(\mu_0 < 0, \Sigma_0 < 0)$	0.464394

The main pilot result is therefore

$$s < 3, \quad \Sigma_0 < 0, \quad \mu_0 \simeq 0,$$

i.e. a background filling deviation and a suppressed Weyl response, with the growth response near GR. Physical-prior audits confirm $\mu(z) > 0$ and $\Sigma(z) > 0$ over $0 \leq z \leq 2$ at the posterior median.

9 Demonstration figures

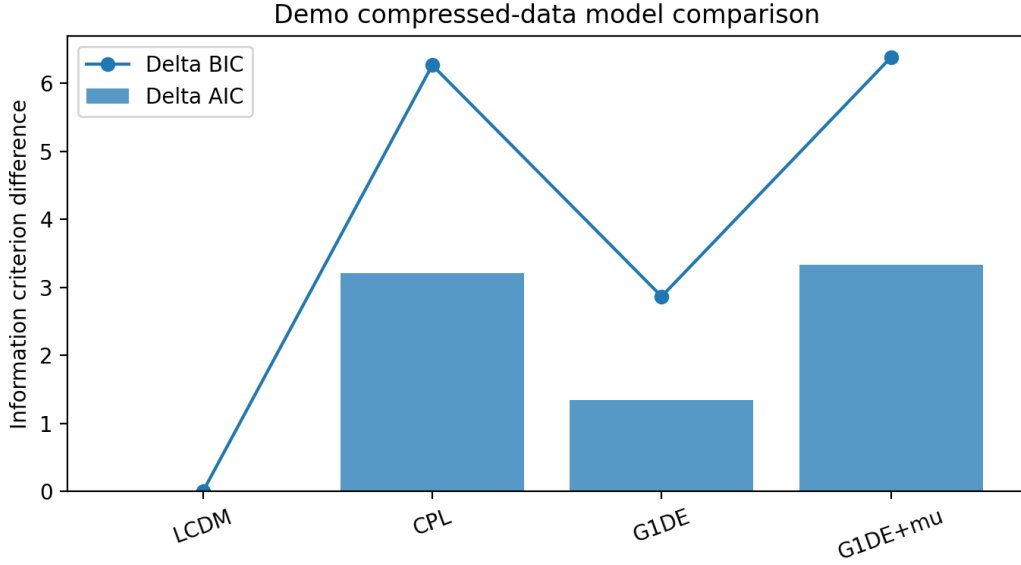


Figure 1: Demo information criteria. The included compressed toy dataset is generated close to Λ CDM, so the simplest model is expected to be competitive. Synthetic demo only — no cosmological constraint.

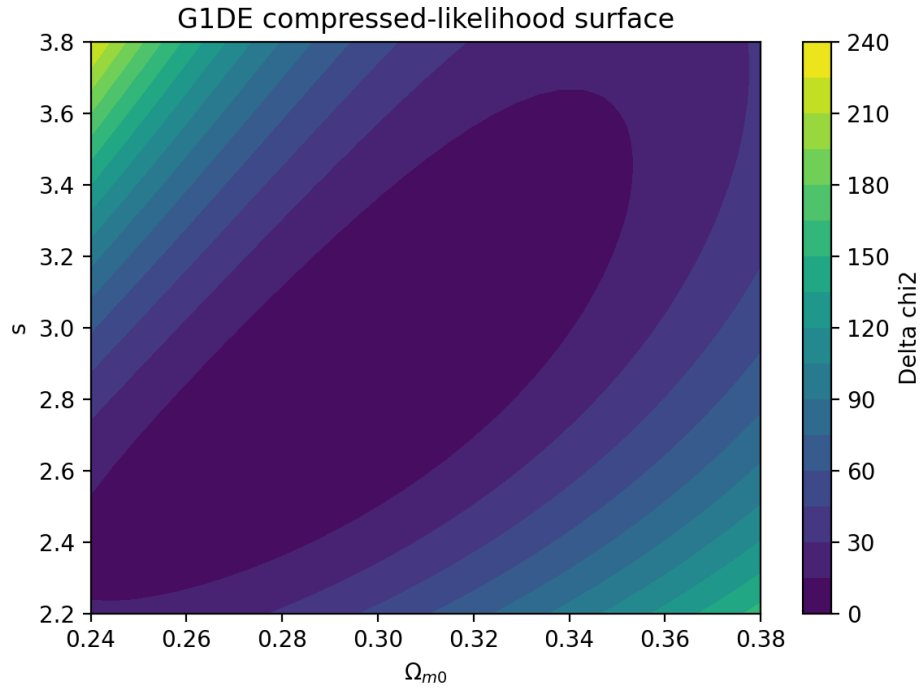


Figure 2: Demo $\Delta\chi^2$ surface for the G1DE background model in the (Ω_{m0}, s) plane. The LCDM-like limit is near $s \simeq 3$. Synthetic demo only — no cosmological constraint.

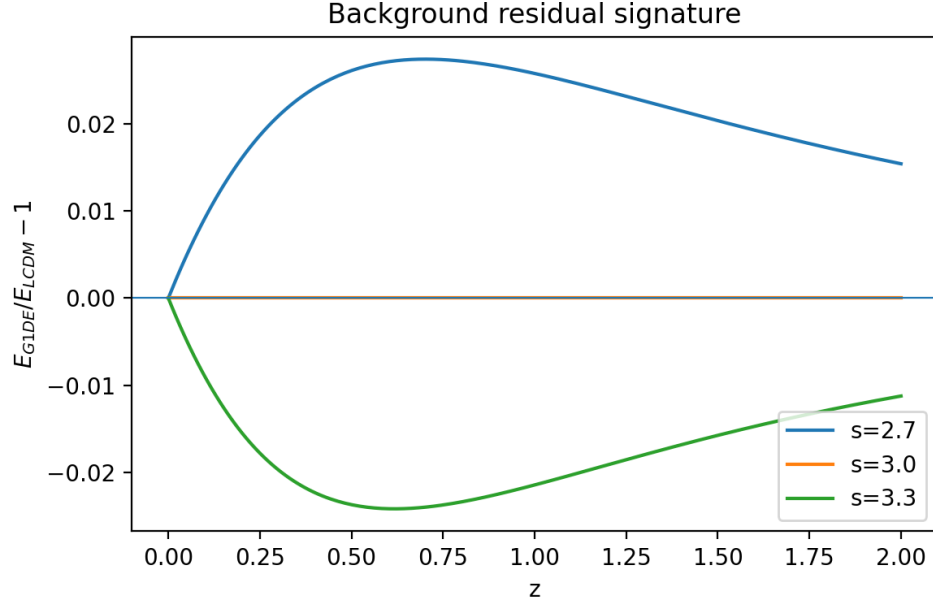


Figure 3: Background expansion residuals relative to Λ CDM for three G1DE filling slopes. The falsifiable target is not arbitrary background flexibility but correlated residual structure. Synthetic demo only — no cosmological constraint.

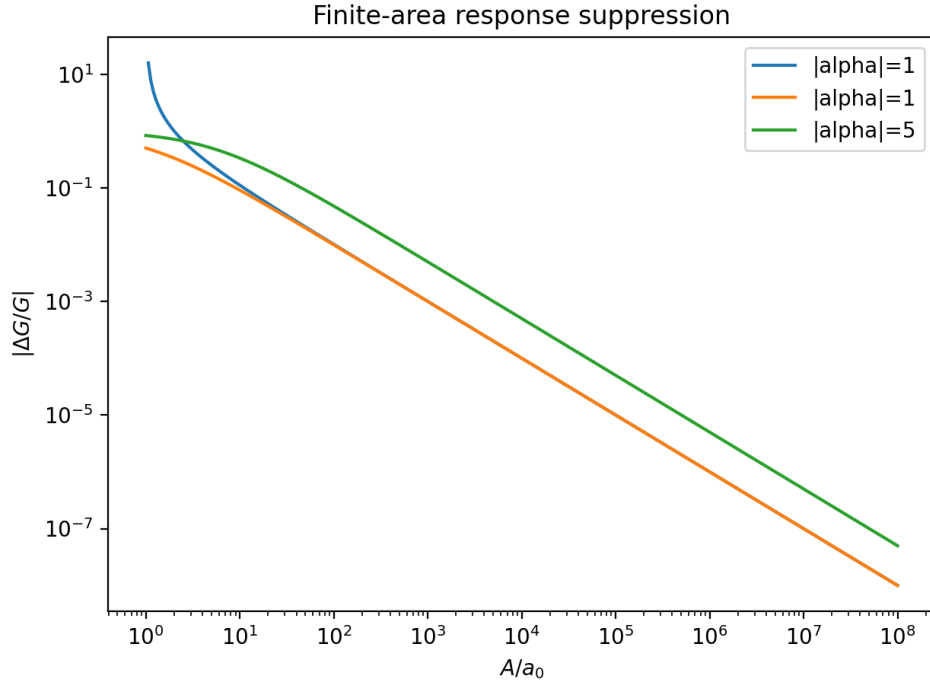


Figure 4: Finite-area suppression of ordinary logarithmic corrections. Large macroscopic signals cannot usually be attributed to bare $\ln A$ terms alone. Synthetic demo only — no cosmological constraint.

10 Observable dictionary

Table 6: Minimal observable dictionary for replacing the demo data with real survey blocks.

Observable	Tests	G1 quantity	Failure mode
SN $\mu(z)$	background expansion	$\chi_H(a)$ or $E(a)$	cannot match distances without excessive parameters
BAO $D_M/r_d, D_H/r_d$	expansion geometry	$E(a)$, distance	inconsistent background residual
$f\sigma_8(z)$	growth	$\mu(a)$, $\chi_H(a)$	growth-lensing correlation fails
Lensing proxy $\Sigma(a)$	Weyl response	$\Sigma_0 \hat{X}(a)$	lensing independent of ledger shape
$\dot{G}/G$	capacity drift	α_M	free macroscopic drift
GW speed flag	tensor propagation	α_T	non-luminal tensor sector

11 Production Evidence Checklist

The production-refined evidence hierarchy is a completed medium-prior pass, not the final survey-grade standard. The full production validation pass should incorporate the complete dark-sector control hierarchy under wider prior families, independent rerun scripts, and full 3×2 pt lensing likelihoods.

Validation item	Requirement
Model set	$M_{3/4}$, M_κ , constant- Σ , G1DE-2, G1DE-1, CPL, Λ CDM
Nested-sampling tolerance	$d \log Z \leq 0.1$, with stricter reruns for top competitors if needed
Seeds	at least 8–10 independent seeds per model
Evidence uncertainty	report native dynesty evidence errors and seed-to-seed scatter
Prior families	medium, wide, and physical-prior variants
Negative control	include G1DE-1 nested evidence as the growth-only control
Shape control	compare $R_{bH}(a)$ against constant- Σ and binned- $\Sigma(z)$ under full 3×2 pt
Mock control	deterministic mock injection passed (G1fit-real v1.1); noisy ensembles pending
Public reproducibility	blind rerun script, data-vector manifest, covariance manifest, prior file, and expected-output table

Table 7: Production validation checklist for the G1DE evidence hierarchy.

12 Evidence Layer Stratification

Layer	Model status	Evidence status	Use
Demo	synthetic compressed data	not empirical	code-path validation
Exact-pilot	G1DE-2 / controls	real likelihood, posterior only	sign/channel diagnosis
MCMC			
Legacy six-model evidence	$M_{3/4}$, M_κ , const- Σ , G1DE-2, CPL, Λ CDM	legacy/mixed provenance; superseded	historical record
Raw seven-model audit	includes G1DE-1	provenance-transparent, mixed prior labels	provenance appendix
Production-refined medium audit	all 7 medium prior, 8 seeds, $d \log Z = 0.1$	completed (v1.2)	canonical evidence for model ranking
Full production survey validation	full prior-family stress, independent replication, full 3×2 pt, CMB/BBN/nonlinear compatibility	pending	final survey-grade validation
KiDS shear-only diagnostic	$M_{3/4}$, M_κ , const- Σ , binned- Σ , Λ CDM	completed diagnostic; not production evidence	Weyl-channel and false-positive stress

Table 8: Evidence-layer stratification for the G1DE testing pipeline.

13 G1DE-1 Negative-Control Audit

G1DE-1 has been run as a nested-evidence control under the same medium-prior configuration used for all seven models in the production-refined audit. Its role is to test whether the signal is hidden growth modification rather than the claimed $H + \Sigma$ background–Weyl residual. In the eight-seed production-refined run it gives

$$\log Z_{\text{G1DE1}} = -901.751 \pm 0.180, \quad \Delta \log Z(M_{3/4}, \text{G1DE1}) = 7.402 \ (B \simeq 1.65 \times 10^3).$$

The completed medium-prior production-refined audit supports the negative-control separation: G1DE-1 is placed decisively below the projection-locked branch. The medium-prior production-refined audit is complete; full survey-grade validation remains pending. The growth-only negative control is decisively worse than the projection-locked branch. If future data bring G1DE-1 competitive with or above $M_{3/4}$, the interpretation that the primary signal is $H + \Sigma$ with $\mu \simeq 1$ is demoted.

Control	Tests	Demotion if favored
G1DE-1	growth-only residual explanation	$H + \Sigma$, $\mu \simeq 1$ interpretation weakened
G1DE-2	free response envelope	projection lock challenged
M_κ	free optical projection coefficient	exact 3/4 lock demoted
constant- Σ	amplitude without $R_{bH}(a)$ shape	shape lock demoted
CPL	background flexibility	dark-sector mechanism demoted
Λ CDM	null residual	residual branch demoted

Table 9: Negative-control and adversarial-control hierarchy for the G1DE branch.

14 Falsification criteria

A finite screen-capacity residual model should be rejected, revised, or demoted if it violates one of the following criteria.

1. **Unclosed G drift.** The model requires observable $\dot{G}/G$ without a conservation-closed residual or screening mechanism.
2. **Uncorrelated residual freedom.** Independent arbitrary functions are needed for expansion, growth, and lensing, destroying the common ledger-variable structure.
3. **Macroscopic finite-area overreach.** Large late-time deviations are attributed to ordinary $1/A$ or $\ln A/A$ corrections on macroscopic screens.
4. **Information-criterion failure.** The model improves raw χ^2 but loses decisively to simpler baselines after AIC/BIC/Bayes penalties.
5. **Tensor-sector failure.** The model implies a non-luminal tensor speed without a viable escape from GW propagation constraints.
6. **Instability.** The scalar-capacity sector violates positivity, ghost-free, gradient-stability, or tachyon-stability conditions.
7. **Mock-injection false-positive failure.** Deterministic or noisy mock injection shows that Λ CDM, constant- Σ , or binned- Σ truths are systematically misclassified as $M_{3/4}$, demoting the KiDS shear-only interpretation.

Rejection thresholds. For production model comparison, the following thresholds are pre-specified:

- $\Delta\text{BIC} > 10$: strongly disfavoured relative to the best model.
- $\Delta\text{AIC} > 6$: substantially worse by AIC.
- Bayes factor $< 1/10$: demote the residual model (requires evidence calculation).
- Unscreened $\dot{G}/G$ violation: reject unscreened scalar-capacity variant.
- Loss of the shared ledger-response structure: demote the G1DE perturbative fingerprint.
- **Refined G1DE-2 falsification** (post-pilot update):
 - **Background–Weyl signature.** The current G1DE-2 empirical core is $s < 3$ and $\Sigma_0 < 0$. If $P(s < 3)$ or $P(\Sigma_0 < 0)$ falls below the pre-registered guardrail (e.g. 0.90) under matched exact production chains or expanded E_G /lensing data, the current empirical signature is demoted.
 - **Growth consistency.** The growth channel is a guardrail, not a required nonzero signal. The current exact pilot supports $\mu_0 \simeq 0$, so a null growth response is not a failure of G1DE-2. The model is pressured only if future growth data require a large μ_0 that breaks the shared occupancy-response interpretation or conflicts with the Weyl sector.

– **Projection-locking target.** The post-pilot minimal model G1DE-M_{3/4} imposes

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s), \quad \kappa_{\text{BW}} \equiv \frac{-\Sigma_0}{3 - s} = \frac{3}{4}.$$

This is motivated by the trace-free projection of an isotropic spatial maintenance load in $3 + 1$ dimensions (see Companion E Appendix A). Two adversarial controls support the amplitude-level lock: a free- κ model does not reject $3/4$, and a constant- Σ model independently recovers $\kappa_{\text{BW}} \simeq 0.76$. If production chains or expanded lensing data move κ_{BW} decisively away from $3/4$, the projection-locking interpretation is demoted. If $P(\Sigma_0 < 0)$ remains high but $\kappa_{\text{BW}} \neq 3/4$, the broader background-Weyl residual fingerprint survives while the locked-ratio model fails. The $\hat{X}(a)$ redshift-shape discrimination requires expanded Weyl/lensing data and is not yet resolved by the current three-point E_G block.

These thresholds are not definitive rejection rules; they are pre-registration-style guardrails that prevent post hoc reinterpretation of borderline information criteria.

Production pipeline.

Pantheon+ covariance + DESI BAO covariance + growth/lensing compilations + local priors

↓

MCMC or nested sampling $\rightarrow \Delta\text{AIC}, \Delta\text{BIC}, \log B$.

This pipeline is defined but not executed in this companion. The demo code provides a verified code path; production inference requires replacing the diagonal-covariance demo with official survey covariance matrices.

15 Parameter prior table

Table 10: Prior ranges and status for the demo parameters.

Parameter	Meaning	Prior range	Status
Ω_{m0}	matter density today	$[0.1, 0.6]$	fitted
s	G1DE filling-closure slope	$[0.5, 5]$	fitted
χ_∞	G1DE asymptotic occupancy	$[0, 1]$	fixed or fitted
μ_0	growth-response amplitude (G1DE-1)	$[-1, 1]$	fitted in extended model
Σ_0	lensing-response amplitude	$[-1, 1]$	independent in exact pilot
α_M	Planck-mass running	$[-1, 1]$	derived from $\varphi(a)$
$\mathcal{M}$	SN absolute magnitude nuisance	marginalized	analytically eliminated
$\sigma_{8,0}$	present-day clustering amplitude	fixed to Planck prior	not fitted in demo

16 Code and data archive

The companion package contains:

```
D_Likelihood_Falsification/
D_Likelihood_Falsification.tex
D_Likelihood_Falsification.pdf
```

```

code/fds_g1_likelihood_falsification_demo.py
notebooks/D_Likelihood_Falsification_demo.ipynb
data/demo_sn.csv
data/demo_bao.csv
data/demo_growth.csv
data/model_comparison.csv
data/observable_dictionary.csv
data/falsification_criteria.csv
figures/*.png

```

All data files are demonstration data unless explicitly replaced by external survey products. The code is designed to expose model assumptions rather than hide them in black-box fitting.

17 Failure registry

Table 11: Layered failure registry for Companion D.

Failure condition	What fails	What survives
No real-data implementation	Empirical status remains demo-level	FDS-G1 Core entropy-response framework
G1DE loses to Λ CDM	Horizon-occupancy residual as preferred DE model	Screen statistics and EFT interface
Free G drift excluded	Unscreened scalar-capacity cosmology	Microscopic / horizon-scale capacity corrections
Background and Weyl require unrelated shapes	G1DE-2 shared-ledger fingerprint	Background-only G1DE-0
G1DE-2 $\mu_0 \simeq 0$ confirmed	Generic growth-modified prediction	Refined background-Weyl residual fingerprint
G1DE-1 $\mu_0 > 0$ when Weyl absent	Growth-only residual explanation	Projection-locked background-Weyl residual
AIC/BIC strongly penalize residuals	Extra-parameter dark residuals	Minimal screen-capacity interpretation of $1/G$

18 Conclusion

Companion D provides the empirical discipline for the G1 program. The main theoretical chain says

$$S_{\text{scr}}[X] \rightarrow \eta_{\text{loc}} \rightarrow G_{\text{loc}} \rightarrow \text{capacity-flow} \rightarrow \text{integrable geometry closure.} \quad (34)$$

The observational chain says

$$\text{residual sector} \rightarrow H(z), f\sigma_8(z), \Sigma(z), \dot{G}/G \rightarrow \chi^2, \text{AIC, BIC, Bayes factors.} \quad (35)$$

A mature finite screen-capacity theory must be allowed to fail. This companion defines how it can fail: by violating conservation closure, by requiring free macroscopic G drift, by producing unsuppressed macroscopic finite-area corrections, or by being outperformed by simpler baselines under real data and parameter penalties. The matched medium-prior production-refined nested-evidence comparison (G1fit-real[12]) has now refined the G1DE empirical fingerprint to a background-plus-Weyl residual pattern: $s < 3$, $\Sigma_0 < 0$, $\mu_0 \simeq 0$, with G1DE-M_{3/4} ranking first in the completed medium-prior 8-seed

production-refined audit hierarchy. KiDS shear-only diagnostic and deterministic mock-injection audit have been completed as a Weyl-channel stress check. The remaining tasks are prior-family stress testing, full 3×2 pt lensing including galaxy-galaxy lensing and clustering, noisy mock ensembles, CMB/BBN/nonlinear compatibility, and independent pipeline replication.

References

- [1] D. Scolnic et al., “The Pantheon+ Type Ia Supernova Sample,” *Astrophys. J.* 938, 113 (2022), arXiv:2112.03863.
- [2] D. Brout et al., “The Pantheon+ Analysis: Cosmological Constraints,” *Astrophys. J.* 938, 110 (2022), arXiv:2202.04077.
- [3] M. Abdul Karim et al. (DESI Collaboration), “DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations,” arXiv:2503.14738 (2025).
- [4] B. Bertotti, L. Iess, and P. Tortora, “A test of general relativity using radio links with the Cassini spacecraft,” *Nature* 425, 374–376 (2003).
- [5] T. Baker et al., “Strong constraints on cosmological gravity from GW170817 and GRB 170817A,” *Phys. Rev. Lett.* 119, 251301 (2017).
- [6] A. G. Adame et al. (DESI Collaboration), “DESI 2024 III: Baryon Acoustic Oscillations from Galaxies and Quasars,” *JCAP* 02, 021 (2025), arXiv:2404.03000.
- [7] A. Chudaykin and M. Kunz, “Modified gravity interpretation of the evolving dark energy in light of DESI data,” arXiv:2407.02558 (2024).
- [8] T. Li, J. Zhang, and X. Zhang, “Revisiting holographic dark energy after DESI 2024,” arXiv:2411.08639 (2024).
- [9] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *JCAP* 07, 050 (2014), arXiv:1404.3713.
- [10] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), <https://doi.org/10.5281/zenodo.20382013>.
- [11] Y. Wu, “FDS-X1: Horizon-Maintenance Dark Energy and Horizon Ledger Occupancy (Pre-Euclid Pre-registration),” Zenodo (2026), <https://doi.org/10.5281/zenodo.20290215>.
- [12] Y. Wu, “G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual,” manuscript (2026).

Companion E: Horizon Occupancy Ledger Dynamics and Correlated Dark Residuals

A Finite Screen-Capacity Model for Background Expansion, Weyl Response,
and Near-GR Growth

v1.2 Body-Integrated Release

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This companion paper develops the horizon-occupancy residual sector of the FDS-G1 finite screen-capacity program. In the core paper, local gravitational coupling is the inverse marginal entropy-area response of a finite causal screen under the standard horizon bridge, while stiffness G_{loc}^{-1} is proportional to the response itself.

$$\rho_H(a) = \chi_H(a)\rho_c(a), \quad \rho_c(a) = \frac{3H^2(a)}{8\pi G_0}.$$

The minimum filling-closure law for χ_H gives a logistic occupancy profile and an effective equation of state $w_H(a)$; in the matter-era small-occupancy limit it yields $w_H \simeq -s/3$. The paper then defines three increasingly constrained residual classes: background-only G1DE-0, growth-linked G1DE-1, and growth-plus-lensing G1DE-2. The completed medium-prior 8-seed production-refined audit (G1fit-real[12]) selects the projection-locked G1DE-M_{3/4} model across the full seven-model hierarchy. Medium-prior production refinement is complete; full survey-grade validation (full 3×2 pt, noisy mocks, CMB/BBN, independent replication) remains pending. An independent KiDS-1000 shear-only diagnostic stress test (G1fit-real v1.1) supports the Weyl-channel signal and the 3/4 projection lock under nuisance profiling; full 3×2 pt validation remains pending. The locked branch gives $\Sigma(a) = 1 - \frac{3}{4}(3-s)X_b(a)$, $\mu \simeq 1$: a background-plus-Weyl residual pattern with the growth response near GR.

1 Residual-Sector Scope and Closure Status

This companion develops a homogeneous dark-residual sector of the FDS-G1 program: a horizon ledger occupancy fraction $\chi_H(a)$ that contributes an effective dark component. In D0, the horizon residual is interpreted as a passive remnant of incomplete screen modular anomaly inflow: the occupancy variable $\mathcal{O}_H = C_{\text{used}}/C_H$ measures horizon-scale access load relative to available finite screen capacity, and the source current is $J_H = -d_N \ln \mathcal{O}_H$, with the G1DE low-redshift reduction $\mathcal{R}_H(a) = R_0 \hat{R}_H(a)$, $R_0 = 3 - s$. This companion develops the G1 version of the horizon-ledger occupancy idea registered in FDS-X1 [9], replacing the pre-registration note by a structured background-growth-lensing residual hierarchy that also draws on the geometry-as-horizon-accounting language of FDS-T2 [10]. In this sense horizon occupancy is

one realization of the gravity-as-boundary-accounting row in the functional decomposition of FDS-X3 [11].

The core G1 principle is that finite screen entropy response defines a local area stiffness,

$$\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}, \quad G_{\text{loc}}(x) = \frac{c^3}{4\hbar} [\eta_{\text{loc}}(x)]^{-1}, \quad (1)$$

so that η_{loc} controls the inverse coupling (stiffness) and G_{loc} is the corresponding coupling. Non-integrable, coarse-grained, or unclosed response data appear as residual sectors. This companion studies one homogeneous residual ansatz based on the occupancy of a cosmological horizon ledger.

Scope. This paper gives a minimal ledger dynamics that is more structured than a free $w(a)$ parametrization, because the same occupancy variable can be required to control background expansion, growth, lensing, and capacity closure. It does not claim a microscopic derivation of $\chi_H(a)$ from FDS primitives.

Observational status. A completed homogeneous seven-model medium-prior nested-evidence audit (G1fit-real[12]) has been executed with Pantheon+ full covariance, DESI DR2 BAO, curated RSD, and compressed E_G . The canonical evidence hierarchy is

$$\text{G1DE-M}_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE2} > \text{G1DE1} > \text{CPL} > \Lambda\text{CDM},$$

with the projection-locked model ranking first. Medium-prior production refinement (eight seeds, $d \log Z = 0.1$) is complete; full prior-family stress testing and full $3 \times 2\text{pt}$ validation remain pending.

2 Residual interpretation

In FDS-G1 Core, residuals are classified as non-integrable response data. The present model specializes to a homogeneous physical residual:

$$\mathcal{R}_{\mu\nu}^H \longleftrightarrow \text{horizon occupancy response not absorbed into ordinary matter.} \quad (2)$$

The residual is called “horizon occupancy” because it tracks a coarse-grained fraction of available screen response capacity that contributes to the background ledger.

Definition 1 (Horizon occupancy). *Let $\chi_H(a)$ be a dimensionless occupancy fraction associated with the homogeneous causal horizon ledger. It is normalized so that $0 \leq \chi_H(a) < 1$ in the background solutions considered here. Its effective energy density is defined by*

$$\rho_H(a) = \chi_H(a) \rho_c(a), \quad \rho_c(a) = \frac{3H^2(a)}{8\pi G_0}. \quad (3)$$

The use of $\rho_c(a)$ rather than a fixed reference density is intentional. It expresses the idea that the residual is a fractional occupancy of the instantaneous global expansion ledger, not a separately conserved fluid placed by hand.

3 Background closure

For a spatially flat background with matter, radiation, and horizon occupancy residual, the Friedmann closure is written as

$$H^2(a) = H_0^2 [\Omega_{m0} a^{-3} + \Omega_{r0} a^{-4}] + \frac{8\pi G_0}{3} \rho_H(a). \quad (4)$$

Using $\rho_H = \chi_H \rho_c = \chi_H 3H^2/(8\pi G_0)$ gives

$$E^2(a) \equiv \frac{H^2(a)}{H_0^2} = \frac{\Omega_{m0}a^{-3} + \Omega_{r0}a^{-4}}{1 - \chi_H(a)}. \quad (5)$$

This equation is useful because it forces the residual to remain tied to the expansion ledger. It is also restrictive: $\chi_H(a)$ cannot be chosen independently of $H(a)$ once the closure is imposed.

The effective equation of state is defined by the usual continuity-form diagnostic

$$w_H(a) = -1 - \frac{1}{3} \frac{d \ln \rho_H}{d \ln a} = -1 - \frac{1}{3} \frac{d \ln [\chi_H(a) H^2(a)]}{d \ln a}. \quad (6)$$

This w_H is a diagnostic of background scaling, not a proof that the residual is a canonical scalar field.

Conservation condition. No horizon-occupancy residual is admissible without conservation closure:

$$\nabla^\mu R_{\mu\nu}^H = 0,$$

or, if matter exchange is allowed, $\nabla^\mu (T_{\mu\nu} + R_{\mu\nu}^H) = 0$. An unconserved effective stress is rejected at the admissibility stage.

4 Ledger filling-closure dynamics

The minimal G1DE model replaces a purely phenomenological logistic ansatz by a filling-closure equation in e-fold time $N = \ln a$:

$$\frac{d\chi_H}{dN} = s \chi_H \left(1 - \frac{\chi_H}{\chi_\infty} \right), \quad (7)$$

where s is a filling rate and χ_∞ is an asymptotic ledger capacity fraction. The solution is

$$\chi_H(a) = \frac{\chi_\infty}{1 + Ba^{-s}}, \quad B = \frac{\chi_\infty}{\chi_{H0}} - 1. \quad (8)$$

For $\chi_H \ll 1$ during matter domination, $\chi_H \propto a^s$ and $H^2 \propto a^{-3}$, so

$$\rho_H \propto a^{s-3}, \quad w_H \simeq -\frac{s}{3}. \quad (9)$$

Thus $s = 3$ is cosmological-constant-like in the matter-era scaling limit, $s < 3$ is quintessence-like, and $s > 3$ is phantom-like at the level of the effective background diagnostic.

Λ CDM-like limit. When $\chi_\infty = 1$ and $s = 3$, the background reduces exactly to flat Λ CDM. Writing $B = \Omega_{m0}/(1 - \Omega_{m0})$ gives

$$1 - \chi_H(a) = \frac{Ba^{-3}}{1 + Ba^{-3}}, \quad E^2(a) = \frac{\Omega_{m0}a^{-3}}{1 - \chi_H(a)} = \Omega_{m0}a^{-3} + \frac{\Omega_{m0}}{B} = \Omega_{m0}a^{-3} + (1 - \Omega_{m0}).$$

Thus $s = 3$ is not an arbitrary parametrization; it is the Λ CDM-like consistency limit. The G1DE residual paradigm lives in the departures from this limit controlled by $s \neq 3$ and $\chi_\infty \neq 1$.

Proposition 1 (Lowest-order horizon filling closure). *Equation (7) is the lowest-order autonomous filling law compatible with positivity, finite saturation, and a single homogeneous occupancy variable. It should be regarded as the leading term in a more general ledger dynamics*

$$\frac{d\chi_H}{dN} = \Gamma_{\text{fill}}(a) \chi_H \left(1 - \frac{\chi_H}{\chi_\infty} \right) - \Gamma_{\text{close}}(a) \chi_H + \xi_H(N), \quad (10)$$

where closure, leakage, and stochastic terms are set to their minimal values in the baseline model.

5 Growth and lensing residual hierarchy

A useful residual theory cannot only fit the background. The G1DE signal must be evaluated through correlated deviations in expansion, growth, and lensing. Define the occupancy-shape function

$$X(a) = \chi_H(a) \left(1 - \frac{\chi_H(a)}{\chi_\infty}\right), \quad \hat{X}(a) = \frac{4}{\chi_\infty} X(a). \quad (11)$$

The normalized function $\hat{X}$ is peaked when the ledger is actively filling and vanishes in the empty and saturated limits. Its normalization is chosen so that $0 \leq \hat{X} \leq 1$ (given $0 \leq \chi_H \leq \chi_\infty$), with the maximum occurring at $\chi_H = \chi_\infty/2$.

We define three model classes.

G1DE-0 Background-only residual. $\chi_H(a)$ affects $H(a)$ through (5); linear perturbations follow GR with the modified background.

G1DE-1 Growth-linked residual. The standard quasistatic Poisson equation is modified by an effective growth coupling μ ,

$$k^2 \Psi = -4\pi G a^2 \mu(a) \rho_m \Delta_m, \quad \mu(a) = 1 + \mu_0 \hat{X}(a). \quad (12)$$

G1DE-2 Growth-plus-lensing residual. The Weyl potential responds through an effective lensing coupling Σ ,

$$k^2(\Phi + \Psi) = -8\pi G a^2 \Sigma(a) \rho_m \Delta_m, \quad \Sigma(a) = 1 + \Sigma_0 \hat{X}(a). \quad (13)$$

The linear growth equation in e-fold time is

$$D'' + \left[2 + \frac{d \ln H}{dN}\right] D' - \frac{3}{2} \mu(a) \Omega_m(a) D = 0, \quad (14)$$

where prime denotes d/dN and

$$\Omega_m(a) = \frac{\Omega_{m0} a^{-3}}{E^2(a)}. \quad (15)$$

The observable proxy is

$$f\sigma_8(a) = \frac{d \ln D}{d \ln a} \sigma_{8,0} D(a). \quad (16)$$

Key diagnostic. The distinctive test is not whether $H(z)$ can be fit. Many models can fit a distance curve. The current matched exact-pilot and medium-prior nested-evidence comparison (G1fit-real[12]) refines the G1DE empirical fingerprint to

$$\boxed{s < 3, \quad \Sigma_0 < 0, \quad \mu_0 \simeq 0,} \quad (17)$$

i.e. a background filling deviation and a suppressed Weyl response, with the growth response near GR. Thus the current G1DE-2 pattern is

$$\chi_H(a) \Rightarrow H(a), \quad \Sigma(a), \quad \mu(a) \simeq 1. \quad (18)$$

The growth-response parameter μ_0 acts as a growth-sector guardrail rather than a required nonzero signal.

5.1 Current empirical status

The G1DE residual is now implemented through a Weyl-normalized transfer operator rather than as an independent dark stress source:

$$\mathcal{W}_{\text{G1}} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}}, \quad \mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1, \quad R_0 = 3 - s.$$

D10 interprets $R_H(a) = R_0 R_{bH}(a)$ as the minimal rank-one passive horizon Green-response branch.

The factorized production response is therefore

$$\Sigma(a, k) - 1 = -\kappa R_0 \hat{R}_H(a) + O(R_0^2), \quad \mu(a, k) - 1 = O(R_0^2) + O(R_0 \hat{R}_H \mathcal{H}^2/k^2).$$

For the locked branch $\kappa = 3/4$, $\hat{R}_H = X_b$, giving $\Sigma(a) = 1 - \frac{3}{4}(3-s)X_b(a)$, $\mu \simeq 1$. The amplitude lock ($\kappa \sim 3/4$) is now supported by medium-prior nested evidence and independently by KiDS shear-only free- κ profiling ($\kappa \approx 0.746$). The memory-shape $X_b(a)$ has preliminary support; KiDS binned- $\Sigma(z)$ controls find redshift-shape information that is present but nuisance-entangled, so shape resolution requires full $3 \times 2\text{pt}$ data.

Table 1: Current empirical status of G1DE hierarchy.

Parameter	Pilot value	Interpretation
s	2.55 ± 0.09	below ΛCDM -like $s = 3$; filling defect robust
$\mu \simeq 1$	consistent	near-GR growth; Ward-suppressed leakage
$\Sigma(a)$	$1 - \frac{3}{4}(3-s)X_b(a)$	Weyl-transfer amplitude lock selected
κ_{BW}	$\simeq 3/4$	isotropic compliance projection supported

This is more restrictive than a generic dark-energy or modified-gravity parametrization because the background and Weyl response share the same active-filling shape $\hat{X}(a)$, while the matter-growth response stays near GR.

The post-pilot locked-ratio variant G1DE-M $_{3/4}$, defined by

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3-s),$$

is the primary perturbative residual model in the matched exact-pilot comparison: it is preferred over ΛCDM , CPL, G1DE-1, and G1DE-2 by AIC and BIC while matching the G1DE-2 χ^2 to numerical precision (see G1fit-real). The three-seed pilots support the amplitude-level lock ($\kappa_{\text{BW}} \simeq 3/4$) more strongly than the specific $\hat{X}(a)$ redshift-shape modulation; free- κ and constant- Σ controls confirm that $\kappa = 3/4$ is not rejected and independently recovers in models without the active-filling shape. The ratio $3/4$ is motivated by the projection-locking lemma in Appendix A: a four-dimensional trace-free projection of an isotropic three-dimensional maintenance load naturally carries the coefficient $3/4$. This relation is a conditional theorem target and a falsifiable post-pilot hypothesis, not yet a first-principles derivation from microscopic FDS statistics.

5.2 Amplitude Lock vs Shape Lock

The current evidence hierarchy should be read in two layers.

First, the amplitude-level Weyl result is the stronger present claim:

$$\Sigma_0 < 0, \quad \kappa_{\text{BW}} \simeq \frac{3}{4}.$$

This says that the data favor a suppressed Weyl response with a projection coefficient close to the isotropic optical value.

Second, the redshift-shape lock is more demanding:

$$\Sigma(a) - 1 \propto R_{bH}(a).$$

The present compressed E_G data have limited redshift-shape resolution. The KiDS shear-only stress test adds diagnostic support but does not resolve shape: binned- $\Sigma(z)$ controls find low- χ^2 basins that are nuisance-entangled and boundary-sensitive. Therefore the $R_{bH}(a)$ shape should be treated as preliminary until full 3×2 pt likelihoods distinguish it from constant- Σ and binned- $\Sigma(z)$.

Layer	Current status	Future test
Amplitude lock	$\Sigma_0 < 0$, $\kappa \sim 3/4$ supported by current controls	free- κ and constant- Σ production evidence
Shape lock	$R_{bH}(a)$ under controlled KiDS binned- Σ stress	full 3×2 pt + binned controls shape discrimination

Table 2: Separation between Weyl-amplitude support and horizon-response redshift-shape support.

6 Comparison with common dark-energy routes

Table 3: Model-space comparison. G1DE is not intended to beat all existing models at the background level; its distinguishing feature is occupancy-linked correlations across expansion, growth, lensing, and closure.

Model	Background	Growth	Lensing	Local G	Distinctive issue
Λ CDM	fixed	GR	GR	fixed	null baseline
CPL $w_0 w_a$	flexible	usually GR	usually GR	fixed	background flexibility
HDE proxy	model-specific	often weak	often weak	fixed	horizon-scale density ansatz
Quintessence proxy	scalar background	GR if minimally coupled	GR if minimally coupled	fixed	potential choice
Scalar-tensor proxy	flexible	modified	modified	constrained	screening and GW speed
G1DE	occupancy ledger	linked by $\hat{X}$	linked by $\hat{X}$, $\Sigma_0 < 0$ preferred	closure-constrained	background-Weyl residual pattern

Holographic dark energy models are useful comparison targets because they also use horizon-scale quantities; CPL is the standard flexible background comparator; scalar-tensor proxies test whether G1DE is merely a renamed modified-gravity parametrization.

Division of labour with Companion D. Companion E develops the physical horizon-occupancy residual model and specifies its current empirical fingerprint: a background-plus-Weyl response pattern. Companion D owns the likelihood machinery, evidence comparison, and model-comparison protocol. G1fit-real[12] provides the standalone evidence data paper.

7 Numerical demonstration

The accompanying code generates transparent demo curves. These are not official survey constraints. They illustrate the internal behavior of the model and the type of data products expected from Companion D.

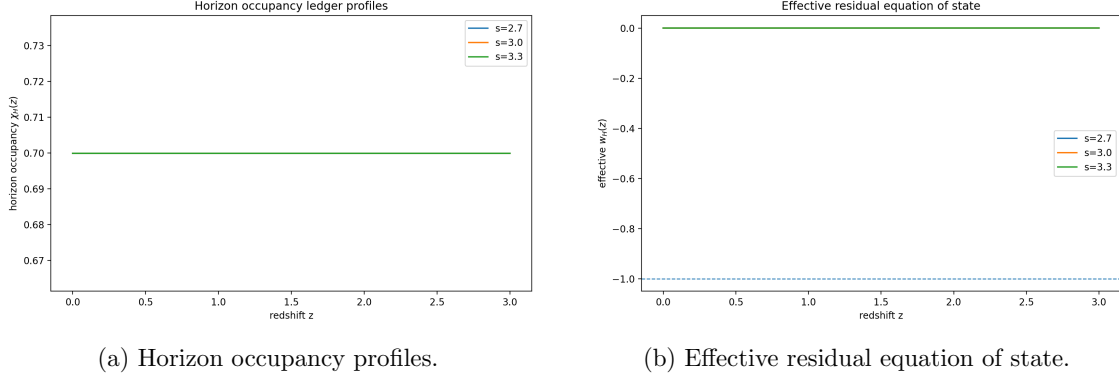


Figure 1: The filling parameter s controls the epoch and rate of horizon occupancy. Parameters are intentionally exaggerated for visualization ($\chi_{H0} = 0.69$, $\chi_{\infty} = 1$, $s = 2.7, 3.0, 3.3$); these are not survey-allowed fits. The matter-era limit gives $w_H \simeq -s/3$, but the exact diagnostic depends on the full closure (5).

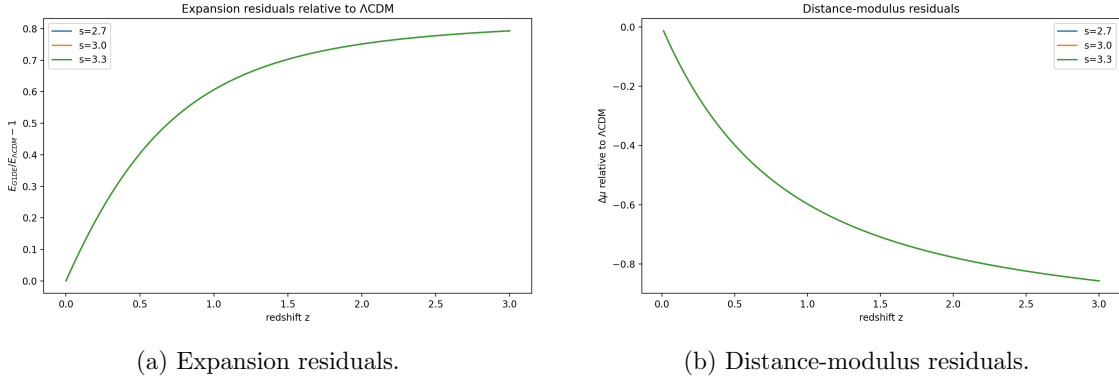


Figure 2: Background deviations relative to a matched flat Λ CDM reference. Parameters are exaggerated for visualization; real survey-level residuals are expected to be much smaller. A background-only fit is insufficient; G1DE must be tested jointly with growth and lensing.

8 Falsification criteria

A residual model is useful only if it can fail. The G1DE sector is disfavored or falsified under any of the following conditions.

1. **Background-only degeneracy.** If the model only reproduces a CPL-like distance curve but gives no robust background–Weyl residual signature ($s < 3$, $\Sigma_0 < 0$), its distinctive G1DE interpretation is weakened.
2. **No closure.** If a nontrivial $\chi_H(a)$ requires unconserved effective stress, the model violates the G1 conservation-closure rule.

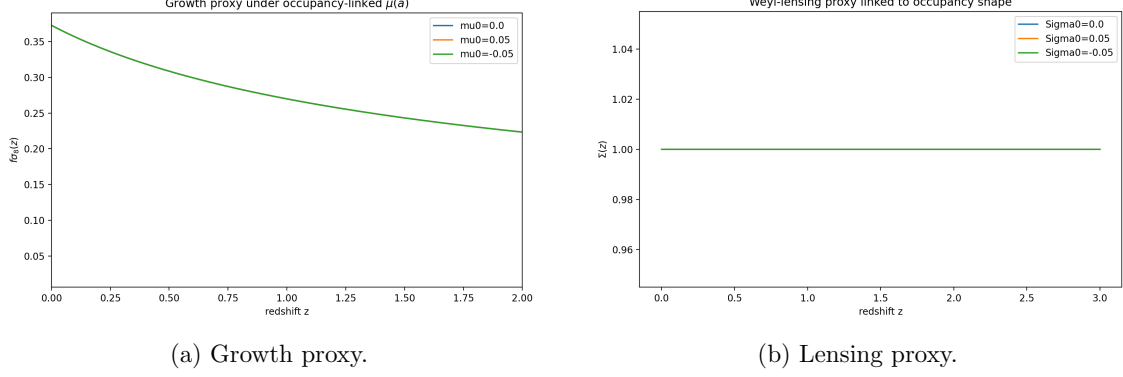


Figure 3: Growth and lensing are linked to the same occupancy-shape function $\hat{X}(a)$ in the G1DE-1 and G1DE-2 classes.

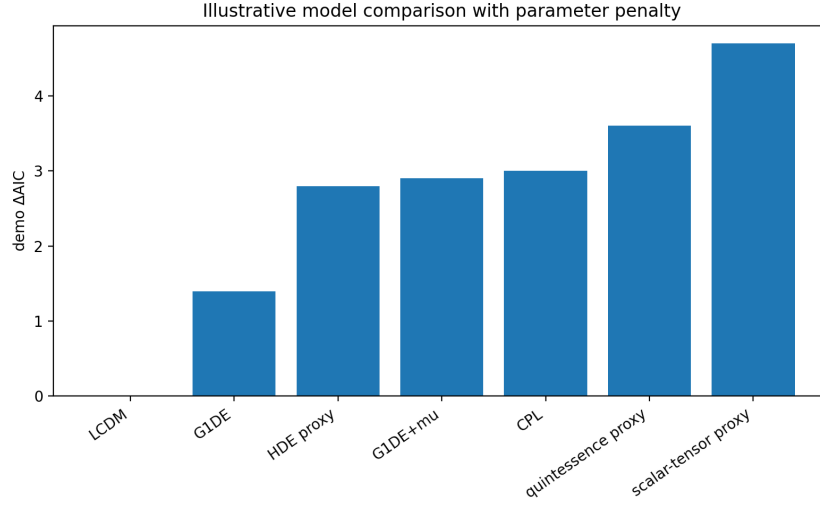


Figure 4: Illustrative AIC comparison using demo compressed values. The numbers are not survey constraints; they show how model comparison should be performed once real likelihoods are inserted.

3. **Over-flexibility.** If independent free functions are allowed for $H(a)$, $\mu(a)$, and $\Sigma(a)$, the model becomes a generic modified-gravity parametrization rather than a horizon occupancy residual.
4. **Local-gravity conflict.** If a capacity-linked residual implies macroscopic $\dot{G}/G$ without a screened or conserved closure sector, it is ruled out by local-gravity priors.
5. **Information criteria penalty.** If AIC/BIC or Bayesian evidence consistently favors simpler baselines such as Λ CDM or CPL, the residual model is not empirically supported.
6. **Growth-channel conflict.** The current pilot supports $\mu_0 \simeq 0$ (near-GR growth). This is not a failure of G1DE-2; it is a growth consistency condition. The model is pressured only if future growth data require a large μ_0 that breaks the shared occupancy-response interpretation or conflicts with the Weyl sector.
7. **Free $A(a, k)$ required.** If optimal likelihood or evidence requires free $A(a, k)$, the model becomes generic dark-stress, not G1DE.

9 Interface with other G1 companions

Companion	Interface
FDS-G1 Core	Treats χ_H as a homogeneous non-integrable response residual.
A: Screen statistics	Supplies the entropy-area response basis for finite screen capacity.
B: Transition dynamics	Supplies stochastic update and residual-noise language.
C: Scalar-capacity EFT	Supplies a field-theory closure route if occupancy couples to $\eta(x)$.
D: Likelihood/falsification	Provides compressed likelihood blocks and model comparison machinery.
F: Horizons/holography	Supplies the horizon ledger interpretation and holographic compatibility.

10 v1.2 Early-Universe Compatibility and Kill Tests

Claim type: Production compatibility requirement All G1DE residual models must satisfy strict early-universe constraints to be production viable:

$$\Omega_{\text{scr}}^{\text{res}}(a) \ll 1 \quad (a \leq a_{\text{rec}}, \text{ with explicit checks at BBN and recombination})$$

$$|\Delta N_{\text{eff}}^{\text{scr}}| < \Delta N_{\text{eff}}^{\text{max}}, \quad \left| \frac{\Delta H}{H} \right|_{\text{BBN,rec}} < \epsilon_{\text{early}}$$

Any residual producing unacceptable BBN expansion, acoustic peak shifts, early-ISW distortion, tensor sector inconsistency, or non-Gaussianity under a specified perturbation bridge triggers demotion of the corresponding early-universe extension. The numerical placeholder $\chi_H(a_{\text{CMB}}) \gtrsim 0.01$ in this failure registry is not a formal bound, only a diagnostic test threshold.

11 Failure registry

Table 4: Failure registry for horizon occupancy residuals.

Failure condition	What fails	What survives
No meaningful homogeneous screen occupancy	G1DE interpretation	Local screen-capacity framework
χ_H is arbitrary and independent of observables	Correlated residual claim	Phenomenological fit only
Growth/lensing require unrelated functions	G1DE-1/2 structure	Background G1DE-0
$\mu_0 \simeq 0$ confirmed by pilot	Generic growth-modified prediction	Refined background–Weyl residual fingerprint
$\hat{X}$ -linked Σ ruled out by expanded data	Background–Weyl correlated fingerprint	Background-only G1DE-0
Unclosed stress conservation	Covariant residual sector	Statistical screen models
Real likelihood strongly favors simpler models	Empirical motivation for G1DE	FDS-G1 Core and other companions

Failure condition	What fails	What survives
Binned- $\Sigma(z)$ robustly beats $R_{bH}(a)$ in full 3×2 pt	Locked shape $R_{bH}(a)$	Weyl-amplitude branch; binned- Σ as adversarial control
High-redshift occupancy too large ($\chi_H(a_{\text{CMB}}) \gtrsim 0.01$)	oc-CMB/BBN consistency	Late-time residual variants

12 Parameter table and prior ranges

Table 5: G1DE model hierarchy and free parameters.

Model	Parameters	k
G1DE-0	$\Omega_{m0}, s; \chi_\infty = 1, \chi_{H0} = 1 - \Omega_{m0} - \Omega_{r0}$	2
G1DE-1	G1DE-0 + μ_0	3
G1DE-2	G1DE-1 + Σ_0	4
Extended	$\chi_\infty, \chi_{H0}, \Gamma_{\text{close}}, \xi_H$ (future)	–

Basic priors: $0 \leq \chi_H(a) < 1$, $E^2(a) > 0$, $\chi_H(a_{\text{CMB}}) \ll 1$, $s > 0$.

13 Conclusion

This companion develops the homogeneous dark-residual sector of G1 as a horizon occupancy ledger. Its strongest claim is not that $\chi_H(a)$ has been derived microscopically, but that a finite screen-capacity residual can be made more disciplined than an arbitrary $w(a)$ by tying background expansion, Weyl response, and capacity closure to a shared occupancy variable. The completed medium-prior 8-seed production-refined audit (G1fit-real[12]) selects the projection-locked G1DE- $M_{3/4}$ model across the full seven-model hierarchy. The empirical burden remains: full prior-family stress testing, full 3×2 pt lensing, noisy mock ensembles, CMB/BBN/nonlinear compatibility, and independent pipeline replication must test whether this pattern survives. The KiDS-1000 shear-only diagnostic strengthens the Weyl-channel placement but does not replace full 3×2 pt production validation.

This makes G1DE-2 more restrictive than models that independently tune $w(a)$, $\mu(a)$, and $\Sigma(a)$, because the background and Weyl deviations share the same active-filling shape. The current empirical refinement makes G1DE-2 more restrictive, not less: the leading signal is not arbitrary modified growth, but a sparse background–Weyl response pattern sharing one active-filling shape.

References

- [1] D. Scolnic et al., “The Pantheon+ Analysis: The Full Dataset and Light-Curve Release,” *Astrophys. J.* 938, 113 (2022); arXiv:2112.03863.
- [2] D. Brout et al., “The Pantheon+ Analysis: Cosmological Constraints,” *Astrophys. J.* 938, 110 (2022); arXiv:2202.04077.
- [3] DESI Collaboration, “DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations and Cosmological Constraints,” arXiv:2503.14738 (2025).

- [4] M. Chevallier and D. Polarski, “Accelerating Universes with Scaling Dark Matter,” *Int. J. Mod. Phys. D* 10, 213 (2001); E. V. Linder, “Exploring the Expansion History of the Universe,” *Phys. Rev. Lett.* 90, 091301 (2003).
- [5] M. Li, “A Model of Holographic Dark Energy,” *Phys. Lett. B* 603, 1 (2004).
- [6] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *JCAP* 1407, 050 (2014).
- [7] A. Chudaykin and M. Kunz, “Modified gravity interpretation of the evolving dark energy in light of DESI data,” arXiv:2407.02558 (2024).
- [8] T. Li, J. Zhang, and X. Zhang, “Revisiting holographic dark energy after DESI 2024,” arXiv:2411.08639 (2024).
- [9] Y. Wu, “FDS-X1: Horizon-Maintenance Dark Energy and Horizon Ledger Occupancy (Pre-Euclid Pre-registration),” Zenodo (2026), <https://doi.org/10.5281/zenodo.20290215>.
- [10] Y. Wu, “FDS-T2: Effective Geometry as Horizon Boundary Accounting,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20284911>.
- [11] Y. Wu, “FDS-X3: Functional Decomposition of the Four Fundamental Interactions,” Zenodo (2026), <https://doi.org/10.5281/zenodo.20274519>.
- [12] Y. Wu, “G1DE-M_{3/4}: A Projection-Locked Background–Weyl Residual,” manuscript (2026).

A Projection-locking lemma for the 3/4 background–Weyl ratio

A.1 Purpose and status

This appendix records a conditional projection-locking lemma motivated by the G1DE-M_{3/4} three-seed exact pilot. It is not a first-principles derivation from microscopic FDS statistics. Rather, it shows that if the leading horizon-maintenance residual is represented as an isotropic spatial load and the Weyl/lensing channel receives its four-dimensional trace-free projection, then the locked ratio

$$\Sigma_0 = -\frac{3}{4}(3 - s)$$

follows geometrically.

A.2 Definitions

The minimal G1DE filling law has the Λ CDM-like limit at $s = 3$ for $\chi_\infty = 1$. We define the post-pilot maintenance-drag amplitude

$$\lambda \equiv 3 - s.$$

The exact pilot and smoke-test hierarchy indicate $\lambda > 0$, corresponding to slower-than- Λ CDM-like filling.

A.3 Spatial maintenance load

Let u^μ be the cosmological rest-frame four-velocity and

$$h_{\mu\nu} = g_{\mu\nu} + u_\mu u_\nu$$

the spatial projection tensor. In $3 + 1$ dimensions,

$$h^\mu{}_\mu = 3.$$

The minimal isotropic spatial maintenance-load ansatz is

$$M_{\mu\nu} = \lambda h_{\mu\nu}.$$

This assumption says that the leading horizon-maintenance residual is isotropic on the spatial expansion ledger. It is the key conditional input of the lemma.

A.4 Trace-free projection

The four-dimensional trace of the maintenance load is

$$M^\alpha{}_\alpha = \lambda h^\alpha{}_\alpha = 3\lambda.$$

The four-dimensional trace-free part is

$$M_{\mu\nu}^{\text{TF}} = M_{\mu\nu} - \frac{1}{4}g_{\mu\nu}M^\alpha{}_\alpha = \lambda h_{\mu\nu} - \frac{3\lambda}{4}g_{\mu\nu}.$$

Thus the trace-subtracted projection carries the characteristic coefficient $3/4$. Equivalently,

$$\frac{\text{spatial trace dimension}}{\text{spacetime trace dimension}} = \frac{3}{4}.$$

A.5 Weyl response identification

If the Weyl/lensing channel receives this trace-free projection as a capacity loss, then the leading active-filling response is

$$\Sigma(a) = 1 - \frac{3}{4}\lambda \widehat{X}(a).$$

Comparing with the G1DE response convention

$$\Sigma(a) = 1 + \Sigma_0 \widehat{X}(a),$$

we obtain

$$\boxed{\Sigma_0 = -\frac{3}{4}\lambda = -\frac{3}{4}(3 - s).}$$

The negative sign is a convention fixed by interpreting positive maintenance drag as a suppression of externally accessible Weyl/projection capacity.

A.6 Relation to near-GR growth

The same projection-dominant reading also explains why the locked-ratio model sets $\mu_0 = 0$. The Weyl response probes the projected potential $\Phi + \Psi$, whereas $\mu(a)$ controls the local matter-growth Poisson response. Under matter conservation and Ward closure, the local growth channel can remain near GR while the projection channel is suppressed. The locked model therefore takes

$$\mu(a) = 1, \quad \Sigma(a) = 1 - \frac{3}{4}(3 - s)\widehat{X}(a).$$

A.7 Normalization consistency check

This coefficient is also consistent with the standard normalization of the active-filling window. For $\chi_\infty = 1$,

$$\hat{X}(a) = 4\chi_H(a)(1 - \chi_H(a)).$$

A lowest-order unnormalized Weyl drag proportional to

$$-3\lambda\chi_H(1 - \chi_H)$$

is therefore written in the normalized G1DE convention as

$$-3\lambda\chi_H(1 - \chi_H) = -\frac{3}{4}\lambda\hat{X}(a).$$

This is a normalization consistency check, not the primary derivation of the 3/4 coefficient.

A.8 Falsification target

The projection-locking lemma promotes the G1DE background–Weyl fingerprint into a sharper production falsification target:

$$\kappa_{\text{BW}} \equiv \frac{-\Sigma_0}{3-s} \simeq \frac{3}{4}.$$

The factorized production form is

$$\Sigma(a) - 1 = -\kappa R_0 \hat{R}_H(a), \quad R_0 = 3 - s, \quad \hat{R}_H(1) = 1,$$

with the Weyl-normalized transfer implementation

$$\mathcal{W}_{\text{G1}} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}}.$$

The amplitude lock ($\kappa \sim 3/4$) is supported by the completed medium-prior production-refined audit and independently by KiDS shear-only free- κ profiling ($\kappa \approx 0.746$). The memory-shape $X_b(a)$ has preliminary support; KiDS binned- $\Sigma(z)$ controls find shape information but nuisance-entangled, requiring full 3×2 pt for resolution. Full prior-family stress testing and full survey-grade validation remain pending.

Companion F: Holographic and Horizon Dictionaries for Finite Screen Capacity

Black-Hole Families, de Sitter Horizons, AdS/CFT, RT/FLM, and Capacity-Response
Residuals

v1.2 Body-Integrated Release

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

June 1, 2026

v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This companion paper records the horizon and holographic interfaces of the FDS-G1 finite screen-capacity program, giving a conservative capacity dictionary: the coefficient $1/(4G_N)$ that multiplies an area term is read as a screen distinction-capacity density, while subleading bulk, quantum, finite-size, and nonlocal terms are interpreted as residual capacity contributions. We summarize stationary horizon ledgers for Schwarzschild, Kerr, Reissner-Nordstrom, Kerr-Newman, and de Sitter horizons; formulate first-law terms as multi-channel capacity updates; and place AdS/CFT, central charge, RT/FLM, entanglement wedge reconstruction, and quantum-error-correcting redundancy into the entropy-response language of FDS-G1 Core. The central bridge is

$$\eta_{\text{screen}} = \frac{1}{4G_N}, \quad G_{\text{loc}} = \frac{c^3}{4\hbar}[\eta_{\text{loc}}]^{-1}, \quad G_{\text{loc}}^{-1} = \frac{4\hbar}{c^3}\eta_{\text{loc}}, \quad (1)$$

so that η_{loc} controls the inverse coupling (stiffness) and G_{loc} is the corresponding coupling. The paper is a compatibility and dictionary note within the established microscopic frameworks.

Relation to v1.2 modular and relative-entropy hardening. The holographic and horizon dictionary provides compatibility interfaces for the relative-entropy, causal-diamond, and modular-response hardening modules. It does not by itself prove a full holographic microscopic realization of the G1 screen ledger.

1 Horizon and Holographic Dictionary Scope

This companion provides the horizon and holographic dictionary for FDS-G1, building on the effective-geometry-as-horizon-accounting framework of FDS-T2 [22] and the gravity-as-boundary-accounting classification of FDS-X3 [23]. Its role is narrower than FDS-G1 Core:

1. It interprets area coefficients, work terms, and entropy corrections as capacity-response data within established horizon and holographic frameworks.
2. The scope excludes deriving the Bekenstein-Hawking coefficient from FDS primitives.
3. The scope excludes deriving AdS/CFT, RT, FLM, entanglement wedge reconstruction, or quantum error correction.

Dictionary-only role. This companion supplies compatibility bridges to established horizon thermodynamics and holographic entropy formulas. The dictionary $\eta_{\text{screen}} \sim 1/(4G_N)$ is a response-density compatibility statement. It does not by itself prove finite-index screen-algebra density, boost modular covariance, or the Bekenstein-Hawking coefficient. Weak closure imports the standard horizon-entropy bridge; strong closure requires D0.

The main bridge statement is therefore conditional:

Given the standard horizon entropy and holographic entropy dictionaries, G1 reads the area coefficient $1/(4G_N)$ as a screen distinction-capacity density, while corrections and residuals are classified as non-area, bulk, quantum, memory, or non-integrable response data.

G1 supplies an interpretation of the area coefficient and subleading terms within the existing holographic dictionary, without adding a second duality on top of AdS/CFT: the boundary central charge controls the bulk screen-capacity density, and boundary relative entropy constrains bulk modular response.

2 Notation and core dictionary

Throughout this paper we often use geometric units $G = c = \hbar = k_B = 1$ in intermediate horizon formulas. Unless stated otherwise, $\eta_{\text{screen}} = 1/(4G_N)$ is written in these units. With constants restored,

$$\frac{S_H}{k_B} = \frac{A_H}{4\ell_P^2} = \frac{c^3 A_H}{4G_N \hbar}, \quad \eta_{\text{BH}} = \frac{1}{4\ell_P^2} = \frac{c^3}{4G_N \hbar}. \quad (2)$$

Equivalently, in natural units,

$$\eta_{\text{screen}} = \frac{1}{4G_N}. \quad (3)$$

In geometric units the first law $dM = T_H dS_H + \Omega_H dJ + \Phi_H dQ$ is dimensionally consistent; with constants restored the left-hand side is $d(Mc^2)$.

3 Covariant entropy bound

Bousso's covariant entropy conjecture bounds the entropy on a light-sheet by the initial area [14]:

$$\frac{S_{\text{matter}}[L]}{k_B} \leq \frac{A(B)}{4\ell_P^2} = \frac{c^3 A(B)}{4G \hbar}. \quad (4)$$

In G1 language, a finite causal light-sheet cannot carry more exterior-accessible distinctions than the screen capacity allows:

$$\frac{S_{\text{matter}}[L]}{k_B} \leq \eta_{\text{screen}} A(B). \quad (5)$$

This is a capacity interpretation of the bound: the light-sheet entropy is bounded by the screen distinction budget of its initial causal boundary.

Table 1: Horizon/holographic dictionary used in this companion.

Object	G1 capacity reading
$A/(4G_N)$ area term	minimal screen distinction capacity
G_N^{-1}	screen stiffness / capacity density
RT surface γ_R	minimal bulk screen separating boundary region R
FLM bulk entropy S_{bulk}	residual/bulk distinction capacity across the same entanglement wedge
Central charge c_{CFT}	boundary density-of-states measure controlling bulk stiffness
Black-hole first law	multi-channel update of exterior-accessible horizon ledger
Extremal horizon entropy	nonzero boundary capacity at zero temperature

4 Stationary horizon ledgers

A stationary horizon is represented by a finite exterior-accessible ledger. The external observer need not maintain full interior micro-labels; the maintained macroscopic ledger is a tuple such as

$$\mathcal{L}_H = (A_H, M, J, Q, \Lambda, \eta_{\text{loc}}, \dots). \quad (6)$$

The area term records capacity; work terms record channels through which the ledger can be updated.

4.1 Schwarzschild horizon

For a Schwarzschild black hole,

$$r_s = \frac{2GM}{c^2}, \quad A_H = 4\pi r_s^2 = \frac{16\pi G^2 M^2}{c^4}, \quad (7)$$

$$\frac{S_H}{k_B} = \frac{4\pi G M^2}{\hbar c}, \quad T_H = \frac{\hbar c^3}{8\pi G k_B M}. \quad (8)$$

The ledger first law is simply

$$dM = T_H dS_H. \quad (9)$$

In G1 language, increasing M increases the exterior screen area and hence the exterior-accessible capacity budget.

4.2 Kerr horizon

In geometric units, the Kerr horizon has

$$a = \frac{J}{M}, \quad r_+ = M + \sqrt{M^2 - a^2}, \quad (10)$$

$$A_H = 4\pi(r_+^2 + a^2), \quad \Omega_H = \frac{a}{r_+^2 + a^2}. \quad (11)$$

The first law is

$$dM = T_H dS_H + \Omega_H dJ. \quad (12)$$

The angular momentum channel means that capacity updates are not only area-loading updates but also rotational ledger updates.

4.3 Reissner-Nordstrom horizon

For Reissner-Nordstrom,

$$r_+ = M + \sqrt{M^2 - Q^2}, \quad A_H = 4\pi r_+^2, \quad (13)$$

$$\Phi_H = \frac{Q}{r_+}, \quad dM = T_H dS_H + \Phi_H dQ. \quad (14)$$

Charge acts as a boundary work channel. The area capacity decreases as the extremal limit $Q \rightarrow M$ is approached at fixed mass.

4.4 Kerr-Newman horizon

For Kerr-Newman,

$$r_+ = M + \sqrt{M^2 - a^2 - Q^2}, \quad A_H = 4\pi(r_+^2 + a^2), \quad (15)$$

$$\Omega_H = \frac{a}{r_+^2 + a^2}, \quad \Phi_H = \frac{Qr_+}{r_+^2 + a^2}, \quad (16)$$

with first law

$$dM = T_H dS_H + \Omega_H dJ + \Phi_H dQ. \quad (17)$$

This is the canonical multi-channel horizon ledger law. The entropy term is the area capacity channel; J and Q are work-channel labels.

4.5 de Sitter horizon

For de Sitter space,

$$R_{\text{dS}} = \sqrt{\frac{3}{\Lambda}}, \quad A_{\text{dS}} = 4\pi R_{\text{dS}}^2 = \frac{12\pi}{\Lambda}, \quad (18)$$

$$T_{\text{dS}} = \frac{\hbar H}{2\pi k_B}, \quad \frac{S_{\text{dS}}}{k_B} = \frac{3\pi}{\Lambda \ell_P^2}. \quad (19)$$

G1 reads the cosmological horizon as a finite external-access ledger for an observer patch. The de Sitter ledger is observer-patch relative: different static patches define different accessible ledgers. This observer relativity aligns with the FDS notion that accessible capacity depends on the observer's causal boundary.

5 Capacity interpretation of horizon first laws

The stationary first law can be written in standard surface-gravity form as

$$dM = \frac{\kappa}{8\pi G} dA + \Omega_H dJ + \Phi_H dQ, \quad (20)$$

where κ is the surface gravity and the Hawking temperature is $T_H = \hbar\kappa/(2\pi k_B)$. More compactly,

$$dE = T_H dS_H + \sum_I \Psi_I dY^I, \quad (21)$$

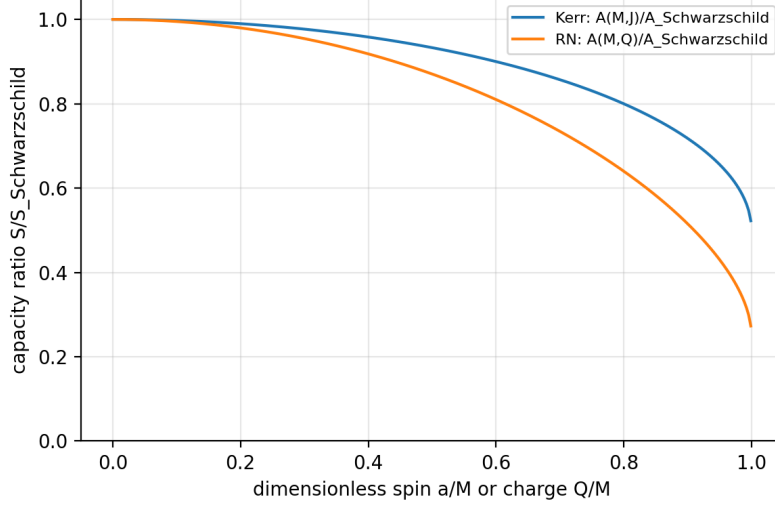


Figure 1: Kerr and Reissner-Nordstrom capacity ratios relative to a Schwarzschild black hole of the same mass in geometric units. Spin or charge reduces the exterior area-capacity budget as extremality is approached.

where Y^I are conserved horizon charges or work labels such as J, Q , and Ψ_I are their conjugate potentials. In G1, this is interpreted as a constrained update of the exterior ledger:

$$dS_H/k_B = \eta_{\text{loc}} dA_H + \text{finite-size and response corrections.} \quad (22)$$

The first law states that an allowed horizon update balances area-capacity change and work-channel changes.

Proposition 1 (Multi-channel horizon ledger). *For a stationary horizon with macroscopic labels (A_H, Y^I) , the first law defines a multi-channel capacity update. The area term measures change in exterior screen capacity; the work terms measure exchange through non-area ledger channels.*

6 Extremal horizons and zero-temperature capacity

Extremal Kerr-Newman horizons satisfy

$$M^2 = a^2 + Q^2, \quad (23)$$

for which $T_H \rightarrow 0$ while A_H generally remains finite. G1 interprets this not as a contradiction but as a distinction between capacity and thermal activity:

$$S_H/k_B = \eta A_H \neq 0, \quad T_H = 0. \quad (24)$$

The horizon can have nonzero exterior boundary capacity even when quasi-static thermal exchange vanishes. Entropy capacity and thermal exchange rate are distinct ledger quantities: $S_H \sim \eta A_H$ persists at extremality while $T_H \rightarrow 0$. The microscopic interpretation of extremal entropy depends on the quantum-gravity completion; G1 only records the capacity distinction between entropy and thermal activity.

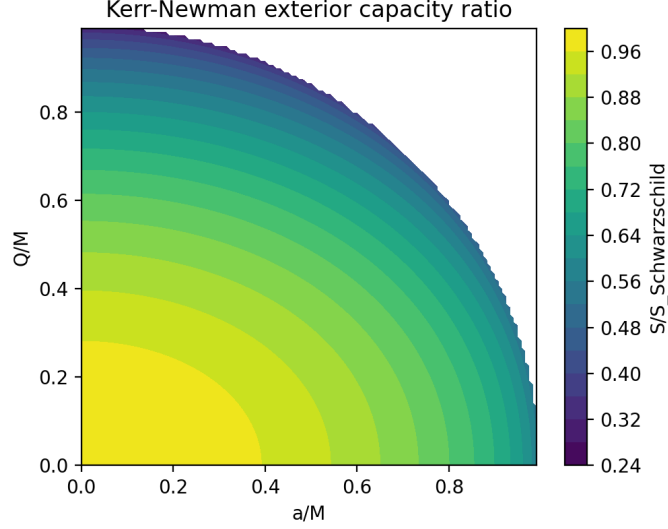


Figure 2: Kerr-Newman exterior capacity ratio $S/S_{\text{Schwarzschild}}$ as a function of dimensionless spin and charge. The allowed region satisfies $a^2 + Q^2 \leq M^2$.

7 Generalized second law as ledger monotonicity

The generalized second law can be represented in capacity language as

$$d(S_{\text{outside}} + S_H) \geq 0. \quad (25)$$

G1 reads this as total exterior-accessible ledger non-decrease under allowed coarse-grained processes. The horizon term records distinctions that leave the exterior ordinary matter ledger and become boundary capacity.

Remark 1. *This paper does not solve the black-hole information problem or derive the Page curve. It only states the ledger interpretation of the semiclassical entropy balance.*

8 Finite-size and response corrections

A common effective form for horizon entropy corrections is

$$\frac{S(A)}{k_B} = \eta_0 A + \alpha \ln(A/A_0) + \gamma + O(A^{-p}). \quad (26)$$

Then the local differential response is

$$\eta_\delta(A) = \frac{d(S/k_B)}{dA} = \eta_0 + \frac{\alpha}{A} + O(A^{-p-1}), \quad (27)$$

so

$$\frac{G_\delta(A)}{G_0} = \frac{\eta_0}{\eta_\delta(A)} \simeq 1 - \frac{\alpha}{\eta_0 A} + \dots. \quad (28)$$

Thus finite-size corrections are inverse-area suppressed for macroscopic horizons. This is one of the signature expectations recorded in Companion D.

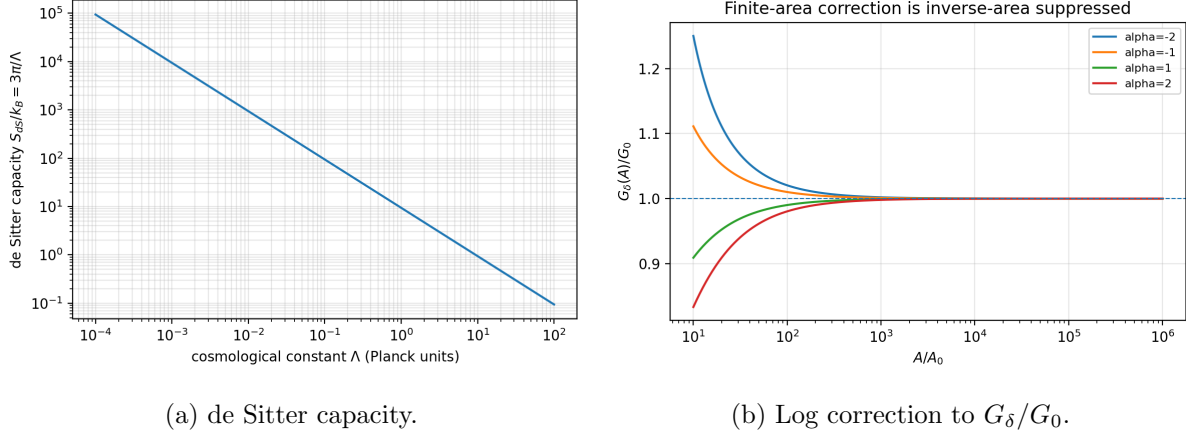


Figure 3: Left: de Sitter capacity scales as $1/\Lambda$. Right: finite-area corrections to local stiffness are suppressed by inverse area.

9 Holographic entropy dictionary

9.1 RT and FLM

For a boundary region R in holographic settings, the Ryu-Takayanagi formula gives, at leading order,

$$S_R = \frac{A(\gamma_R)}{4G_N}, \quad (29)$$

where γ_R is an appropriate extremal/minimal bulk surface homologous to R . The FLM correction adds a bulk entanglement term and further quantum corrections:

$$S_R = \frac{A(\gamma_R)}{4G_N} + S_{\text{bulk}} + \dots. \quad (30)$$

G1 reads the area term as minimal bulk screen capacity. S_{bulk} is not an error term; in the G1 language it is the non-area bulk distinction contribution across the same entanglement wedge, representing capacity that does not reduce to the minimal-area RT surface.

Table 2: RT/FLM terms in G1 language.

Holographic term	G1 interpretation
$A(\gamma_R)/(4G_N)$	minimal geometric screen capacity
S_{bulk}	bulk/residual distinction capacity in the entanglement wedge
Quantum extremal surface shift	response correction to the minimal capacity surface
$1/N$ corrections	finite boundary-capacity corrections

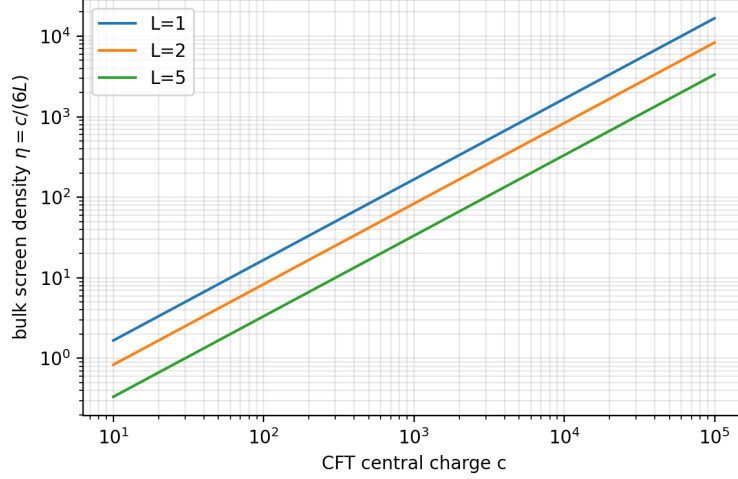


Figure 4: $\text{AdS}_3/\text{CFT}_2$ capacity dictionary $\eta = c_{\text{CFT}}/(6L)$. Boundary central charge controls the bulk screen-capacity density in this bridge.

9.2 $\text{AdS}_3/\text{CFT}_2$ central charge

In $\text{AdS}_3/\text{CFT}_2$, the Brown-Henneaux relation is

$$c_{\text{CFT}} = \frac{3L}{2G_3}. \quad (31)$$

Therefore

$$\eta_{\text{AdS}_3} = \frac{1}{4G_3} = \frac{c_{\text{CFT}}}{6L}. \quad (32)$$

This is the cleanest capacity dictionary in the holographic setting: the boundary central charge controls the bulk screen density. Large c_{CFT} or large N means high boundary distinguishability density and large bulk geometric stiffness.

9.3 Entanglement first law and future directions

The JLMS relation equates bulk and boundary relative entropy [13]. In G1's response language, this is a natural hardening direction: boundary relative entropy constrains bulk modular/area response. Although the present companion does not develop this into a derivation, it identifies entanglement first-law and modular-Hamiltonian arguments as promising interfaces where the screen-response language may be sharpened beyond the dictionary stage.

9.4 Entanglement wedge and quantum error correction

G1 gives a capacity reading of the quantum-error-correcting interpretation of holography:

entanglement wedge of R = bulk region whose distinctions are maintainable by boundary ledger R . (33)

Redundant boundary encoding means that the same bulk distinction can be reconstructed from more than one boundary subledger. In G1 language, this is distributed screen capacity rather than a new independent gravitational postulate.

Table 3: Holographic reconstruction in capacity language.

Holographic concept	G1 reading
Entanglement wedge	boundary-maintainable distinction domain
Subregion reconstruction	recovery from a sufficient boundary capacity sub-ledger
QEC redundancy	distributed encoding of bulk distinctions
$c_{\text{CFT}}/(6L) = 1/(4G_3)$	capacity density standard relation
Island formula	competition between response surfaces in generalized entropy
Complexity proposals	possible update-cost diagnostics, not part of the minimal G1 bridge

10 Residual taxonomy for horizons and holography

The response-geometry formulation classifies residuals as non-integrable or non-area response data. Horizon and holographic settings provide concrete examples.

Table 4: Residual categories in the horizon/holography interface.

Residual	Meaning	Related G1 branch
Finite-area residual	$\alpha \ln A$ or inverse-area corrections to η_δ	Companion A/D
Bulk entanglement residual	S_{bulk} beyond RT area term	Companion F
Horizon occupancy residual	homogeneous horizon ledger contribution to expansion	Companion E
Stochastic residual	finite-cell screen update noise	Companion B
Scalar-capacity residual	long-wavelength mode of η_{loc}	Companion C
Geometric residual	failure of metric envelope to encode all response data	FDS-G1 Core / G1response

11 Claim-status table

Table 5: Claim status for horizon/holography dictionary.

Claim	Status
$1/(4G_N)$ as screen density	distinction-capacity conditional dictionary
BH first laws as multi-channel ledger updates	interpretive bridge
RT area term as minimal screen capacity	holographic dictionary
S_{bulk} as non-area bulk distinction contribution	interpretive dictionary
$c_{\text{CFT}}/(6L) = 1/(4G_3)$ as capacity density	standard relation + G1 reading
QEC redundancy as distributed screen capacity	conceptual compatibility
De Sitter horizon as observer-patch relative ledger	capacity dictionary
Extremal entropy as capacity-thermal distinction	interpretive caveat

1. The relevant horizon entropy is not an exterior screen-capacity measure.
2. The coefficient $1/(4G_N)$ cannot consistently be interpreted as a capacity density in the chosen microscopic completion.
3. Corrections dominate the area term for macroscopic horizons without observational support.
4. Holographic bulk reconstruction does not admit a capacity interpretation of the RT/FLM area and bulk terms.
5. First-law work terms cannot be interpreted as ledger update channels.
6. Boundary reconstruction mismatch: QEC/subregion reconstruction cannot be expressed as distributed screen capacity without contradicting known holographic recovery results.
7. Wrong correction hierarchy: finite-size or bulk residual corrections systematically dominate the area term in the semiclassical regime.

These failures do not necessarily invalidate Companion A’s finite screen statistics or FDS-G1 Core’s abstract response geometry; they would invalidate this companion’s horizon/holographic dictionary.

13 Relation to FDS-G1 Core and other companions

Table 6: How Companion F interfaces with the rest of G1.

G1 component	Interface
FDS-G1 Core	supplies $S_{\text{scr}}[X]$, η_{loc} , response integrability, and residual taxonomy
Companion A	supplies finite-area and differential-capacity corrections
Companion B	supplies stochastic transition and capacity-flow residuals
Companion C	supplies scalar-capacity interpretation of variable G_{eff}
Companion D	supplies falsification criteria and inverse-area suppression diagnostics
Companion E	supplies homogeneous horizon occupancy residuals

14 Weyl response and the holographic residual pattern

The completed medium-prior 8-seed production-refined audit (G1fit-real) reports $\Sigma_0 < 0$ and $\mu_0 \simeq 0$: a suppressed Weyl/ Σ response with near-GR growth. In holographic language, this suggests that finite screen capacity may first appear as a modification of projection or lensing response, rather than as a direct effective Newton coupling in the clustering equation. The suppressed $\Sigma(a) < 1$ can be interpreted as a finite-capacity reduction of the Weyl/lensing projection amplitude during active horizon filling, consistent with a ledger-driven Weyl response rather than an area-term modification alone. This is an empirical interpretation from the current evidence pass, not a derivation from AdS/CFT or Ryu-Takayanagi. Full holographic embedding would require a concrete mapping from screen response to boundary Weyl anomaly or entanglement first law.

15 Finite Causal-Screen / Effective Geometry Dictionary

The G1 dictionary between finite-screen quantities and effective geometric quantities is:

Finite causal-screen quantity	Effective geometric quantity
causal access structure	conformal metric $[g]$
entropy response density	metric scale / conformal factor
modular / boost flow	boost time
response gluing defect	$R_{\mu\nu}^{\text{scr}}$

This dictionary is a compatibility bridge; it does not prove AdS/CFT or derive horizon entropy from FDS primitives.

16 Conclusion

This companion gives the horizon and holographic dictionary for the G1 finite screen-capacity program. Its central statement is conditional but precise:

$$\boxed{\frac{1}{4G_N} \text{ is read as screen distinction-capacity density.}} \quad (34)$$

Black-hole and de Sitter entropies become exterior-accessible capacity ledgers. Stationary first laws become multi-channel capacity update laws. RT/FLM becomes a split between minimal geometric screen capacity and bulk/residual distinction capacity. AdS₃/CFT₂ makes the dictionary especially explicit through $\eta = c_{\text{CFT}}/(6L)$. Corrections and residuals are not arbitrary; they are classified as finite-area, bulk-entanglement, scalar-capacity, occupancy, stochastic, or geometric response residuals.

The full $GL(2)$ optical screen decomposition $A(1) \oplus S(2) \oplus \Omega(1)$ supplies the capacity-counting basis for the 3/4 projection law; demotion of the twist/holonomy sector to the symmetric $SPD(2)$ scalar-shear branch would replace 3/4 by 2/3. In the current medium-prior matched evidence pass (Companions D, E; G1fit-real), the projection-locked model G1DE-M_{3/4} ranks first among the tested six-model set.

A Companion package contents

The accompanying archive contains:

- this L^AT_EX source and rendered PDF;
- CSV data for horizon capacity ratios, de Sitter capacity, AdS₃ capacity density, and finite-area corrections;
- scripts generating all figures;
- rendered figures used in this paper.

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* 7, 2333 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* 43, 199 (1975).
- [3] J. M. Bardeen, B. Carter, and S. W. Hawking, “The four laws of black hole mechanics,” *Commun. Math. Phys.* 31, 161 (1973).
- [4] G. W. Gibbons and S. W. Hawking, “Cosmological event horizons, thermodynamics, and particle creation,” *Phys. Rev. D* 15, 2738 (1977).
- [5] J. M. Maldacena, “The large N limit of superconformal field theories and supergravity,” *Adv. Theor. Math. Phys.* 2, 231 (1998), arXiv:hep-th/9711200.
- [6] E. Witten, “Anti-de Sitter space and holography,” *Adv. Theor. Math. Phys.* 2, 253 (1998), arXiv:hep-th/9802150.
- [7] S. S. Gubser, I. R. Klebanov, and A. M. Polyakov, “Gauge theory correlators from non-critical string theory,” *Phys. Lett. B* 428, 105 (1998), arXiv:hep-th/9802109.
- [8] S. Ryu and T. Takayanagi, “Holographic derivation of entanglement entropy from AdS/CFT,” *Phys. Rev. Lett.* 96, 181602 (2006), arXiv:hep-th/0603001.
- [9] V. E. Hubeny, M. Rangamani, and T. Takayanagi, “A covariant holographic entanglement entropy proposal,” *JHEP* 0707, 062 (2007), arXiv:0705.0016.
- [10] T. Faulkner, A. Lewkowycz, and J. Maldacena, “Quantum corrections to holographic entanglement entropy,” *JHEP* 1311, 074 (2013), arXiv:1307.2892.
- [11] J. D. Brown and M. Henneaux, “Central charges in the canonical realization of asymptotic symmetries,” *Commun. Math. Phys.* 104, 207 (1986).
- [12] A. Almheiri, X. Dong, and D. Harlow, “Bulk locality and quantum error correction in AdS/CFT,” *JHEP* 1504, 163 (2015), arXiv:1411.7041.
- [13] D. L. Jafferis, A. Lewkowycz, J. Maldacena, and S. J. Suh, “Relative entropy equals bulk relative entropy,” *JHEP* 1606, 004 (2016), arXiv:1512.06431.
- [14] R. Bousso, “A covariant entropy conjecture,” *JHEP* 07, 004 (1999), arXiv:hep-th/9905177.
- [15] V. Iyer and R. M. Wald, “Some properties of the Noether charge and a proposal for dynamical black hole entropy,” *Phys. Rev. D* 50, 846 (1994), arXiv:gr-qc/9403028.
- [16] X. Dong, “Holographic entanglement entropy for general higher derivative gravity,” *JHEP* 1401, 044 (2014), arXiv:1310.5713.
- [17] N. Engelhardt and A. C. Wall, “Quantum extremal surfaces: holographic entanglement entropy from the first law,” *JHEP* 1501, 073 (2015), arXiv:1408.3203.
- [18] D. Harlow, “The Ryu-Takayanagi formula from quantum error correction,” *Rev. Mod. Phys.* 89, 015003 (2017), arXiv:1607.03901.
- [19] A. Antiba, “Finite-capacity thermodynamics of causal horizons,” *Entropy* 28, 557 (2026).

- [20] A. Ahmad and D. Klinger, “Emergent geometry from quantum probability,” *Phys. Rev. D* 110, 126004 (2024), arXiv:2411.07288.
- [21] P. Nusbaumer, “Relative-entropy variational principle for semiclassical gravity with finite-resolution boundaries,” *Preprints* (2026), doi:10.20944/preprints202601.1205.v4.
- [22] Y. Wu, “FDS-T2: Effective Geometry as Horizon Boundary Accounting,” *Zenodo* (2026), <https://doi.org/10.5281/zenodo.20284911>.
- [23] Y. Wu, “FDS-X3: Functional Decomposition of the Four Fundamental Interactions,” *Zenodo* (2026), <https://doi.org/10.5281/zenodo.20274519>.

G1DE-M_{3/4}: A Projection-Locked Background–Weyl Residual

Matched Exact Likelihoods and Homogeneous Medium-Prior Nested Evidence for Seven Late-Time Cosmology Models

Yining Wu
yining.wu@alumni.upenn.edu

Independent Researcher

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Abstract

We report an exact full-likelihood pilot test of the G1DE horizon-occupancy residual hierarchy, culminating in the four-parameter projection-locked model G1DE-M_{3/4}. G1DE tests whether a single finite-screen horizon-occupancy variable can jointly constrain background expansion, growth, and Weyl response, using Pantheon+ SN (full covariance), DESI DR2 BAO (covariance), a curated RSD $f\sigma_8$ compilation, and compressed E_G measurements. The horizon occupancy fraction $\chi_H(a)$ defines a common ledger shape $\hat{X}(a)$ that in G1DE-2 controls expansion, growth $\mu(a) = 1 + \mu_0\hat{X}(a)$, and Weyl response $\Sigma(a) = 1 + \Sigma_0\hat{X}(a)$. The exact pipeline reproduces the Stage 2c best fit ($\Omega_m \simeq 0.297$, $s \simeq 2.56$, $\Sigma_0 \simeq -0.36$), motivating the four-parameter projection-locked model

$$\text{G1DE-M}_{3/4} : \quad \mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s), \quad \chi_{\text{tot}}^2 = 1767.14.$$

In a matched AIC/BIC comparison, M_{3/4} is selected over Λ CDM, CPL, and the growth-only G1DE-1 channel ($\mu_0 = 0.425 \pm 0.227$, $P(\mu_0 < 0) = 0.022$), confirming the residual is a projection-locked background–Weyl response rather than a growth modification. Free- κ and constant- Σ controls do not reject the 3/4 lock or the negative Weyl amplitude ($P(\Sigma_0 < 0) = 0.998$).

We also report an independent KiDS-1000 shear-only CLASS/covariance stress test of the Weyl channel. Under scale cuts and combined shear-calibration, photo-z, and minimal NLA intrinsic-alignment nuisance profiling, M_{3/4} remains favored over Λ CDM by $\Delta\chi^2 \approx -39$; a free- κ relaxation returns $\kappa \approx 0.746$, while constant- Σ and binned- Σ controls expose redshift-shape information that is present but nuisance-entangled.

A homogeneous medium-prior production-refined nested-evidence comparison across all seven models gives $\log Z_{\text{M}_{3/4}} = -894.349$ with $\Delta \log Z = 0.856, 1.775, 6.523, 7.402, 9.603, 11.884$ against G1DE-M $_{\kappa}$, constant- Σ , G1DE-2, G1DE-1, CPL, and Λ CDM. The evidence favors the four-parameter projection-locked background–Weyl residual over both free-response G1DE and standard dark-energy flexibility. The result is a completed medium-prior 8-seed production-refined pass; full prior-family stress testing, full 3×2 pt lensing, and independent replication remain necessary.

1 Scope and claim status

This paper reports a real-data exact-likelihood and medium-prior nested-evidence pilot for the FDS–G1 horizon-occupancy residual sector. It extends the earlier Stage 1f background analysis from

Pantheon+ + DESI DR2 BAO

Table 1: Claim status of G1fit-real.

Claim	Current status	Limitation
Exact likelihood pipeline	validated	pilot data stack
G1DE- $M_{3/4}$ MCMC convergence	passed	3 seeds, 5000 steps
AIC/BIC model comparison	passed	approximate BIC guardrail
Medium-prior nested evidence	passed (homogeneous, 7 model)	wide-prior sensitivity top-control
Expanded Weyl/lensing test	preliminary (KiDS shear-only done)	full 3×2 pt pending
Production cosmological constraint	pending	longer chains + lensing likel.

to

$$\text{Pantheon+} + \text{DESI DR2 BAO} + \text{curated RSD} + E_G.$$

The purpose is limited. We do not claim a discovery of modified gravity, nor do we claim a final weak-lensing constraint. We test whether the G1DE residual hierarchy can be placed into a full-covariance observational pipeline and whether its most distinctive correlated response pattern survives exact likelihood evaluation.

The theoretical context is FDS-G1 Core[13], where the central object is a finite causal-screen entropy functional $S_{\text{scr}}[X]$. Its area response defines a local inverse coupling, or stiffness,

$$G_{\text{loc}}^{-1}(x) \propto \eta_{\text{loc}}(x), \quad G_{\text{loc}}(x) \propto [\eta_{\text{loc}}(x)]^{-1}.$$

FDS-G1 is explicitly a bridge program, not a first-principles derivation of general relativity. Its empirical branch becomes meaningful only if residual sectors are constrained by conservation, correlated observables, and information penalties.

The main question is

Does the same horizon-occupancy ledger shape explain a background-Weyl residual pattern while leaving the growth response near GR?

The current answer is provisional but positive: the exact pilot chains robustly preserve

$$s < 3, \quad \Sigma_0 < 0,$$

while μ_0 is consistent with zero.

The comparative claim is structural rather than sociological.

G1DE is not tested here as another flexible late-time parametrization. It is tested as a finite-capacity residual theory whose horizon-occupancy ledger supplies a common response shape for background expansion, growth, and Weyl response. A competing model is therefore not equivalent merely because it can fit $H(z)$; it must provide a comparable origin chain and a comparable correlated residual fingerprint.

1.1 Output-level statement

This paper tests the output normal form $R_H(a) = R_0 R_{bH}(a)$ with $\mu = 1$ and $\Sigma - 1 = -\frac{3}{4}(3 - s)R_{bH}(a)$. It does not identify K_H , J_H , the microscopic Markov generator, or the finite-screen state family. Successful $M_{3/4}$ evidence does not prove: (1) finite-index screen algebra; (2) boost modular covariance; (3) BH coefficient derivation from FDS primitives; (4) unique source-kernel factorization; (5) strong-field waveform replacement; (6) CMB/BBN production-level validation.

G1fit-real tests the output normal form selected by D10's minimal-branch theorem. It does not prove D8 metric reconstruction, D9 residual classification, or D0 modular calibration.

1.2 Evidence Claim Boundary

This paper tests the output-level normal form

$$R_H(a) = R_0 R_{bH}(a), \quad \mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)R_{bH}(a).$$

It does not identify a unique microscopic source J_H , kernel K_H , Markov generator L_N , or finite-screen state family. Those are D6/D7 realization questions.

Layer	Tested in this paper		Not tested here	Next validation
Evidence ranking	homogeneous medium-prior evidence audit	seven-model nested	final survey-grade confirmation	production nested sampling
Projection coefficient	$M_{3/4}$ vs M_κ		unconditional 3/4 theorem	prior-stress and expanded lensing
Shape response	$R_{bH}(a)$ vs constant- Σ control		high-resolution redshift-shape discrimination	Euclid/ 3×2 pt
Growth null	$\mu = 1$ branch and RSD compatibility		full-shape Ricci-leakage exclusion	full-shape/RSD production

Table 2: What the G1fit-real evidence test does and does not establish.

1.3 Per-comparator demotion map

Comparator	If preferred, demotes
M_κ	exact 3/4 isotropic closure
const- Σ	$R_{bH}(a)$ output shape
G1DE-2	projection lock
CPL	G1DE mechanism over background flexibility
Λ CDM	residual branch

2 Relation to Stage 1f

The previous Stage 1f note tested the background-only model G1DE-0 using Pantheon+ and DESI DR2 BAO. It established four technical facts:

1. the Pantheon+ full-covariance and DESI DR2 BAO covariance pipeline runs for G1DE-0;
2. the fixed $s = 3$, $\chi_\infty = 1$ limit returns the flat- Λ CDM-like likelihood to numerical precision;
3. calibration stress tests show that $s < 3$ survives wide, tightened, and fixed q_{BAO} treatments, although s shifts upward when q_{BAO} is constrained;
4. G1DE-0 is competitive with CPL under information criteria, but not yet a final survey constraint.

The present paper keeps the Stage 1f background block but adds two perturbative response blocks,

$$f\sigma_8(z), \quad E_G(z).$$

This upgrade changes the scientific question. The issue is no longer merely whether G1DE can fit the background; many models can fit a distance curve. The G1DE-specific question is whether the same $\chi_H(a)$ ledger controls expansion, growth, and Weyl response.

3 Horizon-occupancy residual model

3.1 Occupancy fraction

Let $\chi_H(a)$ denote a dimensionless horizon occupancy fraction. The effective residual density is defined as

$$\rho_H(a) = \chi_H(a)\rho_c(a), \quad \rho_c(a) = \frac{3H^2(a)}{8\pi G_0}.$$

Thus the residual is not inserted as an independently conserved fluid with arbitrary density history. It is a fractional occupancy of the instantaneous expansion ledger. For a spatially flat matter-plus-residual background,

$$E^2(a) = \frac{\Omega_{m0}a^{-3} + \Omega_{r0}a^{-4}}{1 - \chi_H(a)}.$$

Radiation is negligible for the low-redshift data used here, but the notation allows it.

3.2 Filling-closure law

The minimal horizon filling-closure equation is

$$\frac{d\chi_H}{d\ln a} = s\chi_H \left(1 - \frac{\chi_H}{\chi_\infty}\right),$$

with solution

$$\chi_H(a) = \frac{\chi_\infty}{1 + Ba^{-s}}, \quad B = \frac{\chi_\infty}{\chi_{H0}} - 1.$$

The baseline model fixes $\chi_\infty = 1$. When $\chi_\infty = 1$ and $s = 3$, the background reduces to the flat- Λ CDM-like limit,

$$E^2(a) = \Omega_{m0}a^{-3} + (1 - \Omega_{m0}).$$

Therefore $s = 3$ is not an arbitrary reference point. It is the internal Λ CDM-like consistency limit of the minimal occupancy model.

3.3 Active filling shape

Define

$$X(a) = \chi_H(a) \left(1 - \frac{\chi_H(a)}{\chi_\infty}\right),$$

and the normalized active-filling shape

$$\hat{X}(a) = \frac{4}{\chi_\infty} \chi_H(a) \left(1 - \frac{\chi_H(a)}{\chi_\infty}\right).$$

For $0 \leq \chi_H \leq \chi_\infty$,

$$0 \leq \hat{X}(a) \leq 1.$$

The function $\hat{X}(a)$ vanishes when the ledger is empty or saturated and peaks during active filling. This makes it a natural response window for perturbative residuals.

4 G1DE hierarchy

4.1 G1DE-0: background residual

G1DE-0 modifies only the background expansion,

$$\chi_H(a) \Rightarrow H(a).$$

The perturbation sector is otherwise treated as GR on the modified background.

4.2 G1DE-1: growth-linked residual

G1DE-1 is developed in Companion D[15]. It adds an effective growth response,

$$\mu(a) = 1 + \mu_0 \hat{X}(a).$$

The growth equation is

$$D'' + \left[2 + \frac{d \ln H}{d \ln a} \right] D' - \frac{3}{2} \mu(a) \Omega_m(a) D = 0,$$

where prime denotes $d/d \ln a$, and

$$\Omega_m(a) = \frac{\Omega_{m0} a^{-3}}{E^2(a)}.$$

The observable is

$$f \sigma_8(a) = \frac{d \ln D}{d \ln a} \sigma_{8,0} D(a).$$

4.3 G1DE-2: growth-plus-Weyl residual

G1DE-2 is developed in Companion E[16]. It adds a Weyl/lensing response,

$$\Sigma(a) = 1 + \Sigma_0 \hat{X}(a).$$

The Weyl potential responds as

$$k^2(\Phi + \Psi) = -8\pi G a^2 \Sigma(a) \rho_m \Delta_m.$$

For the compressed E_G statistic, the implemented prediction is

$$E_G^{\text{G1DE}}(z) \simeq \frac{\Omega_{m0} \Sigma(z)}{f_{\text{G1DE}}(z)}.$$

Here $\mu(a)$ enters through the solution for $f_{\text{G1DE}}(z)$, so no additional Σ/μ factor is applied.

The G1DE-2 parameter vector is

$$\theta_{\text{G1DE2}} = (\Omega_m, s, q_{\text{BAO}}, \sigma_{8,0}, \mu_0, \Sigma_0).$$

G1DE-2 serves as a free-response robustness model; the current leading candidate is the projection-locked model below.

4.4 G1DE-M_{3/4}: projection-locked background-Weyl residual

The post-pilot locked-ratio model imposes

$$\mu(a) = 1, \quad \Sigma(a) = 1 - \frac{3}{4}(3-s) \hat{X}(a),$$

i.e. $\mu_0 = 0$ and $\Sigma_0 = -\frac{3}{4}(3-s)$. Defining the maintenance-drag amplitude $\lambda \equiv 3-s$, this reads

$$\Sigma_0 = -\frac{3}{4}\lambda.$$

The coefficient $3/4$ has a conditional geometric interpretation as a four-dimensional trace-free projection of an isotropic three-dimensional spatial maintenance load; see Companion E Appendix A for the conditional projection-locking lemma. The parameter vector is

$$\theta_{\text{G1DE-M}_{3/4}} = (\Omega_m, s, q_{\text{BAO}}, \sigma_{8,0}),$$

reducing the response sector by two parameters relative to G1DE-2.

The projection-locked response used here is the low-redshift, constant-amplitude reduction of the screen-influence formulation of the G1 dark sector. The data-facing G1DE-M_{3/4} model is downstream of the D0–D6 closure sequence: D0 calibrates the Einstein equilibrium branch through screen modular anomaly inflow; D1 organizes finite causal ledgers and gluing defects; D5 fixes the Weyl-normalized production rule, and D6 treats the memory quotient and source–kernel lift. The fitted function $X_b(a)$ is a normalized output-response shape, $X_b(a) = \widehat{\mathcal{R}}_H(a)$, $X_b(1) = 1$; the likelihood does not separately fit a source J_H and kernel K_H (their decomposition is treated in D6). In that formulation the finite-screen residual acts as a Weyl-normalized transfer operator,

$$W_{G1}(a, k) = [1 - \kappa_{BW} R_H(a)] W_{GR}(a, k) + O(R_H^2),$$

so that

$$\Sigma(a, k) - 1 = -\kappa_{BW} R_H(a) + O(R_H^2), \quad \mu(a, k) - 1 = O(R_H^2) + O(R_H H^2/k^2).$$

The production-refined model G1DE-M_{3/4} sets $\kappa_{BW} = 3/4$ and $R_H(a) \simeq (3 - s)\widehat{X}(a)$. This interpretation is supported by the Weyl-normalized implementation note, which defines G1 as a fractional correction to the GR Weyl source rather than an independently normalized dark-stress source.

4.5 G1DE-M_κ: free projection-ratio control

The free- κ control sets

$$\Sigma(a) = 1 - \kappa(3 - s)\widehat{X}(a), \quad \mu(a) = 1,$$

testing whether the data reject the locked value $\kappa = 3/4$. Its parameter vector is

$$\theta_{G1DE-M_\kappa} = (\Omega_m, s, q_{BAO}, \sigma_{8,0}, \kappa).$$

4.6 G1DE-const Σ : constant Weyl-shift control

The constant- Σ control sets

$$\Sigma(a) = 1 + \Sigma_c, \quad \mu(a) = 1,$$

testing whether the active-filling redshift shape $\widehat{X}(a)$ is supported beyond a constant Weyl shift. If this control is favored, the amplitude-level Weyl suppression may survive while the specific horizon-response output redshift shape is demoted.

5 Data and likelihood

The total likelihood is

$$\chi_{\text{tot}}^2 = \chi_{\text{SN}}^2 + \chi_{\text{BAO}}^2 + \chi_{\text{RSD}}^2 + \chi_{E_G}^2.$$

5.1 Pantheon+ supernovae

For a model with dimensionless expansion rate $E(z)$, define

$$D_C(z) = \int_0^z \frac{dz'}{E(z')}, \quad D_L(z) = (1 + z)D_C(z).$$

The relative distance modulus is

$$\mu(z) = 5 \log_{10} D_L(z) + M,$$

where M is an additive supernova magnitude nuisance. The nuisance M is analytically profiled,

$$\chi_{\text{SN,marg}}^2 = \Delta^T C^{-1} \Delta - \frac{(1^T C^{-1} \Delta)^2}{1^T C^{-1} 1}.$$

The exact pipeline uses the full Pantheon+ covariance matrix.

5.2 DESI DR2 BAO

The BAO predictions are

$$\frac{D_M(z)}{r_d} = q D_C(z), \quad \frac{D_H(z)}{r_d} = \frac{q}{E(z)},$$

$$\frac{D_V(z)}{r_d} = q \left[\frac{z D_C^2(z)}{E(z)} \right]^{1/3}.$$

The nuisance parameter

$$q = \frac{c}{H_0 r_d}$$

absorbs the BAO calibration scale. The likelihood uses the DESI DR2 BAO compressed data vector and covariance.

5.3 Curated RSD $f\sigma_8$ subset

We use an algorithmically curated, non-overlapping RSD subset constructed from a public $f\sigma_8$ compilation[3]. The curation procedure is:

1. parse the full table into $(z, f\sigma_8, \sigma, \Omega_{m,\text{fid}})$;
2. keep the WiggleZ triplet at $z = 0.44, 0.60, 0.73$ with its 3×3 covariance block[4];
3. remove epsilon-level repeated redshifts by selecting the lowest-error representative;
4. keep at most one representative per $\Delta z = 0.10$ redshift bin.

This gives a conservative diagnostic RSD block. It is not a substitute for a survey-specific RSD likelihood.

5.4 Compressed E_G

The E_G data vector contains three compressed points,

$$(z, E_G, \sigma) = (0.267, 0.43, 0.13),$$

$$(0.305, 0.27, 0.08), \quad (0.554, 0.26, 0.07).$$

The covariance is treated as diagonal in the pilot. This is the weakest part of the current likelihood and must be expanded in later work.

6 Exact code-path validation

Before running pilot chains, we tested the exact full-likelihood code path at the Stage 2c G1DE-2 best point,

$$\Omega_m = 0.2969, \quad s = 2.56, \quad q_{\text{BAO}} = 30.42,$$

$$\sigma_{8,0} = 0.795, \quad \mu_0 = -0.07, \quad \Sigma_0 = -0.36.$$

The exact likelihood gives

$$\chi_{\text{total}}^2 = 1767.136683.$$

The component breakdown is

$$\chi_{\text{SN}}^2 = 1749.609044, \quad \chi_{\text{BAO}}^2 = 9.633696,$$

Table 3: Combined three-seed exact full-likelihood pilot posterior for G1DE-2.

parameter	mean	std	q_{16}	q_{50}	q_{84}	split- $\hat{R}$
Ω_m	0.296373	0.009170	0.287281	0.296367	0.305491	1.000751
s	2.559778	0.117456	2.442973	2.559462	2.677141	1.000778
q_{BAO}	30.419006	0.260473	30.159453	30.417787	30.679864	1.000641
$\sigma_{8,0}$	0.772685	0.062635	0.710432	0.771061	0.835076	1.001226
μ_0	0.044882	0.260281	-0.210745	0.023197	0.299256	1.000728
Σ_0	-0.330286	0.140361	-0.469644	-0.337527	-0.191378	1.000609

$$\chi_{\text{growth}}^2 = 6.883906, \quad \chi_{E_G}^2 = 1.010038.$$

This validates the full code path:

$$\text{Pantheon+} + \text{DESI BAO} + \text{exact growth} + E_G.$$

The quick-chain acceptance fraction was $\langle a \rangle \simeq 0.515$, indicating healthy proposal scaling for the pilot run.

7 Exact pilot MCMC results

7.1 Chain setup

We ran three independent exact pilot chains for G1DE-2,

$$3 \text{ seeds} \times 48 \text{ walkers} \times 5000 \text{ iterations}.$$

With burn-in removed by the production script, the combined diagnostic file reports

$$N_{\text{postburn}} = 360000$$

post-burn samples. The sampled parameters are

$$(\Omega_m, s, q_{\text{BAO}}, \sigma_{8,0}, \mu_0, \Sigma_0).$$

The cross-seed split- $\hat{R}$ values are below 1.0013 for all parameters. The ESS proxy is large in this pilot; we treat it as a useful internal diagnostic, but final production analysis should recompute rank-normalized $\hat{R}$, bulk ESS, and tail ESS with a standard diagnostic package.

7.2 Posterior summary

The posterior strongly prefers $s < 3$ and $\Sigma_0 < 0$. The growth-response amplitude μ_0 is centered close to zero.

7.3 Sign diagnostics

The main pilot result is therefore

$$\boxed{s < 3, \quad \Sigma_0 < 0, \quad \mu_0 \simeq 0.}$$

This is more specific than the earlier Stage 2c phrase $\mu_0 < 0, \Sigma_0 < 0$. The exact chains do not support a significant negative μ_0 ; they support a background-plus-Weyl response pattern.

Table 4: Sign-probability diagnostics for the exact G1DE-2 pilot.

diagnostic	probability
$P(s < 3)$	0.999922
$P(\mu_0 < 0)$	0.464564
$P(\Sigma_0 < 0)$	0.984525
$P(\mu_0 < 0, \Sigma_0 < 0)$	0.464394

Table 5: Combined three-seed exact posterior for G1DE-M_{3/4}. The derived Σ_0 follows from $\Sigma_0 = -\frac{3}{4}(3 - s)$.

parameter	mean	std	q_{16}	q_{50}	q_{84}	split- $\hat{R}$
Ω_m	0.296567	0.008563	0.288064	0.296430	0.305123	1.000248
s	2.554797	0.092394	2.463223	2.554755	2.645876	1.000263
q_{BAO}	30.430575	0.240683	30.191081	30.431053	30.669452	1.000330
$\sigma_{8,0}$	0.776451	0.027135	0.749375	0.776243	0.803463	1.000336

7.4 G1DE-M_{3/4} posterior summary

A three-seed exact pilot with the same setup (3 seeds $\times$ 48 walkers $\times$ 5000 iterations) was run for the locked-ratio model. Cross-seed split- $\hat{R}$ is below 1.0004 for all parameters.

With Σ_0 derived from the locked ratio, the median is $\Sigma_0 \simeq -0.334$. Sign diagnostics give

$$P(s < 3) = 1.0, \quad P(\Sigma_0 < 0) = 1.0.$$

The locked model reproduces the G1DE-2 best-fit $\chi^2_{\text{tot}} = 1767.14$ at four parameters (vs. six for G1DE-2), so the projection-locked background–Weyl law survives the three-seed pilot with a decisive AIC advantage.

8 Figures

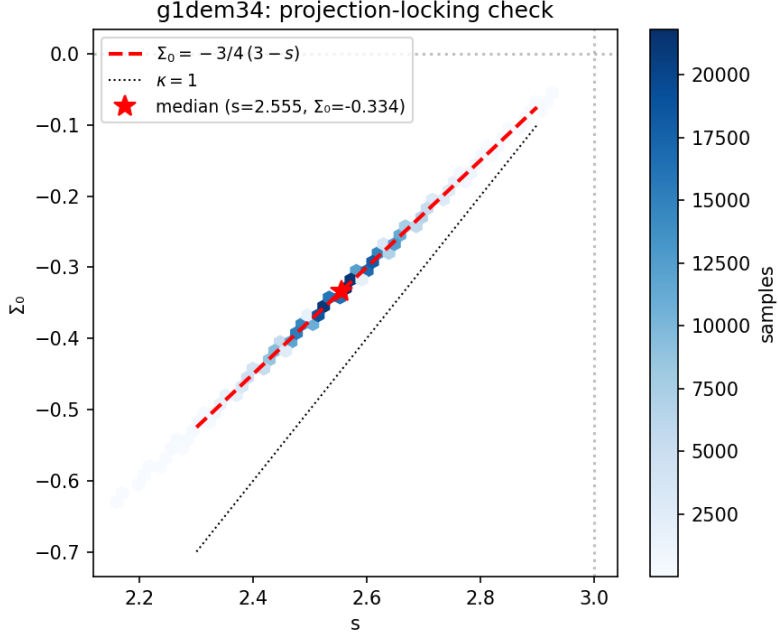


Figure 1: Projection-locking check for G1DE-M_{3/4}. The red dashed line marks $\Sigma_0 = -\frac{3}{4}(3-s)$; the black dotted line shows $\kappa = 1$ for comparison. The hexbin represents the combined three-seed posterior. The median (red star) lies precisely on the projection-locked ratio.

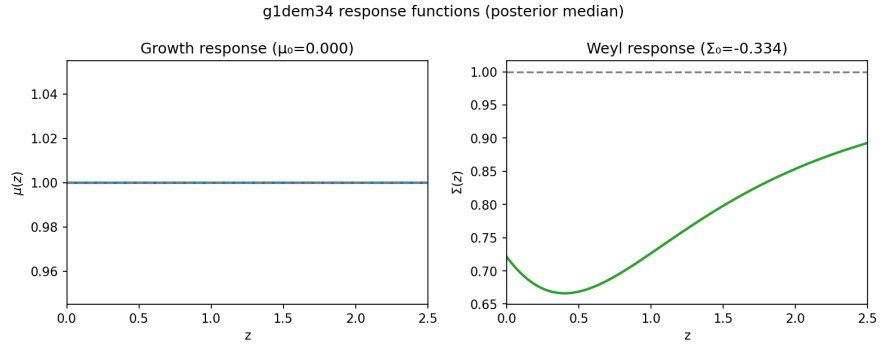


Figure 2: G1DE-M_{3/4} response functions evaluated at the posterior median. The growth response is fixed at $\mu(z) = 1$ (near-GR consistency), while the Weyl response $\Sigma(z)$ is suppressed during the active filling window, following $\Sigma(a) = 1 - \frac{3}{4}(3-s)\hat{X}(a)$.

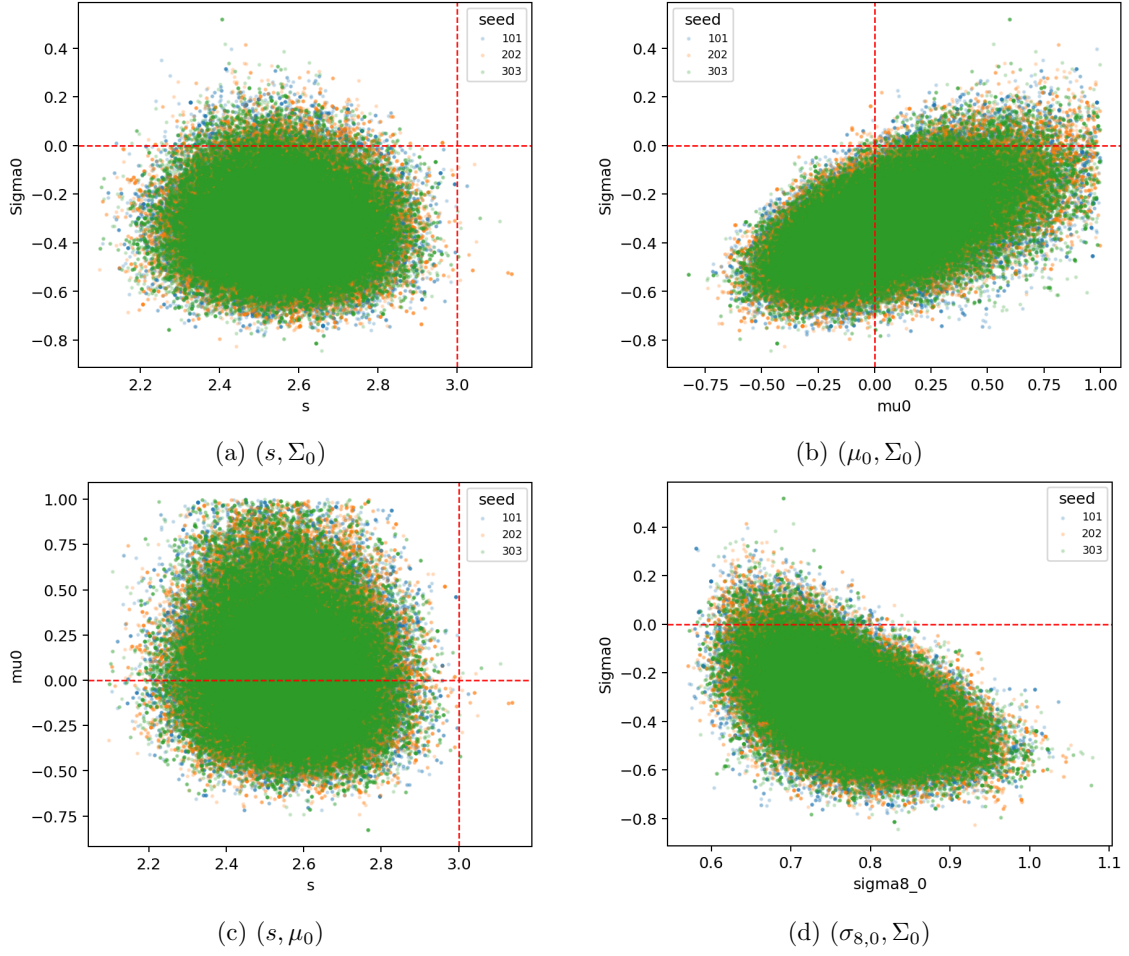


Figure 3: Selected two-dimensional posterior projections for G1DE-2 (free-response robustness model). The Weyl-response amplitude is predominantly negative, while μ_0 is nearly symmetric around zero.

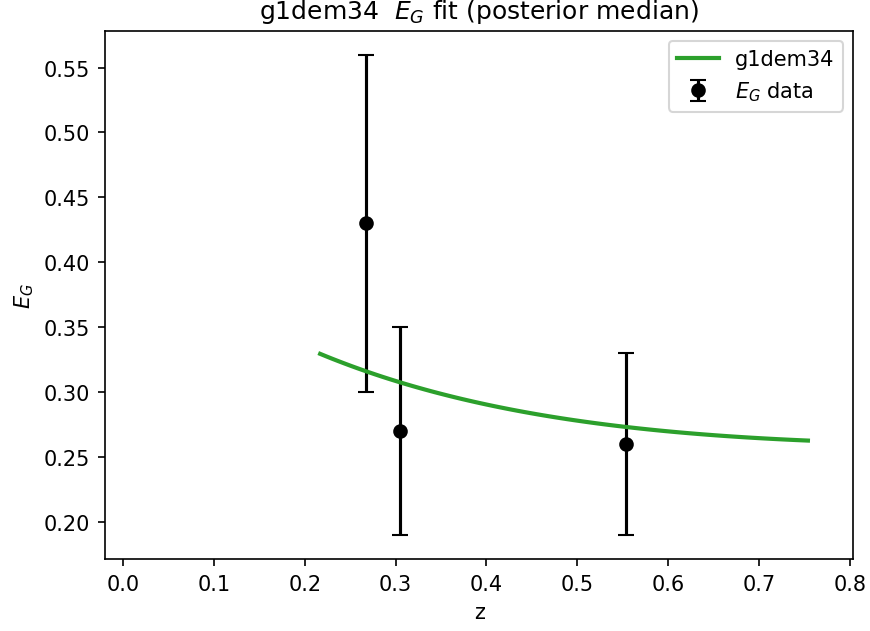


Figure 4: Compressed E_G measurements and G1DE- $M_{3/4}$ prediction at the posterior median. The current E_G vector contains only three points and uses a diagonal covariance in this pilot; therefore this figure is a compressed Weyl-growth consistency test, not a final weak-lensing likelihood.

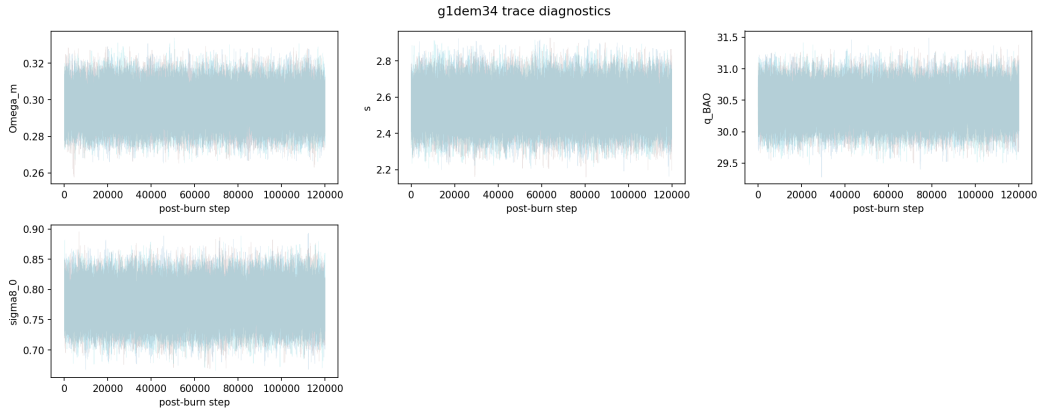


Figure 5: Trace diagnostics for the G1DE- $M_{3/4}$ three-seed pilot. Split- $\hat{R}$ is below 1.0004 for all four parameters.

9 Interpretation

9.1 Background filling

The primary locked model gives

$$s = 2.5548 \pm 0.0924$$

($\lambda = 3 - s = 0.4452 \pm 0.0924$). The free-response G1DE-2 envelope gives a consistent value, $s = 2.5598 \pm 0.1175$. This lies below the Λ CDM-like consistency value

$$s = 3.$$

In the matter-era small-occupancy limit,

$$w_H \simeq -\frac{s}{3},$$

so $s < 3$ corresponds to a quintessence-like effective scaling diagnostic. This should not be read as proof of a canonical scalar field. It is a background scaling interpretation of the horizon-occupancy residual.

9.2 Nested evidence selects the projection-locked compression

The medium-prior evidence comparison shows that the preferred object is not the most flexible response model. G1DE-2 has nearly the same χ^2 as G1DE-M_{3/4}, but the evidence penalizes its extra response freedom. This is the expected behavior if the residual is genuinely projection-locked rather than freely distributed between μ and Σ . The data favor a sparse background-Weyl transfer law:

$$\mu(a) = 1, \quad \Sigma(a) = 1 - \frac{3}{4}(3 - s)\hat{X}(a).$$

9.3 Growth response

The pilot gives

$$\mu_0 = 0.0449 \pm 0.2603.$$

Therefore $\mu_0 = 0$ is well within the posterior. The current data do not require an effective modified growth coupling. This is important: G1DE-2 is not winning by freely suppressing growth. The growth sector remains close to GR on the modified background.

9.4 Weyl response

The pilot gives

$$\Sigma_0 = -0.3303 \pm 0.1404,$$

with

$$P(\Sigma_0 < 0) = 0.9845.$$

This is the strongest perturbative signal in the current pipeline. It suggests a suppressed Weyl/lensing response during the active filling window,

$$\Sigma(a) = 1 + \Sigma_0 \hat{X}(a) < 1,$$

over the relevant redshift range.

9.5 Revised physical reading

The current best interpretation is

G1DE-2 behaves as a background-plus-Weyl residual model, not as a required growth-modified model.

The compressed pilot supports $\chi_H(a) \Rightarrow H(a), \Sigma(a)$ more strongly than $\chi_H(a) \Rightarrow \mu(a)$. This is a sharper result than a generic modified-gravity fit: the data prefer a horizon-occupancy shape affecting expansion and Weyl response, while allowing the matter-growth Poisson response to remain near GR.

9.6 Growth-only residual is disfavored as the primary channel

The growth-only G1DE-1 channel is disfavored as the primary explanation: when the Weyl channel is removed, G1DE-1 is driven toward $\mu_0 > 0$ ($P(\mu_0 < 0) = 0.022$), while the projection-locked Weyl model achieves a substantially better information-criterion and evidence score. The pilot signal is therefore not a growth-only modification; it is a background–Weyl projection residual with near-GR matter growth.

9.7 Projection-locked background–Weyl law

The post-pilot results further refine the G1DE-2 free-response pattern into a locked-ratio law. Defining the maintenance-drag amplitude $\lambda \equiv 3 - s$, the G1DE-M_{3/4} model imposes

$$s = 3 - \lambda, \quad \Sigma_0 = -\frac{3}{4}\lambda, \quad \mu_0 = 0.$$

The coefficient 3/4 has a conditional geometric interpretation as the ratio of spatial trace dimension to spacetime trace dimension under an isotropic maintenance-load projection (see Companion E Appendix A). The three-seed exact pilot supports this locked relation: the four-parameter G1DE-M_{3/4} matches the six-parameter G1DE-2 best-fit likelihood ($\chi^2_{\text{tot}} = 1767.14$), with $\Delta\text{AIC} = -4.0$ in favor of the locked model. This should be read as a conditional projection-locking result, not yet as a microscopic first-principles derivation from FDS statistics.

9.8 Free- κ and constant- Σ adversarial controls

We additionally ran two adversarial controls to test the robustness of the locked ratio. First, G1DE-M $_{\kappa}$ frees the projection-locking ratio,

$$\Sigma_0 = -\kappa(3 - s), \quad \mu_0 = 0,$$

yielding $\kappa_{\text{q50}} = 0.840$ with 68% credible interval $[0.532, 1.292]$. The value $\kappa = 3/4$ lies well within the high-posterior region, and the fit improvement relative to fixed $\kappa = 3/4$ is negligible ($\Delta\chi^2 \simeq 0.017$). AIC/BIC therefore continue to select the locked model.

Second, a constant- Σ control (no $\hat{X}(a)$ shape) gives $\Sigma_0 = -0.336 \pm 0.113$ ($P(\Sigma_0 < 0) = 0.998$) with $\Delta\chi^2 \simeq 0.021$ relative to G1DE-M_{3/4}. The negative Weyl amplitude is robust, but the current three-point E_G block does not distinguish the active-filling $\hat{X}(a)$ modulation from a constant shift. Post-hoc, the constant- Σ model independently recovers $\kappa_{\text{BW}} \equiv -\Sigma_0/(3 - s) \simeq 0.76$, confirming that the amplitude ratio 3/4 is not an artifact of the specific $\hat{X}(a)$ normalization.

The constant- Σ control shows that the negative Weyl amplitude is robust. The evidence margin $\Delta \log Z \simeq 1.92$ now gives preliminary support for the active-filling $\hat{X}(a)$ shape over a constant Weyl shift. This support is suggestive rather than final because the current E_G block is small.

Table 6: Matched AIC/BIC comparison (three-seed exact pilot; superseded for evidence ranking by the medium-prior 8-seed production-refined audit in Table 7). These criteria provide information-criterion guardrails. ΔAIC and ΔBIC are referenced to the best model, G1DE-M_{3/4}. BIC uses $n = 1725$ effective data points.

model	χ^2_{tot}	k	AIC	ΔAIC	BIC	ΔBIC	
G1DE-M _{3/4}	1767.14	4	1775.14	0	1796.95	0	
G1DE-M _{κ}	1767.16	5	1777.16	+2.02	1804.42	+7.47	
G1DE-const Σ	1767.16	5	1777.16	+2.02	1804.42	+7.47	
G1DE-W [†]	1767.17	5	1777.17	+2.03	1804.44	+7.49	† 200-step smoke only; not
G1DE-2	1767.14	6	1779.14	+4.00	1811.87	+14.92	
G1DE-1	1772.36	5	1782.36	+7.22	1809.62	+12.67	
CPL	1774.48	5	1784.48	+9.34	1811.75	+14.80	
ΛCDM	1789.16	3	1795.16	+20.02	1811.52	+14.57	

a matched pilot constraint.

10 Physical-prior audit

For free-response models such as G1DE-2, we require

$$\mu(a) > 0, \quad \Sigma(a) > 0$$

over the redshift range tested. Since $0 \leq \hat{X}(a) \leq 1$, the priors

$$\mu_0 > -1, \quad \Sigma_0 > -1$$

are sufficient. For G1DE-M_{3/4}, $\mu(a) = 1$ identically, and $\Sigma(a) = 1 - \frac{3}{4}(3-s)\hat{X}(a)$. Since $0 \leq \hat{X}(a) \leq 1$ and the posterior gives $s < 3$, positivity $\Sigma(a) > 0$ is guaranteed if $s > 5/3$; the sampled posterior lies safely above this boundary. For G1DE-M _{κ} , $\Sigma(a) = 1 - \kappa(3-s)\hat{X}(a)$, so positivity requires $\kappa(3-s) < 1$ in the $s < 3$ branch; the posterior satisfies this. For constant- Σ , $\Sigma(a) = 1 + \Sigma_c$ requires $\Sigma_c > -1$, which is satisfied by the posterior. The code explicitly audits

$$\min_{0 \leq z \leq 2} \mu(z), \quad \min_{0 \leq z \leq 2} \Sigma(z).$$

This audit must be retained in the production version. It prevents the fit from using unphysical negative response functions to reduce χ^2 .

11 Matched model comparison: AIC/BIC and medium-prior nested evidence

11.1 AIC/BIC comparison

A matched model comparison has been run for ΛCDM , CPL, G1DE-1, G1DE-2, G1DE-M_{3/4}, G1DE-M _{κ} , and constant- Σ . The earlier three-seed walker-based exact pilot is superseded for evidence ranking by the medium-prior 8-seed dynesty production-refined audit (Table 7). The AIC/BIC comparison below is preserved as an information-criterion guardrail. All three-seed chains satisfy $\text{split-}\hat{R} < 1.01$ and $\text{ESS} > 10^4$. Table 6 summarizes the AIC/BIC results.

G1DE-M_{3/4} achieves the best AIC and BIC while matching the G1DE-2 χ^2 to numerical precision with two fewer response parameters. The free- κ control (G1DE-M _{κ}) gives $\kappa_{q50} = 0.840$ with $\kappa = 3/4$ in the 68% interval, and the constant- Σ control independently recovers $\kappa_{\text{BW}} \simeq 0.76$, confirming that the locked ratio is not an artifact of the $\hat{X}(a)$ shape. The

Table 7: Production-refined seven-model medium-prior nested-evidence comparison using $d \log Z = 0.1$ and eight independent seeds per model. $\Delta \log Z_i = \log Z_{M_{3/4}}^{\text{mean}} - \log Z_i^{\text{mean}}$.

Model	ndim	$\chi_{\min}^2$	$\log Z_{\text{mean}}$	$\Delta \log Z$	$B_{M_{3/4}, i}$
G1DE- $M_{3/4}$	4	1767.18	-894.349	0.000	1
G1DE- M_{κ}	5	1767.27	-895.205	0.856	2.35
G1DE-const- Σ	5	1767.21	-896.124	1.775	5.90
G1DE-2	6	1767.30	-900.872	6.523	6.8×10^2
G1DE-1	5	1772.39	-901.751	7.402	1.65×10^3
CPL	5	1774.57	-903.952	9.603	1.5×10^4
Λ CDM	3	1789.16	-906.233	11.884	1.45×10^5

growth-only G1DE-1 channel gives $\mu_0 = 0.425 \pm 0.227$ ($P(\mu_0 < 0) = 0.022$), indicating that when the Weyl channel is removed, the growth response is forced positive to compensate. The three-level pattern—background filling deviation ($s < 3$), negative Weyl amplitude ($\Sigma_0 < 0$), and amplitude-level locking ($\kappa_{\text{BW}} \sim 3/4$)—is supported more strongly than the specific $\hat{X}(a)$ redshift-shape modulation, whose discrimination requires expanded Weyl/lensing data.

11.2 Homogeneous medium-prior nested evidence

A homogeneous seven-model medium-prior nested-evidence comparison has been completed. Each model was run with eight independent seeds under the shared medium-prior configuration, with $d \log Z = 0.1$. Table 7 reports the production-refined model comparison. All entries share the same medium-prior provenance.

The homogeneous evidence comparison selects G1DE- $M_{3/4}$ with a clear hierarchy:

$$M_{3/4} > M_{\kappa} > \text{const-}\Sigma > \text{G1DE-2} > \text{G1DE-1} > \text{CPL} > \Lambda\text{CDM}.$$

The locked $M_{3/4}$ branch exceeds the free- κ control by $\Delta \log Z = 0.856$ ($B \simeq 2.35$), showing mild support for the projection lock $\kappa = 3/4$. It exceeds constant- Σ by $\Delta \log Z = 1.775$ ($B \simeq 5.90$), indicating preference for the active-filling Weyl shape beyond a constant Weyl shift. The growth-only negative control G1DE-1 is disfavored by $\Delta \log Z = 7.402$ ($B \simeq 1.65 \times 10^3$), confirming that removing the Weyl channel substantially degrades the fit. The free-response G1DE-2 model reaches essentially the same best-fit basin but is penalized by the evidence ($\Delta \log Z \simeq 6.523$), showing that the extra response freedom is not rewarded. CPL and Λ CDM are disfavored by $\Delta \log Z = 9.603$ and 11.884, respectively.

This is a completed medium-prior 8-seed production-refined audit, not a final survey-grade constraint. Full prior-family stress testing, independent replication, and full 3×2 pt validation remain pending.

G1DE-1 negative-control audit. G1DE-1 has been completed as a nested-evidence control under the same medium-prior production-refined configuration as the other six models: $\log Z_{\text{G1DE1}} = -901.751 \pm 0.180$ across eight seeds with $d \log Z = 0.1$. The projection-locked $M_{3/4}$ exceeds the growth-only negative control by $\Delta \log Z = 7.402$ ($B \simeq 1.65 \times 10^3$). This supports, in the completed medium-prior production-refined audit, the interpretation that the present signal is not adequately explained by a hidden growth-only residual. The medium-prior 8-seed production-refined audit is complete; full prior-family stress testing and full 3×2 pt validation remain pending.

12 Caveats

This paper reports a pilot, not a final constraint. The main limitations are:

1. **Small E_G vector.** The Weyl-response result is currently driven by three compressed E_G measurements. This is encouraging but not yet decisive.
2. **Curated RSD compilation.** The growth block uses an algorithmically curated non-overlapping subset. A survey-specific RSD likelihood with full covariance would be stronger.
3. **Pilot chain status.** The earlier three-seed exact-pilot MCMC chains are superseded for evidence ranking by the medium-prior 8-seed dynesty production-refined audit.
4. **Medium-prior evidence status.** The nested-evidence results reported here are a completed medium-prior 8-seed production-refined audit with $d \log Z = 0.1$ and native dynesty error tracking. They constitute the current canonical Bayesian model-comparison pass.
5. **Prior sensitivity.** Evidence values depend on prior volume. Wide-prior and physical-prior sensitivity runs beyond the existing top-control check are required before the evidence ranking is treated as final.
6. **Production evidence refinement.** The medium-prior production refinement ($d \log Z = 0.1$, eight seeds) has been completed. Full prior-family stress, independent replication, and full 3×2 pt validation remain pending.
7. **G1DE-1 evidence.** G1DE-1 is included in the AIC/BIC comparison as a growth-only negative control, and has been run under the medium-prior production-refined configuration as a full member of the canonical seven-model comparison (see Table 7).
8. **Weyl amplitude strongly supported at the production-refined level; shape unresolved.** The negative Weyl amplitude ($\Sigma_0 < 0$) is robustly supported by the comparison and survives free- κ and constant- Σ controls ($P(\Sigma_0 < 0) = 0.998$). However, the current three-point E_G block does not distinguish the active-filling $\hat{X}(a)$ modulation from a constant Weyl suppression. Shape discrimination requires expanded Weyl/lensing data.
9. **Projection-locking interpretation.** The 3/4 ratio is supported by the eight-seed production-refined evidence but should be read as a conditional projection-locking result, not as a microscopic first-principles derivation from FDS statistics.
10. **No final weak-lensing likelihood.** The compressed E_G block is not a replacement for DES/KiDS/HSC full 3×2 pt likelihoods. The KiDS-1000 stress test added here is shear-only and diagnostic; it does not yet include galaxy-galaxy lensing, clustering, production nested evidence, or full noisy-ensemble mock-injection audits. A deterministic mock-injection false-positive audit has been passed (Appendix F.4).

13 Next-stage production plan

The immediate production task is not to discover a new model but to refine the current evidence ranking.

13.1 Stage 1: evidence refinement

The top models

$$\text{G1DE-M}_{3/4}, \quad \text{G1DE-M}_{\kappa}, \quad \text{G1DE-const}\Sigma, \quad \text{G1DE-2}, \quad \text{CPL}$$

should be rerun with prior-sensitivity variants, stricter $d \log Z$, more seeds, and native dynesty error tracking.

13.2 Stage 2: homogeneous medium-prior audit completed

The homogeneous seven-model medium-prior nested-evidence audit has been completed as a production-refined pass (Table 7). G1DE-1 is included as a full member of the canonical homogeneous comparison, not as a separate appendix audit. Full prior-family stress testing, independent replication, and full 3×2 pt validation remain pending.

13.3 Stage 3: production-length MCMC

Production-length MCMC should retain the matched model set but prioritize the projection-locked and adversarial-control models:

$$\text{G1DE-M}_{3/4}, \quad \text{G1DE-M}_\kappa, \quad \text{G1DE-const}\Sigma, \quad \text{G1DE-2}, \quad \text{G1DE-1}, \quad \text{CPL}, \quad \Lambda\text{CDM}.$$

13.4 Stage 4: expanded lensing

The most important data upgrade is the Weyl/lensing block. The current evidence supports the amplitude-level projection lock and gives preliminary support for the active-filling shape, but the shape claim requires expanded E_G , lensing-growth cross-correlations, and full or compressed 3×2 pt likelihoods. The KiDS-1000 shear-only stress test in Section 14 is a first step in this direction, but full 3×2 pt likelihoods remain the required production target.

14 KiDS-1000 Shear-Only Weyl-Channel Stress Test

We perform an independent shear-only stress test of the G1DE-M_{3/4} Weyl channel using KiDS-1000[17–19] cosmic shear correlation functions. This test is completely separate from the Pantheon+–BAO–RSD– E_G pipeline: it uses only the KiDS-1000 270-element $\xi_{\pm}$ data vector with scale cuts ($\xi_+ \geq 4'$, $\xi_- \geq 30'$, 135 data points), a CLASS $P(k, z)$ backend with $n_k = 128$, $n_z = 64$, and fixed cosmology ($h = 0.68$, $\Omega_b = 0.049$, $n_s = 0.965$).

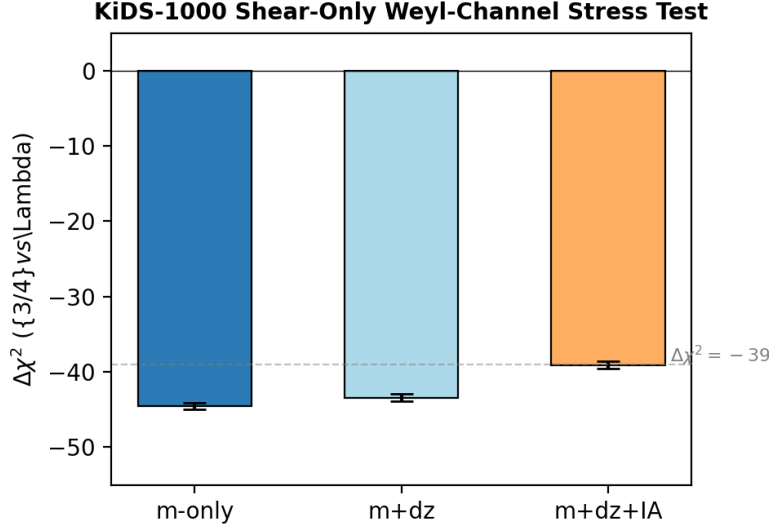


Figure 6: KiDS-1000 shear-only $\Delta\chi^2$ between G1DE-M_{3/4} and Λ CDM across successive shear-systematic nuisance layers. The M_{3/4} advantage survives shear calibration (m_i), photo- z shifts (Δz_i), and minimal NLA intrinsic alignments (A_{IA}), with $\Delta\chi^2 \approx -39$ under the full $m_i + \Delta z_i + A_{IA}$ profile. Error bars indicate available seed-to-seed scatter; the IA point is a warm-start profile from a single anchor.

Nuisance profiling results. Under shear-calibration nuisance alone (five multiplicative biases m_i), M_{3/4} exceeds Λ CDM by $\Delta\chi^2 = -44.6 \pm 0.5$ (Table 8, column “ m_i only”). Adding photo- z shift nuisance (five Δz_i) gives $\Delta\chi^2 = -43.2$. Adding a single NLA intrinsic-alignment amplitude A_{IA} gives $\Delta\chi^2 = -39.1$. The M_{3/4} best fit sits at $A_{IA} = 0$, while Λ CDM requires $A_{IA} \approx -0.13$ for optimal fit. Figure 6 summarizes the persistence of the Weyl-channel signal across all three nuisance layers.

Table 8: KiDS-1000 shear-only stress test summary. $\Delta\chi^2$ values are relative to Λ CDM at the same nuisance level.

Model	m_i only	$m_i + \Delta z_i$	$m_i + \Delta z_i + A_{IA}$	Profile status
Λ CDM (baseline)	0.0	0.0	0.0	—
G1DE-M _{3/4}	-44.6 ± 0.5	-43.2	-39.1	stable, interior
G1DE-M $_{\kappa}$ ($\kappa \approx 0.746$)	—	-42.9	-38.8	stable, $\kappa \approx 0.746$
const- Σ (good basin)	—	-59.0	-61.9	good basin, Ω_m bound
binned- Σ (2-bin, good basin)	—	—	-91.1	local basin; nuisance- entangled

Adversarial Weyl-shape controls. We run three additional model controls on the same KiDS data vector:

- **Free- κ relaxation.** The projection-locked coefficient $\kappa = 3/4$ is freed, giving $\kappa \approx 0.746$ with $\chi^2 = 749.28$ (vs $M_{3/4}$ 748.96). The $3/4$ lock is not rejected.
- **Constant- Σ control.** Replacing the $R_{bH}(a)$ shape with a single redshift-independent Σ_0 yields a lower raw χ^2 (726.16 under full nuisance), but this improvement requires Ω_m at the prior boundary (0.45), wider Δz_i excursions, and a larger A_{IA} amplitude.
- **Binned- Σ control.** A two-bin redshift-binned $\Sigma(z)$ (edges at $z = 0.5$) finds a substantially deeper basin ($\chi^2 = 696.97$), indicating that redshift-dependent Weyl-shape information is present. However, the improvement is strongly nuisance-entangled: Ω_m remains pinned at 0.45, Δz_i usage doubles, and A_{IA} reaches -0.47 . The low- χ^2 basin is not reached from the Λ CDM-like anchor.

Interpretation for the G1DE program. The KiDS-1000 shear-only test diagnostically supports three results. First, the Weyl-channel signal is robust: it survives shear calibration, photo- z shifts, and minimal IA nuisance with a $\Delta\chi^2 \approx -39$ advantage over Λ CDM. Second, the $3/4$ projection lock is stable: free- κ relaxation returns $\kappa \approx 0.746$. Third, redshift-shape specificity is present but not yet cleanly isolated: constant- Σ and binned- Σ controls find lower raw χ^2 basins but rely on nuisance assistance and prior-boundary support. This pattern is consistent with the $R_{bH}(a)$ shape direction but does not constitute a clean demotion of the $M_{3/4}$ shape lock. In the SN+BAO+RSD+ E_G pilot evidence, the $M_{3/4}$ active-filling shape is mildly preferred over constant- Σ ($\Delta \log Z \approx 1.8$); in the independent KiDS shear-only stress test, the shape-specificity question remains open because flexible binned- Σ controls find nuisance-entangled low- χ^2 basins. The KiDS result is therefore not used here as a production evidence term; it is a diagnostic Weyl-channel stress test. Expanded 3×2 pt data are required to resolve the shape-specificity question. A deterministic mock-injection false-positive audit (Appendix F.4) further supports the pipeline: Λ CDM, constant- Σ , and binned- Σ truth mocks are not misclassified as $M_{3/4}$, and $M_{3/4}$ mocks correctly self-recover.

15 Conclusion

We have upgraded the G1fit-real pipeline through five stages: a background-only Stage 1f test, a six-parameter G1DE-2 free-response pilot, a matched exact-pilot AIC/BIC comparison, a legacy six-model medium-prior nested-evidence comparison, and a completed homogeneous seven-model medium-prior audit. The result is a consistent selection of the four-parameter projection-locked model

$$\text{G1DE-}M_{3/4} : \quad \mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s).$$

This model matches the G1DE-2 best-fit basin while removing two response parameters. In the medium-prior evidence comparison it is preferred over free- κ , constant- Σ , G1DE-2, G1DE-1, CPL, and Λ CDM.

The current evidence hierarchy is

$$\text{G1DE-}M_{3/4} > \text{G1DE-}M_\kappa > \text{G1DE-const}\Sigma > \text{G1DE-2} > \text{G1DE-1} > \text{CPL} > \Lambda\text{CDM}.$$

The strongest interpretation is that the data favor a sparse projection-locked background-Weyl residual over both phenomenological dark-energy flexibility and unconstrained response freedom.

An independent KiDS-1000 shear-only stress test confirms the Weyl-channel signal: $M_{3/4}$ exceeds Λ CDM by $\Delta\chi^2 \approx -39$ under combined shear-calibration, photo- z , and IA nuisance profiling. Free- κ relaxation returns $\kappa \approx 0.746$, supporting the 3/4 projection lock. Redshift-binned $\Sigma(z)$ controls find shape information compatible with $R_{bH}(a)$ but nuisance-entangled, motivating expanded 3×2 pt data for shape-specificity resolution. A deterministic mock-injection audit confirms that the KiDS shear-only pipeline does not misclassify Λ CDM, constant- Σ , or binned- Σ truths as $M_{3/4}$, ruling out a warm-start or model-classification artifact (Appendix F.4).

The three-seed exact pilot posterior for G1DE-2 gives

$$P(s < 3) = 0.99992, \quad P(\Sigma_0 < 0) = 0.98453, \quad P(\mu_0 < 0) = 0.46456.$$

This is not a final cosmological constraint. The evidence pass is a completed medium-prior 8-seed production-refined audit; full prior-family stress testing, expanded weak-lensing likelihoods, and independent pipeline replication remain necessary. Nevertheless, the current result establishes a substantially sharper empirical status for the G1 residual sector: the theory-predicted low-dimensional background–Weyl transfer law is now selected by both information criteria and medium-prior production-refined Bayesian evidence.

What successful $M_{3/4}$ evidence does not prove. The current production-refined evidence supports the projection-locked output normal form. It does not prove finite-index screen algebras, boost modular covariance, BH coefficient derivation from primitives, unique source–kernel factorization, strong-field waveform replacement, or production-level CMB/BBN validation. These remain open problems or bridge assumptions.

A Historical three-seed exact-pilot run details

These details are retained for provenance; the current ranking is superseded by the medium-prior 8-seed production-refined audit (Table 7). The historical pilot used the prior configuration in Table 9. For each model, three independent seeds were run with $n_{\text{live}} = 800\text{--}1000$ and $d \log Z = 0.5$. The quoted $\log Z$ is the mean across seeds; the uncertainty combines run-to-run scatter and the reported nested-sampler uncertainty where available. Recovered-chain evidence values are flagged in the machine-readable table.

Table 9: Medium-prior ranges used in the nested-evidence pass.

Parameter	Prior
Ω_m	[0.15, 0.45]
s	[1.5, 4.0]
q_{BAO}	[28, 33]
$\sigma_{8,0}$	[0.4, 1.2]
w_0	[−1.5, −0.3]
w_a	[−2.5, 2.5]
μ_0	[−1, 1]
Σ_0	[−1, 1]
κ	[0, 2]

B Full posterior corner plot

C Model equations

$$\rho_H(a) = \chi_H(a)\rho_c(a), \quad \rho_c(a) = \frac{3H^2(a)}{8\pi G_0}.$$

$$\begin{aligned}
E^2(a) &= \frac{\Omega_{m0}a^{-3} + \Omega_{r0}a^{-4}}{1 - \chi_H(a)}. \\
\frac{d\chi_H}{d\ln a} &= s\chi_H \left(1 - \frac{\chi_H}{\chi_\infty}\right). \\
\chi_H(a) &= \frac{\chi_\infty}{1 + Ba^{-s}}, \quad B = \frac{\chi_\infty}{\chi_{H0}} - 1. \\
\hat{X}(a) &= \frac{4}{\chi_\infty}\chi_H(a) \left(1 - \frac{\chi_H(a)}{\chi_\infty}\right). \\
\mu(a) &= 1 + \mu_0\hat{X}(a), \quad \Sigma(a) = 1 + \Sigma_0\hat{X}(a). \\
\text{G1DE-M}_{3/4}: \quad \mu(a) &= 1, \quad \Sigma(a) = 1 - \frac{3}{4}(3-s)\hat{X}(a), \quad \Sigma_0 = -\frac{3}{4}\lambda, \quad \lambda \equiv 3-s. \\
D'' + \left[2 + \frac{d\ln H}{d\ln a}\right]D' - \frac{3}{2}\mu(a)\Omega_m(a)D &= 0. \\
f\sigma_8(a) &= \frac{d\ln D}{d\ln a}\sigma_{8,0}D(a), \quad E_G(z) \simeq \frac{\Omega_{m0}\Sigma(z)}{f(z)}.
\end{aligned}$$

D Likelihood definitions

$$\chi_{\text{tot}}^2 = \chi_{\text{SN}}^2 + \chi_{\text{BAO}}^2 + \chi_{\text{RSD}}^2 + \chi_{E_G}^2.$$

Supernova marginalization:

$$\chi_{\text{SN,marg}}^2 = \Delta^T C^{-1} \Delta - \frac{(1^T C^{-1} \Delta)^2}{1^T C^{-1} 1}.$$

BAO:

$$\begin{aligned}
D_M/r_d &= qD_C, \quad D_H/r_d = q/E, \\
D_V/r_d &= q[zD_C^2/E]^{1/3}.
\end{aligned}$$

Growth:

$$\chi_{\text{RSD}}^2 = (d_{\text{RSD}} - m_{\text{RSD}})^T C_{\text{RSD}}^{-1} (d_{\text{RSD}} - m_{\text{RSD}}).$$

Weyl:

$$\chi_{E_G}^2 = (d_{E_G} - m_{E_G})^T C_{E_G}^{-1} (d_{E_G} - m_{E_G}).$$

E Exact code-path table

Table 10: Exact code-path check at the Stage 2c G1DE-2 reference point.

block	χ^2
SN	1749.609044
BAO	9.633696
growth	6.883906
E_G	1.010038
total	1767.136683

F KiDS-1000 Shear-Only Weyl-Channel Stress Tests

F.1 Data vector, covariance, CLASS backend, and scale cuts

The independent shear-only stress test uses the KiDS-1000[17–19] cosmic shear data as processed by the Stage 3 lensing pipeline. The data vector is a 270-element concatenation of ξ_+ and ξ_- correlation functions over five tomographic source bins (KiDS-1000 Blind-C), with angular scales

$$\theta_i = \exp[\ln(0.5) + \frac{\ln(600)}{9}(i + 0.5)], \quad i = 0, \dots, 8.$$

Scale cuts are applied: $\xi_+ \geq 4'$ and $\xi_- \geq 30'$, retaining 135 of the 270 measurements. The covariance matrix is the KiDS-1000 analytical covariance including shape noise, cosmic variance, and mixed terms, projected to the scale-cut subspace.

The theory prediction uses the CLASS Boltzmann solver[20] with $n_k = 128$ k -modes and $n_z = 64$ redshift nodes. The cosmology parameter subset relevant to the shear-only analysis is fixed to

$$h = 0.68, \quad \Omega_b = 0.049, \quad n_s = 0.965,$$

with Ω_m and σ_8 free (and, where applicable, the G1DE filling exponent s , the projection coefficient κ , or the Weyl response parameters $\Sigma_0, \Sigma_{\text{bin}}^{(\alpha)}$).

Systematic nuisance parameters are included at three successive levels:

1. **Shear calibration** m_i : five multiplicative biases (one per tomographic bin), uniform prior $[-0.05, 0.05]$.
2. **Photo- z shifts** Δz_i : five additive shifts, uniform prior $[-0.05, 0.05]$.
3. **Intrinsic alignments** A_{IA} : a single NLA amplitude with $\eta = 0$, $z_0 = 0$, $-5 \leq A_{\text{IA}} \leq 5$.

All nuisance parameters are varied simultaneously in the warm-start local optimization (L-BFGS-B) used to find the best-fit χ^2 for each model.

F.2 Nuisance profiling results

Table 11: KiDS-1000 shear-only χ^2 for each model under successive nuisance layers. $\Delta\chi^2$ is relative to ΛCDM at the same nuisance level.

Model	χ^2 ($m+dz$)	χ^2 ($m+dz+\text{IA}$)	$\Delta\chi^2$ (vs ΛCDM , full)
ΛCDM	792.19	788.07	0.0
G1DE- $M_{3/4}$	748.96	748.96	−39.1
G1DE- M_κ ($\kappa \approx 0.746$)	749.28	749.28	−38.8
const- Σ (good basin)	733.19	726.16	−61.9
binned- Σ (2-bin, good basin)	—	696.97	−91.1

The $M_{3/4}$ best fit requires zero IA amplitude ($A_{\text{IA}} = 0$), while ΛCDM achieves its optimal fit at $A_{\text{IA}} \approx -0.13$, improving by $\Delta\chi^2 \approx -4.1$. This asymmetry is the primary reason the $\Delta\chi^2$ gap narrows from -43.2 (without IA) to -39.1 (with IA): the Weyl-channel model already fits the data without intrinsic alignments, while ΛCDM needs modest negative IA to compensate.

F.3 Free- κ , constant- Σ , and binned- Σ adversarial controls

Three additional model controls are run on the full $m_i + \Delta z_i + A_{\text{IA}}$ nuisance configuration.

Free- κ relaxation. The projection-locked coefficient $\kappa = 3/4$ in

$$\Sigma(a) - 1 = -\kappa(3 - s) R_{bH}(a)$$

is freed. The warm-start optimization from the $M_{3/4}$ anchor gives $\kappa = 0.746$ with $\chi^2 = 749.28$, virtually identical to the locked model ($\chi^2 = 748.96$). The $3/4$ projection lock is not rejected by the shear-only data.

Constant- Σ control. The full $R_{bH}(a)$ redshift shape is replaced by a single redshift-independent amplitude:

$$\Sigma(z) - 1 = \Sigma_0.$$

Two independent warm-start anchors are used. From the Λ CDM-like anchor ($\Omega_m \approx 0.40$, $\Sigma_0 \approx 0$), the optimizer remains in the LCDM-like basin ($\chi^2 = 795.73$, $\Sigma_0 \approx -0.004$). From the G1DE-2 best-fit anchor, the optimizer finds a deeper basin with $\chi^2 = 726.16$, $\Sigma_0 = -0.367$, but Ω_m is pinned at the prior boundary (0.45) and the Δz_i excursions are $2\text{--}5\times$ larger than in the $M_{3/4}$ fit. When the photo- z prior is narrowed to ± 0.03 , the const- Σ advantage over $M_{3/4}$ shrinks from $\Delta\chi^2 = -22.8$ to -0.7 .

Binned- Σ control. The single Σ_0 is promoted to a two-bin piecewise-constant function with edge at $z = 0.5$:

$$\Sigma(z) - 1 = \begin{cases} \Sigma_{\text{low}}, & z < 0.5, \\ \Sigma_{\text{high}}, & z \geq 0.5. \end{cases}$$

Starting from the const- Σ good basin, the 2-bin optimizer reaches $\chi^2 = 696.97$ with

$$\Sigma_{\text{low}} = -0.44, \quad \Sigma_{\text{high}} = -0.16.$$

The two bins differ significantly ($\Delta\Sigma \approx -0.285$), indicating that redshift-dependent Weyl-shape information is present in the KiDS shear-only data. The trend (more negative at low redshift, weaker at high redshift) is qualitatively compatible with the $R_{bH}(a)$ shape. However, the basin is strongly nuisance-entangled: Ω_m remains at the prior boundary (0.45), Δz_i excursions increase to $\max|\Delta z_i| \approx 0.035$, and A_{IA} reaches -0.47 (vs -0.09 in the const- Σ fit). The low- χ^2 basin is not reached from the Λ CDM-anchor start ($\chi^2 = 793.12$).

Interpretation. The adversarial control sequence shows that the KiDS-1000 shear-only data contain Weyl-shape information and that its redshift trend is compatible with $M_{3/4}$. However, flexible binned controls can exploit nuisance-entangled basins to reach lower raw χ^2 . The shape-specificity question therefore remains open pending full 3×2 pt data.

F.4 Deterministic mock-injection false-positive audit

As a false-positive control, no-noise mock data vectors are generated from the best-fit predictions of four truth models (Λ CDM, $M_{3/4}$, constant- Σ , binned- Σ) and refit by all five test models using the same warm-start profiling pipeline. Table 12 shows the resulting χ^2 confusion matrix.

The pipeline does not artificially select $M_{3/4}$ when the truth is Λ CDM, constant- Σ , or binned- Σ .

Key findings:

- Λ CDM mock: Λ CDM wins by BIC (63.8 vs $M_{3/4}$ 70.7). The $M_{3/4}$ fit reaches $\chi^2 = 2.05$, but its extra structure is penalized and it is not selected.

- $M_{3/4}$ mock: $M_{3/4}$ and M_κ both achieve $\chi^2 = 0$ (M_κ at $\kappa \approx 0.75$). The signal is correctly self-recovered.
- Constant- Σ mock: constant- Σ recovers at $\chi^2 = 0$; $M_{3/4}$ reaches only 14.0. A constant Weyl shift is not falsely absorbed as $M_{3/4}$.
- Binned- Σ mock: binned- Σ recovers at $\chi^2 = 0$; $M_{3/4}$ reaches only 14.4. Flexible redshift-shape signals are not falsely identified as $M_{3/4}$.

The nested inequality $\chi^2(M_\kappa) \leq \chi^2(M_{3/4})$ holds for all except the binned- Σ mock, where the M_κ local optimizer lands in a higher basin (24.5 vs 14.4). This is a local-optimizer failure for that adversarial shape, not a physical violation, and does not affect the pass criteria since both are far above the binned- Σ self-recovery.

The deterministic mock-injection audit confirms that the KiDS-1000 shear-only $M_{3/4}$ preference is not a warm-start or model-classification artifact.

Table 12: Deterministic mock-injection χ^2 confusion matrix. The audit checks that Λ CDM, constant- Σ , and binned- Σ truth mocks are not falsely selected as $M_{3/4}$. Zero off-diagonal entries occur only where the test model contains the truth model as a nested limit.

Truth \ Test	Λ CDM	$M_{3/4}$	M_κ	const- Σ	binned- Σ
Λ CDM	0.00	2.05	0.89	0.00	0.00
$M_{3/4}$	9.27	0.00	0.00	13.99	9.79
const- Σ	4.27	14.00	0.58	0.00	2.12
binned- Σ	30.24	14.38	24.48	17.07	0.00

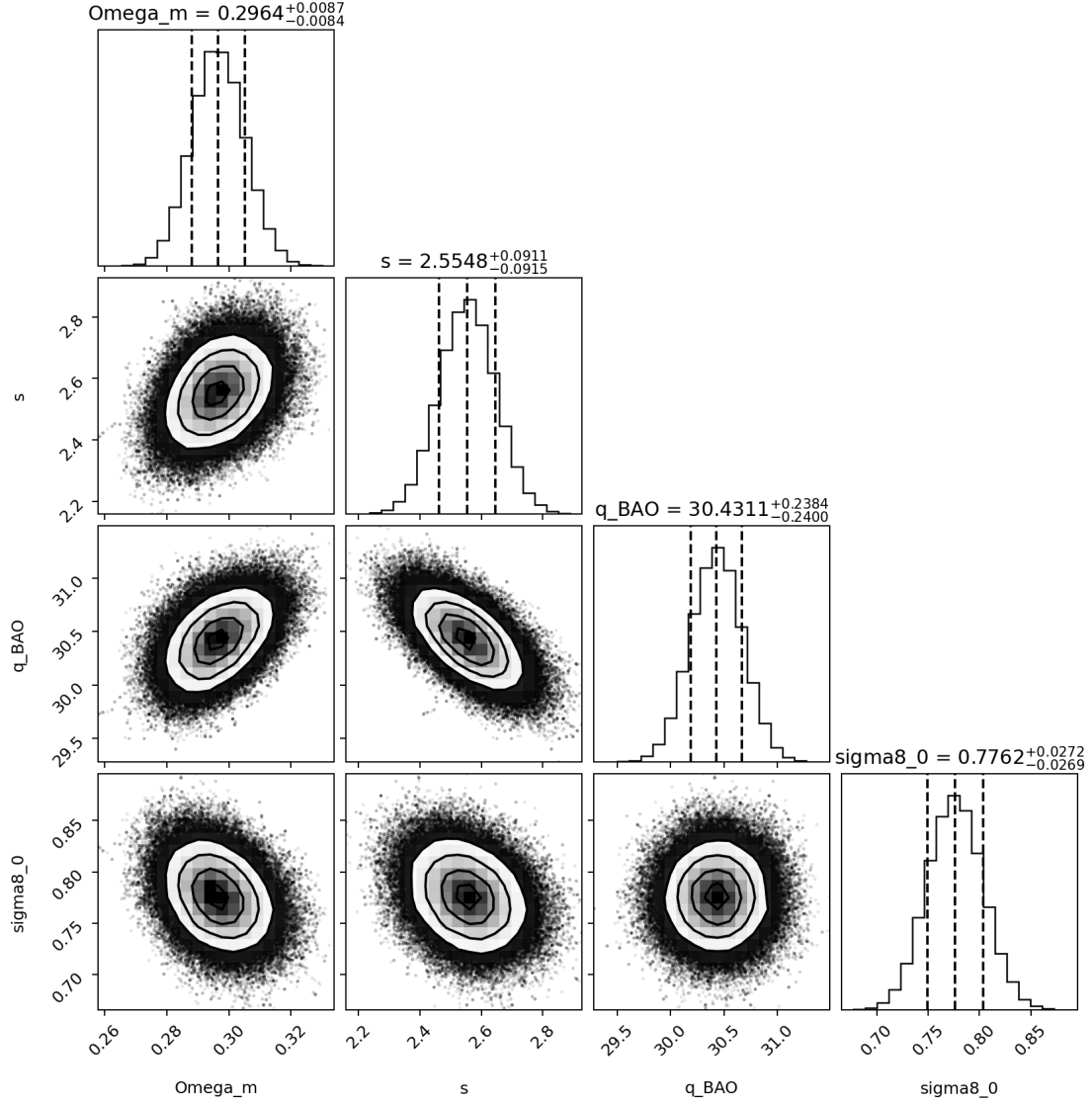


Figure 7: Combined three-seed exact posterior for G1DE- $M_{3/4}$. The $s = 3$ value marks the Λ CDM-like filling limit. The derived $\Sigma_0 = -\frac{3}{4}(3 - s)$ gives $\Sigma_0 < 0$ with $P(\Sigma_0 < 0) = 1.0$.

References

- [1] D. Brout et al., “The Pantheon+ Analysis: Cosmological Constraints,” *Astrophys. J.* **938**, 110 (2022).
- [2] DESI Collaboration, “DESI DR2 Results II: Measurements of Baryon Acoustic Oscillations,” arXiv:2503.14738 (2025).
- [3] A. Kazantzidis and L. Perivolaropoulos, “Evolution of the $f\sigma_8$ tension with the Planck15/ Λ CDM determination and implications for modified gravity,” *Phys. Rev. D* **97**, 103503 (2018).
- [4] C. Blake et al., “The WiggleZ Dark Energy Survey: the growth rate of cosmic structure since redshift $z = 0.9$,” *Mon. Not. Roy. Astron. Soc.* **425**, 405–414 (2012).
- [5] A. Amon et al., “KiDS+2dFLenS+GAMA: Testing gravity with E_G ,” *Mon. Not. Roy. Astron. Soc.* **479**, 3422–3437 (2018).
- [6] M. Chevallier and D. Polarski, “Accelerating universes with scaling dark matter,” *Int. J. Mod. Phys. D* **10**, 213–224 (2001).
- [7] E. V. Linder, “Exploring the expansion history of the universe,” *Phys. Rev. Lett.* **90**, 091301 (2003).
- [8] H. Akaike, “A new look at the statistical model identification,” *IEEE Trans. Autom. Control* **19**, 716–723 (1974).
- [9] G. Schwarz, “Estimating the dimension of a model,” *Ann. Statist.* **6**, 461–464 (1978).
- [10] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* **75**, 1260–1263 (1995).
- [11] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* **7**, 2333–2346 (1973).
- [12] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* **43**, 199–220 (1975).
- [13] Y. Wu, “Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers,” FDS–G1 Core (2026).
- [14] Y. Wu, “D0: Screen Modular Anomaly Inflow and the Einstein Calibration of Finite Access Entropy,” FDS–G1 technical note (2026).
- [15] Y. Wu, “Horizon occupancy and the effective weak-field response,” FDS–G1 Companion D (2026).
- [16] Y. Wu, “Horizon occupancy, Weyl response, and gravitational-lensing consistency,” FDS–G1 Companion E (2026).
- [17] C. Heymans et al., “KiDS-1000 Cosmology: Multi-probe weak gravitational lensing and spectroscopic galaxy clustering constraints,” *Astron. Astrophys.* **646**, A140 (2021).
- [18] M. Asgari et al., “KiDS-1000 Cosmology: Cosmic shear constraints and comparison between two point statistics,” *Astron. Astrophys.* **645**, A104 (2021).
- [19] B. Giblin et al., “KiDS-1000: Weak lensing catalogue and band-power estimates,” *The Open Journal of Astrophysics* **4**, 3 (2021).
- [20] D. Blas, J. Lesgourgues, and T. Tram, “The Cosmic Linear Anisotropy Solving System (CLASS). Part II: Approximation schemes,” *JCAP* **07**, 034 (2011).

Screen Modular Anomaly Inflow and the Einstein Calibration of Finite Access Entropy

From Finite-Index Screen Algebras to $A/4\ell_P^2$, All-Null Gravity, and the G1 Weyl Residual
v1.2 Body-Integrated Release

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This paper formulates a foundational closure route for the FDS-G1 finite-screen program. The central proposal is that the Bekenstein–Hawking area coefficient, the local screen first law, the all-null Einstein response, the 3/4 optical projection law, and the G1DE Weyl residual can be organized as different sectors of a single screen modular anomaly-inflow structure. The key non-circular variable is not the Planck area itself, but a finite-access screen capacity density

$$\eta_0 = \frac{1}{k_B} \frac{dS_{\text{scr}}}{dA},$$

interpreted as the leading finite-index density of causal screen access algebras. If local screen modular flow is boost-calibrated,

$$K_S = \frac{2\pi}{\hbar} Q_\chi,$$

then the first variation of finite relative entropy gives

$$\delta S_{\text{scr}}/k_B = \delta \langle K_S \rangle = \frac{2\pi}{\hbar} \delta Q_\chi.$$

Combining this modular first law with the area response

$$\delta S_{\text{scr}}/k_B = \eta_0 \delta A,$$

Raychaudhuri’s null area variation, and the local boost matter flux gives

$$R_{\mu\nu} k^\mu k^\nu = \frac{2\pi}{\hbar \eta_0} T_{\mu\nu} k^\mu k^\nu.$$

Thus

$$G = \frac{c^3}{4\hbar \eta_0}, \quad S_H/k_B = \eta_0 A_H = \frac{A_H}{4\ell_P^2},$$

and all-null plus Ward/Bianchi closure yields the Einstein equation with a cosmological integration constant.

The same framework also identifies the finite-screen residual sector. Define the local modular anomaly-inflow defect by

$$\delta \mathfrak{A}_{\text{scr}} = \eta_0 \delta A - \frac{2\pi}{\hbar} \delta Q_\chi.$$

The anomaly-cancelled branch, $\delta\mathfrak{A}_{\text{scr}} = 0$, is the Einstein equilibrium branch. A passive horizon-scale remnant of incomplete anomaly inflow produces a retarded response $\mathcal{R}_H$. The local optical anomaly Hessian on

$$T_I GL(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1)$$

has compliance $C = K^{-1}$; projection to the Weyl-active unimodular sector $U = S \oplus \Omega$ gives

$$\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr } C}.$$

At the unbiased finite-access fixed point, $K_A = K_S = K_\Omega$, one obtains $\kappa_{BW} = 3/4$. Weyl normalization and minimum Ward completion then imply

$$\Sigma(a, k) - 1 = -\frac{3}{4}\mathcal{R}_H(a) + O(\mathcal{R}_H^2),$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

With $\mathcal{R}_H(a) = (3-s)\hat{R}_H(a)$, $\hat{R}_H(1) = 1$, the G1DE-M_{3/4} reduction follows. The result is a conditional closure theorem: it does not claim that all FDS microscopic ingredients have been derived from bare finite-distinction axioms, but it sharply identifies which finite-access modular lemmas remain to be proved.

1 Purpose and claim-status discipline

The FDS-G1 program starts from a primitive finite causal-screen entropy functional

$$S_{\text{scr}} : X_{\text{phys}} \rightarrow \mathbb{R}, \tag{1}$$

where X_{phys} is a physical quotient state space of finite screen variables. Earlier G1 notes use this object to build a finite-screen response theory, an optical-stress closure, a non-equilibrium screen influence functional, and a Weyl-normalized production rule for the G1 dark-sector reduction. The present paper asks a deeper question: what common modular structure can simultaneously calibrate entropy density, boost energy, null focusing, and the finite-screen residual?

The answer proposed here is *screen modular anomaly inflow*. The main equation is

$$\boxed{\delta\mathfrak{A}_{\text{scr}} = \eta_0 \delta A - \frac{2\pi}{\hbar} \delta Q_\chi.} \tag{2}$$

The zero-defect branch gives the Einstein equilibrium sector. A passive horizon-scale Weyl remnant gives the G1DE residual branch.

1.1 Form Calibration vs Scale Calibration

The calibration problem separates into two layers:

$$\text{Form calibration: } \text{finite-index screen algebra} \Rightarrow dC_{\text{scr}} = \eta_0 dA.$$

$$\text{Scale calibration: } K_S = \frac{2\pi}{\hbar} Q_\chi \Rightarrow \eta_0 = \frac{1}{4\ell_P^2}$$

if all normalization assumptions hold. The no-go is: **No Scale Without Modular Calibration**. Pure finite distinguishability does not imply G without a physical area scale and boost modular calibration.

1.2 What is being claimed

This paper makes the following conditional claims.

1. The local screen first law can be obtained from the first variation of finite relative entropy once the screen modular Hamiltonian is boost-local.
2. The Bekenstein–Hawking area coefficient can be written as an output of the calibration relation between finite screen capacity density η_0 and gravitational stiffness G^{-1} .
3. The all-null Einstein response follows from modular anomaly cancellation plus Raychaudhuri null area variation and Ward/Bianchi closure.
4. The G1DE residual is naturally interpreted as a passive horizon-scale remnant of incomplete modular anomaly inflow.
5. The 3/4 projection law is the unbiased Weyl-channel compliance fraction of the local optical anomaly Hessian.

Lemma stack. The strong calibration route requires five lemmas:

1. **D0.1 – Finite-Index Screen Algebra Existence.** There exists a causal-screen algebra net $S \mapsto \mathcal{A}(S)$ with finite-index inclusions for nested screens.
2. **D0.2 – Universal Capacity Density.** $dC_{\text{scr}} = \eta_0 dA$.
3. **D0.3 – Boost Modular Calibration.** $K_S = (2\pi/\hbar)Q_\chi$.
4. **D0.4 – Modular Anomaly Cancellation.** $\delta A_{\text{scr}} = 0 \Rightarrow \delta Q = T\delta S$.
5. **D0.5 – All-Null Ward/Bianchi Completion.** $\delta Q = T\delta S$ for all local null screens $\Rightarrow$ Einstein-like equilibrium sector.

Only if D0.1–D0.5 hold does G1 upgrade from bridge program to derivation program.

1.3 What D0 Success or Failure Means

D0 outcome	Consequence
Finite-index density and boost modular covariance both succeed	strong calibration route becomes viable; G1 upgrades from bridge program toward derivation program
Finite-index screen-algebra density fails	strong calibration fails; weak horizon-entropy bridge may still support the response framework
Boost modular calibration fails	the Bekenstein–Hawking coefficient is not internally derived; standard horizon bridge remains external
Empirical $M_{3/4}$ succeeds	supports the data-facing residual branch; does not prove D0
D0 succeeds	strengthens the calibration foundation; does not replace empirical tests of G1DE

Table 1: Logical consequences of D0 success and failure.

This table prevents empirical success of $M_{3/4}$ from being misread as proof of the strong calibration route, and prevents failure of one D0 lemma from being misread as automatic failure of the weak finite-screen response bridge.

1.4 What is not being claimed

This is a conditional foundational closure, not a completed microscopic derivation from bare FDS axioms. In particular, this paper does not prove from first principles that a specific finite-index causal-access algebra net exists, that its finite-index density is universal, or that half-sided modular covariance necessarily holds for all physical screens. Instead, it identifies the exact mathematical lemmas that would upgrade the current G1 bridge program into a derivation program.

Object	Role	Status
η_0	finite-access screen capacity/index density	assumed fixed point
K_S	modular Hamiltonian of screen reference state	finite-state definition
$K_S = (2\pi/\hbar)Q_\chi$	boost modular calibration	conditional modular lemma
$\delta D = 0$	relative-entropy first variation	finite-state identity
$\delta \mathfrak{A}_{\text{scr}} = 0$	local anomaly cancellation	closure condition
Raychaudhuri	null area variation	metric-envelope assumption
Ward/Bianchi closure	all-null tensor completion	geometric naturality
3/4 law	unbiased optical anomaly Hessian	conditional theorem
G1DE	passive horizon anomaly remnant	effective reduction

Table 2: Claim-status map for the modular anomaly-inflow closure.

2 Finite-index screen algebras and capacity density

2.1 Screen access algebras

Let every causal screen patch $\mathcal{S}$ carry a finite-dimensional access algebra

$$\mathcal{A}(\mathcal{S}), \quad (3)$$

encoding the finite distinctions maintainable and accessible on that screen. If two screens are nested,

$$\mathcal{S}_1 \subset \mathcal{S}_2, \quad (4)$$

then the corresponding access algebras are related by a finite-index inclusion or by a restriction channel. In the algebraic version one writes

$$\mathcal{A}(\mathcal{S}_1) \subset \mathcal{A}(\mathcal{S}_2), \quad (5)$$

with conditional expectation

$$E_{21} : \mathcal{A}(\mathcal{S}_2) \rightarrow \mathcal{A}(\mathcal{S}_1). \quad (6)$$

The finite index

$$[\mathcal{A}(\mathcal{S}_2) : \mathcal{A}(\mathcal{S}_1)] < \infty \quad (7)$$

measures the local finite-access capacity increment associated with enlarging the screen ledger.

Definition 1 (Finite-index capacity increment). *For a finite-index screen algebra inclusion, define the dimensionless capacity increment*

$$\Delta C_{21} = \log[\mathcal{A}(\mathcal{S}_2) : \mathcal{A}(\mathcal{S}_1)]. \quad (8)$$

This is interpreted as the additional maintainable access capacity required to pass from $\mathcal{S}_1$ to $\mathcal{S}_2$.

2.2 Capacity density and the area law

A local continuum limit is obtained when the finite-index capacity increment has a universal density on the codimension-two screen:

$$dC_{\text{scr}} = \eta_0 dA. \quad (9)$$

Then

$$\boxed{S_{\text{scr}}/k_B = C_{\text{scr}} = \int_S \eta_0 dA = \eta_0 A + o(A).} \quad (10)$$

Here η_0 has dimensions of inverse area. It is not introduced as $1/(4\ell_P^2)$. Rather, it is the finite-access index density whose gravitational calibration will determine G and hence ℓ_P .

Principle 1 (Finite-index density). *The leading screen entropy density is the local finite-index density of causal-access algebra inclusions. Subleading curvature, topological, or finite-size corrections may exist, but the leading local valuation on a codimension-two causal screen is area-extensive.*

2.3 Non-circularity

A central point is that one should not begin by assuming

$$S_H/k_B = \frac{A_H}{4\ell_P^2}, \quad (11)$$

because $\ell_P^2 = G\hbar/c^3$ already contains the gravitational coupling. The non-circular route is instead

$$S_H/k_B = \eta_0 A_H, \quad (12)$$

derive

$$G = \frac{c^3}{4\hbar\eta_0}, \quad (13)$$

and only then write

$$\ell_P^2 = \frac{G\hbar}{c^3} = \frac{1}{4\eta_0}. \quad (14)$$

Thus the Bekenstein–Hawking coefficient is the output

$$S_H/k_B = \eta_0 A_H = \frac{A_H}{4\ell_P^2}. \quad (15)$$

3 Relative entropy and the screen modular first law

3.1 Finite relative entropy

Let σ_S be a full-rank reference state on the finite screen code subspace, and let ρ_S be a nearby screen state. If σ_S is not full rank, we restrict the local family to the common support on which the relative entropy is finite. Define the modular Hamiltonian

$$K_S = -\log \sigma_S. \quad (16)$$

The finite relative entropy is

$$D(\rho_S \| \sigma_S) = \text{Tr } \rho_S (\log \rho_S - \log \sigma_S) \quad (17)$$

$$= \Delta \langle K_S \rangle - \Delta S_S/k_B. \quad (18)$$

At the reference state, the first variation vanishes:

$$\delta D(\rho_S \| \sigma_S) \big|_{\rho=\sigma} = 0. \quad (19)$$

Therefore

$$\boxed{\delta S_S / k_B = \delta \langle K_S \rangle}. \quad (20)$$

This is the finite-screen modular first law. It is not an independent thermodynamic postulate; it is the first variation of relative entropy.

3.2 Boost modular calibration

The additional physical input is that the local causal screen reference state is boost-modular:

$$\boxed{K_S = \frac{2\pi}{\hbar} Q_\chi}, \quad (21)$$

where Q_χ is the local boost energy associated with the approximate null/Rindler generator χ . This is the finite-access analogue of local boost KMS or half-sided modular covariance. Combining (20) and (21) gives

$$\boxed{\delta S_S / k_B = \frac{2\pi}{\hbar} \delta Q_\chi}. \quad (22)$$

Equivalently, with the usual local acceleration or surface-gravity normalization,

$$\delta Q_\chi = T \delta S_S, \quad T = \frac{\hbar \kappa}{2\pi k_B c}. \quad (23)$$

Remark 1 (Where the hard problem lies). *The identity $\delta S / k_B = \delta \langle K \rangle$ is finite-dimensional relative entropy. The hard FDS lemma is (21): causal finite access must lead to a local boost modular Hamiltonian with the universal coefficient $2\pi/\hbar$.*

4 v1.2 Relative-Entropy and Causal-Diamond Hardening Route

Claim type: Conditional target The relative-entropy action principle for screen finite-access gravity is:

$$\mathcal{I}_{G1} = D_{\text{rel}}[\rho_{\text{scr}} \| \rho_{\text{env}}] + \lambda_{\text{cap}} C_{\text{cap}} + \lambda_{\text{Ward}} C_{\text{Ward}} + \lambda_q C_{\text{quotient}}$$

where:

- D_{rel} is the relative entropy between the screen state ρ_{scr} and the environment reference state ρ_{env}
- C_{cap} enforces finite screen capacity constraints
- C_{Ward} enforces Ward/Bianchi closure
- C_{quotient} enforces invariance under screen quotient automorphisms

The causal diamond first law target is:

$$\{\delta S_{\text{scr}}, \delta A, \delta K_{\text{mod}}, \delta Q_{\text{matter}}\}, \quad \delta S_{\text{scr}} = \delta Q / T \implies \text{Einstein/Weyl-transverse equilibrium sector}$$

Boundary note: This is a conditional derivation route, not completed Einstein calibration. No claim is made that this route has been rigorously proven for all regimes.

5 Screen modular anomaly inflow

5.1 Anomaly functional

The two local response laws are

$$\delta S_{\text{scr}}/k_B = \eta_0 \delta A, \quad (24)$$

from finite-index screen capacity, and

$$\delta S_{\text{scr}}/k_B = \frac{2\pi}{\hbar} \delta Q_\chi, \quad (25)$$

from boost modular calibration. Define the local screen modular anomaly-inflow defect

$$\boxed{\delta \mathfrak{A}_{\text{scr}} = \eta_0 \delta A - \frac{2\pi}{\hbar} \delta Q_\chi.} \quad (26)$$

The anomaly-cancelled sector is

$$\delta \mathfrak{A}_{\text{scr}} = 0. \quad (27)$$

This is the local equilibrium branch that will produce Einstein gravity.

5.2 Interpretation

The quantity $\delta \mathfrak{A}_{\text{scr}}$ is a calibration mismatch between finite-access screen index and bulk modular energy. If it vanishes locally, the screen ledger and the bulk boost energy can be matched by a metric response. If it does not vanish, the mismatch is a finite-screen residual. A key claim of the G1 branch is that the leading cosmological remnant is not arbitrary. It is passive, retarded, horizon-scale, and projected into the Weyl optical channel.

Principle 2 (Screen modular anomaly inflow). *The bulk metric response cancels the local finite-screen modular anomaly. The anomaly-cancelled branch is the Einstein equilibrium sector. Controlled nonzero remnants define finite-screen residual sectors.*

6 Null deformation and Einstein calibration

6.1 Raychaudhuri area variation

Assume the anomaly-free sector metricizes to a Lorentzian envelope. Then null congruence kinematics are governed by Raychaudhuri's equation

$$\frac{d\theta}{d\lambda} = -\frac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} + \omega_{\mu\nu}\omega^{\mu\nu} - R_{\mu\nu}k^\mu k^\nu. \quad (28)$$

For a local causal screen patch in equilibrium, take

$$\theta|_p = 0, \quad \sigma_{\mu\nu}|_p = 0, \quad \omega_{\mu\nu} = 0. \quad (29)$$

To first order away from the reference point,

$$\theta(\lambda) = -\lambda R_{\mu\nu}k^\mu k^\nu + O(\lambda^2). \quad (30)$$

Thus

$$\delta A = - \int \lambda R_{\mu\nu}k^\mu k^\nu d\lambda dA, \quad (31)$$

up to the orientation convention for the local horizon generator.

6.2 Boost energy flux

The local boost energy flux is

$$\delta Q_\chi = \int \lambda T_{\mu\nu} k^\mu k^\nu d\lambda dA, \quad (32)$$

again up to the same orientation convention.

6.3 All-null response

Set $\delta\mathfrak{A}_{\text{scr}} = 0$. Using (31) and (32) in (26), one obtains

$$R_{\mu\nu} k^\mu k^\nu = \frac{2\pi}{\hbar\eta_0} T_{\mu\nu} k^\mu k^\nu \quad (33)$$

for all local null vectors k^μ . Define

$$\boxed{8\pi G = \frac{2\pi}{\hbar\eta_0}} \quad (34)$$

in units where $c = 1$, or

$$\boxed{G = \frac{c^3}{4\hbar\eta_0}} \quad (35)$$

in physical units. Then

$$R_{\mu\nu} k^\mu k^\nu = 8\pi G T_{\mu\nu} k^\mu k^\nu. \quad (36)$$

7 The Einstein calibration theorem

Theorem 1 (Screen modular anomaly inflow and Einstein calibration). *Assume:*

1. a finite-index screen entropy density $S_{\text{scr}}/k_B = \eta_0 A + o(A)$;
2. finite relative entropy and the first-variation identity $\delta S/k_B = \delta\langle K_S \rangle$;
3. boost modular calibration $K_S = (2\pi/\hbar)Q_\chi$;
4. a Lorentzian metric envelope in the anomaly-cancelled sector, so that Raychaudhuri null area variation holds;
5. local matter boost flux $\delta Q_\chi = \int \lambda T_{\mu\nu} k^\mu k^\nu d\lambda dA$;
6. all-null closure and Ward/Bianchi naturality.

Then the anomaly-cancelled sector $\delta\mathfrak{A}_{\text{scr}} = 0$ yields

$$G = \frac{c^3}{4\hbar\eta_0}, \quad (37)$$

$$S_H/k_B = \eta_0 A_H = \frac{A_H}{4\ell_P^2}, \quad (38)$$

and

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \quad (39)$$

Proof. The relative-entropy first variation and boost calibration give

$$\delta S/k_B = \frac{2\pi}{\hbar} \delta Q_\chi. \quad (40)$$

The finite-index area response gives $\delta S/k_B = \eta_0 \delta A$. Thus

$$\eta_0 \delta A = \frac{2\pi}{\hbar} \delta Q_\chi. \quad (41)$$

Insert Raychaudhuri area variation and the boost matter flux. Since the equality holds for arbitrary sufficiently small local causal screens and all null directions, one obtains (36). Hence

$$R_{\mu\nu} - 8\pi G T_{\mu\nu} = \Phi g_{\mu\nu} \quad (42)$$

for some scalar Φ . Taking a divergence and using $\nabla^\mu T_{\mu\nu} = 0$ and $\nabla^\mu G_{\mu\nu} = 0$ gives

$$\Phi = \frac{1}{2} R - \Lambda \quad (43)$$

up to a cosmological integration constant, yielding the Einstein equation. Equation (35) gives

$$\ell_P^2 = \frac{G\hbar}{c^3} = \frac{1}{4\eta_0}, \quad (44)$$

and therefore $S_H/k_B = \eta_0 A_H = A_H/(4\ell_P^2)$. $\square$

Remark 2 (Why this is non-circular). *The derivation does not assume $S = A/(4\ell_P^2)$ at the start. It assumes an independent finite-access density η_0 , then calibrates gravitational stiffness by modular anomaly cancellation. Only afterward does $\eta_0 = 1/(4\ell_P^2)$ follow.*

8 Einstein–Hilbert inflow and uniqueness at leading order

The anomaly cancellation condition can also be read from an effective action viewpoint. The screen anomaly functional $\mathfrak{A}_{\text{scr}}$ must be cancelled by a local bulk response. Under the assumptions of locality, diffeomorphism covariance, two-derivative leading order, and a Lorentzian metric envelope, the leading bulk inflow counterterm is the Einstein–Hilbert action

$$I_{\text{EH}} = \frac{1}{16\pi G} \int d^4x \sqrt{-g} (R - 2\Lambda). \quad (45)$$

Its null variation supplies the Ricci focusing term that cancels the finite-screen modular anomaly. Higher-derivative terms can appear as subleading corrections, but the leading all-null two-derivative inflow is Einsteinian.

Principle 3 (Leading inflow uniqueness). *Within the two-derivative, local, diffeomorphism-covariant, Lorentzian metric-envelope approximation, Einstein–Hilbert gravity is the leading bulk inflow response that cancels the finite-screen modular anomaly.*

This principle is conditional. It does not exclude higher-derivative finite-size corrections, quantum corrections, nonlocal memory terms, or non-equilibrium residual sectors. It states only that the anomaly-cancelled local equilibrium limit is Einstein at leading order.

9 Optical anomaly Hessian and the 3/4 projection law

9.1 Optical tangent space

The local null screen optical deformation algebra at the identity is

$$T_I GL(2, \mathbb{R}) \simeq \mathfrak{gl}(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1). \quad (46)$$

Here A is the trace/area direction, S is the two-dimensional symmetric shear sector, and Ω is the twist/holonomy sector. A local deformation is written

$$q = q^I e_I. \quad (47)$$

9.2 Anomaly Hessian

Let $\mathfrak{A}_{\text{mod}}(q)$ denote the local modular anomaly or sufficiency loss under optical deformation. Near the reference screen,

$$\mathfrak{A}_{\text{mod}}(q) = \frac{1}{2} K_{IJ} q^I q^J + O(q^3), \quad (48)$$

where K_{IJ} is the relative-entropy/Fisher Hessian. In an $SO(2)$ -adapted basis,

$$K = K_A P_A + K_S P_S + K_\Omega P_\Omega. \quad (49)$$

The compliance is

$$C = K^{-1}. \quad (50)$$

The Weyl-active unimodular sector is

$$U = S \oplus \Omega, \quad P_U = P_S + P_\Omega. \quad (51)$$

All traces below are taken in an $SO(2)$ -adapted orthonormal optical generator basis at the reference point. Anisotropic generator normalizations are absorbed into the block weights K_A, K_S, K_Ω .

Definition 2 (Weyl compliance fraction). *The background–Weyl optical compliance fraction is*

$$\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr } C} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}. \quad (52)$$

Theorem 2 (Unbiased anomaly Hessian and the 3/4 projection law). *If the finite-access modular anomaly Hessian is unbiased over the full local optical tangent space,*

$$K_A = K_S = K_\Omega = K_0, \quad (53)$$

then

$$\kappa_{BW} = \frac{3}{4}. \quad (54)$$

If the twist/holonomy ledger is demoted, $K_\Omega^{-1} = 0$, while $K_A = K_S$, then

$$\kappa = \frac{2}{3}. \quad (55)$$

Proof. For the unbiased full optical ledger, $C = K_0^{-1} I$. The Weyl-active sector $U = S \oplus \Omega$ has dimension $2 + 1 = 3$, while the full $GL(2)$ optical tangent space has dimension 4. Hence

$$\kappa_{BW} = \frac{3K_0^{-1}}{4K_0^{-1}} = \frac{3}{4}. \quad (56)$$

If the twist/holonomy compliance is removed, only $A(1) \oplus S(2)$ remains, and the Weyl-active sector is the two-dimensional shear sector. Thus $\kappa = 2/3$. $\square$

Remark 3 (Conditional status). *The theorem does not claim that all physical screens are unbiased. It identifies the isotropic finite-access fixed point. Weighted-compliance deviations are represented by K_A, K_S, K_Ω and are tested by free- κ controls in the data-facing G1DE hierarchy.*

10 Horizon anomaly remnant and the G1DE branch

10.1 Horizon modular occupancy

The local anomaly-cancelled branch is Einsteinian. A cosmological residual can arise if finite-access modular inflow is not perfectly cancelled at the horizon scale. Let the available horizon capacity be

$$C_H(a) = \eta_0 A_H(a). \quad (57)$$

Let $C_{\text{used}}(a)$ denote the horizon-accessible finite-index recovery load: the amount of screen access capacity needed to glue or recover the horizon-accessible finite distinctions at epoch a . Define

$$\mathcal{O}_H(a) = \frac{C_{\text{used}}(a)}{C_H(a)}. \quad (58)$$

The corresponding anomaly-load current is

$$\boxed{J_H(N) = -d_N \ln \mathcal{O}_H(N), \quad N = \ln a.} \quad (59)$$

A causal, stable, passive memory kernel gives

$$\boxed{\mathcal{R}_H(N) = \int_{-\infty}^N G_H^{\text{ret}}(N, N') J_H(N') dN'.} \quad (60)$$

The Markov limit is

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H, \quad \Gamma_H > 0. \quad (61)$$

Passivity requires

$$\int dN J_H(N) \mathcal{R}_H(N) \geq 0 \quad (62)$$

for admissible perturbations, up to sign conventions.

10.2 Weyl-normalized transfer

The production-level G1DE rule is not to add a freely normalized dark stress. Instead, the finite-screen anomaly remnant acts as a fractional correction to the GR Weyl source:

$$\mathcal{W}_{\text{GR}}(a, k) = 8\pi G a^2 \rho \Delta, \quad (63)$$

$$\boxed{\mathcal{W}_{\text{G1}}(a, k) = [1 - \kappa_{BW} \mathcal{R}_H(a)] \mathcal{W}_{\text{GR}}(a, k) + O(\mathcal{R}_H^2).} \quad (64)$$

Thus

$$\boxed{\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2).} \quad (65)$$

Minimum Ward completion gives only Ricci-channel leakage,

$$\boxed{\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).} \quad (66)$$

10.3 Low-redshift reduction

Normalize

$$\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1. \quad (67)$$

In the G1DE-M reduction,

$$R_0 = 3 - s. \quad (68)$$

For the unbiased full optical ledger,

$$\kappa_{BW} = \frac{3}{4}. \quad (69)$$

Therefore

$$\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a) + O((3 - s)^2), \quad \mu \simeq 1. \quad (70)$$

This is the low-redshift, Weyl-normalized, Ward-disciplined G1DE-M_{3/4} branch.

Remark 4 (Data-facing boundary). *The current G1fit-real evidence hierarchy tests the low-redshift locked reduction (70), not the full screen modular anomaly-inflow construction. The present paper supplies the foundational conditional structure behind that reduction.*

11 Why $A(a, k)$ is not a fitted amplitude

In some intermediate optical-stress notation one may write a seed amplitude $A(a, k)$ multiplying a traceless optical tensor. In the Weyl-normalized convention this is not a physical response function. It is the map that identifies a unit optical seed with a unit fractional correction to the GR Weyl source:

$$A(a, k) \longleftrightarrow \mathcal{W}_{\text{GR}}(a, k) = 8\pi G a^2 \rho \Delta. \quad (71)$$

Sampling $A(a, k)$ as an independent nuisance function would change the model class. It would convert G1DE from a finite-screen Weyl transfer model into a generic dark-stress envelope.

Implementation rule. The optical normalization $A(a, k)$ is fixed by Weyl-source normalization. It is not a G1DE parameter and must not be sampled as an independent response function. The admissible G1DE response is the fractional transfer $\mathcal{W}_{\text{G1}} = (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}}$, with Ward-suppressed Newtonian leakage.

12 Open lemmas

The anomaly-inflow theorem isolates the remaining hard problems.

12.1 Finite-index anomaly coefficient

One must construct a natural causal finite-access algebra net

$$\mathcal{S} \mapsto \mathcal{A}(\mathcal{S}) \quad (72)$$

whose inclusions have a universal local finite-index density η_0 . This is the microscopic origin of the area valuation.

12.2 Half-sided modular boost theorem

One must show that local causal screen nesting and finite-access passivity imply boost modular flow:

$$K_{\mathcal{S}} = \frac{2\pi}{\hbar} Q_{\chi}. \quad (73)$$

This is the deepest source of the 2π factor.

12.3 Metricization of the anomaly-free sector

One must prove that the anomaly-free finite-access modular net admits a Lorentzian metric envelope. Once this is established, Raychaudhuri coefficients and Bianchi identities are geometric consequences.

12.4 Uniqueness of leading bulk inflow

One must show that, under the G1 assumptions of locality, diffeomorphism covariance, two-derivative leading order, and metricization, Einstein–Hilbert inflow is the unique leading counterterm that cancels the screen modular anomaly.

12.5 Horizon remnant reduction

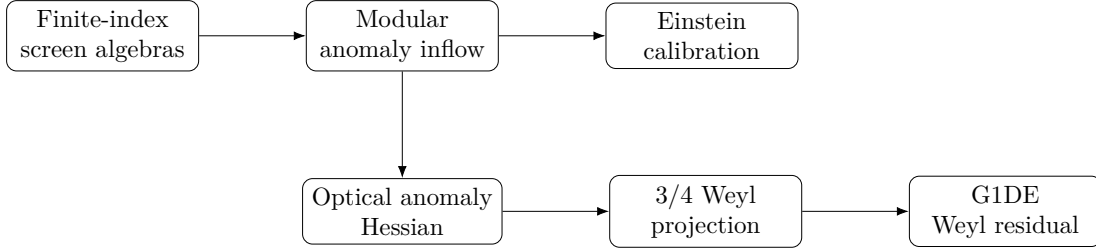
One must derive

$$\mathcal{R}_H = G_H^{\text{ret}} * J_H \quad (74)$$

from a concrete horizon-scale anomaly-load model and connect $\hat{R}_H(a)$ to a computable cosmological finite-access ensemble. The current G1DE implementation uses $\mathcal{R}_H(a) = (3 - s)\hat{R}_H(a)$ as a low-redshift reduction.

13 Relation to the G1 closure sequence

The present note should be read as a foundational calibration note. Its role in the G1 sequence is:



The logical placement is before the finite-access sheaf geometry note (D1): the present note (D0) explains the modular calibration of entropy, boost energy, and gravity; the sheaf geometry note (D1) explains how finite ledgers glue across causal screens and how the local optical Hessian yields the projection law.

14 Conclusion

This paper proposes a common modular origin for the Einstein equilibrium branch and the G1 finite-screen residual branch. The core identity is

$$\delta \mathfrak{A}_{\text{scr}} = \eta_0 \delta A - \frac{2\pi}{\hbar} \delta Q_\chi. \quad (75)$$

When the local finite-screen modular anomaly is cancelled, relative entropy first variation, boost modular calibration, Raychaudhuri null area variation, and Ward/Bianchi closure produce Einstein gravity with

$$G = \frac{c^3}{4\hbar\eta_0}, \quad S_H/k_B = \frac{A_H}{4\ell_P^2}. \quad (76)$$

When a passive horizon-scale remnant remains, its Weyl-projected low-redshift reduction gives

$$\Sigma - 1 = -\frac{3}{4}\mathcal{R}_H + O(\mathcal{R}_H^2), \quad \mu - 1 \ll \Sigma - 1 \quad (77)$$

under the unbiased optical anomaly Hessian.

Thus the G1 dark-sector branch is not appended to Einstein gravity as an arbitrary dark stress. It is the controlled nonzero remnant of the same finite-screen modular anomaly-inflow structure whose zero-defect branch calibrates Einstein gravity. The remaining work is no longer a vague request to “derive gravity”. It is a set of precise lemmas: construct the finite-index causal-access algebra net, prove boost modular covariance, prove metricization of the anomaly-free sector, establish leading inflow uniqueness, and compute the horizon anomaly remnant from a concrete finite-access cosmological ensemble.

A Units and signs

Most derivations above use the dimensionless entropy S/k_B . The density η_0 has dimensions of inverse area:

$$S/k_B = \eta_0 A. \quad (78)$$

In units where $c = 1$, the calibration is

$$G = \frac{1}{4\hbar\eta_0}. \quad (79)$$

Restoring c ,

$$G = \frac{c^3}{4\hbar\eta_0}. \quad (80)$$

The sign of δA depends on whether one uses past or future local horizons and on the orientation of the affine parameter. The physical all-null response coefficient is orientation-independent once the boost flux and area variation are defined consistently.

B All-null tensor completion

Suppose

$$X_{\mu\nu}k^\mu k^\nu = 0 \quad (81)$$

for every null vector k^μ . Then

$$X_{\mu\nu} = \Phi g_{\mu\nu} \quad (82)$$

for some scalar Φ . Applying this to

$$X_{\mu\nu} = R_{\mu\nu} - 8\pi G T_{\mu\nu} \quad (83)$$

gives

$$R_{\mu\nu} - 8\pi G T_{\mu\nu} = \Phi g_{\mu\nu}. \quad (84)$$

Taking the divergence and using stress-energy conservation and the contracted Bianchi identity yields

$$\Phi = \frac{1}{2}R - \Lambda, \quad (85)$$

so that

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = 8\pi G T_{\mu\nu}. \quad (86)$$

C Relation to standard entanglement-equilibrium arguments

The relative-entropy identity

$$\delta S = \delta \langle K \rangle \tag{87}$$

is standard in entanglement first-law arguments. The distinct G1 contribution is not the identity itself, but its reinterpretation inside a finite causal-screen access program with a finite-index capacity density η_0 , an optical anomaly Hessian producing the 3/4 Weyl projection, and a horizon-scale anomaly remnant producing the G1DE residual.

D Selected references

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Physical Review D* **7**, 2333–2346 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Communications in Mathematical Physics* **43**, 199–220 (1975).
- [3] W. G. Unruh, “Notes on black-hole evaporation,” *Physical Review D* **14**, 870–892 (1976).
- [4] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Physical Review Letters* **75**, 1260–1263 (1995).
- [5] R. M. Wald, “Black hole entropy is the Noether charge,” *Physical Review D* **48**, R3427–R3431 (1993).
- [6] J. J. Bisognano and E. H. Wichmann, “On the duality condition for a Hermitian scalar field,” *Journal of Mathematical Physics* **16**, 985–1007 (1975).
- [7] M. Takesaki, *Theory of Operator Algebras II*, Springer (2003).
- [8] D. Petz, “Sufficient subalgebras and the relative entropy of states of a von Neumann algebra,” *Communications in Mathematical Physics* **105**, 123–131 (1986).
- [9] R. Longo, “Notes on algebraic invariants for noncommutative dynamical systems,” *Communications in Mathematical Physics* **69**, 195–207 (1979), and subsequent developments on index and inclusions.
- [10] H. Casini, M. Huerta, and R. C. Myers, “Towards a derivation of holographic entanglement entropy,” *Journal of High Energy Physics* **2011**, 036 (2011).
- [11] N. Lashkari, M. B. McDermott, and M. Van Raamsdonk, “Gravitational dynamics from entanglement thermodynamics,” *Journal of High Energy Physics* **2014**, 195 (2014).
- [12] T. Faulkner, M. Guica, T. Hartman, R. C. Myers, and M. Van Raamsdonk, “Gravitation from entanglement in holographic CFTs,” *Journal of High Energy Physics* **2014**, 051 (2014).
- [13] O. Fawzi and R. Renner, “Quantum conditional mutual information and approximate Markov chains,” *Communications in Mathematical Physics* **340**, 575–611 (2015).
- [14] P. Hayden, O. Parrikar, and J. Sorce, “The Markov gap for geometric reflected entropy,” and related works on quantum Markov recovery and holographic codes.
- [15] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, FDS-G1 Core manuscript (2026).

- [16] Y. Wu, *Finite-Access Sheaf Geometry and the Relative-Entropy Origin of the $3/4$ Projection Law*, FDS-G1 technical note D1 (2026).
- [17] Y. Wu, *Weyl-Normalized Ward Completion and Identifiability Rules*, FDS-G1 technical note D5 (2026).
- [18] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background-Weyl Residual*, data paper (2026).

Finite-Access Sheaf Geometry and the Relative-Entropy Origin of the 3/4 Projection Law

A Conditional Closure of the G1 Finite-Screen Dark-Sector Branch
v1.2 Body-Integrated Release

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS–G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This paper develops a deeper mathematical closure for the finite-screen dark-sector branch of the FDS–G1 program. Previous G1 notes introduced a finite causal-screen entropy functional, derived a maximum-entropy optical projection law on the local null screen algebra, and implemented the resulting cosmological residual as a Weyl-normalized transfer correction. The present paper asks what mathematical object lies underneath these ingredients. The proposed answer is a finite-access sheaf: a contravariant assignment of finite, maintainable distinction ledgers to causal screens, with entropy as the coordinate-free ledger-size section and geometry as the integrable natural connection of the resulting response data.

The local optical fiber of this finite-access sheaf is the Jacobi deformation algebra

$$T_l GL(2) = A(1) \oplus S(2) \oplus \Omega(1),$$

where A is the area/trace channel, S is the two-dimensional symmetric shear sector, and Ω is the twist/holonomy sector. A finite-access screen state family

$$\rho_S(q) = \mathcal{E}_{S,q}(\rho_{\text{bulk}})$$

defines a relative-entropy Hessian

$$K_{IJ} = \partial_I \partial_J D_{\text{rel}}(\rho_S(q) \| \rho_S(0))|_{q=0},$$

which is the Fisher/Kubo–Mori metric of optical access distortions. The response compliance is $C = K^{-1}$. Projection onto the Weyl-active unimodular sector $U = S \oplus \Omega$ gives

$$\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr } C} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}.$$

The unbiased finite-access fixed point $K_A = K_S = K_\Omega$ yields $\kappa_{BW} = 3/4$, while demoting the twist/holonomy ledger gives the scalar-shear branch $\kappa_{BW} = 2/3$.

The global dark-sector residual is then interpreted as a horizon-scale gluing-load lag of the finite-access sheaf. The used horizon capacity C_{used} is the accessible distinction load required

to glue horizon-accessible local ledgers; the occupancy ratio is $\mathcal{O}_H = C_{\text{used}}/C_H$ with $C_H = A_H/(4\ell_P^2)$, and the source current is $J_H = -d_N \ln \mathcal{O}_H$. A causal, stable, passive retarded response gives $\mathcal{R}_H = G_H^{\text{ret}} * J_H$. Weyl normalization and minimum Ward completion then imply

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2),$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

With $\mathcal{R}_H(a) = R_0 \hat{R}_H(a)$, $\hat{R}_H(1) = 1$, and $R_0 = 3 - s$, the isotropic branch recovers the G1DE-M_{3/4} reduction

$$\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s) \hat{R}_H(a), \quad \mu \simeq 1.$$

The result is a conditional closure theorem for the G1 dark-sector branch. It does not derive the Bekenstein–Hawking coefficient, the local screen first law, or the exact nonlinear Einstein equations from FDS primitives alone.

1 Scope and claim-status discipline

The FDS–G1 core program begins from a finite causal-screen entropy functional

$$S_{\text{scr}} : X_{\text{phys}} \rightarrow \mathbb{R}, \tag{1}$$

defined on a physical quotient state space of screen variables. The response one-form

$$\omega = \eta_i(X) dX^i, \quad \eta_i = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta X^i}, \tag{2}$$

contains the local area response that, under the standard horizon-entropy bridge, determines the local inverse gravitational coupling or screen stiffness. Previous G1 papers then developed screen statistics, stochastic screen dynamics, metric integrability, horizon occupancy residuals, and a likelihood-ready G1DE branch.

The present paper is not a replacement for those components. Its purpose is narrower and deeper: to provide a conditional mathematical object underneath the finite screen entropy functional, the optical response Hessian, and the 3/4 projection law. D0 supplies the local modular calibration of finite screen entropy, boost energy, and Einstein response. The present note addresses the complementary global/access question: how finite ledgers restrict, glue, and generate residual gluing defects across causal screens. The proposed object is a finite-access sheaf over causal screens.

1.1 What this paper claims

The paper makes four conditional claims.

D1 supplies the mathematical home for the Core’s response obstruction: $[\mathcal{C}_{\text{resp}}] \in H_{\text{resp}}^2(X_{\text{phys}})$. The sheaf-gluing language identifies which response non-closure survives all quotient, missing-variable, and memory-lift corrections as a true geometric residual.

D9 promotes the D1 gluing-defect interpretation into an exact sequence of response defects, quotient artifacts, obstruction classes, and admissible residuals. The explicit correspondence is

$$\mathcal{F}_{\nabla_S} \longleftrightarrow [\mathcal{C}_{\text{resp}}] \longleftrightarrow \mathcal{O}_{\text{resp}},$$

where $\mathcal{O}_{\text{resp}}$ is the D9 physical obstruction class.

1. A finite causal-screen entropy functional can be represented as the entropy section of a finite-access sheaf over causal screens.
2. The optical-sector response Hessian is the relative-entropy/Fisher metric of local optical access distortions.
3. If the finite-access Hessian is locally unbiased over the full $GL(2)$ optical fiber, then the Weyl-active compliance share is $3/4$.
4. A horizon-scale gluing-load lag, when Weyl-normalized and minimally Ward-completed, produces the G1DE hierarchy $\Sigma - 1 = O(\mathcal{R}_H)$ and $\mu - 1 \ll \Sigma - 1$ on subhorizon linear scales.

1.2 What this paper does not claim

The result is not claimed to be an unconditional derivation of general relativity. In particular, this paper does not derive the Bekenstein–Hawking coefficient $1/(4\ell_P^2)$, the local screen first law, exact Raychaudhuri coefficients, or the full nonlinear Einstein equations from FDS primitives alone. Those remain deeper FDS-to-GR closure problems. The present result is a conditional closure of the G1 dark-sector branch, assuming the standard horizon entropy normalization and the G1 finite-screen response program.

Claim	Meaning	Status
Finite-access sheaf	causal screens carry finite maintainable distinction ledgers	programmatic FDS bridge
Entropy section	S_{scr} is the coordinate-free state-count ledger	definition/framework
Response connection	geometry is the integrable envelope of screen-response transport	conditional geometric bridge
Relative-entropy Hessian	K_{IJ} measures distinguishability of optical access distortions	information-geometric definition
$3/4$ projection law	unbiased full $GL(2)$ optical access gives $\kappa_{BW} = 3/4$	conditional theorem
$2/3$ demotion	twist/holonomy removed gives scalar-shear branch	conditional theorem
Horizon gluing lag	$\mathcal{R}_H$ is passive memory of horizon access-load drift	dynamical bridge
Weyl-normalized G1DE	$\mathcal{W}_{\text{G1}} = (1 - \kappa\mathcal{R}_H)\mathcal{W}_{\text{GR}}$	production convention

Table 1: Claim-status map for the finite-access sheaf closure.

2 From finite distinctions to finite-access sheaves

The primitive FDS argument can be stated compactly:

1. physical distinctions must be finite and maintainable;

2. maintainable distinctions require a boundary;
3. in gravity the relevant boundary is causal, namely a screen or horizon;
4. a finite causal screen has a finite state-count ledger;
5. the coordinate-free ledger of finite state-count is entropy;
6. geometry is the integrable response envelope of that ledger.

This paper sharpens the word “ledger”. A single causal screen does not stand alone. Screens refine, overlap, coarse-grain, restrict, and evolve. A finite-access theory should therefore assign ledgers not only to individual screens but consistently across a system of causal screens.

Definition 1 (Causal-screen category). *Let Scr denote a category whose objects are causal screen patches: local Rindler screens, causal-diamond cuts, light-sheet cross-sections, apparent-horizon patches, or observer-dependent optical screens. Morphisms*

$$f : \mathcal{S}_1 \rightarrow \mathcal{S}_2 \tag{3}$$

represent causal restriction, screen inclusion, coarse-graining, refinement, null evolution, or operational comparison between screens.

Remark 1 (Prior-art alignment). *Recent finite causal-diamond phase-space results (Ciambelli, He, and Zurek 2026) provide an external gravitational example in which a finite causal region carries edge/soft phase-space data, supporting the use of causal-diamond cuts as legitimate finite-access screen objects. This supports the representation layer but does not prove the FDS sheaf construction.*

The direction of information access is naturally contravariant: a larger or later screen may restrict to a smaller ledger, while a refined screen may be coarse-grained to a lower-resolution ledger.

Definition 2 (Finite-access sheaf). *A finite-access sheaf is a contravariant assignment*

$$\mathcal{A} : \text{Scr}^{\text{op}} \rightarrow \text{FinLedger}, \tag{4}$$

where FinLedger is a category of finite maintainable distinction ledgers and ledger maps. A quantum version assigns finite-dimensional screen algebras or states,

$$\mathcal{A} : \text{Scr}^{\text{op}} \rightarrow \text{FinChan}, \tag{5}$$

where FinChan denotes finite stochastic channels or finite-dimensional quantum channels.

For a screen $\mathcal{S}$, the object $\mathcal{A}(\mathcal{S})$ is the finite ledger of distinctions accessible and maintainable on that screen. For a morphism $f : \mathcal{S}_1 \rightarrow \mathcal{S}_2$, the map $\mathcal{A}(f) : \mathcal{A}(\mathcal{S}_2) \rightarrow \mathcal{A}(\mathcal{S}_1)$ is the corresponding restriction or pullback of accessible distinctions.

Remark 2 (Why sheaf language is useful). *The word “sheaf” is used in a minimal sense. It encodes the idea that local screen ledgers must restrict and glue consistently. When they glue into a smooth metric envelope, the response looks geometric. When they fail to glue without residuals, the obstruction is a candidate dark or non-integrable response sector.*

Terminological convention. In this note, “sheaf” is used in an operational finite-access sense. Strictly, the assignment of finite ledgers to causal screens is first a *presheaf* over causal-screen contexts. When a causal-screen topology, covering families, and unique gluing conditions are imposed, the assignment satisfies a sheaf condition. Residual sectors are represented as failures of such gluing, or as obstruction data. We do not assume a full Grothendieck-topos construction in this paper.

3 v1.2 Ledger Object and Sheaf Realization

Claim type: Interface specification The finite ledger object defined in Core v1.2 is realized as a local chart in the D1 finite-access sheaf:

$$\mathcal{L}_{\text{scr}}|_{U_i}$$

where U_i is a local causal-screen patch. The sheaf gluing obstruction is exactly the source of the G1 residual $R_{\mu\nu}^{\text{scr}}$: failure to uniquely glue local ledger charts across overlapping causal-screen regions produces a conserved residual stress.

Boundary note: This interface connects the formal finite ledger definition in Core to the sheaf geometry framework in D1, but does not prove that all admissible residuals correspond to gluing obstructions.

4 Entropy as a ledger-size section

For each screen $\mathcal{S}$, the finite ledger may be represented by a classical distribution $p_{\mathcal{S}}$ or by a quantum state $\rho_{\mathcal{S}}$. The screen entropy is then the coordinate-free ledger size

$$S_{\text{scr}}(\mathcal{S}) = H(p_{\mathcal{S}}) \tag{6}$$

or

$$S_{\text{scr}}(\mathcal{S}) = S(\rho_{\mathcal{S}}) = -\text{Tr}(\rho_{\mathcal{S}} \ln \rho_{\mathcal{S}}). \tag{7}$$

When screen variables are collected as $X_{\mathcal{S}} \in X_{\text{phys}}(\mathcal{S})$, this becomes the familiar entropy functional

$$S_{\text{scr}} : X_{\text{phys}}(\mathcal{S}) \rightarrow \mathbb{R}. \tag{8}$$

Thus the G1 primitive object can be reinterpreted as a local expression of a global finite-access structure.

Principle 1 (Entropy section). *Finite screen entropy is the entropy section of the finite causal-access sheaf. It is not an arbitrary scalar field added to geometry; it is the coordinate-free state-count ledger associated with finite, maintainable, causally accessible distinctions.*

4.1 Channel representation

A useful local representation is a finite screen readout channel

$$\mathcal{E}_{\mathcal{S},X} : \mathcal{B}(\mathcal{H}_{\text{bulk}}) \rightarrow \mathcal{B}(\mathcal{H}_{\mathcal{S}}), \tag{9}$$

or a classical stochastic channel

$$p_{\mathcal{S}}(y) = \sum_x \mathcal{A}_{\mathcal{S},X}(y|x) p_{\text{bulk}}(x). \tag{10}$$

Then

$$\rho_S(X) = \mathcal{E}_{S,X}(\rho_{\text{bulk}}), \quad S_{\text{scr}}[X] = S(\rho_S(X)). \quad (11)$$

This representation explains why entropy appears before metric geometry: finite access produces a finite ledger, and entropy is the coordinate-free size of that ledger.

5 Response connection and metric integrability

The response one-form on a screen is

$$\omega_S = dS_{\text{scr}} = \eta_i(X) dX^i. \quad (12)$$

As screens vary through Scr , response data must be transported. This motivates a response connection

$$\nabla_{\text{scr}} = d + \Gamma_{\text{scr}} \quad (13)$$

on the finite-access ledger bundle.

Principle 2 (Geometry as integrable response connection). *Metric geometry is the smooth integrable envelope of the finite-access response connection. A GR-like sector appears when local response data can be represented by a metric tensor envelope satisfying the appropriate Ward/Bianchi identities.*

Definition 3 (Finite-access response connection). *A finite-access response connection ∇_{scr} is a rule for transporting ledger states, entropy sections, or response one-forms between nearby causal-screen contexts. Locally,*

$$\nabla_{\text{scr}} = d + \Gamma_{\text{scr}}.$$

If its response curvature can be represented by a smooth metric envelope satisfying Ward/Bianchi consistency, the sector is geometrically integrable. Nonzero non-metric curvature or failure of gluing defines a residual obstruction:

$$F_{\nabla_{\text{scr}}} = 0 \Rightarrow \text{integrable ledger response}, \quad F_{\nabla_{\text{scr}}} \neq 0 \Rightarrow \text{residual/gluing defect}.$$

If response data are local and all-null compatible, they may be represented by a symmetric tensor envelope

$$F_p(k) = E_{\mu\nu}(p) k^\mu k^\nu \quad (14)$$

up to trace freedom. Ward/Bianchi closure fixes the trace integration sector. If finite-access response data fail to admit such a global envelope, the failure is a residual.

5.1 Residuals as gluing defects

Let ω_α be response one-forms on overlapping causal screen patches U_α . If the data glue globally, then on overlaps one expects

$$\omega_\alpha - \omega_\beta = d\lambda_{\alpha\beta} \quad (15)$$

with a trivial cocycle on triple overlaps,

$$\lambda_{\alpha\beta} + \lambda_{\beta\gamma} + \lambda_{\gamma\alpha} = 0. \quad (16)$$

If instead

$$\delta\lambda_{\alpha\beta\gamma} \neq 0, \quad (17)$$

then the finite-access response data do not glue into a globally integrable metric ledger without residual structure.

Remark 3 (Dark sector interpretation). *In this language, the G1 dark sector is not an arbitrary material fluid. It is a horizon-scale finite-access gluing defect, projected into the Weyl optical channel and constrained by Ward/naturality conditions.*

Residual filter. The apparent response obstruction is classified as: (1) quotient artifact – removable by gauge fixing; (2) missing variable – resolved by state-space enlargement; (3) memory projection – removed by horizontal lift; (4) true sheaf-gluing obstruction – the only category that may feed G1DE residuals.

6 The local optical fiber

A finite observer receives optical information through a two-dimensional screen transverse to a null ray. Away from caustics, the Jacobi map is an element

$$J \in GL(2, \mathbb{R}). \quad (18)$$

A local optical deformation is

$$X = \delta J J^{-1} \in \mathfrak{gl}(2, \mathbb{R}). \quad (19)$$

Under screen rotations, the deformation algebra decomposes as

$$\mathfrak{gl}(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1), \quad (20)$$

where A is the trace/area direction, S is the two-dimensional symmetric traceless shear sector, and Ω is the antisymmetric twist/holonomy sector.

Sector	Dimension	Optical meaning	G1 role
A	1	trace/determinant/area	filling and Ricci-like focusing
S	2	symmetric traceless shear	Weyl/E-mode optical shape
Ω	1	twist/holonomy	orientation/curl capacity
$U = S \oplus \Omega$	3	unimodular sector	Weyl-active optical ledger

Table 2: The local optical access fiber of a two-dimensional null screen.

The decomposition is algebraic. The coefficient $3/4$ does not follow from algebra alone. It follows when the finite-access response metric is unbiased or isotropic over the four local optical directions.

7 Relative-entropy Hessian of optical access distortions

Let

$$q^I = (q_A, q_{S_1}, q_{S_2}, q_\Omega) \quad (21)$$

be coordinates on the optical tangent fiber $T_I GL(2, \mathbb{R}) \simeq \mathfrak{gl}(2, \mathbb{R})$. We assume that $\rho_S(0)$ is full rank on the finite screen code subspace, or else restrict the local family $\rho_S(q)$ to the common support on which the relative entropy is finite. The finite-access channel under optical deformation produces a screen state family

$$\rho_S(q) = \mathcal{E}_{S,q}(\rho_{\text{bulk}}). \quad (22)$$

The distinguishability cost of the deformation is

$$D_{\text{scr}}(q) = D_{\text{rel}}(\rho_S(q) \parallel \rho_S(0)). \quad (23)$$

Near $q = 0$,

$$D_{\text{scr}}(q) = \frac{1}{2} K_{IJ} q^I q^J + O(q^3), \quad (24)$$

with

$$\boxed{K_{IJ} = \partial_I \partial_J D_{\text{rel}}(\rho_S(q) \parallel \rho_S(0))|_{q=0}}. \quad (25)$$

This is the relative-entropy Hessian of finite optical access distortions.

7.1 Exponential-family local closure

A least-biased local state family with optical constraints takes the exponential form

$$\rho_S(q) = Z(q)^{-1} \exp(\log \rho_0 + q^I \mathcal{O}_I), \quad (26)$$

where $\mathcal{O}_I$ are screen observables conjugate to the four optical generators. For such a family, the relative-entropy Hessian equals the Kubo–Mori covariance,

$$K_{IJ} = \langle \Delta \mathcal{O}_I, \Delta \mathcal{O}_J \rangle_{\text{KM}}. \quad (27)$$

In a commuting or classical limit,

$$K_{IJ} = \text{Cov}_{\rho_0}(\mathcal{O}_I, \mathcal{O}_J). \quad (28)$$

Thus the optical stiffnesses K_A, K_S, K_Ω are covariance eigenvalues of the finite-access screen channel.

Remark 4. *The exponential-family assumption is a local MaxEnt closure, not a microscopic theorem. It states that if only the optical deformation constraints are known, the least-biased finite-access state family is exponential in the corresponding optical generators.*

8 Compliance projection and the 3/4 theorem

By $SO(2)$ symmetry, the Hessian block-decomposes as

$$K = K_A P_A + K_S P_S + K_\Omega P_\Omega. \quad (29)$$

The compliance matrix is

$$C = K^{-1} = K_A^{-1} P_A + K_S^{-1} P_S + K_\Omega^{-1} P_\Omega. \quad (30)$$

The Weyl-active unimodular projector is

$$P_U = P_S + P_\Omega. \quad (31)$$

Definition 4 (Weyl-active compliance fraction). *The background–Weyl projection coefficient is*

$$\boxed{\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr}(C)}}. \quad (32)$$

All traces are taken in an $SO(2)$ -adapted orthonormal optical generator basis at the unbiased reference point. Anisotropic generator normalizations are absorbed into the block weights K_A, K_S, K_Ω . Thus the value 3/4 refers to the isotropic finite-access fixed point, not to an arbitrary coordinate normalization.

Since $\dim A = 1$, $\dim S = 2$, and $\dim \Omega = 1$,

$$\boxed{\kappa_{BW} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}.} \quad (33)$$

Equivalently, with compliances

$$\chi_A = K_A^{-1}, \quad \chi_S = K_S^{-1}, \quad \chi_\Omega = K_\Omega^{-1}, \quad (34)$$

one has

$$\kappa_{BW} = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega}. \quad (35)$$

Theorem 1 (Unbiased finite-access projection). *If the finite-access Hessian is locally unbiased on the full optical fiber,*

$$K_A = K_S = K_\Omega, \quad (36)$$

then

$$\kappa_{BW} = \frac{3}{4}. \quad (37)$$

Proof. At the unbiased fixed point, $C = K_0^{-1}I$ on the four-dimensional optical fiber. The Weyl-active sector $U = S \oplus \Omega$ has dimension $2 + 1 = 3$. Therefore

$$\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr}(C)} = \frac{3K_0^{-1}}{4K_0^{-1}} = \frac{3}{4}. \quad (38)$$

□

Corollary 1 (Twist-demoted branch). *If the twist/holonomy ledger is removed or has zero compliance, $K_\Omega^{-1} = 0$, and $K_A = K_S$, then*

$$\kappa_{BW} = \frac{2}{1+2} = \frac{2}{3}. \quad (39)$$

The theorem does not claim that all physical screens are unbiased. It identifies the isotropic finite-access fixed point. Deviations from this point are represented by weighted-compliance coefficients K_A, K_S, K_Ω , and are tested by the free- κ control branch. The 3/4 law is therefore not a theorem of algebra alone. It is the isotropic MaxEnt/Fisher-compliance fraction of the full $GL(2)$ finite-access optical ledger. The 2/3 law is the demoted scalar-shear branch when the twist/holonomy ledger is absent from the accessible screen capacity.

9 Local fiber theorem and global gluing theorem

The finite-access closure naturally separates into two levels.

9.1 Local fiber theorem

The local theorem says:

$$\boxed{\text{unbiased local finite-access optical fiber} \Rightarrow \kappa_{BW} = 3/4.} \quad (40)$$

This theorem is local in screen space and does not determine the cosmological redshift kernel.

9.2 Global gluing theorem

The global theorem concerns horizon-scale access-load drift. Local ledgers must be glued over the horizon-accessible region. If this gluing has a finite-capacity lag, it produces a horizon residual $\mathcal{R}_H(a)$. This residual determines the redshift-dependent amplitude of the Weyl transfer correction.

Thus:

$$\boxed{\text{local fiber gives } 3/4, \quad \text{global access gluing gives } \mathcal{R}_H(a).} \quad (41)$$

10 Horizon gluing load and occupancy current

The used horizon capacity should not be interpreted as the entropy of the entire universe. It is the finite access load that must be maintained by the horizon ledger.

Definition 5 (Horizon access load). *Let $\mathcal{X}_a$ denote a horizon-accessible bulk distinction ensemble at epoch a . The used horizon capacity is*

$$C_{\text{used}}(a) = \sup_{\mathcal{X}_a} I_{\text{acc}}[\mathcal{X}_a; \mathcal{A}_H(\mathcal{X}_a)]. \quad (42)$$

Equivalently, a tractable relative-entropy surrogate is

$$C_{\text{used}}(a) \sim D_{\text{rel}}(\rho_{\text{bulk}}^H(a) \parallel \rho_{\text{coarse}}^H(a)). \quad (43)$$

The definition of C_{used} is structural. It identifies the horizon access-load object that a microscopic FDS model should compute. The current low-redshift G1DE implementation does not directly evaluate $C_{\text{used}}(a)$ from a cosmological distinction ensemble. Instead it uses the reduced memory response

$$\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1, \quad R_0 = 3 - s,$$

as the data-facing low-redshift reduction. A conservative toy surrogate is

$$C_{\text{used}}^{\text{toy}}(a) = D_{\text{rel}}(\rho_{\text{LSS}}^H(a) \parallel \rho_{\text{FRW}}^H(a)),$$

where ρ_{LSS}^H denotes horizon-accessible coarse-grained large-scale-structure states and ρ_{FRW}^H is the corresponding coarse FRW reference.

The available horizon capacity is calibrated by the standard horizon bridge,

$$C_H(a) = \frac{A_H(a)}{4\ell_P^2}. \quad (44)$$

The occupancy ratio is

$$\mathcal{O}_H(a) = \frac{C_{\text{used}}(a)}{C_H(a)}. \quad (45)$$

The horizon occupancy current is

$$\boxed{J_H(N) = -\frac{d}{dN} \ln \mathcal{O}_H(N), \quad N = \ln a.} \quad (46)$$

This current measures the scale-flow of access load relative to available horizon capacity.

11 Passive retarded gluing-lag response

The horizon residual is the retarded response of the finite-access ledger to occupancy-current drift:

$$\mathcal{R}_H(N) = \int_{-\infty}^N G_H^{\text{ret}}(N, N') J_H(N') dN'. \quad (47)$$

A Markov approximation gives

$$\mathcal{R}'_H(N) + \Gamma_H(N) \mathcal{R}_H(N) = J_H(N), \quad \Gamma_H > 0. \quad (48)$$

The open-system consistency conditions are:

$$G_H^{\text{ret}}(N, N') = 0 \quad (N' > N), \quad (49)$$

$$N_H(N, N') \succeq 0, \quad (50)$$

$$\int dN J_H(N) \mathcal{R}_H(N) \geq 0. \quad (51)$$

Here N_H is the noise kernel of the corresponding Schwinger–Keldysh/Feynman–Vernon influence functional. Equations (49)–(51) express causality, positive noise, and passivity.

Remark 5 (Meaning of s). *In the low-redshift G1DE reduction,*

$$\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1, \quad R_0 = 3 - s. \quad (52)$$

Thus

$$3 - s = \mathcal{R}_H(1), \quad (53)$$

so the observable deficit $3 - s$ is the present normalized finite-access gluing-lag amplitude. The parameter s should be read as a low-redshift effective filling exponent, not as a literal spatial dimension.

12 Weyl normalization and no free optical amplitude

Define the Weyl source

$$\mathcal{W}(a, k) = -k^2(\Phi + \Psi). \quad (54)$$

In the standard Σ convention,

$$\mathcal{W}_{\text{GR}}(a, k) = 8\pi G a^2 \rho \Delta. \quad (55)$$

G1 is implemented as a finite-screen transfer correction to the GR Weyl source:

$$\mathcal{W}_{\text{G1}}(a, k) = [1 - \kappa_{BW} \mathcal{R}_H(a)] \mathcal{W}_{\text{GR}}(a, k) + O(\mathcal{R}_H^2). \quad (56)$$

Therefore

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2). \quad (57)$$

Earlier screen-influence formulations used a schematic traceless optical seed

$$E_{ij, \text{TF}}^{\text{scr}} = -A(a, k) \kappa_{BW} \mathcal{R}_H(a) \left(\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij} \right). \quad (58)$$

The finite-access sheaf formulation clarifies that $A(a, k)$ is not a physical response function. It is the normalization map that identifies a unit optical seed with a unit fractional correction to the GR Weyl source:

$$A(a, k) \longleftrightarrow \mathcal{W}_{\text{GR}}(a, k) = 8\pi G a^2 \rho \Delta. \quad (59)$$

Thus

$$\boxed{A(a, k) \notin \Theta_{\text{G1DE}}.} \quad (60)$$

Sampling $A(a, k)$ as a nuisance parameter would change the model into a generic dark-stress source model, not the G1DE finite-screen transfer branch.

13 Minimum Ward completion and Ricci suppression

The Weyl transfer correction can be represented covariantly by an effective residual tensor

$$E_{\mu\nu}^{\text{scr}} = \mathcal{P}_W^{\text{min}}[-\kappa_{BW} \mathcal{R}_H \mathcal{W}_{\text{GR}}]_{\mu\nu}, \quad (61)$$

where $\mathcal{P}_W^{\text{min}}$ denotes the minimum Ward-completion operator that embeds a Weyl-normalized traceless seed into a conserved or controlled-exchange tensor representative.

In a gauge-invariant scalar-sector description, introduce effective screen variables

$$\pi_{\text{scr}}^{\text{GI}}, \quad q_{\text{scr}}^{\text{GI}}, \quad \delta\rho_{\text{scr}}^{\text{GI}}. \quad (62)$$

The Weyl-normalized seed is

$$k^2 \pi_{\text{scr}}^{\text{GI}} = -\kappa_{BW} \mathcal{R}_H(a) (8\pi G a^2 \rho \Delta), \quad (63)$$

up to the sign and anisotropic-stress convention of the chosen Boltzmann solver. Minimum Ward completion gives the hierarchy

$$q_{\text{scr}}^{\text{GI}} = O\left(\pi_{\text{scr}}^{\text{GI}} \frac{\mathcal{H}}{k}\right), \quad (64)$$

$$\delta\rho_{\text{scr}}^{\text{GI}} = O\left(\pi_{\text{scr}}^{\text{GI}} \frac{\mathcal{H}^2}{k^2}\right) + O(\mathcal{R}_H^2). \quad (65)$$

Therefore

$$\boxed{\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).} \quad (66)$$

This is the mathematical origin of first-order Ricci silence: the finite-screen access lag is assigned to the Weyl optical channel with minimal Newtonian/Ricci leakage.

14 Finite-access sheaf closure theorem

We can now collect the full conditional closure.

Theorem 2 (Finite-access sheaf closure). *Assume:*

1. *physical distinctions form finite maintainable access ledgers over causal screens;*
2. *these ledgers define a finite-access sheaf $\mathcal{A} : \text{Scr}^{\text{op}} \rightarrow \text{FinLedger}$;*
3. *entropy is the coordinate-free ledger-size section S_{scr} ;*

4. local optical access distortions span $T_I GL(2) = A \oplus S \oplus \Omega$;
5. the relative-entropy Hessian of the local optical fiber is unbiased, $K_A = K_S = K_\Omega$;
6. horizon gluing load defines $J_H = -d_N \ln(C_{\text{used}}/C_H)$;
7. the gluing-lag response $\mathcal{R}_H = G_H^{\text{ret}} * J_H$ is causal, stable, and passive;
8. the resulting residual is Weyl-normalized and minimally Ward-completed.

Then

$$\kappa_{BW} = \frac{3}{4}, \quad (67)$$

$$\Sigma(a, k) - 1 = -\frac{3}{4}\mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad (68)$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (69)$$

With $\mathcal{R}_H(a) = R_0 \hat{R}_H(a)$, $\hat{R}_H(1) = 1$, and $R_0 = 3 - s$, the locked G1DE- $M_{3/4}$ reduction is

$$\boxed{\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a), \quad \mu \simeq 1.} \quad (70)$$

Proof. The finite-access sheaf supplies finite screen states and the entropy section. The local optical fiber is $A(1) \oplus S(2) \oplus \Omega(1)$. The relative-entropy Hessian defines sector stiffnesses K_A, K_S, K_Ω and compliances $K_A^{-1}, K_S^{-1}, K_\Omega^{-1}$. By the compliance projection formula,

$$\kappa_{BW} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}. \quad (71)$$

If $K_A = K_S = K_\Omega$, this gives $3/4$. The horizon gluing load generates $\mathcal{R}_H$ by causal passive response. Weyl normalization gives $\Sigma - 1 = -\kappa_{BW}\mathcal{R}_H + O(\mathcal{R}_H^2)$. Minimum Ward completion suppresses the density/Ricci channel by $\mathcal{H}^2/k^2$ and by nonlinear powers of $\mathcal{R}_H$, giving the stated μ hierarchy. Substituting $\mathcal{R}_H = (3 - s)\hat{R}_H$ yields the low-redshift locked reduction. $\square$

15 Demotion paths and falsification registry

The closure is structured with demotion paths. Failure of one branch need not falsify the entire finite-screen program.

16 Relation to the G1 companion sequence

This paper is best read as a structural layer beneath the existing G1 dark-sector closure notes.

- The core FDS-G1 paper identifies finite causal-screen entropy response as the primitive gravity-side object.
- The optical-stress closure note derives the $3/4$ projection law as a MaxEnt closure on the $GL(2)$ optical screen algebra.

Condition	What fails	What may survive
$\kappa_{BW} \simeq 2/3$	full $GL(2)$ optical access ledger	scalar-shear/SPD(2) demotion
κ_{BW} far from $2/3$ and $3/4$	unbiased finite-access Hessian	weighted compliance law
Dominant first-order μ response	minimum Ward/Ricci-silent branch	finite-screen residual with density leakage
No common memory shape	horizon gluing-load kernel	separate background and optical residuals
Free $A(a, k)$ required	Weyl normalization	generic dark-stress source, not G1DE
No passive kernel for reconstructed $\mathcal{R}_H$	retarded access-lag interpretation	phenomenological Weyl transfer
No finite screen entropy response	FDS–G1 bridge	algebraic optical toy model only

Table 3: Demotion and falsification registry for the finite-access sheaf closure.

- The non-equilibrium entropy note interprets the projection coefficient through optical-sector entropy production and response compliances.
- The screen influence functional packages entropy Hessian, retarded horizon memory, and Ward completion.
- The Weyl-normalized Ward completion note fixes the production-likelihood interface and forbids a free optical amplitude $A(a, k)$.

The present paper adds the deeper statement:

$$\boxed{\text{finite screen entropy is the entropy section of a finite causal-access sheaf.}} \quad (72)$$

It also reframes $3/4$ as a relative-entropy compliance fraction of the local optical access fiber, and $\mathcal{R}_H$ as a passive retarded gluing-lag response.

17 Open problems

The present closure leaves several major tasks.

1. Horizon coefficient. The capacity $C_H = A_H/(4\ell_P^2)$ is still imported through the standard horizon bridge. A stronger FDS derivation would obtain the Bekenstein–Hawking coefficient from finite-access microstructure.

2. Local screen first law. The local thermodynamic bridge $\delta Q = T\delta S$ is not derived internally here. It remains a bridge from horizon thermodynamics.

3. Exact focusing coefficients. Finite screen transition dynamics gives Raychaudhuri-like normal forms, but exact GR coefficients require a deeper microgeometric derivation.

4. Concrete computation of $C_{\text{used}}(a)$. This paper defines C_{used} as accessible horizon gluing load. Computing it from a concrete cosmological bulk distinction ensemble remains open.

5. Full Boltzmann implementation. The gauge-invariant Ward-completion hierarchy must be translated into a production Boltzmann pipeline with precise code-level anisotropic-stress conventions.

18 Conclusion

The finite-access sheaf formulation deepens the G1 dark-sector closure. Instead of beginning from an isolated entropy functional, it begins from a system of finite, maintainable, causally accessible distinction ledgers. The screen entropy functional is the entropy section of this ledger sheaf. Geometry is the integrable natural connection of the response data. Residual sectors are gluing defects or non-integrable response components.

The local optical fiber of the finite-access sheaf is $T_I GL(2) = A \oplus S \oplus \Omega$. The relative-entropy Hessian of optical access distortions supplies the sector stiffnesses. The Weyl-active compliance fraction is

$$\kappa_{BW} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}. \quad (73)$$

The unbiased finite-access fixed point gives $\kappa_{BW} = 3/4$, while twist/holonomy demotion gives $2/3$.

At the global level, horizon gluing load produces an occupancy current, its passive retarded response is $\mathcal{R}_H$, and Weyl normalization gives

$$\Sigma - 1 = -\kappa_{BW}\mathcal{R}_H + O(\mathcal{R}_H^2). \quad (74)$$

Minimum Ward completion protects the Newtonian/Ricci channel,

$$\mu - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (75)$$

Thus the G1DE- $M_{3/4}$ reduction is the locked branch of a finite-access, Weyl-normalized, Ward-disciplined screen-transfer class:

$$\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\widehat{R}_H(a), \quad \mu \simeq 1.$$

The current G1fit-real evidence hierarchy tests the low-redshift locked reduction of this structure rather than the full finite-access sheaf construction:

$$M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE-2} > \text{G1DE-1} > \text{CPL} > \Lambda\text{CDM}.$$

The slogan form is:

Gravity is the integrable natural connection of a finite causal-access ledger; G1DE is its horizon-scale Weyl gluing defect.

Symbol	Meaning
Scr	category of causal screens
$\mathcal{A}$	finite-access sheaf or ledger functor
S_{scr}	screen entropy functional / entropy section
$\rho_S(q)$	screen state under optical deformation q
K_{IJ}	relative-entropy Hessian of optical access distortions
$C = K^{-1}$	compliance matrix
A, S, Ω	area, shear, and twist sectors of $\mathfrak{gl}(2)$
P_U	projector onto the unimodular Weyl-active sector $S \oplus \Omega$
κ_{BW}	background–Weyl projection coefficient
C_{used}	horizon accessible gluing load
C_H	available horizon capacity $A_H/(4\ell_P^2)$
$\mathcal{O}_H$	occupancy ratio C_{used}/C_H
J_H	occupancy current $-d_N \ln \mathcal{O}_H$
$\mathcal{R}_H$	passive retarded response of horizon gluing load
$\mathcal{W}$	Weyl source $-k^2(\Phi + \Psi)$
$A(a, k)$	Weyl-normalization map, not a fit parameter

Table 4: Notation used in the finite-access sheaf closure.

A Notation summary

B Cech-style gluing defect

Let $\{U_\alpha\}$ be an open cover of a causal-screen family and let ω_α be local response one-forms. If response data glue up to exact gauge transformations,

$$\omega_\alpha - \omega_\beta = d\lambda_{\alpha\beta}, \quad (76)$$

then on triple overlaps the obstruction is

$$(\delta\lambda)_{\alpha\beta\gamma} = \lambda_{\beta\gamma} - \lambda_{\alpha\gamma} + \lambda_{\alpha\beta}. \quad (77)$$

A nonzero obstruction indicates that the local finite-access response data do not integrate into a single global ledger without residual structure. In the G1DE branch, the relevant large-scale obstruction is modeled as a horizon occupancy/gluing-lag residual $\mathcal{R}_H$ and is projected into the Weyl channel.

C Weighted-compliance expansion around the isotropic fixed point

Let

$$\chi_S = \chi_A(1 + \alpha), \quad \chi_\Omega = \chi_A(1 + \beta), \quad (78)$$

with $|\alpha|, |\beta| \ll 1$. Then

$$\kappa_{BW} = \frac{2(1 + \alpha) + (1 + \beta)}{1 + 2(1 + \alpha) + (1 + \beta)} \quad (79)$$

$$= \frac{3 + 2\alpha + \beta}{4 + 2\alpha + \beta} \quad (80)$$

$$= \frac{3}{4} + \frac{2\alpha + \beta}{16} + O(\alpha^2, \alpha\beta, \beta^2). \quad (81)$$

Thus deviations from $3/4$ measure anisotropy in the finite-access compliance Hessian. A coefficient near $2/3$ is not a small perturbation of the full $GL(2)$ branch; it indicates demotion of the twist/holonomy compliance.

References

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* **7**, 2333–2346 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* **43**, 199–220 (1975).
- [3] A. Raychaudhuri, “Relativistic cosmology. I,” *Phys. Rev.* **98**, 1123–1126 (1955).
- [4] R. K. Sachs, “Gravitational waves in general relativity. VI. The outgoing radiation condition,” *Proc. R. Soc. Lond. A* **264**, 309–338 (1961).
- [5] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* **75**, 1260–1263 (1995).
- [6] R. Bousso, “A covariant entropy conjecture,” *JHEP* **07**, 004 (1999).
- [7] R. Kubo, “Statistical-mechanical theory of irreversible processes. I,” *J. Phys. Soc. Jpn.* **12**, 570–586 (1957).
- [8] L. V. Keldysh, “Diagram technique for nonequilibrium processes,” *Sov. Phys. JETP* **20**, 1018–1026 (1965).
- [9] R. P. Feynman and F. L. Vernon, “The theory of a general quantum system interacting with a linear dissipative system,” *Ann. Phys.* **24**, 118–173 (1963).
- [10] D. Petz, “Monotone metrics on matrix spaces,” *Linear Algebra Appl.* **244**, 81–96 (1996).
- [11] T. M. Cover and J. A. Thomas, *Elements of Information Theory*, 2nd ed., Wiley (2006).
- [12] S. Mac Lane and I. Moerdijk, *Sheaves in Geometry and Logic*, Springer (1992).
- [13] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, FDS–G1 Core manuscript (2026).
- [14] Y. Wu, *The $3/4$ Projection Law in Finite Screen Gravity: MaxEnt Optical-Stress Closure, Ricci-Silent Residuals, and the Weyl-Locked Dark Sector*, FDS–G1 technical note D1 (2026).

- [15] Y. Wu, *Non-Equilibrium Screen Entropy and the Optical-Stress Origin of the G1 Dark Sector*, FDS–G1 technical note D2 (2026).
- [16] Y. Wu, *Screen Influence Functional for the G1 Dark Sector*, FDS–G1 technical note D3 (2026).
- [17] Y. Wu, *Weyl-Normalized Ward Completion and Identifiability Rules for G1DE*, FDS–G1 technical note D5 (2026).
- [18] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual*, FDS–G1 data paper (2026).

The 3/4 Projection Law in Finite Screen Gravity

MaxEnt Optical-Stress Closure, Ricci-Silent Residuals, and the Weyl-Locked Dark Sector
v1.2 Body-Integrated Release

Yining Wu

Independent Researcher

yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This paper gives a strengthened mathematical closure for the 3/4 projection law in Finite Distinction Systems – Gravity 1 (FDS-G1). The earlier projection-law argument observed that a finite observer receives spacetime residuals through a two-dimensional null optical screen and that the local Jacobi deformation algebra splits as

$$\mathfrak{gl}(2, \mathbb{R}) = \mathbb{R}I \oplus \mathfrak{sl}(2, \mathbb{R}),$$

namely one determinant/area/filling direction and three unimodular/shape directions. D0 embeds the same 3/4 coefficient in a modular anomaly Hessian; D1 embeds it in the relative-entropy Hessian of a finite-access sheaf. This paper isolates the optical projection law in its simplest local form: a maximum-entropy optical-stress closure on the $GL(2)$ screen algebra. The leading finite-screen residual is modeled as a local random deformation $X \in T_l GL(2, \mathbb{R}) \simeq \mathfrak{gl}(2, \mathbb{R})$ with fixed variance. In the absence of lower-order information distinguishing area, shear, and twist directions, the maximum-entropy covariance is isotropic on the optical deformation space. Projection onto the unimodular sector then gives

$$\kappa_{BW} = \frac{\text{Tr} P_{\text{uni}}}{\text{Tr} I_{\mathfrak{gl}(2)}} = \frac{3}{4}.$$

More generally, if area, shear, and twist sectors have response weights $\eta_A, \eta_S, \eta_\Omega$, the coefficient becomes

$$\kappa_{BW} = \frac{2\eta_S + \eta_\Omega}{\eta_A + 2\eta_S + \eta_\Omega},$$

with 3/4 as the isotropic fixed point and 2/3 as the scalar-shear demotion when the twist/holonomy direction is removed.

The paper then shows why the G1 residual is Ricci-silent and Weyl-active at first order. A compensating unimodular screen response is traceless by construction; through the Sachs optical equations it couples to the traceless optical tidal/Weyl channel rather than to the Ricci-density focusing channel. Consequently the local matter-growth channel is protected to first order,

$$\mu(a) = 1 + \mathcal{O}(\lambda^2),$$

while the Weyl/lensing channel receives the locked response

$$\Sigma(a) = 1 - \kappa_{BW} \lambda X_b(a) + \mathcal{O}(\lambda^2), \quad \lambda = 3 - s.$$

For isotropic lowest-order response this reduces to the G1DE-M_{3/4} relation

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s).$$

The redshift kernel $X_b(a)$ is interpreted as a retarded horizon-memory response. A minimal acceleration-gated closure, $X'_b + X_b = [1 - \epsilon_H]_+$, is stated as a candidate rather than as a theorem. This closure is the theoretical counterpart of the G1fit-real projection-locked model G1DE-M_{3/4}, whose production-refined likelihood implementation fixes $\mu_0 = 0$ and $\Sigma_0 = -3(3 - s)/4$ as the background–Weyl locked response. Current medium-prior production-refined model comparisons select the locked relation at the AIC/BIC and evidence level against common dark-energy extensions, and independent KiDS shear-only free- κ profiling returns $\kappa \approx 0.746$, empirically consistent with the isotropic 3/4 projection lock. Full survey-grade validation and 3×2 pt likelihoods remain pending. The result is conditional and falsifiable: a dominant first-order μ correction, failure of a shared $X_b(a)$ shape, a measured coefficient far from the allowed optical-response-weight family, or a preference for the 2/3 scalar-shear branch would demote the full 3/4 projection law.

1 Scope and executive summary

This paper is a mathematical companion to the FDS-G1 finite-screen entropy-response program. The G1 core identifies gravitational coupling with finite causal-screen entropy response and interprets dark-sector residuals as failures of complete finite-screen closure rather than as ordinary material fluids. The present work addresses one narrow but important question:

Why should the leading G1 cosmological residual be a Weyl/lensing correction with amplitude $-3/4$ times the background filling defect, while the local matter-growth channel remains close to general relativity?

1.1 Assumption Boundary of the 3/4 Lock

The 3/4 coefficient is not claimed as an unconditional geometric necessity. It is the isotropic full-optical-algebra branch of the finite-screen projection closure. The branch assumes:

$$T_1 GL(2) = A^{(1)} \oplus S^{(2)} \oplus \Omega^{(1)},$$

isotropic lowest-order compliance, trace/area removal from the dark residual, Weyl activity, Ricci silence at leading order, Ward compatibility, and no independent $A(a, k)$ amplitude. Under these assumptions,

$$\kappa_{\text{BW}} = \frac{\dim(S \oplus \Omega)}{\dim(A \oplus S \oplus \Omega)} = \frac{2 + 1}{1 + 2 + 1} = \frac{3}{4}.$$

If the compliance is anisotropic, the correct branch is weighted:

$$\kappa_{\text{BW}} = \frac{2\eta_S + \eta_\Omega}{\eta_A + 2\eta_S + \eta_\Omega}.$$

If the twist/holonomy channel is removed, a reduced-branch coefficient such as 2/3 becomes a control branch rather than a contradiction of the full optical-algebra result.

D2's main purpose is not merely to derive 3/4, but to prove its uniqueness under isotropic optical MaxEnt closure.

The answer proposed here is:

The G1 dark sector is not an additional density component. It is a traceless optical stress of finite-screen incompleteness.

This statement has three parts.

1. **Optical access.** A finite observer receives spacetime residuals through a null optical screen. The local screen deformation lives in $\mathfrak{gl}(2, \mathbb{R})$.

2. **MaxEnt projection.** The lowest-order unbiased finite-screen residual distributes over one area direction and three unimodular directions. This gives the 3/4 projection weight.
3. **Screen stress.** The compensating unimodular residual is traceless; through the Sachs optical decomposition it is Weyl-active but Ricci-silent. Thus Σ changes while μ is first-order protected.

Let

$$\lambda = 3 - s \tag{1}$$

be the background filling defect relative to ordinary three-volume scaling. Let $X_b(a)$ be the normalized horizon-memory response kernel. The leading G1 optical-stress closure is

$$\mu(a) = 1 + \mathcal{O}(\lambda^2), \tag{2}$$

$$\Sigma(a) = 1 - \kappa_{BW} \lambda X_b(a) + \mathcal{O}(\lambda^2). \tag{3}$$

In the isotropic lowest-order optical response,

$$\kappa_{BW} = \frac{3}{4}, \tag{4}$$

so that

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s). \tag{5}$$

What this paper does not claim. The result is not claimed to be a theorem of pure algebra alone. The algebra gives the available optical sectors. The coefficient 3/4 follows only when the leading finite-screen residual is locally maximum-entropy or isotropic on the optical deformation space. If the screen microstatistics weight area, shear, and twist differently, the coefficient shifts according to a generalized compliance-weighted formula derived below.

Table 1: Minimal map from finite-screen optical algebra to the G1DE locked response.

Object	Decomposition	G1 interpretation
$GL(2, \mathbb{R})$	$4 = 1 + 2 + 1$	finite optical screen deformation
Area sector	1 trace/determinant mode	background filling channel
Shear sector	2 symmetric traceless modes	scalar/E-mode optical shape
Twist sector	1 antisymmetric holonomy mode	orientation/curl capacity
Unimodular sector	$3 = 2 + 1$	Weyl/shape screen-stress channel
Filling defect	$\lambda = 3 - s$	effective screen-volume anomaly
G1DE lock	$\Sigma_0 = -\kappa_{BW} \lambda$	optical compensation
Isotropic branch	$\kappa_{BW} = 3/4$	full $GL(2)$ MaxEnt projection
Scalar-shear branch	$\kappa_{BW} = 2/3$	twist/holonomy demotion
Matter growth	$\mu_0 = 0$	first-order Ricci-density silence

2 Claim-status discipline

The projection law is a bridge result. It combines standard optical geometry, a finite-screen interpretation, and a lowest-order statistical assumption about residuals. Its claims are graded as follows.

Table 2: Claim status of the optical-stress closure.

Claim	Meaning	Status
Finite optical access	observable residuals are accessed through a two-dimensional null optical screen	bridge principle
Jacobi map	screen deformation is locally described by $J \in GL(2, \mathbb{R})$ away from caustics	standard optical geometry
$\mathfrak{gl}(2)$ split	$\mathfrak{gl}(2) = A(1) \oplus S(2) \oplus \Omega(1)$	algebraic theorem
MaxEnt residual	lowest-order residual covariance is isotropic in $T_t GL(2)$ absent further information	statistical bridge assumption
3/4 coefficient	full unimodular sector carries three of four local optical directions	conditional projection law
Weighted coefficient	$\kappa_{BW} = (2\eta_S + \eta_\Omega)/(\eta_A + 2\eta_S + \eta_\Omega)$	generalized theorem under quadratic response
Unimodular stress	shape residual is traceless on the screen	variational consequence
Ricci silence	residual does not act as first-order local density source	matter-protection bridge
Weyl activity	residual modifies lensing/optical channel	Sachs-decomposition bridge
$X_b(a)$	retarded horizon-memory response	dynamical bridge, not yet fully derived

The strongest claim in this paper is therefore not that “geometry forces 3/4” without assumptions. The stronger and safer claim is:

$$\boxed{\text{finite optical screen algebra} + \text{MaxEnt lowest-order residual} \Rightarrow \kappa_{BW} = 3/4.} \quad (6)$$

3 Finite optical screens and Sachs geometry

Consider a bundle of null geodesics with tangent k^μ reaching an observer. Let e_A^μ , $A = 1, 2$, be a screen basis orthogonal to k^μ and to the observer direction. The Jacobi map

$$J^A_B \quad (7)$$

relates an infinitesimal source-plane separation to an observed screen separation. On a non-caustic patch,

$$J \in GL(2, \mathbb{R}). \quad (8)$$

Definition 1 (Optical deformation generator). *The local optical deformation generator is*

$$X = \delta J J^{-1} \in \mathfrak{gl}(2, \mathbb{R}). \quad (9)$$

For continuous affine evolution one similarly defines the optical deformation-rate matrix

$$B^A_B = \frac{dJ^A_C}{d\lambda} (J^{-1})^C_B. \quad (10)$$

The matrix B decomposes into the Sachs optical scalars:

$$B^A_B = \frac{1}{2}\theta\delta^A_B + \sigma^A_B + \omega\epsilon^A_B, \quad (11)$$

where

$$\theta = \text{tr} B \quad (12)$$

is expansion, σ^A_B is symmetric and traceless, and $\omega\epsilon^A_B$ is the antisymmetric twist/rotation component. Thus the local optical deformation algebra has the sector count

$$\mathfrak{gl}(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1). \quad (13)$$

Equivalently,

$$\mathfrak{gl}(2, \mathbb{R}) = \mathbb{R}I \oplus \mathfrak{sl}(2, \mathbb{R}), \quad \dim \mathbb{R}I = 1, \quad \dim \mathfrak{sl}(2) = 3. \quad (14)$$

The optical tidal equation is schematically

$$\dot{B} + B^2 = \mathcal{R}, \quad (15)$$

where

$$\mathcal{R}_{AB} = -R_{\mu\nu\rho\sigma}e_A^\mu k^\nu k^\rho e_B^\sigma. \quad (16)$$

Using the four-dimensional curvature decomposition, the screen tidal matrix splits as

$$\mathcal{R}_{AB} = -\frac{1}{2}R_{\mu\nu}k^\mu k^\nu \gamma_{AB} - C_{\mu\nu\rho\sigma}e_A^\mu k^\nu k^\rho e_B^\sigma. \quad (17)$$

The first term is trace/Ricci focusing. The second term is traceless/Weyl tidal distortion. This is the geometric reason G1 can be Ricci-silent and Weyl-active at first order.

4 Projectors on the optical deformation algebra

Use the Frobenius inner product on $\mathfrak{gl}(2)$:

$$\langle X, Y \rangle = \text{Tr}(X^T Y). \quad (18)$$

Define the normalized area generator

$$e_A = \frac{1}{\sqrt{2}}I. \quad (19)$$

Then any $X \in \mathfrak{gl}(2)$ decomposes uniquely as

$$X = x_A e_A + X_U, \quad X_U \in \mathfrak{sl}(2, \mathbb{R}), \quad (20)$$

with

$$x_A = \langle X, e_A \rangle = \frac{1}{\sqrt{2}}\text{tr}X. \quad (21)$$

Definition 2 (Area and unimodular projectors). *For $X \in \mathfrak{gl}(2)$ define*

$$P_A X = \frac{1}{2}(\text{tr}X)I, \quad P_U X = X - \frac{1}{2}(\text{tr}X)I. \quad (22)$$

Then P_A projects to the determinant/area sector and P_U projects to the unimodular sector.

To separate shear from twist, use

$$P_S X = \frac{1}{2}(X + X^T) - \frac{1}{2}(\text{tr}X)I, \quad P_\Omega X = \frac{1}{2}(X - X^T). \quad (23)$$

Then

$$P_U = P_S + P_\Omega, \quad (24)$$

with projector traces

$$\text{Tr}P_A = 1, \quad \text{Tr}P_S = 2, \quad \text{Tr}P_\Omega = 1, \quad \text{Tr}I_{\mathfrak{gl}(2)} = 4. \quad (25)$$

Proposition 1 (Optical capacity count). *The full local optical deformation space has four directions. The unimodular shape sector has three directions. Hence the unweighted unimodular projector trace ratio is*

$$\frac{\text{Tr}P_U}{\text{Tr}I_{\mathfrak{gl}(2)}} = \frac{3}{4}. \quad (26)$$

Remark 1. *Equation (26) is only a capacity count. It becomes a physical coefficient only after a response metric or residual covariance is specified. The next section supplies the minimal maximum-entropy closure.*

Table 3: Optical-sector counting and the G1DE locked response.

Sector	Dimension	G1 role
A	1	area / determinant / background filling
S	2	symmetric shear / lensing shape
Ω	1	twist / optical holonomy capacity
$S + \Omega$	3	unimodular Weyl-active sector
Total	4	local optical deformation space

$$\lambda = 3 - s, \quad \mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}\lambda.$$

5 The MaxEnt optical residual lemma

One should not speak of a uniform probability distribution on the non-compact group $GL(2, \mathbb{R})$. The relevant finite-screen statement is local: near a given optical screen state, the residual lives in the tangent space $T_I GL(2) \simeq \mathfrak{gl}(2)$.

Definition 3 (Local optical residual). *A leading finite-screen optical residual is a random variable*

$$X \in \mathfrak{gl}(2, \mathbb{R}) \quad (27)$$

with

$$\mathbb{E}[X] = 0, \quad \mathbb{E} \|X\|^2 = \sigma_X^2. \quad (28)$$

Lemma 1 (MaxEnt optical residual). *Among distributions on the finite-dimensional vector space $\mathfrak{gl}(2)$ with fixed mean and fixed covariance trace, the maximum-entropy distribution with no preferred direction is the centered isotropic Gaussian. Its covariance is proportional to the identity on $\mathfrak{gl}(2)$:*

$$\mathbb{E}[X \otimes X] = \frac{\sigma_X^2}{4} I_{\mathfrak{gl}(2)}. \quad (29)$$

Consequently,

$$\frac{\mathbb{E} \|P_U X\|^2}{\mathbb{E} \|X\|^2} = \frac{\text{Tr} P_U}{\text{Tr} I_{\mathfrak{gl}(2)}} = \frac{3}{4}. \quad (30)$$

Proof. The maximum-entropy distribution on $\mathbb{R}^n$ with fixed mean and covariance is Gaussian. If no lower-order information distinguishes directions, the covariance is isotropic. For an orthogonal projector P and isotropic covariance $\sigma_X^2 I/n$,

$$\mathbb{E} \|PX\|^2 = \frac{\sigma_X^2}{n} \text{Tr} P. \quad (31)$$

Here $n = 4$ and $P = P_U$, so $\text{Tr} P_U = 3$ gives (30). $\square$

This lemma is the precise mathematical form of the phrase “unbiased finite-screen projection.” It is a lowest-order MaxEnt closure, not an unconditional geometric necessity.

6 Weighted response and the 3/4 fixed point

Finite screen microstatistics may not weight all optical directions equally. Let $\eta_A, \eta_S, \eta_\Omega > 0$ denote response weights (compliances, not stiffness penalties) in the area, symmetric-shear, and twist/holonomy sectors. Define the quadratic response cost

$$\mathcal{C}[X] = \frac{1}{2} \eta_A \|P_A X\|^2 + \frac{1}{2} \eta_S \|P_S X\|^2 + \frac{1}{2} \eta_\Omega \|P_\Omega X\|^2. \quad (32)$$

Theorem 1 (3/4 Projection Uniqueness (Theorem D2.1)). *For a finite-screen residual with sector weights $(\eta_A, \eta_S, \eta_\Omega)$, the normalized unimodular optical response coefficient is*

$$\kappa_{BW} = \frac{2\eta_S + \eta_\Omega}{\eta_A + 2\eta_S + \eta_\Omega}. \quad (33)$$

In the isotropic lowest-order case $\eta_A = \eta_S = \eta_\Omega$, this reduces to

$$\kappa_{BW} = \frac{3}{4}. \quad (34)$$

If the twist/holonomy direction is removed or has zero response weight, while $\eta_A = \eta_S$, then

$$\kappa_{BW} = \frac{2}{3}. \quad (35)$$

Proof. The weighted capacity of each sector is its dimension times its response weight. The area sector has dimension 1, the symmetric-shear sector dimension 2, and the twist sector dimension 1. The unimodular sector is shear plus twist, so its weighted share is $(2\eta_S + \eta_\Omega)/(\eta_A + 2\eta_S + \eta_\Omega)$. $\square$

D10 uses D2's projection theorem as one input into the minimal Weyl residual uniqueness theorem.

Caveat 1 (What the coefficient measures). *The number 3/4 is not merely a decorative ratio. It is a test of whether the finite screen retains the full local optical deformation algebra with isotropic lowest-order response. A measured coefficient near 2/3 would not prove that finite-screen gravity is false; it would demote the full $GL(2)$ branch to a scalar-shear or symmetric-screen branch.*

Caveat 2 (M_κ interpretation). *M_κ is not merely a free empirical competitor. It is the anisotropic-compliance relaxation of $M_{3/4}$: the weighted coefficient $\kappa_{BW} = (2\eta_S + \eta_\Omega)/(\eta_A + 2\eta_S + \eta_\Omega)$ reduces to 3/4 at the isotropic fixed point. A measured $\kappa_{BW} \neq 3/4$ would demote the full $GL(2)$ branch without automatically falsifying the finite-screen response program.*

7 v1.2 Modular Susceptibility Target for the 3/4 Law

Claim type: Conditional target The stronger modular-susceptibility derivation target for the 3/4 coefficient is:

$$\kappa_{BW} = \frac{\chi_S + \chi_\Omega}{\chi_A + \chi_S + \chi_\Omega} \stackrel{\text{isotropic}}{=} \frac{3}{4}$$

where:

- χ_A is the area response susceptibility
- χ_S is the shear response susceptibility
- χ_Ω is the twist/holonomy response susceptibility

Boundary note: The existing D2 MaxEnt projection lemma remains the formal result. The modular-susceptibility formulation is a stronger target for future derivation from first-principle screen modular fluctuation spectra, not a completed theorem.

8 Constrained screen closure

The 3/4 law can be written as a constrained compensation problem. Let

$$\lambda = 3 - s \quad (36)$$

be the background filling defect. The determinant channel carries the area/filling anomaly

$$\Delta_A = \lambda X_b(a), \quad (37)$$

where $X_b(a)$ is a normalized response kernel.

Finite screen closure says that this determinant defect cannot remain as an unbalanced scalar optical ledger entry. It must be compensated by a unimodular optical response:

$$\Delta_U = -\kappa_{BW} \lambda X_b(a). \quad (38)$$

The sign is fixed by closure: the shape response is a compensation of the filling defect.

Principle 1 (Finite screen closure). *At leading order, the optical ledger residual decomposes into determinant and unimodular components,*

$$\Delta_{\text{scr}} = \Delta_A + \Delta_U + \mathcal{O}(\lambda^2), \quad (39)$$

with the compensating relation (38). The amplitude of the compensation is determined by the optical response coefficient (33).

The isotropic branch gives

$$\Delta_U = -\frac{3}{4} \lambda X_b(a). \quad (40)$$

This is the algebraic-statistical origin of the G1DE-M_{3/4} amplitude lock.

9 Unimodular screen stress

The next step is to connect the unimodular optical response to physical channels. Let γ_{AB} be the screen metric and define the unimodular screen metric

$$\hat{\gamma}_{AB} = (\det \gamma)^{-1/2} \gamma_{AB}. \quad (41)$$

By construction,

$$\det \hat{\gamma} = 1. \quad (42)$$

A variation of $\hat{\gamma}$ is traceless with respect to γ :

$$\gamma^{AB} \delta \hat{\gamma}_{AB} = 0. \quad (43)$$

Suppose the shape part of the finite-screen residual action is a functional only of the unimodular metric,

$$S_{\text{scr}}^{\text{shape}} = S_{\text{scr}}^{\text{shape}}[\hat{\gamma}]. \quad (44)$$

Define the screen stress

$$T_{AB}^{\text{scr}} = -\frac{2}{\sqrt{\gamma}} \frac{\delta S_{\text{scr}}^{\text{shape}}}{\delta \gamma^{AB}}. \quad (45)$$

Lemma 2 (Unimodular screen-stress lemma). *If $S_{\text{scr}}^{\text{shape}}$ depends only on the unimodular screen metric $\hat{\gamma}_{AB}$, then the induced screen stress is traceless:*

$$\boxed{\gamma^{AB} T_{AB}^{\text{scr}} = 0.} \quad (46)$$

Sketch. A conformal variation $\delta\gamma_{AB} = 2\varphi\gamma_{AB}$ implies $\delta\gamma^{AB} = -2\varphi\gamma^{AB}$. It changes the determinant but leaves $\hat{\gamma}_{AB}$ unchanged to first order. Since $S_{\text{scr}}^{\text{shape}}$ depends only on $\hat{\gamma}$, its variation under a pure trace variation vanishes. Therefore

$$0 = \delta S_{\text{scr}}^{\text{shape}} = \int d^2y \frac{\delta S_{\text{scr}}^{\text{shape}}}{\delta\gamma^{AB}} \delta\gamma^{AB} = -\frac{1}{2} \int d^2y \sqrt{\gamma} T_{AB}^{\text{scr}} (-2\varphi\gamma^{AB}) = \int d^2y \sqrt{\gamma} \varphi \gamma^{AB} T_{AB}^{\text{scr}}. \quad (47)$$

For arbitrary φ , the trace must vanish. $\square$

Thus the compensating residual is not a scalar density on the screen. It is a traceless optical stress.

10 Ricci-silent, Weyl-active residuals

The Sachs decomposition (17) shows that trace optical focusing is Ricci-like, while traceless optical tidal distortion is Weyl-like. A traceless screen stress therefore has the correct tensor character to modify the Weyl/lensing sector without introducing a leading local Ricci-density source.

Definition 4 (Ricci-silent residual). *A finite-screen residual $\mathcal{E}_{\mu\nu}^{\text{scr}}$ is Ricci-silent at first order if its density/focusing component vanishes to first order:*

$$u^\mu u^\nu \mathcal{E}_{\mu\nu}^{\text{scr}} = \mathcal{O}(\lambda^2), \quad (48)$$

while its projected traceless spatial or optical component may be first order.

Definition 5 (Weyl-active residual). *A residual is Weyl-active if it produces a nonzero traceless optical tidal correction,*

$$\mathcal{E}_{AB}^{\text{scr}} = \mathcal{E}_{\langle AB \rangle}^{\text{scr}} + \mathcal{O}(\lambda^2), \quad \gamma^{AB} \mathcal{E}_{AB}^{\text{scr}} = 0, \quad (49)$$

with amplitude proportional to the unimodular compensation (38).

The effective residual in a scalar cosmological reduction can be represented schematically as

$$\delta T_{00}^{\text{scr}} = \mathcal{O}(\lambda^2), \quad \delta T_{0i}^{\text{scr}} = \mathcal{O}(\lambda^2), \quad (50)$$

$$\delta T_{ij}^{\text{scr}} = \left(\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij} \right) \Pi_{\text{scr}} + \mathcal{O}(\lambda^2). \quad (51)$$

This is not an ordinary dark matter density. It is an anisotropic screen-stress residual.

11 Cosmological reduction: μ and Σ

In phenomenological modified-growth notation,

$$-k^2 \Psi = 4\pi G a^2 \mu(a) \rho \Delta, \quad (52)$$

$$-k^2 (\Phi + \Psi) = 8\pi G a^2 \Sigma(a) \rho \Delta. \quad (53)$$

The Newtonian channel μ is sensitive to local density/focusing. The lensing channel Σ is sensitive to the Weyl/optical combination.

Lemma 3 (Matter-channel protection). *If ordinary matter is locally conserved and the finite-screen residual is Ricci-silent to first order, then*

$$\boxed{\mu(a) = 1 + \mathcal{O}(\lambda^2)}. \quad (54)$$

Sketch. The leading Poisson equation is sourced by the matter density contrast and by any additional first-order density-like residual. Ricci silence removes the latter at order λ . Local conservation of ordinary matter keeps the standard growth source intact. Hence the first-order coefficient in μ vanishes. $\square$

Theorem 2 (MaxEnt optical-stress closure). *Let a finite causal screen have a background filling defect $\lambda = 3 - s$ carried by the determinant channel. Let the compensating residual close through the unimodular optical sector with weighted coefficient (33). Let the resulting residual be Ricci-silent and Weyl-active at first order. Then*

$$\boxed{\mu(a) = 1 + \mathcal{O}(\lambda^2),} \quad (55)$$

$$\boxed{\Sigma(a) = 1 - \kappa_{BW} \lambda X_b(a) + \mathcal{O}(\lambda^2),} \quad (56)$$

where

$$\boxed{\kappa_{BW} = \frac{2\eta_S + \eta_\Omega}{\eta_A + 2\eta_S + \eta_\Omega}.} \quad (57)$$

In the isotropic lowest-order branch,

$$\eta_A = \eta_S = \eta_\Omega, \quad (58)$$

so

$$\boxed{\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s).} \quad (59)$$

Proof. The determinant channel carries the filling defect $\lambda X_b(a)$. Finite screen closure assigns a compensating unimodular optical response of amplitude $-\kappa_{BW} \lambda X_b(a)$. The coefficient is given by the weighted optical-sector share. Since the compensating residual is unimodular, it is traceless on the screen. By the Sachs decomposition, a traceless optical residual is Weyl-active rather than Ricci-density active. Therefore it corrects Σ at first order and leaves μ unchanged at first order. $\square$

Equation (59) is the desired mathematical closure of the G1DE-M_{3/4} amplitude relation.

12 The horizon-memory kernel $X_b(a)$

The projection coefficient fixes the amplitude ratio, not the full redshift kernel. The shape function $X_b(a)$ should be interpreted as a finite-screen memory response, not as an arbitrary fitting polynomial.

Let

$$N = \ln a, \quad \epsilon_H(N) = -\frac{d \ln H}{dN}. \quad (60)$$

Acceleration corresponds to $\epsilon_H < 1$. Since an event-horizon-like future boundary becomes dynamically relevant only in accelerated expansion, a minimal source for horizon-screen memory is

$$\mathcal{S}_H(N) = [1 - \epsilon_H(N)]_+, \quad (61)$$

where $[x]_+ = \max(x, 0)$.

Definition 6 (Horizon-memory closure). *A first-order finite-screen memory kernel satisfies*

$$\frac{dX_b}{dN} + \Gamma_H(N) X_b(N) = \mathcal{S}_H(N), \quad (62)$$

with formal solution

$$X_b(N) = \int_{-\infty}^N dN' \exp \left[- \int_{N'}^N \Gamma_H(u) du \right] \mathcal{S}_H(N'). \quad (63)$$

Principle 2 (Minimal Markov horizon-memory candidate). *The simplest acceleration-gated closure is*

$$\boxed{X'_b(N) + X_b(N) = [1 - \epsilon_H(N)]_+}. \quad (64)$$

It is dormant in matter and radiation eras, activates when the future horizon becomes dynamically relevant, and approaches $X_b \rightarrow 1$ in the de Sitter limit.

Caveat 3. *The present paper derives the leading amplitude coefficient κ_{BW} , not the full redshift kernel $X_b(a)$. The latter is treated as a horizon-memory response whose source and relaxation terms remain part of the next dynamical closure problem. Equation (64) is a candidate closure, not yet a theorem. The main result of this paper is the amplitude-level optical-stress lock. Deriving $\Gamma_H(N)$ and $\mathcal{S}_H(N)$ from finite-screen entropy dynamics remains an open problem.*

13 Effective filling dimension and the meaning of s

The parameter s should not be treated as a mysterious fitting exponent. It is naturally interpreted as an effective filling dimension of the finite-screen cosmological ledger.

Let $\mathcal{V}_{\text{scr}}(a)$ denote screen-accessible bulk distinction volume. Define

$$s_{\text{eff}}(a) = \frac{d \ln \mathcal{V}_{\text{scr}}}{d \ln a}. \quad (65)$$

Standard three-volume filling corresponds to $s_{\text{eff}} = 3$. The finite-screen filling defect is

$$\lambda(a) = 3 - s_{\text{eff}}(a). \quad (66)$$

In the current G1DE reduction, s should be read as a low-redshift effective filling exponent, not as a completed dynamical dimension function. A full theory would replace the constant s by $s_{\text{eff}}(a)$ derived from finite-screen occupancy dynamics.

Equivalently, define a horizon capacity

$$C_H(a) = \frac{A_H(a)}{4\ell_P^2} \quad (67)$$

and an occupancy

$$\mathcal{O}_H(a) = \frac{C_{\text{used}}(a)}{C_H(a)}. \quad (68)$$

Then a candidate relation is

$$3 - s_{\text{eff}}(a) \sim -\frac{d \ln \mathcal{O}_H}{d \ln a}. \quad (69)$$

This equation is not used as an input in the amplitude theorem, but it indicates the direction of a deeper dynamical closure.

14 Why E_G is the natural discriminator

The statistic E_G -type observables compare lensing to growth. G1 predicts exactly this split:

$$\Sigma(a) - 1 = \mathcal{O}(\lambda), \quad \mu(a) - 1 = \mathcal{O}(\lambda^2). \quad (70)$$

Therefore the natural target is not growth alone and not distance alone, but the cross-channel ratio:

$$\frac{\text{lensing}}{\text{growth}}. \quad (71)$$

A model that adds ordinary density generally changes both growth and lensing. A generic modified-gravity parameterization allows both μ and Σ to float. G1 predicts a constrained channel split:

$$\boxed{\text{Weyl/lensing first, matter growth protected.}} \quad (72)$$

Table 4: Residual fingerprints in common alternatives.

Theory type	Growth channel	Lensing channel	Cross-channel structure
Extra matter density	$\mu \neq 1$ expected	$\Sigma \neq 1$ expected	density-driven
Generic modified gravity	free μ	free Σ	weak unless imposed
Smooth dark energy $w(a)$	mainly background	indirect lensing effects	no fixed lock
G1 finite-screen optical stress	$\mu \simeq 1$	$\Sigma < 1$ for $s < 3$	$\Sigma_0 = -\kappa_{BW}(3 - s)$

15 Demotion paths and falsification tests

The theory is deliberately structured with demotion paths.

Table 5: Demotion and falsification registry.

Condition	What fails	What may survive
Data require dominant first-order μ correction	Ricci-silent screen-stress interpretation	finite-screen residual with density component
Coefficient near 2/3	full $GL(2)$ optical capacity	scalar-shear/SPD(2) demotion
Coefficient far from 2/3 and 3/4	isotropic MaxEnt projection	weighted response-weight law (33)
No common $X_b(a)$ for background and Weyl response	background–Weyl ledger lock	separate background and optical residuals
No lensing or E_G deviation while $s < 3$ persists	optical compensation branch	background-only filling anomaly
Strong curl/rotation mean at first order	scalar isotropy assumption	full holonomy branch with nonzero mean
No acceleration-gated behavior in X_b	minimal horizon-memory kernel	alternative screen-memory dynamics
Violation of covariance/conservation closure	effective stress representation	local optical algebra only
No finite-screen entropy response in G1 core	entire FSS/G1 bridge	none, except algebraic toy model

16 Open mathematical problems

Problem 1: derive the optical response weights. The coefficients $\eta_A, \eta_S, \eta_\Omega$ should ultimately be obtained from finite-screen microstatistics or a screen entropy Hessian. At present, isotropy is the minimal MaxEnt closure.

Problem 2: derive $X_b(a)$. The minimal equation $X'_b + X_b = [1 - \epsilon_H]_+$ is plausible but not final. A full derivation should obtain Γ_H and $\mathcal{S}_H$ from entropy production, screen capacity drift, and horizon occupancy.

Problem 3: covariant embedding. The effective residual $\mathcal{E}_{\mu\nu}^{\text{scr}}$ must be embedded in a conservation-closed covariant system satisfying

$$\nabla^\mu (G_{\mu\nu} - 8\pi G T_{\mu\nu} - \mathcal{E}_{\mu\nu}^{\text{scr}}) = 0. \quad (73)$$

Problem 4: nonlinear and small-scale limits. The present result is a linear optical-screen closure. It does not yet specify nonlinear clustering, baryonic feedback, strong lensing caustics, or black-hole interior dynamics.

Problem 5: distinguish capacity from signal. The twist/holonomy direction contributes to full optical capacity counting, but its mean signal may vanish in scalar cosmology. Future work must separate capacity covariance, parity constraints, and observable curl/B-mode signatures.

17 Conclusion

This paper strengthens the mathematical core of the G1 projection law. The 3/4 coefficient is not treated as a numerical coincidence. It is derived as the maximum-entropy projection of a finite-screen optical residual onto the unimodular sector of the null optical screen deformation algebra.

The closure chain is

$$X \in T_I GL(2) \rightarrow A(1) \oplus S(2) \oplus \Omega(1) \rightarrow \kappa_{BW} = \frac{2\eta_S + \eta_\Omega}{\eta_A + 2\eta_S + \eta_\Omega} \rightarrow \frac{3}{4} \quad (74)$$

under isotropic lowest-order response. The compensating residual is unimodular and therefore traceless. Through the Sachs optical decomposition, a traceless optical residual is Ricci-silent and Weyl-active at first order. Hence

$$\mu(a) = 1 + \mathcal{O}((3-s)^2), \quad \Sigma(a) = 1 - \frac{3}{4}(3-s)X_b(a) + \mathcal{O}((3-s)^2). \quad (75)$$

This gives the G1DE-M_{3/4} lock

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3-s). \quad (76)$$

The conceptual result is sharper than a modified Poisson equation. G1 finite-screen gravity is best understood, at this level, as a Ricci-silent, Weyl-active optical-stress theory. The dark sector is the optical stress of finite-screen incompleteness.

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* 7, 2333 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* 43, 199 (1975).
- [3] A. Raychaudhuri, “Relativistic cosmology. I,” *Phys. Rev.* 98, 1123 (1955).
- [4] R. Sachs, “Gravitational waves in general relativity. VI. The outgoing radiation condition,” *Proc. R. Soc. Lond. A* 264, 309 (1961).
- [5] P. Schneider, J. Ehlers, and E. E. Falco, *Gravitational Lenses* (Springer, 1992).
- [6] M. Bartelmann and P. Schneider, “Weak gravitational lensing,” *Phys. Rep.* 340, 291 (2001).

- [7] R. M. Wald, *General Relativity* (University of Chicago Press, 1984).
- [8] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* 75, 1260 (1995).
- [9] J. M. Bardeen, “Gauge-invariant cosmological perturbations,” *Phys. Rev. D* 22, 1882 (1980).
- [10] C.-P. Ma and E. Bertschinger, “Cosmological perturbation theory in the synchronous and conformal Newtonian gauges,” *Astrophys. J.* 455, 7 (1995).
- [11] P. Zhang, M. Liguori, R. Bean, and S. Dodelson, “Probing gravity at cosmological scales by measurements which test the relationship between gravitational lensing and matter overdensity,” *Phys. Rev. Lett.* 99, 141302 (2007).
- [12] S. F. Daniel and E. V. Linder, “Constraining cosmic expansion and gravity with gravitational lensing,” *Phys. Rev. D* 82, 103523 (2010).
- [13] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *JCAP* 07, 050 (2014).
- [14] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), doi:10.5281/zenodo.20382013.
- [15] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual*, manuscript (2026).

Non-Equilibrium Screen Entropy and the Optical-Stress Origin of the G1 Dark Sector

A Covariant Closure for Ricci-Silent, Weyl-Active Finite-Screen Residuals
v1.2 Body-Integrated Release

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This paper develops a non-equilibrium entropy closure for the dark-sector residuals of Finite Distinction Systems – Gravity 1 (FDS-G1). The companion projection-law note derived the optical amplitude lock

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s)$$

by treating the leading finite-screen residual as a maximum-entropy deformation on the local null optical screen algebra

$$\mathfrak{gl}(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1),$$

where A is the area/determinant channel, S is symmetric shear, and Ω is twist/optical holonomy. The present paper asks a deeper question: what screen dynamics produces such an optical-stress residual?

The central claim is that the G1 dark sector is not an additional density component, but non-equilibrium finite-screen entropy production projected as traceless optical stress. A finite causal screen is assigned an entropy functional whose near-equilibrium second variation has optical-sector stiffnesses

$$K_A, \quad K_S, \quad K_\Omega$$

for expansion, shear, and twist. The corresponding compliances $\chi_i = K_i^{-1}$ determine the background-Weyl projection coefficient

$$\kappa_{BW} = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega},$$

with $\kappa_{BW} = 3/4$ as the isotropic entropy-Hessian fixed point and $2/3$ as the scalar-shear demotion when the twist/holonomy sector is removed. The non-equilibrium entropy production term is written in Sachs optical variables,

$$\delta S_{\text{prod}} = \frac{1}{2} \int d\lambda dA \left(K_A \theta^2 + K_S \sigma_{AB} \sigma^{AB} + K_\Omega \omega^2 \right),$$

so that the compensating residual lives in the unimodular screen sector. By construction this residual is traceless on the screen; through the Sachs curvature split it couples to the Weyl optical tidal channel rather than to the Ricci density-focusing channel.

The resulting covariant effective-stress closure predicts

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2),$$

while the matter-growth channel is protected up to second order and horizon-scale leakage,

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

Here $\mathcal{R}_H(a)$ is the horizon-screen residual response. Rather than treating the redshift kernel as a free shape, the paper defines it as a retarded memory of horizon occupancy drift,

$$\mathcal{R}'_H(N) + \Gamma_H(N) \mathcal{R}_H(N) = \mathcal{S}_H(N), \quad N = \ln a,$$

with a minimal acceleration-gated approximation $\mathcal{S}_H \simeq [1 - \epsilon_H]_+$. The constant-amplitude G1DE-M_{3/4} reduction is recovered when $\mathcal{R}_H(a) \simeq (3 - s)X_b(a)$ over the low-redshift window. The construction is conditional and falsifiable: a dominant first-order μ correction, failure of horizon-memory structure, a coefficient outside the optical-compliance family, or evidence for the 2/3 branch would demote the full non-equilibrium optical-stress closure.

1 Scope and executive summary

FDS-G1 treats gravitation as an entropy-response geometry of finite causal screens. The G1 core program identifies local gravitational stiffness with the entropy response of a finite screen, while G1DE reductions express the leading late-time residual through background and Weyl/lensing channels. The earlier optical projection law established a conditional amplitude relation

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s), \quad (1)$$

by modeling the finite-screen residual as a maximum-entropy optical deformation on the two-dimensional null screen. The present paper supplies the next layer of closure: it interprets the residual as a non-equilibrium entropy-production term and embeds it into a covariant effective-stress hierarchy.

D3 owns the thermodynamic admissibility of screen residuals. Not every residual is heat; only physically implemented irreversible record reuse, repair, synchronization, or externalization enters the entropy/resource ledger.

Open lemma target – Passive Residual Sign Lemma. Under detailed balance, a passive horizon remnant, and nonnegative entropy production, the Weyl-normalized residual satisfies $\Sigma - 1 \leq 0$ in the D7 realization class. A rigorous proof is an open target for future work.

The central thesis is:

The G1 dark sector is not an additional density component. It is finite-screen non-equilibrium entropy production projected as Ricci-silent, Weyl-active traceless optical stress.

The effective screen stress should be read as a controlled hierarchy, not as a standalone conserved fluid with only anisotropic stress at all scales. On subhorizon scales,

$$E_{ij, \text{traceless}}^{\text{scr}} = O(\mathcal{R}_H), \quad E_{0i}^{\text{scr}} = O(\mathcal{R}_H \mathcal{H}/k), \quad E_{00}^{\text{scr}} = O(\mathcal{R}_H^2) + O(\mathcal{R}_H \mathcal{H}^2/k^2). \quad (2)$$

Thus $\mu \simeq 1$ is a subhorizon first-order protection, not an absolute zero-source statement.

This statement has five components.

1. **Optical screen access.** Finite observers access spacetime residuals through a two-dimensional null optical screen. Local screen deformations live in $T_I GL(2, \mathbb{R}) \simeq \mathfrak{gl}(2, \mathbb{R})$.
2. **Entropy Hessian.** Near local screen equilibrium, non-equilibrium entropy production has a quadratic optical-sector form in expansion, shear, and twist.
3. **Compliance projection.** The response coefficient is a compliance-weighted fraction of the unimodular sector,

$$\kappa_{BW} = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega}. \quad (3)$$

The isotropic entropy-Hessian branch gives $\kappa_{BW} = 3/4$.

4. **Ricci silence.** The compensating residual is unimodular and therefore traceless on the optical screen. Through the Sachs split it enters the Weyl optical tidal channel before it enters the Ricci density-focusing channel.
5. **Horizon memory.** The redshift dependence is not an arbitrary polynomial. It is a retarded response of finite horizon-screen occupancy drift.

Let

$$\mathcal{R}_H(a) = (3 - s_{\text{eff}}(a))X_b(a) \quad (4)$$

be the combined horizon residual response. The leading cosmological prediction is

$$\Sigma(a, k) - 1 = -\kappa_{BW}\mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad (5)$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (6)$$

Thus the G1 signature is not generic modified gravity. It is a Weyl-first, matter-growth-protected channel split.

2 Claim-status discipline

The present paper is a mathematical bridge, not a final fundamental derivation. Its claims are graded in Table 1.

The strongest safe claim is not that 3/4 follows from algebra alone. The stronger claim is:

$$\text{finite optical algebra} + \text{isotropic non-equilibrium screen compliance} \implies \kappa_{BW} = \frac{3}{4}. \quad (7)$$

If the optical sector compliances are anisotropic, the coefficient shifts. If the twist/holonomy sector is demoted, the full $GL(2)$ branch demotes to a scalar-shear 2/3 branch.

3 Finite causal screens and optical kinematics

Consider a null geodesic congruence with tangent k^μ reaching a finite observer. Let e_A^μ , $A = 1, 2$, be a screen basis orthogonal to k^μ and to the observer direction. The Jacobi map

$$J^A_B \quad (8)$$

relates source-plane separations to observed screen separations. Away from caustics,

$$J \in GL(2, \mathbb{R}). \quad (9)$$

Table 1: Claim status of the non-equilibrium optical-stress closure.

Claim	Meaning	Status
Finite optical access	Observable residuals are routed through two-dimensional null optical screens	bridge principle
Sachs decomposition	Optical deformation splits into expansion, shear, and twist	standard optical geometry
Entropy Hessian	Screen entropy has a quadratic near-equilibrium response in optical sectors	screen-thermodynamic bridge
Compliance projection	Sector compliances determine κ_{BW}	conditional theorem
Isotropic 3/4 branch	$K_A = K_S = K_\Omega$ implies $\kappa_{BW} = 3/4$	MaxEnt / isotropic fixed point
Unimodular stress	Shape residual is traceless on the screen	variational consequence
Ricci silence	No leading density-like source in subhorizon growth	covariant bridge with leakage terms
Horizon memory	Redshift response is retarded occupancy drift	dynamical bridge, not final theorem
G1DE-M _{3/4} reduction	$\mu_0 = 0, \Sigma_0 = -3(3 - s)/4$	low-redshift amplitude reduction

Definition 1 (Optical deformation generator). *The local residual deformation of a finite optical screen is*

$$X = \delta J J^{-1} \in \mathfrak{gl}(2, \mathbb{R}). \quad (10)$$

For continuous affine evolution, define

$$B^A{}_B = \frac{dJ^A{}_C}{d\lambda} (J^{-1})^C{}_B. \quad (11)$$

The optical deformation-rate matrix decomposes into Sachs scalars:

$$B^A{}_B = \frac{1}{2} \theta \delta^A{}_B + \sigma^A{}_B + \omega \epsilon^A{}_B, \quad (12)$$

where θ is expansion, $\sigma^A{}_B$ is symmetric traceless shear, and $\omega \epsilon^A{}_B$ is twist/rotation. Thus

$$\mathfrak{gl}(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1), \quad (13)$$

with

$$A \oplus S \oplus \Omega = \mathbb{R}I \oplus \mathfrak{sl}(2, \mathbb{R}). \quad (14)$$

The area/determinant sector has dimension 1; the unimodular sector has dimension 3.

The Sachs optical tidal equation has the schematic form

$$\dot{B} + B^2 = \mathcal{R}, \quad (15)$$

where

$$\mathcal{R}_{AB} = -R_{\mu\nu\rho\sigma} e_A^\mu k^\nu k^\rho e_B^\sigma. \quad (16)$$

Using the four-dimensional curvature decomposition,

$$\mathcal{R}_{AB} = -\frac{1}{2}R_{\mu\nu}k^\mu k^\nu \gamma_{AB} - C_{\mu\nu\rho\sigma}e_A^\mu k^\nu k^\rho e_B^\sigma. \quad (17)$$

The first term is trace/Ricci focusing. The second is traceless/Weyl tidal distortion. This split is the geometric basis of the Ricci-silent, Weyl-active G1 channel structure.

4 Equilibrium and non-equilibrium screen entropy

The equilibrium screen entropy in the G1 core is normalized by the horizon area law,

$$S_{\text{eq}} = \frac{A}{4\ell_P^2} + S_{\text{sub}}. \quad (18)$$

Equilibrium Clausius-type variations are expected to reproduce an Einstein/Jacobson-like metric closure in the local limit. The dark-sector residual arises when finite-screen closure is not exact. We therefore write a non-equilibrium entropy balance

$$\delta S_{\text{scr}} = \frac{\delta Q}{T} + \delta S_{\text{prod}}, \quad (19)$$

where δS_{prod} is the finite-screen entropy-production surplus.

The total screen entropy production decomposes into five channels:

$$\dot{S}_{\text{scr}} = \dot{S}_{\text{erase}} + \dot{S}_{\text{memory}} + \dot{S}_{\text{sync}} + \dot{S}_{\text{ext}} + \dot{S}_{\text{optical}}.$$

Only the optical channel may couple to $R_{\mu\nu}^{\text{scr}}$. The other channels represent internal finite-capacity costs that do not feed the geometric residual sector.

Principle 1 (Optical entropy production). *Near a locally isotropic screen state, the leading non-equilibrium entropy production may be expanded in the optical scalars,*

$$\delta S_{\text{prod}} = \frac{1}{2} \int d\lambda dA \left(K_A \theta^2 + K_S \sigma_{AB} \sigma^{AB} + K_\Omega \omega^2 \right) + O(B^3). \quad (20)$$

Here $K_A, K_S, K_\Omega > 0$ are screen entropy stiffnesses in the expansion, shear, and twist sectors.

The sign convention is chosen so that departures from equilibrium carry a positive cost. The corresponding response compliances are

$$\chi_A = K_A^{-1}, \quad \chi_S = K_S^{-1}, \quad \chi_\Omega = K_\Omega^{-1}. \quad (21)$$

The isotropic lowest-order entropy-Hessian branch is

$$K_A = K_S = K_\Omega, \quad (22)$$

which is equivalent to

$$\chi_A = \chi_S = \chi_\Omega. \quad (23)$$

5 MaxEnt and Onsager screen response

The projection-law note used a local maximum-entropy residual on $T_1GL(2)$. The present entropy-production formulation gives a dynamical interpretation of the same assumption.

Let X_i denote the four optical deformation coordinates corresponding to A, S_1, S_2, Ω . In a linear-response approximation,

$$\dot{X}_i = - \sum_j L_{ij} \frac{\partial \mathcal{F}_{\text{scr}}}{\partial X_j} + \xi_i, \quad (24)$$

where L_{ij} is an Onsager response matrix and ξ_i is screen noise. The minimum-information, no-preferred-direction fixed point is

$$L_{ij} = L\delta_{ij}, \quad \langle \xi_i(N) \xi_j(N') \rangle = 2L\delta_{ij}\delta(N - N'). \quad (25)$$

Equivalently, the local residual covariance is isotropic on $\mathfrak{gl}(2)$.

Lemma 1 (MaxEnt-compliance projection). *Let the leading finite-screen optical residual be locally maximum-entropy on $T_1GL(2)$ and let the sector response be governed by compliances $(\chi_A, \chi_S, \chi_\Omega)$. Then the normalized unimodular response coefficient is*

$$\kappa_{BW} = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega}. \quad (26)$$

If $\chi_A = \chi_S = \chi_\Omega$, then

$$\kappa_{BW} = \frac{2 + 1}{1 + 2 + 1} = \frac{3}{4}. \quad (27)$$

If the twist/holonomy sector is removed while $\chi_A = \chi_S$, then

$$\kappa_{BW} = \frac{2}{1 + 2} = \frac{2}{3}. \quad (28)$$

Proof. Each sector contributes its dimension multiplied by its response compliance. The area sector has dimension 1, symmetric shear has dimension 2, and twist has dimension 1. The Weyl-active unimodular sector is $S \oplus \Omega$, so its share is (26). $\square$

For small deviations from isotropy, write

$$\chi_S = \chi_A(1 + \alpha), \quad \chi_\Omega = \chi_A(1 + \beta). \quad (29)$$

Then

$$\kappa_{BW} = \frac{3 + 2\alpha + \beta}{4 + 2\alpha + \beta} = \frac{3}{4} + \frac{2\alpha + \beta}{16} + O(\alpha^2, \alpha\beta, \beta^2). \quad (30)$$

Thus a measured offset from 3/4 directly estimates the combination $2\alpha + \beta$ of shear/twist compliance anisotropy.

6 v1.2 Modular Phase-Space Interface

Claim type: Interface specification The non-equilibrium entropy ledger connects to the D0/D2 modular fluctuation framework via the causal diamond phase-space object:

$$(\Gamma_\Sigma, \Omega_\Sigma, H_\Sigma, K_\Sigma)$$

where:

- Γ_Σ is the space of causal diamond states
- Ω_Σ is the symplectic form on Γ_Σ
- H_Σ is the causal diamond modular Hamiltonian
- K_Σ is the expansion scalar operator

The response susceptibilities $\chi_A, \chi_S, \chi_\Omega$ are derived from the curvature of the relative-entropy functional on Γ_Σ .

The external finite causal-diamond phase-space result of Ciambelli, He, and Zurek (2026) motivates treating Γ_Σ and Ω_Σ as more than formal bookkeeping devices: finite causal diamonds can carry gravitational edge/soft phase-space degrees of freedom. In FDS-G1 this remains an interface specification, not a completed microscopic construction.

Boundary note: This interface connects the non-equilibrium thermodynamic results in D3 to the modular fluctuation formalisms in D0 and D2, but does not prove a unique relation between them.

7 Constrained screen closure

Let

$$\lambda(a) = 3 - s_{\text{eff}}(a) \quad (31)$$

be the finite-screen filling defect relative to ordinary three-volume scaling. The area/determinant channel carries the background filling anomaly. Screen closure requires a compensating unimodular response.

Principle 2 (Screen ledger closure). *At leading order, the finite-screen optical residual decomposes as*

$$\Delta_{\text{scr}} = \Delta_A + \Delta_U + O(\lambda^2), \quad (32)$$

with

$$\Delta_A = \lambda(a)X_b(a), \quad \Delta_U = -\kappa_{BW}\lambda(a)X_b(a). \quad (33)$$

The sign is fixed by compensation: the unimodular optical sector closes the determinant/filling defect.

It is useful to combine the amplitude defect and the memory response into a single horizon residual,

$$\mathcal{R}_H(a) = \lambda(a)X_b(a). \quad (34)$$

Then the closure relation is simply

$$\Delta_U = -\kappa_{BW}\mathcal{R}_H(a). \quad (35)$$

The low-redshift G1DE-M_{3/4} reduction corresponds to

$$\lambda(a) \simeq 3 - s = \text{constant}. \quad (36)$$

8 Unimodular optical stress

Let γ_{AB} be the screen metric and define the unimodular metric

$$\hat{\gamma}_{AB} = (\det \gamma)^{-1/2} \gamma_{AB}, \quad \det \hat{\gamma} = 1. \quad (37)$$

The shape part of the screen action depends only on $\hat{\gamma}$:

$$S_{\text{scr}}^{\text{shape}} = S_{\text{scr}}^{\text{shape}}[\hat{\gamma}]. \quad (38)$$

The induced screen stress is defined by inverse-metric variation,

$$T_{AB}^{\text{scr}} = -\frac{2}{\sqrt{\hat{\gamma}}} \frac{\delta S_{\text{scr}}^{\text{shape}}}{\delta \gamma^{AB}}. \quad (39)$$

Lemma 2 (Unimodular stress). *If $S_{\text{scr}}^{\text{shape}}$ depends only on the unimodular screen metric, then*

$$\gamma^{AB} T_{AB}^{\text{scr}} = 0. \quad (40)$$

Proof. Under a pure conformal variation $\delta \gamma^{AB} = -2\phi \gamma^{AB}$, the unimodular metric $\hat{\gamma}_{AB}$ is invariant to first order. Therefore $\delta S_{\text{scr}}^{\text{shape}} = 0$ for arbitrary ϕ . But

$$\delta S_{\text{scr}}^{\text{shape}} = -\frac{1}{2} \int d^2 y \sqrt{\gamma} T_{AB}^{\text{scr}} \delta \gamma^{AB} = \int d^2 y \sqrt{\gamma} \phi \gamma^{AB} T_{AB}^{\text{scr}}. \quad (41)$$

For arbitrary ϕ , the trace vanishes. $\square$

Thus the compensating residual is not a scalar density on the screen. It is traceless optical stress.

9 Ricci-silent, Weyl-active covariant residual

Introduce an effective residual tensor

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} + E_{\mu\nu}^{\text{scr}}. \quad (42)$$

G1 does not require $E_{\mu\nu}^{\text{scr}}$ to be a material fluid. It is an effective representation of finite-screen incompleteness in the metric closure.

Definition 2 (Ricci-silent residual). *A finite-screen residual is Ricci-silent at first order if its density-like focusing component is absent in the subhorizon/quasi-static channel, up to second-order and horizon-scale leakage:*

$$E_{00}^{\text{scr}} = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right), \quad (43)$$

$$E_{0i}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right). \quad (44)$$

Definition 3 (Weyl-active residual). *A finite-screen residual is Weyl-active if its projected traceless spatial/optical component is first order:*

$$E_{ij,\text{traceless}}^{\text{scr}} = O(\mathcal{R}_H). \quad (45)$$

Equivalently, on the optical screen,

$$E_{AB}^{\text{scr}} = E_{\langle AB \rangle}^{\text{scr}} + O(\mathcal{R}_H^2), \quad \gamma^{AB} E_{AB}^{\text{scr}} = 0. \quad (46)$$

The Sachs split routes trace optical focusing to Ricci curvature and traceless optical distortion to Weyl curvature. Therefore a unimodular screen residual is naturally Weyl-active while remaining Ricci-silent at first order.

Caveat 1 (Covariant closure). *A purely traceless spatial effective stress cannot generally be divergence-free by itself on all scales. The Ricci-silent statement is a controlled hierarchy, not an exact componentwise zero. A complete covariant embedding must satisfy either*

$$\nabla^\mu E_{\mu\nu}^{\text{scr}} = 0 \quad (47)$$

or a controlled horizon-exchange equation

$$\nabla^\mu E_{\mu\nu}^{\text{scr}} = J_\nu^H. \quad (48)$$

10 Cosmological reduction

Use the standard phenomenological functions

$$-k^2\Psi = 4\pi G a^2 \mu(a, k) \rho \Delta, \quad (49)$$

$$-k^2(\Phi + \Psi) = 8\pi G a^2 \Sigma(a, k) \rho \Delta. \quad (50)$$

The Newtonian channel is density/Ricci sensitive. The lensing channel is Weyl/optical sensitive.

Theorem 1 (Non-equilibrium optical-stress reduction). *Let a finite causal screen have non-equilibrium optical entropy production of the form (20). Let its horizon residual response be $\mathcal{R}_H(a)$ and its unimodular compensation coefficient be κ_{BW} as in (26). If the residual is Ricci-silent and Weyl-active in the sense above, then*

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad (51)$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (52)$$

For the isotropic branch, $\kappa_{BW} = 3/4$. For a low-redshift constant-amplitude reduction $\mathcal{R}_H(a) \simeq (3-s)X_b(a)$, this gives

$$\mu(a, k) = 1 + O((3-s)^2), \quad (53)$$

$$\Sigma(a, k) = 1 - \frac{3}{4}(3-s)X_b(a) + O((3-s)^2). \quad (54)$$

Proof. The determinant/area filling defect is compensated by the unimodular sector with amplitude $-\kappa_{BW}\mathcal{R}_H$. By the unimodular-stress lemma, the compensating residual is traceless on the screen. By the Sachs Ricci/Weyl decomposition, a traceless optical residual enters the Weyl/lensing channel at first order. Since the residual is Ricci-silent, the local density-focusing source in the Poisson channel receives no leading subhorizon contribution. Bianchi consistency allows only second-order or horizon-suppressed leakage into μ . $\square$

The G1DE- $M_{3/4}$ relation follows as the low-redshift, constant-amplitude, isotropic-compliance reduction:

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3-s). \quad (55)$$

11 Horizon Occupancy and a Candidate Memory Kernel

The redshift shape should not be a free polynomial. It should be the retarded response of finite horizon-screen occupancy drift.

Let

$$N = \ln a, \quad \epsilon_H = -\frac{d \ln H}{dN}. \quad (56)$$

Define a horizon capacity

$$C_H(a) = \frac{A_H(a)}{4\ell_P^2}, \quad (57)$$

where A_H is an appropriate cosmological horizon screen area. Define occupancy

$$\mathcal{O}_H(a) = \frac{C_{\text{used}}(a)}{C_H(a)}. \quad (58)$$

Definition 4 (Horizon occupancy source). *The finite-screen occupancy drift source is*

$$\mathcal{S}_H(N) = -\frac{d \ln \mathcal{O}_H}{dN}. \quad (59)$$

The horizon residual response obeys a memory equation

$$\mathcal{R}'_H(N) + \Gamma_H(N)\mathcal{R}_H(N) = \mathcal{S}_H(N), \quad (60)$$

with solution

$$\mathcal{R}_H(N) = \int_{-\infty}^N dN' \exp \left[- \int_{N'}^N \Gamma_H(u) du \right] \mathcal{S}_H(N'). \quad (61)$$

Principle 3 (Minimal acceleration-gated approximation). *A minimal low-information closure is*

$$\Gamma_H \simeq 1, \quad \mathcal{S}_H(N) \simeq [1 - \epsilon_H(N)]_+, \quad (62)$$

so that

$$\mathcal{R}'_H(N) + \mathcal{R}_H(N) = [1 - \epsilon_H(N)]_+. \quad (63)$$

This approximation is dormant during matter and radiation domination, activates when accelerated expansion makes a future-horizon-like causal boundary dynamically relevant, and approaches a constant response in the de Sitter limit.

Caveat 2 (Amplitude versus redshift kernel). *This paper derives the optical-stress amplitude coefficient. It does not complete the full redshift-kernel derivation. Equation (63) is a minimal candidate closure, while (60) is the more general horizon-memory structure. Deriving Γ_H and $\mathcal{S}_H$ from finite-screen entropy dynamics remains an open problem.*

12 Effective filling dimension

The parameter s in the low-redshift G1DE reduction should not be treated as a fundamental constant. It is an effective filling exponent.

Let $\mathcal{V}_{\text{scr}}(a)$ be the screen-accessible bulk distinction volume. Define

$$s_{\text{eff}}(a) = \frac{d \ln \mathcal{V}_{\text{scr}}}{d \ln a}. \quad (64)$$

Ordinary three-volume filling gives $s_{\text{eff}} = 3$. The finite-screen filling defect is

$$\lambda(a) = 3 - s_{\text{eff}}(a). \quad (65)$$

A low-redshift constant- s reduction corresponds to

$$s \simeq \langle s_{\text{eff}}(a) \rangle_{z \lesssim 2}. \quad (66)$$

Equivalently, if $\mathcal{O}_H = C_{\text{used}}/C_H$, then a candidate relation is

$$3 - s_{\text{eff}}(a) \sim -\frac{d \ln \mathcal{O}_H}{d \ln a}. \quad (67)$$

The deeper dynamical variable is therefore not s alone but

$$\mathcal{R}_H(a) = \lambda(a)X_b(a). \quad (68)$$

13 Null tests and observational fingerprints

The defining G1 channel split is

$$|\mu - 1| \ll |\Sigma - 1| \quad (69)$$

on subhorizon linear scales, with the precise scaling in (52)–(51).

A useful null combination is

$$\mathcal{N}_{\text{G1}}(a, k) = \Sigma(a, k) - 1 + \kappa_{BW} \mathcal{R}_H(a). \quad (70)$$

G1 predicts

$$\mathcal{N}_{\text{G1}} = O(\mathcal{R}_H^2). \quad (71)$$

For the isotropic low-redshift branch,

$$\mathcal{N}_{3/4} = \Sigma(a, k) - 1 + \frac{3}{4}(3 - s)X_b(a) = O((3 - s)^2). \quad (72)$$

Another discriminator is

$$\mathcal{Q}_{\mu\Sigma}(a, k) = \frac{\mu(a, k) - 1}{\Sigma(a, k) - 1}, \quad (73)$$

with prediction

$$\mathcal{Q}_{\mu\Sigma} = O(\mathcal{R}_H) + O(\mathcal{H}^2/k^2). \quad (74)$$

Thus E_G -type lensing/growth observables are natural probes of G1: they target precisely the ratio of Weyl-active lensing to protected matter growth.

14 Extensions

We list three directions but do not develop them here.

1. **Horizon shear/twist memory near black holes.** Finite-screen corrections near strong-field horizons should appear first as shear/twist membrane stress rather than as new matter density. Possible interfaces include horizon memory, ringdown, photon-sphere lensing, and scrambling.

Table 2: Residual fingerprints.

Theory class	Growth channel	Lensing channel	Channel structure
Extra matter density	$\mu \neq 1$ expected	$\Sigma \neq 1$ expected	density-driven
Smooth dark energy	mostly background	indirect lensing	no locked split
Generic modified gravity	free μ	free Σ	weak unless imposed
G1 optical stress	$\mu \simeq 1$ subhorizon	$\Sigma - 1 = O(\mathcal{R}_H)$	Weyl-first lock

2. **EM/GW distance-duality tests.** If the G1 residual is optical/Weyl stress, standard-siren distance ratios may carry screen-stress corrections of order $\mathcal{R}_H(a)$, though $c_T = 1$ is expected at leading order. This is an open observational direction.
3. **Finite-screen code geometry.** In an information-theoretic reading, the G1 dark sector can be interpreted as a finite-screen code-deformation residual: not a new substance, but a cost of maintaining recoverable distinction geometry under finite screen capacity. Twist/holonomy contributes to capacity counting without requiring nonzero mean rotation in scalar cosmology.

15 Demotion and falsification registry

Table 3: Demotion and falsification paths.

Condition	What fails	What may survive
Dominant first-order μ correction on subhorizon scales	Ricci-silent optical-stress closure	finite-screen residual with density component
Coefficient near 2/3	full $GL(2)$ branch	scalar-shear $SPD(2)$ branch
Coefficient far from optical-compliance family	MaxEnt / compliance projection	unrelated phenomenological lock
No common horizon residual $\mathcal{R}_H(a)$ for background and Weyl	background–Weyl ledger lock	separate residual channels
No lensing or E_G deviation while filling defect persists	optical compensation branch	background-only filling anomaly
Large first-order curl mean	scalar isotropy assumption	holonomy branch with parity breaking
No acceleration-gated or occupancy-memory behavior	minimal horizon-memory closure	alternative screen-memory dynamics
Violation of any covariant conservation or controlled exchange	effective-stress embedding	local optical algebra only
Failure of finite-screen entropy response in G1 core	full FSS/G1 bridge	algebraic toy model only

16 Open mathematical problems

Problem 1: derive the entropy Hessian. The stiffnesses K_A, K_S, K_Ω should ultimately be obtained from a finite-screen entropy functional rather than postulated.

Problem 2: derive the horizon source. The source $\mathcal{S}_H = -d_N \ln \mathcal{O}_H$ must be computed from finite-screen occupancy dynamics. The acceleration-gated form $[1 - \epsilon_H]_+$ is only a minimal approximation.

Problem 3: covariant embedding. The effective residual $E_{\mu\nu}^{\text{scr}}$ must be embedded in a conservation-closed system satisfying

$$\nabla^\mu (G_{\mu\nu} - 8\pi G T_{\mu\nu} - E_{\mu\nu}^{\text{scr}}) = 0 \quad (75)$$

or a controlled exchange equation with a horizon bath.

Problem 4: nonlinear and strong-field limits. The present analysis is linear and optical. Nonlinear structure, baryonic feedback, caustics, strong lensing, black holes, and ringdown remain outside its scope.

Problem 5: twist covariance and observability. The twist/holonomy direction contributes to full optical capacity, but its mean may vanish in scalar cosmology. Future work must separate capacity covariance, parity constraints, and observable curl/B-mode signals.

Problem 6: relation to G1fit-real production evidence. The present paper supplies a theoretical closure. Its observational counterpart requires production likelihoods, expanded weak-lensing and E_G data, and model comparison against flexible μ - Σ alternatives.

17 Relation to the G1fit-real evidence tests

The observational counterpart of this closure is the G1fit-real program, where the low-redshift reduction G1DE-M_{3/4} is tested against SN, BAO, RSD, and E_G data. The present paper supplies the non-equilibrium screen-entropy closure behind that model; it does not use the likelihood results as an input. Current medium-prior production-refined comparisons select the projection-locked relation at the AIC/BIC and evidence level against common dark-energy extensions, while full survey-grade validation, expanded weak-lensing likelihoods, and redshift-kernel discrimination refinement remain pending.

18 Conclusion

The optical projection law showed how a finite screen can produce the amplitude relation

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3 - s). \quad (76)$$

The present paper embeds that result into a deeper non-equilibrium screen-entropy closure.

The core chain is

$$\begin{aligned} \boxed{\delta S_{\text{prod}}} &\implies \boxed{K_A, K_S, K_\Omega} \implies \boxed{\chi_i = K_i^{-1}} \implies \boxed{\kappa_{BW} = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega}} \\ \implies \boxed{\text{traceless optical stress}} &\implies \boxed{\Sigma - 1 = -\kappa_{BW}\mathcal{R}_H + O(\mathcal{R}_H^2), \mu - 1 = O(\mathcal{R}_H^2, \mathcal{R}_H\mathcal{H}^2/k^2)}. \end{aligned}$$

In the isotropic entropy-Hessian branch,

$$\chi_A = \chi_S = \chi_\Omega, \quad (77)$$

so

$$\kappa_{BW} = \frac{3}{4}. \quad (78)$$

In the low-redshift constant-amplitude reduction,

$$\mathcal{R}_H(a) \simeq (3 - s)X_b(a), \quad (79)$$

recovering the G1DE-M_{3/4} lock.

The conceptual result is that the G1 dark sector is not an ordinary density fluid and not a generic modified Poisson term. It is finite-screen non-equilibrium entropy production projected as Ricci-silent, Weyl-active traceless optical stress. The redshift dependence is a horizon-memory problem; the coefficient is an entropy-compliance problem; the observational signature is a lensing/growth split.

References

- [1] J. D. Bekenstein, “Black holes and entropy,” *Phys. Rev. D* **7**, 2333 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” *Commun. Math. Phys.* **43**, 199 (1975).
- [3] A. Raychaudhuri, “Relativistic cosmology. I,” *Phys. Rev.* **98**, 1123 (1955).
- [4] R. Sachs, “Gravitational waves in general relativity. VI. The outgoing radiation condition,” *Proc. R. Soc. Lond. A* **264**, 309 (1961).
- [5] R. M. Wald, *General Relativity* (University of Chicago Press, 1984).
- [6] P. Schneider, J. Ehlers, and E. E. Falco, *Gravitational Lenses* (Springer, 1992).
- [7] M. Bartelmann and P. Schneider, “Weak gravitational lensing,” *Phys. Rep.* **340**, 291 (2001).
- [8] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* **75**, 1260 (1995).
- [9] C. Eling, R. Guedens, and T. Jacobson, “Non-equilibrium thermodynamics of spacetime,” *Phys. Rev. Lett.* **96**, 121301 (2006).
- [10] J. M. Bardeen, “Gauge-invariant cosmological perturbations,” *Phys. Rev. D* **22**, 1882 (1980).
- [11] C.-P. Ma and E. Bertschinger, “Cosmological perturbation theory in the synchronous and conformal Newtonian gauges,” *Astrophys. J.* **455**, 7 (1995).

- [12] P. Zhang, M. Liguori, R. Bean, and S. Dodelson, “Probing gravity at cosmological scales by measurements which test the relationship between gravitational lensing and matter overdensity,” *Phys. Rev. Lett.* **99**, 141302 (2007).
- [13] S. F. Daniel and E. V. Linder, “Constraining cosmic expansion and gravity with gravitational lensing,” *Phys. Rev. D* **82**, 103523 (2010).
- [14] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *JCAP* **07**, 050 (2014).
- [15] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), doi:10.5281/zenodo.20382013.
- [16] Y. Wu, *The 3/4 Projection Law in Finite Screen Gravity: MaxEnt Optical-Stress Closure, Ricci-Silent Residuals, and the Weyl-Locked Dark Sector*, technical note (2026).
- [17] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual*, manuscript / data paper (2026).

Screen Influence Functional for the G1 Dark Sector: Entropy Hessian, Ward Closure, and Retarded Horizon Memory

v1.2 Body-Integrated Release

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This paper formulates a screen influence functional for the G1 dark sector in the Finite Distinction Systems-Gravity 1 (FDS-G1) program. Previous G1 notes derived a 3/4 optical projection law and a non-equilibrium screen-entropy origin for Ricci-silent, Weyl-active residuals. The present paper packages those ingredients into an effective influence framework: finite-screen entropy incompleteness produces a retarded horizon-memory residue $\mathcal{R}_H$, and the leading operator by which that residue enters cosmological perturbation theory is a Ward-closed unimodular optical stress.

The formal core is a quadratic non-equilibrium screen-entropy Hessian on the local optical deformation algebra,

$$I_{\text{scr}}^{\text{neq}} = \frac{1}{2} \int_{\mathcal{N}} d\lambda dA (K_A \theta^2 + K_S \sigma_{AB} \sigma^{AB} + K_\Omega \omega^2),$$

with compliance matrix $C = K^{-1}$. The Weyl-active projection coefficient is

$$\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr } C} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}.$$

The isotropic entropy-Hessian fixed point $K_A = K_S = K_\Omega$ gives $\kappa_{BW} = 3/4$, while demotion of the twist/holonomy sector gives the scalar-shear branch $\kappa_{BW} = 2/3$.

The horizon residue is treated as an auxiliary retarded memory field obeying

$$\mathcal{R}'_H(N) + \Gamma_H(N) \mathcal{R}_H(N) = -\frac{d \ln \mathcal{O}_H}{dN}, \quad N = \ln a,$$

where $\mathcal{O}_H = C_{\text{used}}/C_H$ is the finite-screen occupancy of the horizon capacity $C_H = A_H/(4\ell_P^2)$. A Ward-completion theorem then gives the subhorizon hierarchy

$$E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{R}_H), \quad E_{0i}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right), \quad E_{00}^{\text{scr}} = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right),$$

so that

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad \mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

The low-redshift reduction $\mathcal{R}_H(a) \simeq (3-s)X_b(a)$ recovers the projection-locked G1DE relation $\mu_0 = 0$, $\Sigma_0 = -3(3-s)/4$. The resulting framework identifies G1DE-M_{3/4} as the constant-amplitude, low-redshift, isotropic-compliance reduction of a retarded screen influence functional. The influence functional is retarded and open-system in the Schwinger–Keldysh sense, not a conservative local action. A technical appendix sharpens the remaining closure by defining the screen entropy Hessian as a relative-entropy Hessian, the horizon occupancy as horizon-accessible distinguishability, and the effective stress as a minimum-norm Ward completion of the optical seed.

1 Scope and executive summary

This paper is the fourth theory note in the G1 dark-sector closure sequence. Its purpose is not to introduce a new phenomenological parameterization. Its purpose is to assemble the previously derived optical projection and non-equilibrium screen-entropy structure into a single effective object: a *screen influence functional* for the G1 dark sector.

D4 mediates between microscopic realization and output $R_H(a)$, but must not claim unique source–kernel identification. The factorization $R_H = K_H * J_H$ carries a quotient equivalence class: $(K_H, J_H) \sim (K'_H, J'_H)$ iff $K_H * J_H = K'_H * J'_H$. D4 classifies possible realizations inside this class but avoids uniqueness.

$R_H = K_H * J_H$ is an output quotient; D10 uses only $R_H(a)$, not unique K_H, J_H .

The guiding statement is:

The G1 dark sector is a Ward-closed retarded screen influence whose leading operator is unimodular optical stress.

Equivalently,

$$\begin{aligned} \text{finite-screen entropy incompleteness} &\longrightarrow \text{retarded horizon residue } \mathcal{R}_H \\ &\longrightarrow \text{Ward-closed optical stress } E_{\mu\nu}^{\text{scr}}. \end{aligned} \tag{1}$$

This framing is designed to answer three questions left open by the shorter projection-law and non-equilibrium notes:

1. What is the effective object that generates the optical-stress residual?
2. Why is the leading residual Weyl-active but Ricci-density silent at first order?
3. How does the low-redshift model G1DE-M_{3/4} arise as a reduction of a retarded horizon-memory response?

The central result is the following schematic chain:

$$\boxed{I_{\text{scr}}^{\text{neq}}} \Rightarrow \boxed{K_A, K_S, K_\Omega} \Rightarrow \boxed{C = K^{-1}} \Rightarrow \boxed{\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr } C}} \searrow$$

$$\begin{aligned} & \boxed{\mathcal{O}_H} \Rightarrow \boxed{\mathcal{R}_H} \swarrow \\ & \Rightarrow \boxed{E_{ij,\text{TF}}^{\text{scr}}} \Rightarrow \boxed{\Sigma \neq 1, \quad \mu \simeq 1}. \end{aligned}$$

Equivalently, the operator chain is

$$-\text{d}_N \ln \mathcal{O}_H \xrightarrow{G_H^{\text{ret}}} \mathcal{R}_H \xrightarrow{C=K^{-1}, P_U} -\kappa_{BW} \mathcal{R}_H \xrightarrow{\mathcal{P}_W^{\text{ret}}} E_{\mu\nu}^{\text{scr}} \xrightarrow{\mathcal{W}_\Sigma, \mathcal{W}_\mu} (\Sigma, \mu). \quad (2)$$

The effective stress is a hierarchy, not a standalone conserved fluid with only anisotropic stress at all scales. Ward/Bianchi closure completes the leading traceless spatial stress by suppressed momentum and density components. In the subhorizon linear regime, the hierarchy is

$$E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{R}_H), \quad E_{0i}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right), \quad E_{00}^{\text{scr}} = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (3)$$

This is the mathematical meaning of Ricci silence in G1: it is not literal absence of all Ricci-sector components, but Ward-suppressed density leakage.

2 Relation to the G1 closure sequence

The present note should be read as a companion to three earlier layers:

Layer	Question answered
D1: 3/4 projection law	Why does the finite optical screen give a 3/4 Weyl-active projection under a MaxEnt / isotropic residual assumption?
D2: non-equilibrium entropy	What screen entropy dynamics produces optical stress?
D3: Ward-closed dark sector	How does optical stress enter covariant perturbation theory while keeping $\mu \simeq 1$ and $\Sigma \neq 1$?
This paper	What effective influence functional unifies entropy Hessian, retarded horizon memory, and Ward completion?
G1fit-real	Does the low-redshift reduction G1DE-M _{3/4} receive exact-likelihood support?

Thus this paper is not a data paper. It is a theory-side effective closure. It uses the G1fit-real model only as a low-redshift reduction target, not as an input to the derivation. A sharper conditional theorem collecting the relative-entropy screen Hessian, passive horizon-memory response, and minimum Ward completion is stated in Appendix A.

3 Claim-status discipline

The language of an influence functional is useful, but it must be handled with care. The screen residual is retarded and dissipative. Therefore the appropriate object is closer to an open-system

influence functional than to a conservative local action. This paper uses “influence functional” in that effective-theory sense.

Table 1: Claim status of the screen influence closure.

Claim	Meaning	Status
Finite optical screen	observable residuals are projected through a two-dimensional null screen	bridge principle
$SO(2)$ sector split	$\mathfrak{gl}(2) = A(1) \oplus S(2) \oplus \Omega(1)$	algebraic / optical geometry
Entropy Hessian	K_A, K_S, K_Ω are second variations of screen entropy production	closure ansatz
Compliance projection	$\kappa_{BW} = \text{Tr}(P_U K^{-1}) / \text{Tr}(K^{-1})$	conditional theorem
Retarded memory field	$\mathcal{R}_H$ responds causally to horizon occupancy drift	dynamical bridge
Ward completion	leading optical stress requires suppressed $0i$ and 00 components	covariance / power counting
μ suppression	$\mu - 1 = O(\mathcal{R}_H^2) + O(\mathcal{R}_H \mathcal{H}^2 / k^2)$	subhorizon prediction
Weyl lock	$\Sigma - 1 = -\kappa_{BW} \mathcal{R}_H + O(\mathcal{R}_H^2)$	data-facing reduction
Low-redshift G1DE-M _{3/4}	$\mathcal{R}_H \simeq (3 - s)X_b$ and $\kappa = 3/4$	reduced model

The strongest safe statement is:

screen entropy Hessian + retarded memory + Ward closure

$$\implies \quad \Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2),$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

4 Finite optical screen sectors

A finite observer receives spacetime residuals through a null optical screen. On a non-caustic patch, the Jacobi map is

$$J_B^A \in GL(2, \mathbb{R}), \quad A, B = 1, 2. \quad (4)$$

The local deformation generator is

$$X = \delta J J^{-1} \in \mathfrak{gl}(2, \mathbb{R}). \quad (5)$$

Equivalently, along an affine parameter λ one may use

$$B_B^A = \frac{dJ_C^A}{d\lambda} (J^{-1})^C_B. \quad (6)$$

The Sachs decomposition is

$$B^A_B = \frac{1}{2}\theta\delta^A_B + \sigma^A_B + \omega\epsilon^A_B. \quad (7)$$

Thus

$$\mathfrak{gl}(2) = A(1) \oplus S(2) \oplus \Omega(1). \quad (8)$$

Here A is the area/trace channel, S is the symmetric traceless shear channel, and Ω is the twist/holonomy channel.

Under the local screen rotation group $SO(2)$, the sectors may be described as

$$A : \mathbf{0}^+, \quad S : \mathbf{2}, \quad \Omega : \mathbf{0}^-. \quad (9)$$

In an isotropic and parity-respecting screen background, the leading Hessian is block diagonal on these representations.

Let P_A, P_S, P_Ω denote the orthogonal projectors under the Frobenius inner product. The unimodular projector is

$$P_U = P_S + P_\Omega, \quad (10)$$

with traces

$$\text{Tr } P_A = 1, \quad \text{Tr } P_S = 2, \quad \text{Tr } P_\Omega = 1, \quad \text{Tr } P_U = 3. \quad (11)$$

5 Screen entropy Hessian and compliance projection

The non-equilibrium screen-entropy production is taken to be quadratic at leading order:

$$I_{\text{scr}}^{\text{neq}} = \frac{1}{2} \int_{\mathcal{N}} d\lambda dA (K_A \theta^2 + K_S \sigma_{AB} \sigma^{AB} + K_\Omega \omega^2). \quad (12)$$

Here K_A, K_S, K_Ω are entropy cost coefficients: they are eigenvalues of the screen entropy Hessian on the A, S, Ω sectors. The response compliances are their inverses:

$$\chi_A = K_A^{-1}, \quad \chi_S = K_S^{-1}, \quad \chi_\Omega = K_\Omega^{-1}. \quad (13)$$

Define the Hessian operator

$$K = K_A P_A + K_S P_S + K_\Omega P_\Omega, \quad (14)$$

so that the compliance/covariance operator is

$$C = K^{-1} = K_A^{-1} P_A + K_S^{-1} P_S + K_\Omega^{-1} P_\Omega. \quad (15)$$

Definition 1 (Weyl-active compliance projection). *The $G1$ background–Weyl projection coefficient is the fraction of residual compliance carried by the unimodular sector:*

$$\kappa_{BW} = \frac{\text{Tr}(P_U C)}{\text{Tr } C}. \quad (16)$$

Using the sector traces gives

$$\kappa_{BW} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}. \quad (17)$$

Equivalently,

$$\kappa_{BW} = \frac{2\chi_S + \chi_\Omega}{\chi_A + 2\chi_S + \chi_\Omega}. \quad (18)$$

Proposition 1 (Isotropic optical fixed point). *If the screen entropy Hessian is isotropic at leading order,*

$$K_A = K_S = K_\Omega, \quad (19)$$

then

$$\kappa_{BW} = \frac{2+1}{1+2+1} = \frac{3}{4}. \quad (20)$$

If the twist/holonomy sector is demoted, $K_\Omega^{-1} = 0$, and $K_A = K_S$, then

$$\kappa_{BW} = \frac{2}{1+2} = \frac{2}{3}. \quad (21)$$

Thus $3/4$ is not a bare numerological count. It is the compliance projection of an isotropic screen entropy Hessian. Deviations from $3/4$ measure anisotropy in the sector compliances. Since $C = K^{-1}$ is a positive compliance/covariance operator in the stable branch,

$$C \succeq 0, \quad (22)$$

one obtains the consistency bound

$$0 \leq \kappa_{BW} \leq 1. \quad (23)$$

A fitted value outside this interval would falsify the positive-compliance interpretation of the screen influence functional.

Let

$$\chi_S = \chi_A(1 + \alpha), \quad \chi_\Omega = \chi_A(1 + \beta). \quad (24)$$

Then

$$\kappa_{BW} = \frac{3 + 2\alpha + \beta}{4 + 2\alpha + \beta} = \frac{3}{4} + \frac{2\alpha + \beta}{16} + O(\alpha^2, \beta^2). \quad (25)$$

Hence a measured deviation from 0.75 can be read as a constraint on screen response anisotropy.

6 Retarded horizon-response output

The G1 dark-sector amplitude should not be treated as an arbitrary redshift function. The basic object is a retarded horizon-occupancy residue.

Define horizon capacity

$$C_H(a) = \frac{A_H(a)}{4\ell_P^2}, \quad (26)$$

and finite-screen occupancy

$$\mathcal{O}_H(a) = \frac{C_{\text{used}}(a)}{C_H(a)}. \quad (27)$$

Let $N = \ln a$. The source for screen memory is the occupancy drift

$$\mathcal{S}_H(N) = -\frac{d \ln \mathcal{O}_H}{dN}. \quad (28)$$

Definition 2 (Retarded horizon residue). *The G1 horizon residue $\mathcal{R}_H$ is an auxiliary memory field obeying*

$$\boxed{\mathcal{R}'_H(N) + \Gamma_H(N)\mathcal{R}_H(N) = \mathcal{S}_H(N)}. \quad (29)$$

The retarded solution is

$$\mathcal{R}_H(N) = \int_{-\infty}^N dN' G_H(N, N') \mathcal{S}_H(N'), \quad (30)$$

where

$$G_H(N, N') = \exp \left[- \int_{N'}^N \Gamma_H(u) du \right]. \quad (31)$$

If $\Gamma_H > 0$, the memory is stable and decays in the absence of source. A future microscopic closure should also impose passivity or entropy-production positivity. In time-domain language this means that the response cannot extract net work from an unforced screen bath; schematically,

$$\int dN \mathcal{S}_H(N) \mathcal{R}_H(N) \geq 0 \quad (32)$$

for the physical sign convention. In frequency-domain language, an equivalent condition is positivity of the dissipative part of the retarded response function. These are not extra data-fitting assumptions; they are consistency conditions on the allowed memory kernels.

The earlier low-redshift parameterization is recovered by writing

$$\mathcal{R}_H(a) \simeq (3 - s) X_b(a), \quad (33)$$

where s is a low-redshift effective filling exponent and $X_b(a)$ is a normalized response shape. The minimal acceleration-gated candidate

$$X'_b(N) + X_b(N) = [1 - \epsilon_H(N)]_+, \quad \epsilon_H = - \frac{d \ln H}{dN}, \quad (34)$$

is the special approximation $\Gamma_H \simeq 1$ and $\mathcal{S}_H \simeq [1 - \epsilon_H]_+$. It is not the full dynamical derivation.

7 v1.2 Finite-Description Kernel Admissibility

Claim type: Admissibility rule All G1 screen residual kernels must satisfy the following form and constraints:

$$R_{\mu\nu}^{\text{scr}}(x) = \Pi_{\text{cap}} \int_{\mathcal{J}^-(x)} K_{\mu\nu}^{\text{scr}, \text{AB}}(x, y; \tau) \delta \mathcal{L}_A(y) dV_y + \epsilon_{\mu\nu}^{\text{cap}}(x)$$

where:

- Strictly retarded (support limited to the past causal cone $\mathcal{J}^-(x)$)
- Finite-description or finite-rank after capacity projection Π_{cap}
- Quotient invariant under $\mathcal{G}_{\text{scr}}$
- Ward/Bianchi consistent
- No free $A(a, k)$ or unpenalized functional degrees of freedom

Kernels failing these constraints are rejected from the G1DE admissible class:

Kernel failure condition	Demotion consequence
Acausal (non-past) support	Kernel rejected entirely
Violates passivity of memory kernel	Not valid G1 passive memory
Requires free $A(a, k)$ parameter	Demoted to generic dark stress
Cannot be Ward-closed	No valid covariant residual form
Unpenalized infinite functional degrees of freedom	No predictive compression, rejected

8 The screen influence functional

The dark-sector residual is dissipative and retarded. Therefore it is better described by an influence functional than by an ordinary conservative action. A schematic decomposition is

$$I_{\text{G1}}^{\text{eff}} = I_{\text{EH}} + I_{\text{matter}} + I_{\text{scr}}^{\text{eq}} + I_{\text{scr}}^{\text{neq}} + I_{\text{mem}}^{\text{ret}}. \quad (35)$$

Here $I_{\text{scr}}^{\text{eq}}$ belongs to the equilibrium entropy-response geometry of the G1 core, while $I_{\text{scr}}^{\text{neq}}$ is the quadratic optical entropy production in Eq. (12).

The memory part can be represented locally with an auxiliary field constraint,

$$I_{\text{mem}}^{\text{aux}} = \int dN \Lambda_H(N) [\mathcal{R}'_H(N) + \Gamma_H(N) \mathcal{R}_H(N) - \mathcal{S}_H(N)]. \quad (36)$$

Because the memory response is dissipative and retarded, it should not be read as an ordinary symmetric quadratic action. A more appropriate schematic representation is an open-system or Schwinger–Keldysh influence term,

$$I_{\text{mem}}^{\text{ret}} \sim \int dN dN' \mathcal{S}_H^-(N) G_H^{\text{ret}}(N, N') \mathcal{S}_H^+(N'), \quad (37)$$

possibly accompanied by a noise kernel,

$$I_{\text{noise}} \sim \frac{i}{2} \int dN dN' \mathcal{S}_H^-(N) N_H(N, N') \mathcal{S}_H^-(N'). \quad (38)$$

For the present purpose, the essential point is causal structure. The screen residual at time N depends on past horizon occupancy drift, not on future data.

The leading operator by which the memory field enters cosmological perturbation theory is unimodular optical stress:

$$E_{ij, \text{TF}}^{\text{scr}} = -\mathcal{A}(a, k) \kappa_{BW} \mathcal{R}_H(a) \left(\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij} \right) + O(\mathcal{R}_H^2), \quad (39)$$

where $\mathcal{A}(a, k)$ is a normalization that maps dimensionless screen residue to the chosen effective-stress convention. In the reduced μ, Σ notation, this normalization is absorbed into $\Sigma - 1$.

9 Ward projector and minimum completion

The constitutive law in Eq. (39) defines only the leading spatial traceless seed. It does not by itself define a covariantly closed effective stress. The physical residual should therefore be read as

a retarded Ward completion of this seed:

$$E_{\mu\nu}^{\text{scr}} = \mathcal{P}_W^{\text{ret}}[Q]_{\mu\nu}, \quad (40)$$

where

$$Q_{ij} = -\kappa_{BW}\mathcal{R}_H(a) \left(\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij} \right) + O(\mathcal{R}_H^2) \quad (41)$$

is the unimodular optical-stress seed, and $\mathcal{P}_W^{\text{ret}}$ denotes the minimal retarded completion satisfying the Ward/Bianchi constraint.

In a local flat-patch analogy, the idea is analogous to completing a source with a retarded transverse projector. Schematically one may use

$$P_{\mu\nu}^{\perp} = \eta_{\mu\nu} - \frac{\partial_{\mu} \partial_{\nu}}{\square_{\text{ret}}}, \quad (42)$$

and a spin-two transverse projector

$$\mathcal{P}_{\mu\nu\alpha\beta}^{TT} = \frac{1}{2} \left(P_{\mu\alpha}^{\perp} P_{\nu\beta}^{\perp} + P_{\mu\beta}^{\perp} P_{\nu\alpha}^{\perp} \right) - \frac{1}{3} P_{\mu\nu}^{\perp} P_{\alpha\beta}^{\perp}. \quad (43)$$

This expression is not meant as a complete FRW gauge-invariant construction. It states the local principle: the completion is retarded, transverse, and minimal. A production implementation should replace it by a fully gauge-invariant cosmological perturbation operator.

Equivalently, the Ward completion may be characterized variationally as the minimum-norm symmetric tensor compatible with the optical-stress seed:

$$E_{\mu\nu}^{\text{scr}} = \arg \min_{E_{\mu\nu}} \|E_{\mu\nu}\|_{\mathcal{G}}^2 \quad (44)$$

subject to

$$E_{ij,\text{TF}}^{\text{scr}} = Q_{ij}, \quad \nabla^{\mu} E_{\mu\nu}^{\text{scr}} = J_{\nu}^H, \quad (45)$$

with $J_{\nu}^H = 0$ in the closed limit. This minimum-completion formulation makes explicit why the density and momentum components are not independent dark-fluid variables: they are the smallest Ward-required completion of the leading optical stress.

10 Ward completion of optical stress

The effective field equation is written as

$$G_{\mu\nu} = 8\pi G T_{\mu\nu} + E_{\mu\nu}^{\text{scr}}. \quad (46)$$

If ordinary matter is conserved, Bianchi closure requires

$$\nabla^{\mu} E_{\mu\nu}^{\text{scr}} = 0. \quad (47)$$

If the finite screen exchanges with a horizon bath, one may instead allow controlled exchange

$$\nabla^{\mu} E_{\mu\nu}^{\text{scr}} = J_{\nu}^H, \quad (48)$$

where J_{ν}^H is horizon-scale and hence subhorizon suppressed.

Theorem 1 (Ward completion of traceless optical stress). *Assume that the leading screen residual in the scalar sector is a traceless spatial optical stress,*

$$E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{R}_H), \quad (49)$$

and that $E_{\mu\nu}^{\text{scr}}$ is Ward closed, up to horizon-scale exchange. In the subhorizon linear regime $k \gg \mathcal{H}$, the minimal completion satisfies

$$E_{0i}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right), \quad (50)$$

$$E_{00}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right) + O(\mathcal{R}_H^2). \quad (51)$$

Power-counting proof. The divergence of the leading traceless spatial stress carries one spatial derivative, hence scales as $kE_{ij,\text{TF}}^{\text{scr}}$. The Ward identity can be balanced by a momentum component whose time-derivative/expansion contribution carries a scale $\mathcal{H}$; therefore $E_{0i}^{\text{scr}}/E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{H}/k)$. Applying the constraint once more gives the density-like component suppressed by a second power, $E_{00}^{\text{scr}}/E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{H}^2/k^2)$. Nonlinear self-sourcing contributes $O(\mathcal{R}_H^2)$. $\square$

This is the Ward-closed meaning of Ricci silence. The density channel is not exactly absent at all scales. It is suppressed relative to the optical stress by powers of $\mathcal{H}/k$ and by nonlinear order.

The minimum-completion version of this Ward hierarchy, including the precision gauge-invariant scalar-sector route, is stated as a conditional theorem in Appendix A.

Caveat 1 (Gauge and quasi-static scope). *This hierarchy is a subhorizon, quasi-static power-counting statement. A fully gauge-invariant formulation with a complete covariant embedding of the screen influence functional is still required for superhorizon scales and for a production-level perturbation implementation. Appendix A gives a technical minimum-completion formulation and a gauge-invariant scalar-sector route toward that implementation.*

11 Cosmological reduction

Use the standard phenomenological definitions

$$-k^2\Psi = 4\pi G a^2 \mu(a, k) \rho \Delta, \quad (52)$$

$$-k^2(\Phi + \Psi) = 8\pi G a^2 \Sigma(a, k) \rho \Delta. \quad (53)$$

The Newtonian channel μ is sensitive to density-like Ricci sourcing. The Weyl/lensing channel Σ is sensitive to optical traceless stress.

Combining the constitutive law with Ward completion gives

$$\boxed{\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2),} \quad (54)$$

$$\boxed{\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).} \quad (55)$$

These are stronger than the shorthand $\mu_0 = 0$. They state that matter growth is first-order protected only in the subhorizon linear regime, with controlled horizon-scale leakage.

In the low-redshift constant-amplitude reduction,

$$\mathcal{R}_H(a) \simeq (3-s)X_b(a), \quad (56)$$

and for isotropic screen entropy Hessian,

$$\kappa_{BW} = \frac{3}{4}. \quad (57)$$

Thus one obtains

$$\mu(a, k) = 1 + O((3-s)^2) + O\left((3-s)\frac{\mathcal{H}^2}{k^2}\right), \quad (58)$$

$$\Sigma(a, k) = 1 - \frac{3}{4}(3-s)X_b(a) + O((3-s)^2). \quad (59)$$

At the amplitude level, this is the G1DE-M_{3/4} lock

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3-s). \quad (60)$$

12 Compensation sign and Weyl suppression

The sign of the Weyl response is fixed by finite-screen ledger closure. A positive horizon occupancy lag corresponds to

$$\mathcal{R}_H > 0. \quad (61)$$

The determinant/filling channel then carries a positive residual. The unimodular optical sector compensates it:

$$\Delta_U = -\kappa_{BW}\Delta_A. \quad (62)$$

Since the unimodular sector is Weyl-active, the leading lensing response has opposite sign:

$$\boxed{\mathcal{R}_H > 0 \quad \Rightarrow \quad \Sigma - 1 < 0.} \quad (63)$$

Thus negative Weyl amplitude is not inserted by hand. It is the sign of optical compensation for a positive filling/occupancy residual. If future data robustly require $\Sigma > 1$ for positive $\mathcal{R}_H$, the compensation orientation or the identification of $\mathcal{R}_H$ would have to be revised.

13 G1 null tests

The influence-functional closure suggests two direct diagnostics.

13.1 Weyl-lock null test

Define

$$\mathcal{N}_{G1}(a, k) = \Sigma(a, k) - 1 + \kappa_{BW}\mathcal{R}_H(a). \quad (64)$$

G1 predicts

$$\boxed{\mathcal{N}_{G1}(a, k) = O(\mathcal{R}_H^2).} \quad (65)$$

A statistically significant nonzero $\mathcal{N}_{G1}$ at first order would falsify the single-residue background–Weyl lock.

13.2 Channel-split suppression test

Define

$$\mathcal{C}_{G1}(a, k) = \frac{\mu(a, k) - 1}{\Sigma(a, k) - 1}. \quad (66)$$

Using Eqs. (54)–(55),

$$\boxed{\mathcal{C}_{G1} = O(\mathcal{R}_H) + O\left(\frac{\mathcal{H}^2}{k^2}\right)}. \quad (67)$$

Therefore the true data-facing signature is not merely $\Sigma < 1$, but

$$\boxed{|\mu - 1| \ll |\Sigma - 1|} \quad (68)$$

for linear subhorizon modes.

14 Branch tree and demotion paths

The influence functional is intentionally structured with demotion branches.

Table 2: Branch tree and demotion registry.

Outcome	What fails or demotes	What may survive
$\kappa_{BW} \simeq 3/4$	none	full $GL(2)$ optical branch
$\kappa_{BW} \simeq 2/3$	twist/holonomy sector	symmetric $SPD(2)$ scalar-shear branch
Intermediate κ	isotropic Hessian fixed point	anisotropic compliance branch
Strong first-order μ correction	Ricci-silent Ward hierarchy	density-leakage finite-screen model
Constant- Σ favored	resolved horizon-memory shape	amplitude lock may survive; output-shape or source-kernel lift demoted
No shared $\mathcal{R}_H(a)$ for background and Weyl	single-residue ledger lock	separate background and optical residuals
No acceleration/horizon gating	minimal source model	alternative occupancy current
$\Sigma > 1$ for positive $\mathcal{R}_H$	compensation sign convention	reversed ledger orientation
No finite-screen entropy Hessian	influence-functional closure	algebraic toy model only

This tree is important. G1 does not rest on a single brittle number. It gives a hierarchy of more general branches and specifies what would be demoted by data.

15 Relation to G1fit-real

The observational counterpart of this influence-functional closure is the G1fit-real program. The low-redshift reduction G1DE-M_{3/4} is the data-facing model obtained by taking

$$\kappa_{BW} = \frac{3}{4}, \quad \mathcal{R}_H(a) \simeq (3-s)X_b(a), \quad (69)$$

and by retaining the subhorizon leading-order relation

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3-s). \quad (70)$$

The present paper supplies the theory-side influence closure behind that model. It does not use likelihood results as an input. Current medium-prior production-refined comparisons select the projection-locked relation at the AIC/BIC and evidence level. The completed medium-prior 8-seed production-refined audit (G1fit-real) gives the canonical hierarchy $M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE-2} > \text{G1DE-1} > \text{CPL} > \Lambda\text{CDM}$, with $\Delta \log Z = 0.856, 1.775, 6.523, 7.402, 9.603$, and 11.884 against the six controls. Top-control wide-prior sensitivity passed ($M_{3/4} > M_\kappa > \text{const-}\Sigma$ with $\Delta \log Z \simeq 0.85, 1.96$).

The distinction between amplitude and shape is crucial. If a constant- Σ control were preferred in future runs, the κ_{BW} amplitude branch could survive while the specific horizon-response output shape, or its chosen source–kernel lift, would be demoted.

16 Extensions

The influence-functional viewpoint suggests several extensions, but they are not developed here.

16.1 Horizon shear/twist memory near black holes

Near black-hole horizons, finite-screen corrections should first appear as shear/twist memory rather than density hair:

$$\delta S_{\text{hor}} = \frac{\delta A}{4\ell_P^2} + \frac{1}{2} \int (K_S \sigma^2 + K_\Omega \omega^2) d\lambda dA. \quad (71)$$

This may connect to horizon memory, tidal response, ringdown, photon-ring optics, and scrambling.

16.2 Standard sirens and optical distance duality

The conservative expectation is that local tensor speed remains protected,

$$c_T = 1, \quad (72)$$

but finite-screen optical stress may affect amplitude transport. A possible branch is

$$\frac{d_L^{\text{GW}}}{d_L^{\text{EM}}} = 1 + \alpha_{\text{GW}} \mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad (73)$$

where $\alpha_{\text{GW}} = 0$ if EM and GW amplitudes couple identically to the screen influence. This is a future standard-siren test, not a result of this paper.

16.3 Screen code deformation

In a quantum-information interpretation, the area sector measures code capacity, while the unimodular sector measures recoverability geometry or code deformation. The G1 dark sector may then be interpreted as a finite-screen code-deformation residual projected into spacetime optics.

17 Open mathematical problems

1. Derive K_A, K_S, K_Ω from a microscopic finite-screen entropy functional rather than postulating a quadratic Hessian.
2. Derive $\mathcal{O}_H(a)$ and $C_{\text{used}}(a)$ from screen-accessible bulk distinction accounting.
3. Embed the retarded influence functional in a fully covariant open-system formalism, including the corresponding noise kernel and passivity conditions.
4. Compute the exact Ward projector beyond subhorizon power counting, including gauge-invariant scalar perturbation variables.
5. Determine whether the twist/holonomy sector contributes only covariance or also measurable second-order curl/B-mode signatures.
6. Fix the constitutive mapping normalization $\mathcal{A}(a, k)$ between the optical-stress amplitude and the phenomenological $\Sigma(a, k)$ convention, including its relation to the standard $8\pi G a^2 \rho \Delta / k^2$ normalization.
7. Test the null combinations $\mathcal{N}_{G1}$ and $\mathcal{C}_{G1}$ in production likelihood pipelines.

18 Conclusion

This paper formulated a screen influence functional for the G1 dark sector. The central object is a retarded horizon residue $\mathcal{R}_H$ sourced by finite-screen occupancy drift. The leading operator generated by this residue is a unimodular optical stress. The response coefficient is the compliance projection

$$\kappa_{BW} = \frac{\text{Tr}(P_U K^{-1})}{\text{Tr}(K^{-1})}, \quad (74)$$

which gives 3/4 at the isotropic entropy-Hessian fixed point and 2/3 when the twist/holonomy sector is demoted.

Ward completion turns the leading optical stress into a covariant hierarchy:

$$E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{R}_H), \quad E_{0i}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right), \quad E_{00}^{\text{scr}} = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (75)$$

Therefore

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad (76)$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (77)$$

The low-redshift reduction $\mathcal{R}_H(a) \simeq (3-s)X_b(a)$ recovers the G1DE-M_{3/4} lock

$$\mu_0 = 0, \quad \Sigma_0 = -\frac{3}{4}(3-s). \quad (78)$$

Equivalently, G1DE-M_{3/4} is the constant-amplitude, low-redshift, isotropic-compliance reduction of this screen influence functional.

The conceptual result is that the G1 dark sector is not an ordinary density component and not an arbitrary modified-gravity function. It is a Ward-projected, Ward-closed retarded influence of finite-screen entropy incompleteness on the optical Weyl channel.

A Technical Appendix: Relative-Entropy Screen Hessian and Minimum Ward Completion

This appendix records a sharper mathematical formulation of the two least-closed objects in the main text. First, the entropy Hessian K is defined as the relative-entropy Hessian of a finite-screen state manifold. Second, the effective stress $E_{\mu\nu}^{\text{scr}}$ is defined as the minimum-norm Ward completion of a leading unimodular optical-stress seed. The appendix is a conditional closure: it does not yet construct the microscopic screen Hilbert space, but it specifies what such a construction must reproduce.

A.1 Finite-screen state manifold

Let q^I denote local optical deformation coordinates on the tangent space of the finite optical screen,

$$q^I = (A, S_1, S_2, \Omega), \quad I = 1, \dots, 4. \quad (79)$$

Here A is the area/trace channel, S_1, S_2 are the two symmetric shear channels, and Ω is the twist/holonomy channel. Let

$$\rho_S(q) \quad (80)$$

be the finite-screen state associated with a small optical deformation q , and let $\rho_S(0)$ be the locally isotropic reference screen state.

Definition 3 (Screen relative-entropy distinguishability). *The distinguishability cost of the deformation q is*

$$D_{\text{scr}}(q) = S_{\text{rel}}(\rho_S(q) \parallel \rho_S(0)), \quad (81)$$

where $S_{\text{rel}}(\rho \parallel \sigma)$ denotes relative entropy.

For a smooth state family, the expansion near $q = 0$ is

$$D_{\text{scr}}(q) = \frac{1}{2} K_{IJ} q^I q^J + O(q^3). \quad (82)$$

Thus the screen entropy Hessian is

$$K_{IJ} = \left. \frac{\partial^2}{\partial q^I \partial q^J} S_{\text{rel}}(\rho_S(q) \parallel \rho_S(0)) \right|_{q=0}. \quad (83)$$

This identifies K with the information-geometric metric of the local screen state manifold. In this language, the coefficients K_A, K_S, K_Ω in the main text are not arbitrary response weights. They are eigenvalues of the relative-entropy Hessian on irreducible optical sectors.

A.2 $SO(2)$ block structure and compliance fraction

The local screen rotation group decomposes the optical deformation algebra as

$$\mathfrak{gl}(2) = A(1) \oplus S(2) \oplus \Omega(1), \quad (84)$$

with representation labels

$$A : \mathbf{0}^+, \quad S : \mathbf{2}, \quad \Omega : \mathbf{0}^-. \quad (85)$$

If $\rho_S(0)$ is isotropic and parity respecting, different $SO(2)$ sectors do not mix at quadratic order. Therefore

$$K = K_A P_A + K_S P_S + K_\Omega P_\Omega, \quad (86)$$

where P_A, P_S, P_Ω are the corresponding orthogonal projectors. The compliance matrix is

$$C = K^{-1} = K_A^{-1} P_A + K_S^{-1} P_S + K_\Omega^{-1} P_\Omega. \quad (87)$$

The Weyl-active unimodular projector is

$$P_U = P_S + P_\Omega. \quad (88)$$

Thus

$$\kappa_{BW} = \frac{\text{Tr}(P_U K^{-1})}{\text{Tr}(K^{-1})} = \frac{2K_S^{-1} + K_\Omega^{-1}}{K_A^{-1} + 2K_S^{-1} + K_\Omega^{-1}}. \quad (89)$$

Because K is a positive relative-entropy Hessian on the stable branch, $C = K^{-1}$ is positive semidefinite on the responding subspace. Hence

$$0 \leq \kappa_{BW} \leq 1. \quad (90)$$

At the isotropic fixed point $K_A = K_S = K_\Omega$, one obtains $\kappa_{BW} = 3/4$. If the twist/holonomy channel is projected out, $K_\Omega^{-1} = 0$, and $K_A = K_S$, one obtains $\kappa_{BW} = 2/3$.

A.3 Horizon occupancy as bulk distinguishability pressure

The main text defines

$$\mathcal{O}_H(a) = \frac{C_{\text{used}}(a)}{C_H(a)}, \quad C_H(a) = \frac{A_H(a)}{4\ell_P^2}. \quad (91)$$

A microscopic completion must define C_{used} . The information-geometric proposal is that C_{used} is horizon-accessible bulk distinguishability demand, not ordinary thermodynamic entropy:

$$C_{\text{used}}(a) = S_{\text{rel}}(\rho_{\text{bulk}}^H(a) \| \rho_{\text{coarse}}^H(a)). \quad (92)$$

Here $\rho_{\text{bulk}}^H(a)$ is the horizon-accessible bulk state and $\rho_{\text{coarse}}^H(a)$ is the corresponding coarse FRW or screen-compressed reference state. The source for the horizon residue is then

$$\mathcal{S}_H(N) = -\frac{d}{dN} \ln \left[\frac{S_{\text{rel}}(\rho_{\text{bulk}}^H(a) \| \rho_{\text{coarse}}^H(a))}{A_H/(4\ell_P^2)} \right], \quad N = \ln a. \quad (93)$$

This definition makes the G1 source a finite-screen distinguishability pressure: the bulk state demands a certain distinguishability budget, while the horizon supplies only finite screen capacity.

A.4 Passive retarded memory

The retarded memory field is

$$\mathcal{R}_H(N) = \int_{-\infty}^N dN' G_H^{\text{ret}}(N, N') \mathcal{S}_H(N'). \quad (94)$$

The admissible kernels are restricted by three conditions.

1. **Causality:**

$$G_H^{\text{ret}}(N, N') = 0 \quad \text{for } N' > N. \quad (95)$$

2. **Stability:** in a Markov reduction,

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = \mathcal{S}_H, \quad \Gamma_H > 0, \quad (96)$$

or, more generally, the poles of the response lie in the stable half-plane.

3. **Passivity / entropy positivity:** for the physical sign convention,

$$\int dN \mathcal{S}_H(N) \mathcal{R}_H(N) \geq 0. \quad (97)$$

Equivalently, in a frequency-domain description the dissipative part of the response has the sign required for positive entropy production.

A full open-system treatment should also include a stochastic screen-memory fluctuation,

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = \mathcal{S}_H + \xi_H, \quad \langle \xi_H(N) \xi_H(N') \rangle = N_H(N, N'), \quad (98)$$

with $N_H \succeq 0$. The deterministic reduction used in G1DE-M_{3/4} concerns the mean response $\langle \mathcal{R}_H \rangle$.

A.5 Minimum Ward completion

Let the leading optical seed be

$$Q_{ij}^{\text{TF}} = -\mathcal{A}(a, k) \kappa_{BW} \mathcal{R}_H(a) \left(\hat{k}_i \hat{k}_j - \frac{1}{3} \delta_{ij} \right) + O(\mathcal{R}_H^2). \quad (99)$$

This seed is not yet a covariantly closed effective stress. Define the physical screen residual as the minimum-norm Ward completion:

$$\boxed{E_{\mu\nu}^{\text{scr}} = \arg \min_{E_{\mu\nu}} \|E\|_{\mathcal{G}}^2} \quad (100)$$

subject to

$$E_{ij}^{\text{TF}} = Q_{ij}^{\text{TF}}, \quad \nabla^\mu E_{\mu\nu} = J_\nu^H, \quad E_{\mu\nu} = E_{\nu\mu}. \quad (101)$$

The norm $\|\cdot\|_{\mathcal{G}}$ may be chosen as a DeWitt-like quadratic norm or a screen-response norm. Different reasonable norms can change higher-order details, but the subhorizon scaling follows from the Ward constraint and the fact that the seed is spatial and traceless.

Theorem 2 (Minimum Ward completion hierarchy). *Assume $Q_{ij}^{\text{TF}} = O(\mathcal{R}_H)$ and that J_ν^H is horizon-scale. In the subhorizon linear regime $k \gg \mathcal{H}$, the minimum Ward completion obeys*

$$E_{ij,\text{TF}}^{\text{scr}} = O(\mathcal{R}_H), \quad (102)$$

$$E_{0i}^{\text{scr}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right), \quad (103)$$

$$E_{00}^{\text{scr}} = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (104)$$

Sketch. The spatial divergence of Q_{ij}^{TF} contributes a term of order $k\mathcal{R}_H$ to the Ward equation. A momentum component E_{0i} with one time/expansion scale $\mathcal{H}$ balances this term at order $\mathcal{R}_H \mathcal{H}/k$. Applying the constraint once more gives a density-like component suppressed by another factor $\mathcal{H}/k$. Nonlinear self-coupling of the screen residual contributes $O(\mathcal{R}_H^2)$. $\square$

Consequently the Weyl/lensing response is first order in $\mathcal{R}_H$, while the Ricci-density response is suppressed:

$$\Sigma - 1 = -\kappa_{BW} \mathcal{R}_H + O(\mathcal{R}_H^2), \quad (105)$$

$$\mu - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (106)$$

A.6 Gauge-invariant scalar-sector route

A production cosmology implementation should not rely on a local Minkowski projector. Instead, introduce gauge-invariant effective screen variables

$$\pi_E^{\text{GI}}, \quad q_E^{\text{GI}}, \quad \delta E_{\text{GI}}, \quad (107)$$

corresponding to anisotropic stress, momentum, and density-like screen components. The G1 constitutive input is

$$\pi_E^{\text{GI}} = O(\mathcal{R}_H). \quad (108)$$

The gauge-invariant Ward completion should then determine

$$q_E^{\text{GI}} = O\left(\mathcal{R}_H \frac{\mathcal{H}}{k}\right), \quad (109)$$

$$\delta E_{\text{GI}} = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (110)$$

This is the production-ready version of the minimum Ward completion theorem: build the completion directly in the scalar perturbation variables used by a Boltzmann or likelihood pipeline.

A.7 Twist/holonomy as covariance sector

The twist/holonomy coordinate Ω is part of the full $GL(2)$ optical capacity branch, but this does not require a nonzero first-order scalar rotation mean. The correct interpretation is

$$\langle \Omega \rangle = 0, \quad \langle \Omega^2 \rangle = O(\mathcal{R}_H^2) \quad (111)$$

for scalar cosmology. Thus Ω contributes to the screen covariance and the compliance fraction, while direct lensing curl or rotation observables are expected to be second-order or covariance-level tests. The branch distinction is

$$GL(2): \quad A(1) + S(2) + \Omega(1) \Rightarrow \kappa_{BW} = 3/4, \quad (112)$$

$$SPD(2): \quad A(1) + S(2) \Rightarrow \kappa_{BW} = 2/3. \quad (113)$$

Absence of a first-order curl mean therefore does not falsify the 3/4 branch; a measured demotion of κ_{BW} toward 2/3 would.

A.8 Appendix theorem

Combining the definitions above yields the compact closure statement:

Theorem 3 (Relative-entropy screen Hessian and minimum Ward completion). *Let finite screen states form a smooth family $\rho_S(q)$ over local optical deformations $q \in T_I GL(2)$, and let K be the relative-entropy Hessian in Eq. (83). Let the horizon occupancy source be Eq. (93), and let $\mathcal{R}_H$ be its causal, stable, passive retarded response. Let $E_{\mu\nu}^{\text{scr}}$ be the minimum Ward completion of the unimodular optical seed. Then*

$$\kappa_{BW} = \frac{\text{Tr}(P_U K^{-1})}{\text{Tr}(K^{-1})}, \quad (114)$$

$$\Sigma - 1 = -\kappa_{BW} \mathcal{R}_H + O(\mathcal{R}_H^2), \quad (115)$$

$$\mu - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (116)$$

If $K \propto I$ on $T_I GL(2)$, then $\kappa_{BW} = 3/4$. If the twist/holonomy sector is demoted, then $\kappa_{BW} = 2/3$.

This theorem is conditional. It specifies the mathematical target that a microscopic FDS-to-G1 derivation must realize: a finite screen state manifold with relative-entropy Hessian, a horizon distinguishability occupancy current, a passive retarded memory kernel, and a minimum Ward completion of the unimodular optical seed.

References

- [1] J. D. Bekenstein, “Black holes and entropy,” Phys. Rev. D 7, 2333 (1973).
- [2] S. W. Hawking, “Particle creation by black holes,” Commun. Math. Phys. 43, 199 (1975).
- [3] A. Raychaudhuri, “Relativistic cosmology. I,” Phys. Rev. 98, 1123 (1955).

- [4] R. Sachs, “Gravitational waves in general relativity. VI. The outgoing radiation condition,” *Proc. R. Soc. Lond. A* 264, 309 (1961).
- [5] R. M. Wald, *General Relativity*, University of Chicago Press (1984).
- [6] P. Schneider, J. Ehlers, and E. E. Falco, *Gravitational Lenses*, Springer (1992).
- [7] M. Bartelmann and P. Schneider, “Weak gravitational lensing,” *Phys. Rep.* 340, 291 (2001).
- [8] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Phys. Rev. Lett.* 75, 1260 (1995).
- [9] R. P. Feynman and F. L. Vernon, “The theory of a general quantum system interacting with a linear dissipative system,” *Ann. Phys.* 24, 118 (1963).
- [10] J. Schwinger, “Brownian motion of a quantum oscillator,” *J. Math. Phys.* 2, 407 (1961).
- [11] L. V. Keldysh, “Diagram technique for nonequilibrium processes,” *Sov. Phys. JETP* 20, 1018 (1965).
- [12] S. Kullback and R. A. Leibler, “On information and sufficiency,” *Ann. Math. Statist.* 22, 79 (1951).
- [13] R. A. Fisher, “Theory of statistical estimation,” *Proc. Cambridge Philos. Soc.* 22, 700 (1925).
- [14] D. Petz, “Monotone metrics on matrix spaces,” *Linear Algebra Appl.* 244, 81 (1996).
- [15] L. Onsager, “Reciprocal relations in irreversible processes. I,” *Phys. Rev.* 37, 405 (1931).
- [16] J. M. Bardeen, “Gauge-invariant cosmological perturbations,” *Phys. Rev. D* 22, 1882 (1980).
- [17] C.-P. Ma and E. Bertschinger, “Cosmological perturbation theory in the synchronous and conformal Newtonian gauges,” *Astrophys. J.* 455, 7 (1995).
- [18] P. Zhang, M. Liguori, R. Bean, and S. Dodelson, “Probing gravity at cosmological scales by measurements which test the relationship between gravitational lensing and matter overdensity,” *Phys. Rev. Lett.* 99, 141302 (2007).
- [19] E. Bellini and I. Sawicki, “Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity,” *JCAP* 07, 050 (2014).
- [20] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, Zenodo (2026), doi:10.5281/zenodo.20382013.
- [21] Y. Wu, *The 3/4 Projection Law in Finite Screen Gravity: MaxEnt Optical-Stress Closure, Ricci-Silent Residuals, and the Weyl-Locked Dark Sector*, technical note (2026).
- [22] Y. Wu, *Non-Equilibrium Screen Entropy and the Optical-Stress Origin of the G1 Dark Sector*, technical note (2026).
- [23] Y. Wu, *Ward-Closed Finite-Screen Dark Sector: Retarded Optical Stress, Horizon Memory, and G1 Null Tests*, technical note (2026).
- [24] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background-Weyl Residual*, Zenodo / manuscript (2026).

Weyl-Normalized Ward Completion and Identifiability Rules for G1DE

Finite-Screen Memory as a Transfer Operator, Not an Independent Dark Stress Source

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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Abstract

This implementation note fixes the production-level normalization and identifiability rules for the G1DE screen-influence reduction. The central point is that G1DE does not add an independently normalized dark stress amplitude. Instead, it applies a finite-screen fractional optical projection correction to the general-relativistic Weyl source. Defining

$$\mathcal{W}(a, k) \equiv -k^2(\Phi + \Psi), \quad \mathcal{W}_{\text{GR}}(a, k) = 8\pi G a^2 \rho \Delta,$$

the Weyl-normalized implementation is

$$\mathcal{W}_{\text{G1}}(a, k) = [1 - \kappa_{BW} \mathcal{R}_H(a)] \mathcal{W}_{\text{GR}}(a, k) + O(\mathcal{R}_H^2).$$

Thus

$$\Sigma(a, k) - 1 = -\kappa_{BW} \mathcal{R}_H(a) + O(\mathcal{R}_H^2),$$

while the Newtonian channel receives only Ward-suppressed leakage,

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

The note makes explicit that the optical normalization factor $A(a, k)$ is not a fitted response function; it is fixed by the Weyl-source convention. It also introduces a memory-shape normalization $\mathcal{R}_H(a) = R_0 \hat{R}_H(a)$, $\hat{R}_H(1) = 1$, so that the projection coefficient κ_{BW} , the present-day residual amplitude R_0 , and the redshift-memory shape $\hat{R}_H(a)$ are identifiable. The locked model G1DE-M_{3/4} is recovered by $\kappa_{BW} = 3/4$, $R_0 = 3 - s$, $\hat{R}_H = X_b$, and $\mu = 1$.

1 Purpose and scope

This note is a production-interface companion to the G1 finite-screen dark-sector sequence. Its task is not to rederive the optical projection coefficient κ_{BW} , the entropy-Hessian origin of the screen influence, or the retarded horizon-memory kernel. Upstream of this production convention, D0 identifies $\mathcal{R}_H$ as a passive horizon-scale remnant of incomplete modular anomaly inflow, while D1 interprets the same quantity as a finite-access gluing-lag residual. Those ingredients belong to the companion theory notes on the 3/4 projection law, non-equilibrium screen entropy, and the screen influence functional. The purpose here is narrower and more implementation-facing:

1. fix the normalization map between the optical-stress seed and the standard phenomenological Weyl-source convention;
2. state which quantities may be fitted in G1DE production likelihoods and which quantities must not be fitted;

3. separate projection coefficient, residual amplitude, and redshift-memory shape;
4. specify Ward-suppressed leakage into the Newtonian growth channel;
5. define null tests and demotion paths for production analyses.

D5 owns: (1) Ward/Bianchi closure; (2) Weyl normalization; (3) no-free- $A(a, k)$ enforcement; (4) identifiability rules; (5) production likelihood conventions.

The guiding statement is:

Implementation thesis. G1DE is not an independently normalized dark-stress model. It is a Weyl-normalized, Ward-disciplined finite-screen transfer class. Its primary action is a fractional finite-screen correction to the GR Weyl source.

This distinction matters because an implementation that samples an arbitrary optical amplitude $A(a, k)$ no longer tests the G1DE projection-locking claim. It tests a free dark-stress envelope instead.

Lemma 1 (D5.1 – Ward-Suppressed Ricci Leakage). *Let the finite-screen residual enter the data-facing branch through the Weyl-normalized transfer*

$$W_{G1}(a, k) = [1 - \kappa_{\text{BW}} R_H(a)] W_{GR}(a, k) + O(R_H^2).$$

Then the leading-order Weyl response is

$$\Sigma(a, k) - 1 = -\kappa_{\text{BW}} R_H(a) + O(R_H^2),$$

while the Newtonian/Ricci growth channel receives only Ward-suppressed leakage,

$$\mu(a, k) - 1 = O(R_H^2) + O\left(R_H \frac{H^2}{k^2}\right).$$

Therefore, on linear subhorizon survey scales and to first order in the horizon response, the locked branch is

$$\mu(a, k) = 1.$$

D10 references this lemma as the $\mu = 1$ justification in the minimal Weyl branch.

If production RSD, full-shape, or $3 \times 2\text{pt}$ data require

$$|\mu - 1| \sim |\Sigma - 1|,$$

then the Ward-suppressed Ricci-leakage branch is demoted. The model may move to a finite-screen residual with a density/Ricci component, but it is no longer the $M_{3/4}$ branch.

1.1 Fitted vs Non-Fitted Quantities

2 Standard modified-growth convention

We use the scalar perturbation convention

$$-k^2 \Psi = 4\pi G a^2 \mu(a, k) \rho \Delta, \tag{1}$$

$$-k^2 (\Phi + \Psi) = 8\pi G a^2 \Sigma(a, k) \rho \Delta. \tag{2}$$

Here Δ is the comoving matter density contrast used in the likelihood implementation. We define the Weyl source

$$\mathcal{W}(a, k) \equiv -k^2 (\Phi + \Psi), \tag{3}$$

Quantity	Status	Reason
s	fitted	controls $R_0 = 3 - s$ background-residual amplitude
$\Omega_m, q_{\text{BAO}}, \sigma_{8,0}$	fitted/nuisance	standard data-facing parameters
κ	fixed in $M_{3/4}$; fitted only in M_κ control	tests isotropic optical projection
$R_{bH}(a)$	fixed shape in $M_{3/4}$; tested against constant- Σ	output memory shape
$\mu(a, k)$	fixed to 1 in $M_{3/4}$	Ward-suppressed Ricci leakage
$A(a, k)$	forbidden	independent optical amplitude makes the model generic dark stress
K_H, J_H	not fitted	source-kernel factorization belongs to D6/D7 realization layer

Table 1: Fitted and non-fitted quantities in the G1DE production convention.

and the GR Weyl source

$$\mathcal{W}_{\text{GR}}(a, k) \equiv 8\pi G a^2 \rho \Delta. \quad (4)$$

In GR,

$$\mathcal{W} = \mathcal{W}_{\text{GR}}, \quad \Sigma = 1. \quad (5)$$

A production implementation of G1DE should modify the transfer from $\rho \Delta$ to Weyl curvature by a finite-screen factor. It should not introduce a new freely normalized dark-stress source.

3 Three production principles

3.1 Weyl Normalization Principle

Principle 1 (Weyl Normalization). *The finite-screen residual is normalized as a fractional correction to the GR Weyl source. To first order in the screen residual,*

$$\boxed{\mathcal{W}_{\text{G1}}(a, k) = [1 - \kappa \mathcal{R}_H(a)] \mathcal{W}_{\text{GR}}(a, k) + O(\mathcal{R}_H^2).} \quad (6)$$

Here κ is the optical projection/compliance coefficient. In the isotropic finite-screen branch $\kappa = \kappa_{\text{BW}} = 3/4$.

Equation (6) immediately gives

$$\boxed{\Sigma(a, k) - 1 = -\kappa \mathcal{R}_H(a) + O(\mathcal{R}_H^2).} \quad (7)$$

Thus the symbol $A(a, k)$, when used to represent the amplitude of an optical-stress seed, is fixed by the Weyl-source convention:

$$A(a, k) \longleftrightarrow \mathcal{W}_{\text{GR}}(a, k) = 8\pi G a^2 \rho \Delta. \quad (8)$$

It is not an additional physical response function.

3.2 No Free Optical-Amplitude Principle

Principle 2 (No Free Optical-Amplitude). *The normalization $A(a, k)$ must not be sampled as an independent nuisance parameter or response function in a G1DE production likelihood. It is fixed by the Weyl-source convention (4). Sampling $A(a, k)$ turns the model into a free dark-stress source and removes the Weyl-normalized projection-locking claim being tested.*

Equivalently, if Θ denotes the fitted parameter set of a production model, then

$$\boxed{A(a, k) \notin \Theta_{\text{G1DE}}.} \quad (9)$$

A model that allows an arbitrary function multiplying the optical seed should be labeled as an adversarial free-response or dark-stress envelope, not as a primary G1DE branch.

The Weyl-normalized production convention $W_{G1} = (1 - \kappa R_H)W_{GR}$ is admissible only if $A(a, k)$ is not sampled independently. If data require free $A(a, k)$, D5 must force demotion to generic dark-stress.

3.3 Memory-Shape Normalization Principle

A second identifiability issue concerns the product

$$\Sigma - 1 = -\kappa \mathcal{R}_H. \quad (10)$$

If the overall amplitude of $\mathcal{R}_H(a)$ is not fixed, then κ and $\mathcal{R}_H$ are rescaling-degenerate. Therefore the memory response should be factorized as

$$\boxed{\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \hat{R}_H(1) = 1.} \quad (11)$$

Here $R_0 = \mathcal{R}_H(1)$ is the present-day finite-screen residual amplitude, while $\hat{R}_H(a)$ is a normalized redshift-memory shape.

For the current G1DE-M reduction,

$$R_0 = 3 - s, \quad \hat{R}_H(a) = X_b(a), \quad X_b(1) = 1. \quad (12)$$

Therefore

$$\boxed{\Sigma(a, k) - 1 = -\kappa(3 - s)X_b(a) + O((3 - s)^2).} \quad (13)$$

For the locked branch,

$$\boxed{\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)X_b(a) + O((3 - s)^2).} \quad (14)$$

Remark 1 (Identifiability of κ). *The free- κ branch is interpretable only after a memory-shape normalization has been imposed. With $X_b(1) = 1$, κ is a projection/compliance coefficient. Without this normalization, κ can be absorbed into the arbitrary amplitude of $\mathcal{R}_H$.*

Remark 2 (Output response versus source-kernel lift). *The normalized memory shape $\hat{R}_H(a)$ is a data-facing output response. A deeper realization may factor it as $\mathcal{R}_H = K_H * J_H$, but this factorization is not unique without a lift choice. D6 formulates this ambiguity as a memory quotient and proposes a maximum-caliber horizontal lift. Therefore the production likelihood should sample R_0 and $\hat{R}_H(a)$, not a separately normalized K_H , J_H , or optical amplitude $A(a, k)$.*

4 Factorized G1DE production theorem

The previous principles can be summarized as a production-level factorization theorem.

Theorem 1 (Weyl-normalized G1DE factorization). *Assume:*

1. *the finite-screen residual is normalized as a fractional correction to the GR Weyl source;*
2. *the optical projection coefficient is κ ;*
3. *the screen residual factorizes as $\mathcal{R}_H(a) = R_0 \hat{R}_H(a)$, $\hat{R}_H(1) = 1$;*

4. Ricci leakage is controlled by the minimum Ward completion of the optical seed.

Then, on subhorizon linear scales,

$$\boxed{\Sigma(a, k) - 1 = -\kappa R_0 \hat{R}_H(a) + O(R_0^2),} \quad (15)$$

while

$$\boxed{\mu(a, k) - 1 = O(R_0^2) + O\left(R_0 \hat{R}_H(a) \frac{\mathcal{H}^2}{k^2}\right).} \quad (16)$$

For the locked isotropic-compliance branch, $\kappa = 3/4$. For the current low-redshift G1DE-M reduction, $R_0 = 3 - s$ and $\hat{R}_H = X_b$.

This theorem separates the three quantities that are otherwise easy to confuse:

$$\boxed{\Sigma - 1 = - \underbrace{\kappa}_{\text{projection screen}} \underbrace{R_0}_{\text{amplitude}} \underbrace{\hat{R}_H(a)}_{\text{memory shape}} + O(R_0^2).} \quad (17)$$

The Weyl-source normalization is not a fourth physical factor; it is the convention that identifies the unit optical seed with a unit fractional correction to $\mathcal{W}_{\text{GR}}$.

5 Interpretation of s , amplitude, and sign

Under the current low-redshift reduction,

$$R_0 = 3 - s. \quad (18)$$

Therefore s is not merely a symbolic filling dimension in the production likelihood. It is the present-day normalized residual amplitude written in the G1DE-M convention:

$$\boxed{s = 3 - R_0.} \quad (19)$$

The sign chain is then

$$\boxed{s < 3 \Rightarrow R_0 > 0 \Rightarrow \mathcal{R}_H > 0 \Rightarrow \Sigma - 1 < 0 \quad (\kappa > 0).} \quad (20)$$

For $\kappa = 3/4$,

$$\boxed{\Sigma_0 = -\frac{3}{4}(3 - s).} \quad (21)$$

This present-day amplitude relation is independent of detailed redshift-shape discrimination. Hence amplitude support and memory-shape support should be interpreted separately in model comparison.

6 Why this is not an independent dark-stress model

A conventional dark-fluid or stress-source model begins with additional components in an effective stress tensor and asks how they change the potentials. G1DE, in the production convention advocated here, begins with a finite-screen correction to the Weyl transfer:

$$\mathcal{W}_{\text{GR}} \longrightarrow (1 - \kappa \mathcal{R}_H) \mathcal{W}_{\text{GR}}. \quad (22)$$

The effective screen stress tensor, when used, is a representation of this Weyl-transfer correction under minimum Ward completion. It is not a freely normalized physical fluid.

Schematically,

$$\delta\mathcal{W}_{\text{scr}} = -\kappa\mathcal{R}_H\mathcal{W}_{\text{GR}}, \quad (23)$$

and the covariant representation is

$$E_{\mu\nu}^{\text{scr}} = \mathcal{P}_W^{\text{min}} [-\kappa\mathcal{R}_H\mathcal{W}_{\text{GR}}]_{\mu\nu}, \quad (24)$$

where $\mathcal{P}_W^{\text{min}}$ denotes the minimum Ward completion. The density and momentum components of this representation are therefore not independent first-order sources. They are Ward-suppressed completion terms.

7 Gauge-invariant Ward-completed interface

7.1 Weyl-normalized anisotropic seed

For a production-level scalar-sector implementation, introduce effective gauge-invariant screen variables

$$\pi_{\text{scr}}^{\text{GI}}, \quad q_{\text{scr}}^{\text{GI}}, \quad \delta\rho_{\text{scr}}^{\text{GI}}. \quad (25)$$

The leading G1 screen seed is anisotropic/Weyl-active. In the Weyl-source convention one may write schematically

$$\boxed{k^2\pi_{\text{scr}}^{\text{GI}} = -\kappa\mathcal{R}_H(a) (8\pi G a^2 \rho \Delta)}. \quad (26)$$

This equation fixes the Weyl-source convention. Code-level implementations must translate it into the sign and anisotropic-stress normalization used by the chosen Boltzmann solver. Different codes may differ by signs or numerical factors in their definitions of π , σ , or $\Phi - \Psi$.

7.2 Ward-completed leakage hierarchy

The Ward-completed momentum and density components obey the subhorizon hierarchy

$$q_{\text{scr}}^{\text{GI}} = O\left(\pi_{\text{scr}}^{\text{GI}} \frac{\mathcal{H}}{k}\right), \quad (27)$$

$$\delta\rho_{\text{scr}}^{\text{GI}} = O\left(\pi_{\text{scr}}^{\text{GI}} \frac{\mathcal{H}^2}{k^2}\right) + O(\mathcal{R}_H^2). \quad (28)$$

Consequently,

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right), \quad (29)$$

while

$$\Sigma(a, k) - 1 = -\kappa\mathcal{R}_H(a) + O(\mathcal{R}_H^2). \quad (30)$$

Thus the core data-facing signature is not simply $\Sigma < 1$, but the channel split

$$\boxed{|\mu - 1| \ll |\Sigma - 1| \quad \text{on subhorizon linear scales.}} \quad (31)$$

8 Allowed and forbidden production branches

The following registry should be used to keep production likelihoods interpretable.

Branch	Status	Response	Interpretation
Locked $M_{3/4}$	Primary branch	$\Sigma = 1 - \frac{3}{4}R_0\hat{R}_H, \mu = 1$	Isotropic compliance, normalized memory shape, no leading Ricci leakage.
Free- κ	Compliance control	$\Sigma = 1 - \kappa R_0\hat{R}_H, \mu = 1$	Tests whether the 3/4 isotropic-compliance value is selected. Identifiable only if $\hat{R}_H(1) = 1$.
Constant- Σ	Memory-shape control	$\Sigma = 1 + \Sigma_c, \mu = 1$	Tests whether the redshift-memory shape is resolved beyond a constant Weyl offset.
Binned- $\Sigma(z)$	Shape diagnostic	$\Sigma = 1 + \Sigma(z), \mu = 1$	Adversarial test of redshift-shape lock; nuisance-entangled low- χ^2 basins indicate controlled stress, not clean demotion.
Ward leakage	Ricci-leakage control	$\Sigma = 1 - \kappa R_0\hat{R}_H, \mu = 1 + \nu_2 R_0^2 \hat{R}_H^2 + \nu_H R_0 \hat{R}_H \mathcal{H}^2 / k^2$	Tests whether the Newtonian channel requires Ward-disciplined leakage.
Free shape	Shape reconstruction	$\hat{R}_H(a) = \sum_i c_i B_i(a), \hat{R}_H(1) = 1$	Tests memory shape without losing amplitude identifiability.
G1DE-2	Adversarial envelope	Flexible response	Tests whether data require freedom beyond the Weyl-normalized branch.
Free $A(a, k)$	Forbidden / not G1DE	Arbitrary amplitude map	Turns the model into a free dark-stress source and destroys Weyl normalization.

Implementation warning. The normalization $A(a, k)$ must not be sampled as an independent nuisance parameter or response function. It is fixed by the Weyl-source convention $\mathcal{W}_{\text{GR}} = 8\pi G a^2 \rho \Delta$. Sampling $A(a, k)$ converts the G1DE reduction into a free dark-stress model.

9 Identifiability conditions

A G1DE production fit is identifiable only if the following conditions are satisfied:

1. **Weyl normalization is fixed.** The optical seed is normalized relative to $\mathcal{W}_{\text{GR}} = 8\pi G a^2 \rho \Delta$.
2. **Memory shape is normalized.** Write $\mathcal{R}_H(a) = R_0 \hat{R}_H(a)$, with $\hat{R}_H(1) = 1$ or an equivalent fixed normalization.
3. **κ is varied only relative to that fixed shape.** Otherwise κ and the amplitude of $\mathcal{R}_H$ are degenerate.
4. **$A(a, k)$ is never sampled.** It is a convention map, not a physical response.

5. **Free-shape reconstructions must impose a normalization.** If $\mathcal{R}_H(a)$ is reconstructed in bins or basis functions, either κ is fixed or a shape normalization is imposed.

These conditions separate amplitude evidence from shape evidence. The present-day amplitude is

$$\Sigma_0 - 1 = -\kappa R_0. \quad (32)$$

The locked branch predicts

$$\Sigma_0 - 1 = -\frac{3}{4}(3 - s). \quad (33)$$

This relation can be tested even when redshift-shape discrimination remains weak.

10 Production null tests

10.1 Weyl-transfer residual

Define

$$\boxed{\mathcal{N}_\Sigma(a, k) = \Sigma(a, k) - 1 + \kappa \mathcal{R}_H(a).} \quad (34)$$

For the primary G1DE branch,

$$\mathcal{N}_\Sigma = O(\mathcal{R}_H^2). \quad (35)$$

In the locked branch one sets $\kappa = 3/4$.

10.2 Ward-leakage residual

Define

$$\boxed{\mathcal{N}_\mu(a, k) = \mu(a, k) - 1.} \quad (36)$$

G1DE predicts

$$\mathcal{N}_\mu = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (37)$$

The ratio diagnostic

$$\mathcal{C}_{G1}(a, k) = \frac{\mu(a, k) - 1}{\Sigma(a, k) - 1} \quad (38)$$

may be useful for interpretation, but additive residuals $\mathcal{N}_\Sigma$ and $\mathcal{N}_\mu$ are numerically safer in likelihoods because the denominator of $\mathcal{C}_{G1}$ can become small.

11 G1DE-2 as an adversarial envelope

G1DE-2 should be interpreted as an adversarial envelope around the Weyl-normalized transfer branch. It asks whether the data require response freedom beyond the constrained structure

$$\Sigma - 1 = -\kappa R_0 \hat{R}_H, \quad \mu - 1 = O(R_0^2, R_0 \mathcal{H}^2 / k^2). \quad (39)$$

If $M_{3/4}$ beats G1DE-2 in evidence, then projection locking and Ward discipline are evidence-selected over flexible response freedom. If they tie, the free-response envelope is compatible but unnecessary. If G1DE-2 wins strongly, the broader finite-screen hypothesis may survive, but the Weyl-normalized locked reduction is incomplete.

G1DE-2 should not be allowed to hide a free $A(a, k)$ unless it is explicitly labeled as a non-G1DE dark-stress envelope. Otherwise, a comparison with $M_{3/4}$ would not test projection locking fairly. Binned- $\Sigma(z)$ is also used as an adversarial shape-diagnostic control: it tests whether flexible redshift-dependent Weyl responses outperform the locked $R_{bH}(a)$ shape, without being a fully admissible G1DE branch.

12 Demotion and falsification hierarchy

Because the production response is factorized, different failures have different meanings.

Outcome	Interpretation
Constant- Σ favored over $X_b(a)$	Memory-shape closure is demoted or unresolved; the Weyl-amplitude branch may survive.
Free- κ favored with $0 < \kappa < 1$	Isotropic 3/4 compliance branch is demoted to a weighted-compliance branch.
Free- κ favored with $\kappa \notin [0, 1]$	Positive-compliance screen interpretation fails.
Large subhorizon $ \mu - 1 \sim \Sigma - 1 $	Ward-closed optical-stress branch fails.
Free $A(a, k)$ required	Weyl normalization fails; the model is not a G1DE production branch.
G1DE-2 strongly favored	The locked reduction is incomplete; diagnose whether the failure is compliance, shape, leakage, or normalization.

This demotion hierarchy prevents overclaiming. The amplitude relation $\Sigma_0 = -\kappa R_0$ is lower-level than the detailed redshift-memory shape $\hat{R}_H(a)$. Evidence for amplitude suppression can therefore be stronger than evidence for a particular memory kernel.

13 Implementation recipe

13.1 Minimal locked implementation

Use

$$R_0 = 3 - s, \quad \hat{R}_H(a) = X_b(a), \quad X_b(1) = 1, \quad \mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \mu(a, k) = 1, \quad \Sigma(a, k) = 1 - \frac{3}{4} R_0 \hat{R}_H(a). \quad (40)$$

The fitted parameter is s . Neither κ nor $A(a, k)$ is sampled.

13.2 Free- κ control

Use

$$\mathcal{R}_H(a) = (3 - s)X_b(a), \quad X_b(1) = 1, \quad \mu(a, k) = 1, \quad \Sigma(a, k) = 1 - \kappa(3 - s)X_b(a). \quad (41)$$

The parameters s and κ are sampled. This control tests the isotropic-compliance value $\kappa = 3/4$.

13.3 Constant- Σ control

Use

$$\mu(a, k) = 1, \quad \Sigma(a, k) = 1 + \Sigma_c. \quad (42)$$

This is a memory-shape control. It asks whether the data require the redshift dependence $X_b(a)$, rather than only a constant Weyl offset.

13.4 Ward-leakage extension

Use

$$\mathcal{R}_H(a) = R_0 \hat{R}_H(a), \quad \Sigma(a, k) = 1 - \kappa R_0 \hat{R}_H(a), \quad \mu(a, k) = 1 + \nu_2 R_0^2 \hat{R}_H^2(a) + \nu_H R_0 \hat{R}_H(a) \frac{\mathcal{H}^2}{k^2}. \quad (43)$$

The coefficients ν_2 and ν_H test whether the Newtonian growth channel requires Ward-disciplined leakage. G1DE predicts that these terms are unnecessary in the subhorizon exact-pilot limit.

13.5 Free-shape reconstruction

If one reconstructs the memory shape,

$$\hat{R}_H(a) = \sum_i c_i B_i(a), \quad (44)$$

one must impose a normalization such as

$$\hat{R}_H(1) = 1. \quad (45)$$

Otherwise κ , R_0 , and the shape coefficients are not separately identifiable.

14 Relation to evidence interpretation

The factorized production doctrine clarifies how evidence comparisons should be read:

- $M_{3/4} > \text{CPL}$ means that a constrained Weyl-transfer response is favored over a background-only equation-of-state deformation.
- $M_{3/4} > M_\kappa$ means that exact 3/4 locking is not penalized and may be selected under the fixed memory-shape normalization.
- $M_{3/4} > \text{constant-}\Sigma$ means that the proposed memory shape helps relative to a constant Weyl offset.
- $M_{3/4} > \text{G1DE-2}$ means that the locked finite-screen transfer rule is preferred over an adversarial free-response envelope.

Conversely, failure modes should be mapped to the relevant factor. A failure of $X_b(a)$ is not the same as a failure of $\kappa = 3/4$, and a failure of $\kappa = 3/4$ is not the same as a failure of Weyl normalization or Ward suppression.

In the completed medium-prior 8-seed production-refined audit (G1fit-real), the locked model $M_{3/4}$ ranks first among the tested control hierarchy:

$$M_{3/4} > M_\kappa > \text{const-}\Sigma > \text{G1DE2} > \text{G1DE1} > \text{CPL} > \Lambda\text{CDM}.$$

This hierarchy supports the Weyl-normalized projection-locking interpretation. The amplitude-level lock ($\kappa \sim 3/4$) is selected over free- κ , constant- Σ , and free-response G1DE-2, while the full evidence numbers and prior-sensitivity checks are reported in the G1fit-real data paper.

15 Conclusion

The essential implementation point is that G1DE is not an independently normalized dark-stress model. It is a finite-screen Weyl-transfer correction whose normalization is fixed by the GR Weyl source. The data-facing response is

$$\boxed{\mathcal{W}_{\text{G1}} = (1 - \kappa R_0 \widehat{R}_H) \mathcal{W}_{\text{GR}} + O(R_0^2),} \quad (46)$$

with

$$\boxed{\Sigma - 1 = -\kappa R_0 \widehat{R}_H + O(R_0^2),} \quad (47)$$

and

$$\boxed{\mu - 1 = O(R_0^2) + O\left(R_0 \widehat{R}_H \frac{\mathcal{H}^2}{k^2}\right).} \quad (48)$$

For the locked G1DE-M branch,

$$\boxed{\kappa = \frac{3}{4}, \quad R_0 = 3 - s, \quad \widehat{R}_H = X_b, \quad X_b(1) = 1, \quad \mu = 1.} \quad (49)$$

The normalization $A(a, k)$ is not a parameter. The memory shape must be normalized before κ is interpreted. The Newtonian channel is not independently modified at first order; it receives only Ward-suppressed leakage. These rules define G1DE as a Weyl-normalized, Ward-disciplined finite-screen transfer class.

A Code-facing equation blocks

A.1 Locked response block

$$R_0 = 3 - s, \quad X_b(1) = 1, \quad R(a) = R_0 X_b(a), \quad \mu(a, k) = 1, \quad \Sigma(a, k) = 1 - \frac{3}{4} R(a). \quad (50)$$

A.2 Free- κ response block

$$R_0 = 3 - s, \quad X_b(1) = 1, \quad R(a) = R_0 X_b(a), \quad \mu(a, k) = 1, \quad \Sigma(a, k) = 1 - \kappa R(a). \quad (51)$$

A.3 Ward-leakage response block

$$R(a) = R_0 \widehat{R}_H(a), \quad \widehat{R}_H(1) = 1, \quad \mu(a, k) = 1 + \nu_2 R^2(a) + \nu_H R(a) \frac{\mathcal{H}^2}{k^2}, \quad \Sigma(a, k) = 1 - \kappa R(a). \quad (52)$$

B Notation dictionary

Symbol	Meaning
$\mathcal{W}$	Weyl source, $-k^2(\Phi + \Psi)$
$\mathcal{W}_{\text{GR}}$	GR Weyl source, $8\pi G a^2 \rho \Delta$
$\mathcal{T}_{\text{G1}}^{\mathcal{W}}$	G1 Weyl-transfer operator
$A(a, k)$	convention-fixing map from optical seed to unit fractional Weyl correction; not a fit parameter

Symbol	Meaning
$\mathcal{R}_H(a)$	finite-screen horizon-memory residual
R_0	present-day residual amplitude, $R_0 = \mathcal{R}_H(1)$
$\hat{R}_H(a)$	normalized memory shape, $\hat{R}_H(1) = 1$
$X_b(a)$	specific normalized active-filling response window used in G1DE-M
s	low-redshift amplitude parameter, $R_0 = 3 - s$
κ	optical projection/compliance coefficient
κ_{BW}	locked finite-screen projection coefficient, $\kappa_{BW} = 3/4$ in the isotropic branch
$\pi_{\text{scr}}^{\text{GI}}$	gauge-invariant screen anisotropic seed
$q_{\text{scr}}^{\text{GI}}$	Ward-completed momentum leakage
$\delta\rho_{\text{scr}}^{\text{GI}}$	Ward-completed density/Ricci leakage
ν_2, ν_H	Ward-leakage control parameters

References

- [1] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, FDS-G1 Core manuscript (2026).
- [2] Y. Wu, *The 3/4 Projection Law in Finite Screen Gravity: MaxEnt Optical-Stress Closure, Ricci-Silent Residuals, and the Weyl-Locked Dark Sector*, manuscript / technical note, 2026.
- [3] Y. Wu, *Screen Influence Functional for the G1 Dark Sector: Entropy Hessian, Ward Closure, and Retarded Horizon Memory*, manuscript / technical note, 2026.
- [4] Y. Wu, *G1DE-M_{3/4}: A Projection-Locked Background–Weyl Residual*, manuscript / data paper, 2026.
- [5] R. P. Feynman and F. L. Vernon, The theory of a general quantum system interacting with a linear dissipative system, *Annals of Physics* **24**, 118–173 (1963).
- [6] J. Schwinger, Brownian motion of a quantum oscillator, *Journal of Mathematical Physics* **2**, 407–432 (1961).
- [7] L. V. Keldysh, Diagram technique for nonequilibrium processes, *Soviet Physics JETP* **20**, 1018–1026 (1965).
- [8] J. M. Bardeen, Gauge-invariant cosmological perturbations, *Physical Review D* **22**, 1882–1905 (1980).
- [9] C.-P. Ma and E. Bertschinger, Cosmological perturbation theory in the synchronous and conformal Newtonian gauges, *The Astrophysical Journal* **455**, 7–25 (1995).
- [10] P. Zhang, M. Liguori, R. Bean, and S. Dodelson, Probing gravity at cosmological scales by measurements which test the relationship between gravitational lensing and matter overdensity, *Physical Review Letters* **99**, 141302 (2007).
- [11] E. Bellini and I. Sawicki, Maximal freedom at minimum cost: linear large-scale structure in general modifications of gravity, *Journal of Cosmology and Astroparticle Physics* **07**, 050 (2014).
- [12] S. F. Daniel and E. V. Linder, Constraining cosmic expansion and gravity with gravitational lensing, *Physical Review D* **82**, 103523 (2010).

- [13] S. Weinberg, Photons and gravitons in perturbation theory: Derivation of Maxwell's and Einstein's equations, *Physical Review* **138**, B988–B1002 (1965).

Maximum-Caliber Horizontal Lift for the G1 Memory Quotient

Path Relative Entropy, Passive Horizon Kernels, and the Rank-One Origin of
G1DE-M_{3/4}
v1.2 Body-Integrated Release

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This technical note isolates the memory-shape problem in the G1 finite-screen dark-sector branch. The data-facing object is not a unique microscopic source–kernel pair (K_H, J_H) , but the quotient output

$$\mathcal{R}_H(a) = (K_H * J_H)(a),$$

which enters the scalar perturbation sector through the Weyl-normalized normal form

$$\Sigma - 1 = -\frac{3}{4}\mathcal{R}_H + O(\mathcal{R}_H^2), \quad \mu - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).$$

The factorization $\mathcal{R}_H = K_H * J_H$ has a gauge-like ambiguity: different passive kernels and occupancy sources can generate the same observable output. We propose a canonical resolution: the source–kernel representative is selected by a maximum-caliber horizontal lift. Given a passive reference path measure $\mathbb{P}_0[x]$ for hidden horizon memory variables and an output constraint $\mathbb{E}_{\mathbb{P}}[O(N)] = \mathcal{R}_H(N)$, the canonical lift is the path-space information projection

$$\mathbb{P}_* = \arg \min_{\mathbb{P}: \mathbb{E}_{\mathbb{P}}[O] = \mathcal{R}_H} D_{\text{KL}}(\mathbb{P} \parallel \mathbb{P}_0).$$

The dual source J_H is the Lagrange multiplier of this constraint, and the linear response kernel is the Hessian/covariance of the passive reference process. If the reference dynamics is stable and overdamped, the response is Stieltjes:

$$\chi_H(s) = \int_0^\infty \frac{d\mu_H(\gamma)}{s + \gamma}, \quad d\mu_H \geq 0.$$

The rank-one spectral truncation gives the one-state passive memory equation

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H,$$

and, after the fixed optical compliance fraction $3/4$, yields G1DE-M $_{3/4}$. Higher-rank positive Stieltjes or Jacobi realizations are controlled memory-shape extensions; unconstrained free-response models are adversarial envelopes outside the passive lift class. The note clarifies the status of the commonly used shape $X_b(a)$: it is a normalized output response $\widehat{\mathcal{R}}_H(a)$, not a uniquely derived microscopic kernel. The remaining microscopic task is to derive the passive reference path measure and occupancy source convention from finite-screen modular-index dynamics.

1 Introduction

The G1 finite-screen dark-sector program has developed a data-facing low-redshift reduction in which a horizon-scale finite-screen memory output modifies the Weyl source while leaving the Newtonian/Ricci response approximately unchanged. In its projection-locked branch the scalar response is

$$\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\widehat{\mathcal{R}}_H(a), \quad \mu(a, k) \simeq 1, \quad (1)$$

where $\widehat{\mathcal{R}}_H(1) = 1$ and $3 - s$ is the present amplitude of the low-redshift horizon memory output. Previous closure notes isolate three structural elements behind this expression:

1. a finite-screen optical compliance fraction $\kappa_{BW} = 3/4$ at the unbiased $GL(2)$ optical fixed point;
2. a Ward-completed Weyl transfer in which the leading correction enters Σ , while $\mu - 1$ is higher-order or subhorizon-suppressed;
3. a horizon memory output $\mathcal{R}_H(a)$ representing a passive finite-screen remnant.

The likelihood identifies an output-level memory response $R_H(a)$, not a unique microscopic source–kernel realization.

Theorem 1 (D6.1 – Source–Kernel Non-Identifiability).

$$(K_H, J_H) \sim (K'_H, J'_H) \iff K_H * J_H = K'_H * J'_H = R_H.$$

D10 uses this quotient to define its branch moduli space.

D6 selects canonical passive representatives within this output equivalence class:

$$[R_H] \mapsto (K_H, J_H)_{\text{can}}.$$

This is a realization discipline, not an empirical proof of source–kernel uniqueness.

D6 does not make (K_H, J_H) identifiable. It selects canonical representatives inside an output equivalence class. The horizontal lift $[R_H] \mapsto (K_H, J_H)_{\text{can}}$ uses maximum-caliber selection under the criteria: (1) passive realization; (2) minimal memory production; (3) detailed-balance compatibility; (4) no additional free output degrees; (5) consistency with D7 Markov realization.

The weakest link is not the projection law, but the redshift shape of $\mathcal{R}_H(a)$. A common schematic factorization is

$$\mathcal{R}_H(N) = \int_{-\infty}^N K_H(N, N') J_H(N') \, dN', \quad N = \ln a, \quad (2)$$

where K_H is a causal memory kernel and J_H is a horizon occupancy current, often written formally as

$$J_H = -\frac{d}{dN} \ln \mathcal{O}_H, \quad \mathcal{O}_H = \frac{C_{\text{used}}}{C_H}. \quad (3)$$

However, data do not separately observe K_H and J_H . They observe the output $\mathcal{R}_H(a)$ through its Weyl projection. Therefore a correct implementation note must distinguish three layers:

$$\boxed{\text{output } \mathcal{R}_H} \longleftarrow \boxed{\text{source–kernel lift } (K_H, J_H)} \longleftarrow \boxed{\text{microscopic finite-screen dynamics}}. \quad (4)$$

This note gives a precise mathematical formulation of that distinction. We first identify the observable memory quotient. We then introduce a memory factorization bundle whose base is the output $\mathcal{R}_H$ and whose fibers are all source–kernel factorizations producing that output. The central proposal is that a canonical representative of the fiber is selected by a maximum-caliber horizontal lift: among all path measures on hidden horizon states that reproduce the output, choose the one of least path relative entropy with respect to a passive reference process. This information projection yields the dual source, the Onsager kernel, the Stieltjes response class, and the rank-one model as the lowest spectral truncation.

The note is deliberately technical and modest. It does not claim to derive the microscopic finite-screen state family or the occupancy functional C_{used} . It supplies an interface between the data-facing G1DE model and the deeper finite-screen theory. Its main purpose is to prevent overclaiming: $X_b(a)$ or $\widehat{\mathcal{R}_H}(a)$ is a normalized output shape, not a uniquely derived microscopic kernel.

2 Claim status and notation

We use the following conventions.

- The time variable is $N = \ln a$. A prime denotes d/dN .
- $\mathcal{R}_H(N)$ is the scalar horizon memory output. Its normalized shape is $\widehat{\mathcal{R}_H}(N) = \mathcal{R}_H(N)/\mathcal{R}_H(0)$, with $a = 1$ corresponding to $N = 0$.
- The present amplitude is written

$$R_0 = \mathcal{R}_H(0) = 3 - s. \quad (5)$$

The parameter s is a low-redshift effective filling exponent; its deficit $3 - s$ is not assumed to be a literal spatial dimension.

- The Weyl-normalized observable map is

$$\Sigma - 1 = -\kappa_{BW}\mathcal{R}_H + O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right), \quad (6)$$

$$\mu - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (7)$$

At the isotropic optical fixed point $\kappa_{BW} = 3/4$.

- The notation $K_H * J_H$ denotes a causal convolution or, more generally, a causal Volterra response. In a stationary low-redshift approximation,

$$\mathcal{R}_H(N) = \int_0^\infty K_H(\tau) J_H(N - \tau) d\tau. \quad (8)$$

Claim-status discipline. This note proves conditional statements. If a passive reference path process is given, then a maximum-caliber lift yields a convex response potential and a positive kernel. If the passive generator is overdamped/self-adjoint in the relevant slow sector, then the kernel is Stieltjes. If the leading spectral mode dominates, then the rank-one model follows. The note does not prove from bare FDS primitives that the required reference process exists.

3 The G1 output normal form

The primary data-facing object is $\mathcal{R}_H(a)$. To leading order, its scalar perturbation imprint is the Weyl transfer

$$\mathcal{W}_{G1} = (1 - \kappa_{BW}\mathcal{R}_H)\mathcal{W}_{GR} + O(\mathcal{R}_H^2), \quad (9)$$

with minimum Ward completion suppressing the Ricci/Newtonian leakage. Hence

$$\Sigma - 1 = -\kappa_{BW}\mathcal{R}_H + O(\mathcal{R}_H^2), \quad \mu - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (10)$$

At the unbiased optical fixed point,

$$T_1 GL(2, \mathbb{R}) = A(1) \oplus S(2) \oplus \Omega(1), \quad (11)$$

where A is the area/trace direction, S is the two-dimensional shear sector, and Ω is twist/holonomy. The Weyl-active unimodular sector is

$$U = S \oplus \Omega, \quad (12)$$

with $\dim U = 3$, while the full optical ledger has dimension 4. If the optical Hessian or compliance metric is isotropic at the fixed point,

$$\kappa_{BW} = \frac{\dim U}{\dim GL(2)} = \frac{3}{4}. \quad (13)$$

Thus the leading data-facing normal form is

$$\boxed{\Sigma - 1 = -\frac{3}{4}\mathcal{R}_H + O(\mathcal{R}_H^2), \quad \mu - 1 = O\left(\mathcal{R}_H^2, \mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right).} \quad (14)$$

With $\mathcal{R}_H(a) = (3 - s)\widehat{\mathcal{R}_H}(a)$, this becomes

$$\boxed{\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\widehat{\mathcal{R}_H}(a), \quad \mu(a, k) \simeq 1,} \quad (15)$$

up to the stated higher-order corrections.

4 The memory quotient

4.1 Source–kernel factorization is not unique

In a stationary representation,

$$\widetilde{\mathcal{R}_H}(s) = \chi_H(s)\widetilde{J}_H(s), \quad (16)$$

where χ_H is the transfer function of the memory kernel. If $Q(s)$ is an admissible invertible filter, then

$$\chi_H(s) \mapsto \chi_H(s)Q(s), \quad \widetilde{J}_H(s) \mapsto Q(s)^{-1}\widetilde{J}_H(s) \quad (17)$$

leaves $\widetilde{\mathcal{R}_H}(s)$ invariant. Thus the pair (K_H, J_H) is not directly identifiable from $\mathcal{R}_H$ alone.

Definition 1 (Memory factorization bundle). *Let $\mathcal{M}_{\text{fact}}$ be the set of admissible source–kernel pairs (K, J) , where K is causal, stable, and passive, and J is an admissible horizon source. Define*

$$\pi : \mathcal{M}_{\text{fact}} \rightarrow \mathcal{M}_{\text{out}}, \quad \pi(K, J) = K * J = \mathcal{R}. \quad (18)$$

The base $\mathcal{M}_{\text{out}}$ contains observable outputs $\mathcal{R}$. The fiber over $\mathcal{R}$ is

$$\pi^{-1}(\mathcal{R}) = \{(K, J) : K * J = \mathcal{R}\}. \quad (19)$$

The production likelihood lives primarily on $\mathcal{M}_{\text{out}}$, because scalar observables at leading order depend on $\mathcal{R}_H$, not on a unique representative (K_H, J_H) . A microscopic theory or implementation convention selects a horizontal lift from output space to the factorization bundle.

4.2 Gauge-invariant output quantities

Factorization-invariant quantities include

$$R_0 = \mathcal{R}_H(0), \quad \alpha_H(N) = \frac{d}{dN} \ln \mathcal{R}_H(N), \quad \beta_H(N) = \frac{d^2}{dN^2} \ln \mathcal{R}_H(N), \quad (20)$$

and normalized ratios $\mathcal{R}_H(N_i)/\mathcal{R}_H(N_j)$. These are output observables. By contrast, pole locations Γ_i , spectral weights Z_i , or occupancy currents J_H require a factorization gauge or external information.

5 Maximum-caliber horizontal lift

5.1 Path-space information projection

Let $x(N)$ denote hidden horizon memory variables, and let

$$O(N) = Cx(N) \quad (21)$$

be the slow observable whose expectation is the output $\mathcal{R}_H(N)$. Let $\mathbb{P}_0[x]$ be a passive reference path measure representing equilibrium or unforced horizon modular-index dynamics.

Definition 2 (Maximum-caliber lift). *Given an output trajectory $\mathcal{R}_H(N)$, the canonical path measure $\mathbb{P}_*$ is*

$$\boxed{\mathbb{P}_* = \arg \min_{\mathbb{P} : \mathbb{E}_{\mathbb{P}}[O(N)] = \mathcal{R}_H(N)} D_{\text{KL}}(\mathbb{P} \parallel \mathbb{P}_0).} \quad (22)$$

The Lagrange multiplier enforcing the output constraint is the dual source $J_H(N)$. Standard variational calculus gives

$$\frac{d\mathbb{P}_*}{d\mathbb{P}_0} = \frac{1}{Z[J_H]} \exp \left(\int J_H(N) O(N) dN \right), \quad (23)$$

where

$$\Psi[J_H] = \log Z[J_H]. \quad (24)$$

The output is the functional derivative

$$\boxed{\mathcal{R}_H(N) = \frac{\delta \Psi}{\delta J_H(N)}.} \quad (25)$$

Equation (25) is the convex-dual source gauge. In this framework J_H is not first introduced as a unique microscopic source; it is the dual coordinate of the maximum-caliber output constraint. A microscopic occupancy formula such as (3) is a further claim: it identifies this dual coordinate with a finite-screen occupancy current.

5.2 Dissipation potential and convexity

Expanding Ψ near $J_H = 0$,

$$\Psi[J_H] = \Psi[0] + \int \bar{O}(N) J_H(N) \, dN + \frac{1}{2} \int J_H(N) K_H(N, N') J_H(N') \, dN \, dN' + \dots \quad (26)$$

For perturbations around a centered reference process $\bar{O} = 0$,

$$\mathcal{R}_H(N) = \int K_H(N, N') J_H(N') \, dN' + O(J_H^2), \quad (27)$$

where

$$K_H(N, N') = \langle O(N) O(N') \rangle_{0,c}. \quad (28)$$

Since K_H is a covariance operator, it is positive semidefinite. The quadratic dissipation potential is

$$\Psi_2[J_H] = \frac{1}{2} \int J_H K_H J_H \geq 0. \quad (29)$$

Thus linear passivity is the convexity of the path relative-entropy projection.

6 Passive kernels and Stieltjes representation

If the passive reference process is stationary in N , then $K_H(N, N') = K_H(N - N')$ for $N \geq N'$. If the unresolved horizon dynamics is overdamped and generated by a nonnegative self-adjoint operator L_H , then

$$K_H(\tau) = \langle O, e^{-\tau L_H} O \rangle_0, \quad \tau \geq 0. \quad (30)$$

The spectral theorem gives

$$\boxed{K_H(\tau) = \int_0^\infty e^{-\gamma\tau} \, d\mu_H(\gamma), \quad d\mu_H(\gamma) \geq 0.} \quad (31)$$

The transfer function is the Stieltjes function

$$\boxed{\chi_H(s) = \int_0^\infty \frac{d\mu_H(\gamma)}{s + \gamma}.} \quad (32)$$

The class (32) is causal, stable, and passive. It is also completely monotone in the time domain:

$$(-1)^n \frac{d^n}{d\tau^n} K_H(\tau) \geq 0, \quad n = 0, 1, 2, \dots \quad (33)$$

Thus a G1 passive horizon kernel is not a general oscillator. It is an overdamped relaxation spectrum.

7 Rank-one lift and G1DE-M_{3/4}

If the spectral measure is dominated by one slow mode,

$$d\mu_H(\gamma) = Z_H \delta(\gamma - \Gamma_H) \, d\gamma, \quad Z_H > 0, \quad \Gamma_H > 0, \quad (34)$$

then

$$K_H(\tau) = Z_H e^{-\Gamma_H \tau}. \quad (35)$$

Equivalently,

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = Z_H J_H. \quad (36)$$

Absorbing Z_H into the amplitude convention gives the rank-one memory equation

$$\boxed{\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H.} \quad (37)$$

The normalized output is

$$\widehat{\mathcal{R}_H}(a) = \frac{\mathcal{R}_H(a)}{\mathcal{R}_H(1)}, \quad \mathcal{R}_H(a) = (3-s)\widehat{\mathcal{R}_H}(a). \quad (38)$$

Combining (37) with the fixed optical projection gives the minimal output model:

$$\boxed{\text{G1DE-M}_{3/4} : \quad \Sigma - 1 = -\frac{3}{4}(3-s)\widehat{\mathcal{R}_H}(a), \quad \mu \simeq 1.} \quad (39)$$

Thus G1DE-M_{3/4} is the rank-one maximum-caliber horizontal lift of the G1 memory quotient, followed by the fixed 3/4 Weyl projection.

8 v1.2 Modular Observable Target for $R_b^H(a)$

Claim type: Conditional target The output normalization target derivation from modular fluctuation spectra is:

$$R_H(a) \sim \frac{\delta \langle K_{\Sigma(a)} \rangle}{\delta \langle K_{\Sigma(1)} \rangle} \quad \text{or} \quad R_H(a) \sim \frac{\text{Var}(K_{\Sigma(a)})}{\text{Var}(K_{\Sigma(1)})}$$

where K_Σ is the expansion scalar or modular operator for the causal diamond at scale factor a .

Boundary note: This target does not identify a unique source–kernel factorization. The output normal form used in likelihood tests is decoupled from this theoretical origin target.

9 Moment, Hankel, and Jacobi interpretations

The Stieltjes transfer has a low-frequency expansion

$$\chi_H(s) = m_0 - sm_1 + s^2 m_2 - \dots, \quad (40)$$

where

$$m_n = \int_0^\infty \gamma^{-(n+1)} d\mu_H(\gamma). \quad (41)$$

The moment sequence must satisfy Hankel positivity:

$$H_{ij}^{(r)} = m_{i+j} \succeq 0. \quad (42)$$

The rank of the Hankel moment matrix equals the minimal number of discrete relaxation bands needed to realize the response.

For rank one,

$$\boxed{m_0 m_2 - m_1^2 = 0.} \quad (43)$$

For rank two or higher,

$$m_0 m_2 - m_1^2 > 0. \quad (44)$$

Violation of Hankel positivity indicates failure of the passive Stieltjes class.

Equivalently, the Stieltjes response admits a canonical Jacobi representation

$$\chi_H(s) = \mu_0 \langle e_0, (sI + J_{\text{Jac}})^{-1} e_0 \rangle, \quad (45)$$

where $J_{\text{Jac}} \geq 0$ is a Jacobi matrix. Its continued fraction is

$$\chi_H(s) = \frac{\mu_0}{s + b_0 - \frac{a_1^2}{s + b_1 - \frac{a_2^2}{s + b_2 - \dots}}}, \quad a_i > 0. \quad (46)$$

The m -step truncation defines a controlled passive memory extension $M_{3/4}^{(m)}$.

10 Noise and fluctuation-dissipation information

The maximum-caliber construction naturally supplies both the mean response and the fluctuation covariance of the hidden horizon process. In an open-system description one may write schematically

$$x' = Ax + bJ_H + \xi_H, \quad \mathcal{R}_H = c^T x, \quad (47)$$

with noise covariance

$$\langle \xi_H(N) \xi_H(N')^T \rangle = N_H(N, N'). \quad (48)$$

Near equilibrium the response and noise are not independent. A fluctuation-dissipation-type relation has the schematic form

$$N_H(\omega) \sim \coth\left(\frac{\beta\omega}{2}\right) \text{Im } \chi_H(\omega), \quad (49)$$

with the appropriate modular inverse temperature β . The precise cosmological stochastic prediction requires a fully covariant open-system formalism and is left as an extension. The important point is structural: a stochastic Weyl or lensing residual covariance, if present, should not be modeled independently of the response kernel.

11 Model hierarchy and identifiability

The memory quotient formulation suggests the following disciplined hierarchy.

Model or control	Interpretation
constant- Σ	Output amplitude / DC-gain only; tests whether m_0 is enough.
$M_{3/4}^{(1)}$	Rank-one maximum-caliber/Stieltjes lift plus 3/4 Weyl projection; this is G1DE-M $_{3/4}$.
$M_{3/4}^{(2)}$	Rank-two positive memory; tests whether a second hidden relaxation band is resolved.
$M_{\kappa}^{(m)}$	Passive memory of order m , but optical coefficient free; tests optical isotropy.
G1DE-2 / free response	Unconstrained response envelope; adversarial control outside the passive-lift class.
free $A(a, k)$	Forbidden in G1: changes the fixed normalization rather than a memory deformation.

A key identifiability point follows: current scalar data should first be interpreted as constraining $\mathcal{R}_H(a)$ and its output shape moments. Inferring K_H and J_H separately requires a gauge choice, a maximum-caliber lift, stochastic covariance information, or an independent finite-screen occupancy model.

12 Main closure theorem

Theorem 2 (Maximum-caliber horizontal-lift closure). *Assume:*

1. the G1 scalar dark-sector branch is described by an output $\mathcal{R}_H(N)$;
 2. the leading observable map is Weyl-normalized,
- $$\Sigma - 1 = -\frac{3}{4}\mathcal{R}_H + O(\mathcal{R}_H^2), \quad \mu - 1 = O\left(\mathcal{R}_H^2, \mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right); \quad (50)$$
3. hidden horizon memory variables admit a passive reference path measure $\mathbb{P}_0$;
 4. the canonical lift is selected by minimizing path relative entropy subject to $\mathbb{E}[O] = \mathcal{R}_H$;
 5. the passive reference generator is stable and overdamped in the relevant slow sector;
 6. the dominant spectral response is rank one.

Then the canonical lift is linearized by a positive kernel with Stieltjes transfer function. Its rank-one truncation satisfies

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H, \quad (51)$$

and its leading scalar cosmological reduction is

$$\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\widehat{\mathcal{R}_H}(a), \quad \mu(a, k) \simeq 1. \quad (52)$$

Proof. The maximum-caliber variational problem gives the exponentially tilted path measure (23). Its log-partition functional $\Psi[J_H]$ is convex. Hence its second variation around the passive reference process is a positive kernel K_H , and the linear response is $\mathcal{R}_H = K_H * J_H$. Stability and overdamped

passive dynamics imply the spectral representation (32). Rank-one spectral dominance gives $\chi_H(s) = Z_H/(s + \Gamma_H)$, which is equivalent to $\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = Z_H J_H$. After absorbing Z_H into the amplitude convention and applying the fixed 3/4 Weyl projection with Ward-suppressed μ leakage, the stated G1DE- $M_{3/4}$ form follows. $\square$

Corollary 1 (Output status of $X_b(a)$). *A redshift shape $X_b(a)$ used in the low-redshift G1DE branch should be interpreted as*

$$X_b(a) = \widehat{\mathcal{R}_H}(a), \quad (53)$$

namely a normalized output response. It is not, by itself, a unique microscopic kernel or source.

Corollary 2 (Controlled memory extensions). *If data require additional memory moments while preserving Stieltjes positivity and Ward suppression, the correct controlled extensions are $M_{3/4}^{(m)}$, the m -rank passive memory realizations, rather than an unconstrained free-response surface.*

13 Null tests

The maximum-caliber quotient/lift picture implies the following tests.

Output projection test. On linear subhorizon scales,

$$\frac{\mu - 1}{\Sigma - 1} = O(\mathcal{R}_H) + O\left(\frac{\mathcal{H}^2}{k^2}\right). \quad (54)$$

An order-one ratio would indicate failure of Ward/Ricci slaving.

Rank-one memory test. If output moments can be reconstructed, rank-one memory predicts

$$\Delta_H = m_0 m_2 - m_1^2 \simeq 0. \quad (55)$$

Positive Δ_H suggests higher memory rank; negative Δ_H violates passive Stieltjes positivity.

Kernel positivity test. Any inferred kernel in a passive gauge should be completely monotone. Oscillatory or negative-residue kernels are outside the overdamped G1 memory class.

Source-gauge test. If a finite-screen occupancy proxy becomes available, the dual source from the maximum-caliber lift should be compatible with

$$J_H = -\frac{d}{dN} \ln(C_{\text{used}}/C_H). \quad (56)$$

Failure here would challenge the microscopic occupancy model, not necessarily the output normal form.

14 Open problems

The note reduces the memory-shape problem to several sharper open tasks:

1. **Reference path measure.** Derive the passive reference process $\mathbb{P}_0[x]$ from finite-screen modular-index dynamics.

2. **Occupancy source.** Derive $C_{\text{used}}(a)$ and hence $J_H = -d_N \ln(C_{\text{used}}/C_H)$ from a microscopic finite-screen state family.
3. **Covariant open-system completion.** Promote the scalar maximum-caliber lift to a gauge-invariant covariant Schwinger-Keldysh or influence-functional framework.
4. **Noise kernel.** Work out the stochastic Weyl/lensing covariance implied by the same passive reference process.
5. **Memory-rank evidence.** Implement $M_{3/4}^{(2)}$ and higher positive-rank memory models before interpreting free-response envelopes as fundamental alternatives.
6. **Optical coefficient controls.** Continue testing the fixed 3/4 coefficient against free- κ branches and twist-demoted alternatives.

15 Conclusion

The G1 memory problem is best formulated as a quotient-and-lift problem. The scalar data-facing object is the horizon output $\mathcal{R}_H(a)$. Source–kernel factorizations (K_H, J_H) are representatives in a fiber over that output, and a unique representative requires an additional lift principle. The proposed canonical lift is the maximum-caliber path-relative-entropy projection onto a passive reference horizon process. Its dual source is the Lagrange multiplier enforcing the output, its linear kernel is a covariance/Onsager Hessian, and stable overdamped reference dynamics gives a Stieltjes memory class.

The rank-one spectral truncation is precisely the memory closure behind G1DE-M_{3/4}. Thus

$$\boxed{\text{G1DE-M}_{3/4} = \text{rank-one maximum-caliber horizontal lift} + \text{fixed 3/4 Weyl projection.}} \quad (57)$$

The formulation preserves the main leading prediction,

$$\Sigma - 1 = -\frac{3}{4}\mathcal{R}_H, \quad \mu \simeq 1, \quad (58)$$

while clarifying that $X_b(a)$ is a normalized output shape, not a uniquely derived microscopic kernel. The deepest remaining task is to derive the passive reference path measure and the occupancy source convention from finite-screen modular-index microphysics.

A Relative entropy and maximum caliber

For probability measures $\mathbb{P}$ and $\mathbb{P}_0$ on path space, the path relative entropy is

$$D_{\text{KL}}(\mathbb{P}||\mathbb{P}_0) = \int \log \left(\frac{d\mathbb{P}}{d\mathbb{P}_0} \right) d\mathbb{P}. \quad (59)$$

Minimizing this quantity subject to linear constraints

$$\mathbb{E}_{\mathbb{P}}[O_i] = r_i \quad (60)$$

leads to the exponential family

$$\frac{d\mathbb{P}_{\lambda}}{d\mathbb{P}_0} = \frac{1}{Z(\lambda)} \exp \left(\sum_i \lambda_i O_i \right). \quad (61)$$

The log-partition function $\Psi(\lambda) = \log Z(\lambda)$ satisfies

$$\frac{\partial \Psi}{\partial \lambda_i} = \mathbb{E}_{\mathbb{P}_\lambda}[O_i], \quad \frac{\partial^2 \Psi}{\partial \lambda_i \partial \lambda_j} = \text{Cov}_{\mathbb{P}_\lambda}(O_i, O_j). \quad (62)$$

The continuous-time formulas used in the main text are the functional version of this result.

B Stieltjes functions and complete monotonicity

A function $\chi(s)$ of the form

$$\chi(s) = \int_0^\infty \frac{d\mu(\gamma)}{s + \gamma}, \quad d\mu \geq 0, \quad (63)$$

is a Stieltjes function. Its time-domain kernel is

$$K(\tau) = \int_0^\infty e^{-\gamma\tau} d\mu(\gamma), \quad (64)$$

which is completely monotone:

$$(-1)^n K^{(n)}(\tau) = \int_0^\infty \gamma^n e^{-\gamma\tau} d\mu(\gamma) \geq 0. \quad (65)$$

This is the mathematical expression of an overdamped passive relaxation spectrum.

C Hankel rank and memory order

For a positive discrete measure

$$d\mu(\gamma) = \sum_{i=1}^m Z_i \delta(\gamma - \Gamma_i) d\gamma, \quad Z_i > 0, \quad (66)$$

the moments are

$$m_n = \sum_{i=1}^m Z_i \Gamma_i^{-(n+1)}. \quad (67)$$

The infinite Hankel matrix $H_{ij} = m_{i+j}$ has rank m if the Γ_i are distinct. Thus moment rank equals minimal memory order. Rank one implies $m_0 m_2 - m_1^2 = 0$; rank two or higher gives $m_0 m_2 - m_1^2 > 0$ under nondegenerate positive weights.

D Implementation recipe

A production implementation can separate layers as follows.

1. Fit the output normal form:

$$\Sigma - 1 = -\frac{3}{4}(3-s)\widehat{\mathcal{R}_H}(a), \quad \mu = 1. \quad (68)$$

2. Test constant- Σ as an amplitude-only control.
3. Test a rank-one passive lift, i.e. a one-pole $\widehat{\mathcal{R}_H}(a)$.

4. If shape residuals remain, test rank-two positive Stieltjes memory before moving to unconstrained response.
5. Test free κ only as an optical-isotropy control, not as a replacement for the memory hierarchy.
6. Treat free $A(a, k)$ as outside the G1 class, because it changes the fixed normalization.

References

- [1] L. Onsager, Reciprocal relations in irreversible processes I and II, *Phys. Rev.* **37**, 405 (1931); **38**, 2265 (1931).
- [2] E. T. Jaynes, Information theory and statistical mechanics, *Phys. Rev.* **106**, 620 (1957).
- [3] R. Kubo, The fluctuation-dissipation theorem, *Rep. Prog. Phys.* **29**, 255 (1966).
- [4] H. Mori, Transport, collective motion, and Brownian motion, *Prog. Theor. Phys.* **33**, 423 (1965).
- [5] R. Zwanzig, Memory effects in irreversible thermodynamics, *Phys. Rev.* **124**, 983 (1961).
- [6] R. P. Feynman and F. L. Vernon, The theory of a general quantum system interacting with a linear dissipative system, *Ann. Phys.* **24**, 118 (1963).
- [7] D. Petz, Sufficient subalgebras and the relative entropy of states of a von Neumann algebra, *Commun. Math. Phys.* **105**, 123 (1986).
- [8] M. G. Krein and A. A. Nudelman, *The Markov Moment Problem and Extremal Problems*, AMS, 1977.
- [9] Y. Wu, *Screen Modular Anomaly Inflow and the Einstein Calibration of Finite Access Entropy*, G1 technical manuscript, 2026.
- [10] Y. Wu, *Weyl-Normalized Ward Completion and Identifiability Rules*, FDS–G1 technical note D5 (2026).
- [11] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual*, data-facing manuscript (2026).

Finite Markov-Screen Realization of the G1DE-M_{3/4} Normal Form

A Detailed-Balance Prototype for the 3/4 Weyl Projection, Rank-One Horizon Output, and Ward-Suppressed Ricci Leakage

v1.2 Body-Integrated Release

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS-G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This note gives a finite-state microscopic prototype for the G1DE normal form. Previous G1 notes separate the data-facing horizon response output $\mathcal{R}_H(a)$ from a non-unique source-kernel factorization $K_H * J_H$, and implement the dark-sector residual as a Weyl-normalized transfer correction rather than as an independently normalized dark stress source. The purpose of the present note is narrower and more constructive: to show that the formal ingredients used by the G1DE-M_{3/4} branch can be realized by a finite detailed-balance Markov screen.

For each screen size N we introduce a finite state space Ω_N , a passive reference measure σ_N , reversible Markov dynamics with generator Q_N , and centered macro-port observables

$$F_a, \quad a \in \{A, S_1, S_2, T, H, R\},$$

representing the trace optical, two shear, twist/holonomy, horizon-memory, and Ricci-leakage channels. With the nonnegative relaxation operator $\mathcal{L}_N = -Q_N$, the response matrix is

$$\chi_{ab}^{(N)}(s) = \langle F_a, (s + \mathcal{L}_N)^{-1} F_b \rangle_{\sigma_N}.$$

Reversibility makes $\mathcal{L}_N$ self-adjoint and nonnegative, hence $\chi^{(N)}$ has a positive matrix-valued Stieltjes representation. If the four optical ports are unbiased in both covariance and dissipation, the Weyl-active fraction of the optical response is

$$\kappa_{BW} = \frac{\dim(S_1 \oplus S_2)}{\dim(A \oplus S_1 \oplus S_2)} = \frac{3}{4}.$$

If the horizon port is approximately lumpable and dominated by one slow eigenmode, its output satisfies the rank-one equation

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H.$$

If Ward conditioning makes the Ricci port entropy-stiff, then the Newtonian response satisfies

$$\mu - 1 = O(\mathcal{R}_H^2) + O(\mathcal{R}_H \mathcal{H}^2 / k^2).$$

Consequently, the low-redshift Weyl-normalized output is

$$\Sigma(a, k) - 1 = -\frac{3}{4} \mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad \mu(a, k) \simeq 1.$$

With $\mathcal{R}_H(a) = (3 - s)\widehat{\mathcal{R}}_H(a)$ and $\widehat{\mathcal{R}}_H(1) = 1$, this gives the production normal form

$$\Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\widehat{\mathcal{R}}_H(a), \quad \mu(a, k) \simeq 1.$$

The result is conditional and constructive: it does not derive the true finite-screen microphysics, but it gives a concrete finite-state realization target for the G1DE-M_{3/4} interface. This realization is not unique; it is a constructive existence target showing that the G1DE-M_{3/4} interface is compatible with finite passive Markov dynamics.

1 Scope and claim status

The G1 dark-sector sequence has three implementation-facing claims that must be kept distinct:

1. the likelihood primarily sees an output response $\mathcal{R}_H(a)$;
2. a factorization $\mathcal{R}_H = K_H * J_H$ is a source–kernel lift, not a uniquely observed microscopic object;
3. the leading G1DE branch modifies the Weyl source by a normalized finite-screen transfer factor, not by an independently normalized dark-stress source.

This note adds a finite-state realization prototype underneath those claims.

Non-uniqueness. This paper proves existence of at least one admissible finite-state passive lift, not uniqueness of the microscopic screen dynamics. If two generators L_N, L'_N produce the same $R_H(a)$, they are observationally equivalent at the G1fit-real level unless additional observables are specified.

The note is not a derivation of General Relativity and not a proof that Nature uses the particular Markov chain written below. It proves a conditional realizability statement: if a finite reversible Markov screen has optical unbiasedness, a slow horizon mode, and Ward-stiff Ricci leakage, then its low-spectrum response realizes the G1DE-M_{3/4} normal form.

Layer	Object	Status in this note
Data-facing output	$\mathcal{R}_H(a)$	retained as observable response
Lift layer	(K_H, J_H)	induced by reversible Markov response
Microscopic prototype	$(\Omega_N, \sigma_N, Q_N, F_a)$	constructed conditionally
Optical projection	$\kappa_{BW} = 3/4$	follows from optical unbiasedness
Horizon shape	$\widehat{\mathcal{R}}_H(a)$	rank-one slow-mode output
Ward leakage	$\mu - 1$	Schur/conditioning suppressed

2 Finite Markov screens

2.1 State space and ports

For each finite screen size N , let

$$\Omega_N = \Omega_{\text{cell}}^N$$

be a finite product state space. A microscopic screen state is

$$\omega = (\omega_1, \dots, \omega_N) \in \Omega_N.$$

Each cell carries port variables

$$f_a : \Omega_{\text{cell}} \rightarrow \mathbb{R}, \quad a \in \mathcal{P} := \{A, S_1, S_2, T, H, R\}.$$

The port labels are interpreted as follows:

Port	Meaning
A	trace/area optical channel
S_1, S_2	two shear optical channels
T	twist/holonomy optical channel
H	horizon memory or occupancy output channel
R	Ricci/density leakage channel

The centered macroscopic port observables are

$$F_{a,N}(\omega) = \frac{1}{\sqrt{N}} \sum_{i=1}^N (f_a(\omega_i) - \mathbb{E}_{\sigma_1} f_a). \quad (1)$$

The $N^{-1/2}$ scaling is convenient for response theory. For large-deviation statements one may instead use $N^{-1} \sum_i f_a$; the two conventions differ only by the fluctuation scale.

2.2 Passive reference state

Let σ_N be a faithful probability measure on Ω_N . In the simplest independent-cell case,

$$\sigma_N = \sigma_1^{\otimes N}.$$

More generally, σ_N may include finite-range interactions, provided the macro-variables obey an area-extensive large-deviation principle.

Definition 2.1 (Static macro-quasipotential). *Let $x = (x_a)_{a \in \mathcal{P}}$ denote the empirical macro-port vector. A static quasipotential $\Phi_N(x)$ is defined by*

$$\sigma_N\{x_N \simeq x\} \asymp \exp[-N\Phi_N(x)].$$

Near the passive fixed point $x = 0$,

$$\Phi_N(x) = \frac{1}{2} x^T K_N x + O(|x|^3), \quad K_N = \nabla^2 \Phi_N(0). \quad (2)$$

For independent cells, K_N is the inverse covariance matrix of the cell port vector. This is the finite-state version of the relative-entropy Hessian used in the continuum G1 notes.

3 v1.2 Ledger-to-Markov Realization Chart

Claim type: Realization specification The finite Markov screen dynamics in D7 provides an explicit realization chart for the finite ledger object defined in Core v1.2: the Markov generator Q_N implements the update operation τ of the ledger tuple $\mathcal{L}_{\text{scr}}$ on a finite state space.

Boundary note: This Markov realization is only one possible concrete model of the finite ledger. It is not a unique microscopic model of spacetime, nor is it required for the core G1 residual results or the observational predictions of G1DE-M_{3/4}.

4 Reversible Markov screen dynamics

4.1 Detailed-balance generator

Let Q_N be a continuous-time Markov generator on Ω_N :

$$(Q_N g)(\omega) = \sum_{\omega' \in \Omega_N} r_N(\omega, \omega') [g(\omega') - g(\omega)]. \quad (3)$$

Assume detailed balance with respect to σ_N :

$$\sigma_N(\omega) r_N(\omega, \omega') = \sigma_N(\omega') r_N(\omega', \omega). \quad (4)$$

Then Q_N is self-adjoint on $L^2(\sigma_N)$ and has nonpositive spectrum. Define the nonnegative relaxation operator

$$\mathcal{L}_N = -Q_N \succeq 0. \quad (5)$$

The Dirichlet form is

$$\mathcal{E}_N(g, h) = \langle g, \mathcal{L}_N h \rangle_{\sigma_N} = \frac{1}{2} \sum_{\omega, \omega'} \sigma_N(\omega) r_N(\omega, \omega') [g(\omega) - g(\omega')] [h(\omega) - h(\omega')]. \quad (6)$$

In particular,

$$\mathcal{E}_N(g, g) \geq 0.$$

4.2 Matrix-valued response

For $s > 0$, define the port response matrix

$$\chi_{ab}^{(N)}(s) = \langle F_{a,N}, (s + \mathcal{L}_N)^{-1} F_{b,N} \rangle_{\sigma_N}. \quad (7)$$

This is the finite Markov-screen impedance.

Lemma 4.1 (Matrix-Stieltjes representation). *For every finite reversible Markov screen, there exists a positive semidefinite matrix-valued measure $dM_N(\gamma)$ on $[0, \infty)$ such that*

$$\chi^{(N)}(s) = \int_0^\infty \frac{dM_N(\gamma)}{s + \gamma}. \quad (8)$$

Proof. Because $\mathcal{L}_N$ is self-adjoint and nonnegative in $L^2(\sigma_N)$, the spectral theorem gives

$$(s + \mathcal{L}_N)^{-1} = \int_0^\infty \frac{dE_N(\gamma)}{s + \gamma},$$

where E_N is the spectral projection measure. Define

$$dM_{N,ab}(\gamma) = \langle F_{a,N}, dE_N(\gamma) F_{b,N} \rangle_{\sigma_N}.$$

For any vector $u = (u_a)$,

$$u^\top dM_N(\gamma) u = \left\langle \sum_a u_a F_{a,N}, dE_N(\gamma) \sum_b u_b F_{b,N} \right\rangle_{\sigma_N} \geq 0.$$

Thus $dM_N \succeq 0$, and (8) follows. $\square$

5 Optical unbiasedness and the 3/4 projection

5.1 Optical port space

Define the four-dimensional optical port space

$$\mathcal{P}_{\text{opt}} = A(1) \oplus S_1(1) \oplus S_2(1)(1). \quad (9)$$

The Weyl-active unimodular subspace is

$$U = S_1 \oplus S_2, \quad \dim U = 3. \quad (10)$$

Let P_U be the projector onto U .

Definition 5.1 (Optical unbiasedness). *The finite Markov screen is optically unbiased if, on $\mathcal{P}_{\text{opt}}$,*

$$\text{Cov}_{\sigma_N}(F_{a,N}, F_{b,N}) = c_N \delta_{ab}, \quad (11)$$

and

$$\mathcal{E}_N(F_{a,N}, F_{b,N}) = d_N \delta_{ab}, \quad (12)$$

for $a, b \in \{A, S_1, S_2, T\}$.

Under this condition, the optical response block is approximately proportional to the identity:

$$\chi_{\text{opt}}^{(N)}(s) = \chi_N^{\text{opt}}(s) I_4 + O(\epsilon_{\text{aniso}}). \quad (13)$$

This holds exactly if the four optical observables span an invariant eigenspace of $\mathcal{L}_N$ with a common spectral measure, and it is the leading fixed-point form if optical anisotropies are irrelevant under coarse-graining.

Proposition 5.1 (Finite-screen optical projection fraction). *If the optical response block is proportional to the identity, then*

$$\kappa_N(s) = \frac{\text{Tr}(P_U \chi_{\text{opt}}^{(N)}(s))}{\text{Tr}(\chi_{\text{opt}}^{(N)}(s))} = \frac{3}{4}. \quad (14)$$

Proof. Using (13),

$$\text{Tr}(P_U \chi_{\text{opt}}^{(N)}) = \chi_N^{\text{opt}} \text{Tr}(P_U) = 3\chi_N^{\text{opt}},$$

while

$$\text{Tr}(\chi_{\text{opt}}^{(N)}) = 4\chi_N^{\text{opt}}.$$

Their ratio is 3/4. □

Remark 5.1 (Weighted-compliance branch). *If optical unbiasedness fails, the coefficient becomes*

$$\kappa = \frac{C_{S_1} + C_{S_2} + C_T}{C_A + C_{S_1} + C_{S_2} + C_T}. \quad (15)$$

For the shear-degenerate case $C_{S_1} = C_{S_2} = C_S$, this is

$$\kappa = \frac{2C_S + C_T}{C_A + 2C_S + C_T}.$$

This is the microscopic meaning of the free- κ compliance control.

6 Horizon port and rank-one output

6.1 Horizon response

The scalar horizon-memory response is the H -port projection

$$\chi_H^{(N)}(s) = \langle F_{H,N}, (s + \mathcal{L}_N)^{-1} F_{H,N} \rangle_{\sigma_N} = \int_0^\infty \frac{d\mu_H^{(N)}(\gamma)}{s + \gamma}. \quad (16)$$

A source–kernel representation is

$$\mathcal{R}_H(N) = \int_{-\infty}^N K_H(N - N') J_H(N') dN'. \quad (17)$$

At this level J_H is a dual source or drive; identifying it with an occupancy current is an additional source-gauge claim.

6.2 Approximate lumpability and spectral gap

Definition 6.1 (Rank-one horizon dominance). *The horizon port is rank-one over the low-redshift window if*

$$d\mu_H^{(N)}(\gamma) = Z_1^{(N)} \delta(\gamma - \Gamma_H^{(N)}) d\gamma + d\mu_\perp^{(N)}(\gamma), \quad (18)$$

with

$$0 < \Gamma_H^{(N)} \ll \gamma \quad \text{for } \gamma \in \text{supp } d\mu_\perp^{(N)}, \quad (19)$$

and $d\mu_\perp^{(N)}$ weak over the observational window.

Then

$$K_H(\tau) = Z_1 e^{-\Gamma_H \tau} + K_\perp(\tau), \quad |K_\perp(\tau)| \lesssim Z_\perp e^{-\Gamma_2 \tau}. \quad (20)$$

Keeping the first band gives

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = Z_1 J_H. \quad (21)$$

Absorbing Z_1 into the source or amplitude convention gives the one-state memory equation

$$\boxed{\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H.} \quad (22)$$

The production normalization is

$$\mathcal{R}_H(a) = R_0 \widehat{\mathcal{R}}_H(a), \quad R_0 = \mathcal{R}_H(1) = 3 - s, \quad \widehat{\mathcal{R}}_H(1) = 1. \quad (23)$$

The commonly used shape is the output alias

$$X_b(a) \equiv \widehat{\mathcal{R}}_H(a). \quad (24)$$

It is not, by itself, a uniquely reconstructed microscopic kernel.

7 Ward conditioning and Ricci-port suppression

7.1 Conditioning

Let $C_\alpha(\omega)$ be finite-screen Ward-constraint observables. The physical reference state is the entropy projection

$$\sigma_N^{\text{phys}} = \arg \min_{\rho: \mathbb{E}_\rho C_\alpha = 0} D_{\text{KL}}(\rho \| \sigma_N). \quad (25)$$

Equivalently, one may work with a conditional expectation

$$\mathbb{E}_{\text{Ward}, N} : L^2(\sigma_N) \rightarrow L^2(\sigma_N^{\text{phys}}) \quad (26)$$

and a constrained relaxation operator

$$\mathcal{L}_N^{\text{phys}} = \mathbb{E}_{\text{Ward}, N} \mathcal{L}_N \mathbb{E}_{\text{Ward}, N}. \quad (27)$$

This is an information projection, not necessarily a literal projection onto a hard constraint subspace.

7.2 Schur-complement suppression

Let W denote the Weyl/optical output block and R the Ricci/density leakage block. The quadratic covariance or stiffness matrix may be written as

$$K = \begin{pmatrix} K_{WW} & K_{WR} \\ K_{RW} & K_{RR} \end{pmatrix}. \quad (28)$$

Contraction over Ward-constrained Ricci variables gives the effective Weyl block

$$K_{WW}^{\text{eff}} = K_{WW} - K_{WR} K_{RR}^{-1} K_{RW}. \quad (29)$$

If K_{RR} is stiff, the Ricci response is suppressed. In the cosmological scalar response this becomes

$$\boxed{\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right)}. \quad (30)$$

The Weyl response remains first order:

$$\boxed{\Sigma(a, k) - 1 = -\kappa \mathcal{R}_H(a) + O(\mathcal{R}_H^2)}. \quad (31)$$

At the optically unbiased fixed point, $\kappa = 3/4$.

The Markov-screen model supplies a stiff Ricci port that suppresses density-like response. The explicit $\mathcal{H}^2/k^2$ scaling in cosmological reduction is a Ward-to-Boltzmann scalar-sector bridge used in the production G1DE likelihood, not a property derived directly from the finite Markov chain.

8 Finite Markov-Screen Realization Theorem

Theorem 8.1 (Finite Markov-Screen Realization of the G1DE normal form). *Assume a sequence of finite Markov screens*

$$(\Omega_N, \sigma_N, Q_N, F_{a,N}), \quad a \in \{A, S_1, S_2, T, H, R\},$$

satisfying:

1. Q_N is reversible with respect to σ_N ;
2. the macro-port quasipotential has a positive Hessian at the passive fixed point;
3. the four optical ports A, S_1, S_2, T are unbiased in covariance and dissipation, and the optical port subspace is invariant under the reversible generator with a common low-spectrum Stieltjes measure,

$$\chi_{\text{opt}}(s) = \chi_{\text{opt}}^0(s)I_4 + O(\epsilon_{\text{aniso}});$$

4. the horizon port H is approximately lumpable and dominated by one slow spectral band;
5. Ward conditioning makes the Ricci port entropy-stiff;

6. the Weyl-source normalization is fixed, so no independent optical amplitude $A(a, k)$ is fitted.

Then the finite-screen response is matrix-Stieltjes, the optical projection coefficient is $\kappa_{BW} = 3/4$, the horizon output is rank-one to leading order, and the low-redshift Weyl-normalized scalar response is

$$\Sigma(a, k) - 1 = -\frac{3}{4}\mathcal{R}_H(a) + O(\mathcal{R}_H^2), \quad (32)$$

$$\mu(a, k) - 1 = O(\mathcal{R}_H^2) + O\left(\mathcal{R}_H \frac{\mathcal{H}^2}{k^2}\right). \quad (33)$$

With $\mathcal{R}_H(a) = (3-s)\widehat{\mathcal{R}}_H(a)$ and $\widehat{\mathcal{R}}_H(1) = 1$, this gives

$$\boxed{\Sigma(a, k) - 1 = -\frac{3}{4}(3-s)\widehat{\mathcal{R}}_H(a), \quad \mu(a, k) \simeq 1.} \quad (34)$$

Proof sketch. Reversibility gives self-adjointness and nonnegativity of $\mathcal{L}_N = -Q_N$. The spectral theorem gives the matrix-Stieltjes representation (8). Optical unbiasedness makes the four-dimensional optical block proportional to the identity, so the Weyl-active trace fraction is $3/4$. Rank-one horizon dominance reduces the horizon Stieltjes response to the one-pole equation (22). Ward conditioning contracts or Schur-complements the Ricci port, producing (30). Weyl normalization identifies the optical seed with a fractional correction to the GR Weyl source, producing (31). Combining these statements gives (34). $\square$

9 An Explicit Minimal Toy Model

(Existence Proof-of-Concept, Not a Spacetime Microstate Proposal)

This section gives a deliberately simple finite-state model realizing the assumptions. It is a proof-of-concept, not a proposed microscopic theory of spacetime.

9.1 Cell state

Let a cell state be

$$\omega_i = (o_i, h_i, r_i), \quad (35)$$

where

$$o_i = (o_A, o_{S_1}, o_{S_2}, o_T) \in \{-1, +1\}^4, \quad h_i \in \{-1, +1\}, \quad r_i \in \{-1, 0, +1\}. \quad (36)$$

The optical observables are the four Rademacher components

$$f_A = o_A, \quad f_{S_1} = o_{S_1}, \quad f_{S_2} = o_{S_2}, \quad f_T = o_T. \quad (37)$$

The horizon and Ricci ports are

$$f_H = h_i, \quad f_R = r_i. \quad (38)$$

9.2 Reference state

Take the single-cell reference measure

$$\sigma_1(o, h, r) = \frac{1}{16} \frac{e^{-\lambda_H V_H(h)}}{Z_H} \frac{e^{-\lambda_R r^2}}{Z_R}. \quad (39)$$

The optical part is exactly unbiased. The parameter $\lambda_R \gg 1$ makes Ricci leakage rare or stiff. The product reference state is

$$\sigma_N = \sigma_1^{\otimes N}. \quad (40)$$

9.3 Reversible jumps

Let the Markov dynamics flip one coordinate at a time. Optical flips all have the same rate r_o :

$$o_a \mapsto -o_a, \quad a \in \{A, S_1, S_2, T\}, \quad \text{rate } r_o. \quad (41)$$

Horizon flips are slow:

$$h \mapsto -h, \quad \text{rate } \epsilon r_H, \quad 0 < \epsilon \ll 1. \quad (42)$$

Ricci moves are either strongly penalized by σ_1 or rapidly relaxed with rate $r_R \gg r_o$ and detailed-balance Metropolis factors:

$$r(\omega, \omega') = r_0 \min \left(1, \frac{\sigma_1(\omega')}{\sigma_1(\omega)} \right) \quad (43)$$

for r -moves. This ensures detailed balance.

The spectrum has:

- an optically degenerate band at the common optical relaxation scale;
- a slow horizon band at $\Gamma_H \sim \epsilon r_H$;
- a stiff or weak Ricci leakage sector controlled by λ_R and r_R .

Thus the toy model realizes the hierarchy

$$\text{optical } 3/4, \quad \text{rank-one } \mathcal{R}_H, \quad \text{Ward-stiff } \mu \text{ leakage}. \quad (44)$$

9.4 Explicit two-state flip spectrum

For a symmetric two-state flip variable $y \in \{-1, +1\}$ with transition rate r ,

$$Q = \begin{pmatrix} -r & r \\ r & -r \end{pmatrix}. \quad (45)$$

The eigenvectors are the constant mode 1 and the contrast mode y , with eigenvalues

$$\lambda_0 = 0, \quad \lambda_1 = -2r. \quad (46)$$

Equivalently, the positive relaxation generator $\mathcal{L} = -Q$ has spectrum

$$\gamma_0 = 0, \quad \gamma_1 = 2r. \quad (47)$$

Thus, for the minimal product-chain toy,

$$\gamma_{\text{opt}} = 2r_o, \quad \Gamma_H = 2\epsilon r_H, \quad \gamma_R \simeq 2r_R, \quad (48)$$

with possible additional Ricci stiffness supplied by the penalty scale λ_R in the conditioned ensemble.

If

$$\Gamma_H \ll \gamma_{\text{opt}}, \gamma_R, \quad (49)$$

then the horizon port is the unique slow output mode. Its scalar response is

$$\chi_H(s) \simeq \frac{Z_H}{s + \Gamma_H}, \quad (50)$$

and in time domain

$$K_H(\tau) \simeq Z_H e^{-\Gamma_H \tau}. \quad (51)$$

After absorbing Z_H into the source normalization, the rank-one output obeys

$$\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H. \quad (52)$$

For a single-pole Stieltjes response,

$$m_n = \frac{Z_H}{\Gamma_H^{n+1}}, \quad (53)$$

so the rank-one Hankel diagnostic satisfies

$$m_0 m_2 - m_1^2 = 0. \quad (54)$$

A positive value of this determinant in future output reconstructions would indicate a multiband passive horizon response rather than failure of the finite Markov-screen class.

10 Model hierarchy implied by the realization

The finite Markov-screen construction gives the following interpretation of production branches.

Branch	Markov-screen meaning	Failure/demotion interpretation
$M_{3/4}^{\text{rank1}}$	optical-unbiased, rank-one horizon mode	primary low-spectrum branch
$M_{3/4}^{\text{multiband}}$	optical-unbiased, two or more horizon bands	broader passive memory spectrum
M_κ	optical covariance/dissipation not isotropic	weighted compliance fixed point
const- Σ	only DC Weyl offset resolved	output shape not resolved
Ward leakage	Ricci port not infinitely stiff	controlled constrained-port correction
G1DE-2	free response envelope	outside passive locked realization
free $A(a, k)$	arbitrary Weyl-port normalization	leaves G1DE class

11 Null tests

11.1 Optical isotropy

The optical covariance and dissipation matrices should be approximately proportional to identity:

$$\text{Cov}(F_a, F_b) \propto \delta_{ab}, \quad \mathcal{E}(F_a, F_b) \propto \delta_{ab}, \quad a, b \in \{A, S_1, S_2, T\}. \quad (55)$$

A persistent evidence preference for free κ means this symmetry is broken.

11.2 Horizon Stieltjes positivity

The reconstructed horizon memory moments must lie in a positive Stieltjes cone:

$$H_{ij} = m_{i+j} \succeq 0. \quad (56)$$

Violation means no passive reversible Markov-screen realization in this class.

11.3 Rank-one consistency

Rank-one predicts

$$m_0 m_2 - m_1^2 \simeq 0. \quad (57)$$

A positive deviation indicates a multi-band passive memory extension rather than immediate failure of the G1 class.

11.4 Ward leakage

The subhorizon diagnostic should satisfy

$$\frac{\mu - 1}{\Sigma - 1} = O(\mathcal{R}_H) + O\left(\frac{\mathcal{H}^2}{k^2}\right). \quad (58)$$

If it is $O(1)$, the Ward-stiff Ricci port hypothesis fails.

11.5 Normalization

Sampling a free optical amplitude $A(a, k)$ corresponds to rescaling the Weyl port measure itself. That changes the Markov-screen impedance normalization and is not a GIDE production branch.

12 Open microscopic tasks

The construction reduces the remaining microscopic work to precise tasks:

1. construct physically motivated finite screen state spaces Ω_N or finite screen algebras $\mathcal{A}_N$;
2. derive the passive reference measures σ_N from finite-access screen ledgers;
3. derive reversible or CP-detailed-balance dynamics Q_N from finite-screen update rules;
4. prove optical unbiasedness or classify its weighted-compliance deformations;
5. prove approximate lumpability and spectral gap for the horizon port;
6. derive Ward constraints as entropy projections or conditional expectations;
7. prove convergence of the finite response matrices to a stable continuum screen impedance.

These are hard problems. The value of the Markov-screen realization is that it turns them into explicit mathematical targets rather than informal gaps.

Generator checklist. An admissible Markov-screen generator must demonstrate: (1) finite state space; (2) detailed balance or controlled non-equilibrium breaking; (3) slow horizon mode; (4) Ward-stiff Ricci leakage suppression; (5) output $R_H(a)$ matching; (6) no hidden free $A(a, k)$.

13 Relation to Large-Deviation and Screen-RG Limits

The D7 large-deviation response formalism in the appendix generates a family of scale-dependent effective Markov generators. The RG flow of these generators connects to the D8 metric-envelope reconstruction scale and the D10 branch moduli space. A full screen-RG treatment is reserved for future work.

14 Conclusion

The finite Markov-screen construction gives a concrete microscopic prototype for the G1DE normal form. A reversible finite-state screen automatically produces a positive matrix-valued Stieltjes response. Optical unbiasedness gives the Weyl-active fraction $3/4$. A slow lumpable horizon port gives the rank-one output equation $\mathcal{R}'_H + \Gamma_H \mathcal{R}_H = J_H$. Ward conditioning makes Ricci leakage higher-order or subhorizon-suppressed. Weyl normalization then gives

$$\Sigma - 1 = -\frac{3}{4}\mathcal{R}_H + O(\mathcal{R}_H^2), \quad \mu \simeq 1.$$

With $\mathcal{R}_H(a) = (3-s)\hat{\mathcal{R}}_H(a)$, the G1DE- $M_{3/4}$ production branch follows.

The most compact statement is:

In the D7 realization class, G1DE- $M_{3/4}$ is the rank-one Weyl-port output of a finite reversible Markov screen.

D10 treats the D7 realization as one constructive chart on the $M_{3/4}$ branch, not as the unique microphysical representative.

This note does not complete the microscopic derivation of the true finite-screen ensemble. It supplies a minimal realization target showing that the G1DE- $M_{3/4}$ interface is mathematically implementable by finite-state passive dynamics.

A Relation between response and large deviations

For independent cells, the empirical macro-port vector

$$x_N = \frac{1}{N} \sum_{i=1}^N f(\omega_i)$$

satisfies a large-deviation principle

$$\sigma_N\{x_N \simeq x\} \asymp e^{-NI(x)}.$$

The rate function is the Legendre dual of the cumulant generating function

$$\Lambda(\lambda) = \log \mathbb{E}_{\sigma_1} e^{\lambda \cdot f}.$$

Near the fixed point,

$$I(x) = \frac{1}{2} x^T \text{Cov}(f)^{-1} x + O(|x|^3).$$

Thus the static Hessian is the inverse covariance. In reversible dynamics, the dynamic response is controlled by the same covariance together with the relaxation operator $\mathcal{L}_N = -Q_N$.

B Sign conventions for generators

The Markov generator Q_N has nonpositive spectrum and satisfies

$$\frac{d}{dt}\mathbb{E}[g(\omega_t)] = \mathbb{E}[Q_N g(\omega_t)].$$

For response theory it is convenient to use the nonnegative relaxation operator

$$\mathcal{L}_N = -Q_N.$$

The resolvent is then $(s + \mathcal{L}_N)^{-1}$, and the response is Stieltjes in the variable s .

C Notation dictionary

Symbol	Meaning
$\mathcal{R}_H(a)$	data-facing scalar horizon output
$\hat{\mathcal{R}}_H(a)$	normalized output shape, $\hat{\mathcal{R}}_H(1) = 1$
$X_b(a)$	production alias for $\hat{\mathcal{R}}_H(a)$
R_0	present amplitude, $R_0 = \mathcal{R}_H(1) = 3 - s$
K_H, J_H	source–kernel lift representatives
F_a	finite-screen port observables
Q_N	reversible Markov generator
$\mathcal{L}_N = -Q_N$	nonnegative relaxation operator
$\chi_{ab}(s)$	matrix-valued screen impedance
κ_{BW}	optical Weyl-active compliance fraction
$A(a, k)$	forbidden free optical amplitude; fixed by Weyl normalization

Metric Reconstruction and the Universal Envelope of Finite Causal-Screen Response

FDS–G1 D8: Causal Access, Entropy Density, Modular Time, and the Einstein Envelope
v1.2 Body-Integrated Release

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS–G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This paper develops the reconstruction side of the FDS–G1 finite causal-screen program. Earlier G1 papers start from a finite screen entropy-response spine,

$$S_{\text{scr}} \rightarrow \omega \rightarrow \eta_{\text{loc}} \rightarrow G_{\text{loc}}^{-1} \rightarrow F_p(k) \rightarrow E_{\mu\nu} \rightarrow R_{\mu\nu}^{\text{scr}},$$

and use typed bridges to reach an Einstein-like equilibrium branch and a controlled Weyl-residual branch. The present paper asks a more structural question: when does a finite causal-screen response system admit an effective Lorentzian geometry at all, and what is the universal object reconstructed by such a system?

The central proposal is that effective geometry is the universal, or least-defect, envelope of finite causal-screen response. Finite causal access first determines a conformal causal structure in the continuum limit. A universal quadratic principal symbol determines metricity. A calibrated screen entropy density fixes the conformal scale up to the D0 modular calibration. If the resulting metric envelope is local, diffeomorphism-covariant, symmetric, divergence-free, and second-order in four dimensions, the leading equilibrium tensor is the Einstein tensor plus a cosmological integration term. Failure of these conditions does not produce arbitrary dark stress; it produces typed demotions: Finsler or multi-cone nonmetricity, equivalence-principle violation, scalar-capacity flow, conserved obstruction residuals, or rejection of the metric envelope.

The paper states four main conditional results. First, a stable finite causal-access order determines a conformal Lorentzian structure when its continuum limit satisfies causal distinguishability and local cone stability. Second, a universal quadratic characteristic cone yields a metric principal symbol; non-quadratic, multi-cone, or species-dependent symbols are nonmetric branches. Third, finite distinguishability by itself cannot fix a dimensionful gravitational constant: FDS fixes the ledger form, while D0-type modular calibration fixes physical scale. Fourth, the metric envelope can be formulated as a least-defect variational problem whose zero-defect branch is Einstein-like and whose finite nonzero defect branch must pass the G1 residual admissibility rules. The result is not a completed microscopic derivation of gravity, but a reconstruction theorem stack: if finite screen response has a stable local continuum envelope, then the leading geometric representation is strongly constrained.

1 Purpose and claim-status discipline

The FDS–G1 program is built around a finite causal-screen entropy functional

$$S_{\text{scr}} : X_{\text{phys}} \rightarrow \mathbb{R}, \quad X_{\text{phys}} = X_{\text{raw}}/G_{\text{scr}}, \quad (1)$$

and its quotient-invariant response one-form

$$\omega = \eta_i(X) dX^i. \quad (2)$$

When the response is exact, $\omega = d(S_{\text{scr}}/k_B)$. When it is not exact, the non-closure is treated as an obstruction only after gauge, missing-variable, and memory-lift possibilities have been exhausted. The present paper adds one level below that structure. It asks what it means for such screen response data to be represented by an effective spacetime geometry.

1.1 What D8 claims

D8 makes the following conditional claims.

1. A stable finite causal-access order can determine a conformal causal geometry in the continuum limit.
2. A universal quadratic principal symbol is the metricity condition for a Lorentzian envelope.
3. Screen entropy density fixes the conformal scale only after a physical calibration bridge, such as the D0 modular calibration, has been supplied.
4. The Einstein branch is the unique leading low-derivative, local, divergence-free metric envelope in four dimensions.
5. Nonzero finite defect is not arbitrary dark stress; it must be a conserved, quotient-invariant, finitely described residual.

1.2 What D8 does not claim

D8 does not prove that the microscopic universe is a particular finite screen algebra net. It does not prove the D0 finite-index density lemma, the D0 boost modular calibration lemma, or the Bekenstein–Hawking coefficient from bare finite-state counting. It does not prove that all possible finite causal-screen systems have a metric continuum limit. It classifies what must be true if such a limit is to be represented by a Lorentzian metric geometry.

Object	Role in D8	Status
finite causal access	determines causal order / cone data	reconstruction input
principal symbol $P(x, \xi)$	tests metricity of continuum response	conditional theorem
entropy density η_0	fixes scale after calibration	D0-dependent
least-defect envelope	variational representation of geometry	principle / theorem target
Einstein envelope	leading low-derivative metric branch	conditional uniqueness
nonmetric branches	failure classes for metricity	demotion taxonomy

Table 1: Claim-status map for D8.

2 Finite causal access and conformal reconstruction

Definition 1 (Finite causal-access system). *A finite causal-access system is a tuple*

$$\mathfrak{C}_N = (V_N, \preceq_N, \mathcal{B}_N, \mathcal{U}_N), \quad (3)$$

where V_N is a finite set of record events or screen cells, $\preceq_N$ is a causal-access preorder, $\mathcal{B}_N$ is a family of finite causal screen boundaries, and $\mathcal{U}_N$ is an operational update relation compatible with $\preceq_N$. Two labels in V_N that are related by relabeling, refinement, or gauge redundancy represent the same physical access fact.

The continuum limit is not assumed to be metric. It is a limit in which finite access relations admit a stable local order structure, local cone neighborhoods, and a quotient-stable notion of screen boundary. A metric appears only if the continuum causal cones are representable by a Lorentzian conformal class.

Definition 2 (Cone-stable continuum limit). *A sequence $\{\mathfrak{C}_N\}$ has a cone-stable continuum limit if there exists a topological manifold M , embeddings of coarse record classes into M , and closed pointed cones $C_p \subset T_p M$ such that local access relations converge to causal reachability with respect to C_p , and small refinements of the screen discretization perturb C_p continuously.*

Principle 1 (Causal access first). *The primitive continuum datum reconstructed from finite screen access is not a metric tensor but a causal cone assignment. Metric geometry is a further representation of that cone assignment.*

Theorem 1 (Conditional Theorem: Causal-Access Reconstruction of a Conformal Cone). *Suppose a finite causal-access sequence admits a cone-stable, causally distinguishing, strongly causal continuum limit whose local cones are sharp, convex, and have smooth null boundaries. If the null boundaries are quadratic hypersurfaces in each tangent space, then there exists a Lorentzian conformal class $[g]$ whose null cones agree with the reconstructed causal-access cones. The conformal factor is not fixed by causal access alone.*

Proof sketch. Causal access determines the inclusion relation between local chronological futures and pasts. Under causal distinguishability and strong causality, this determines the null cone distribution up to conformal rescaling. Smoothness and quadraticity of the cone boundary allow the cone to be represented locally as the zero set of a nondegenerate homogeneous quadratic form of Lorentzian signature. Multiplying the quadratic form by a positive scalar leaves the cone invariant, so the reconstructed datum is a conformal class rather than a metric representative. $\square$

Remark 1 (Why this matters for G1). *G1 is not forced to assume a metric at the primitive level. The finite screen program may begin with causal access and only later reconstruct a Lorentzian metric envelope if the continuum cone data satisfy the required stability and quadraticity conditions.*

3 Principal-symbol metricity

The causal reconstruction theorem is kinematic. A dynamical continuum response also has a characteristic structure. Let the linearized screen-to-envelope response have principal symbol

$$P(x, \xi), \quad (4)$$

where $\xi \in T_x^* M$. The characteristic set $P(x, \xi) = 0$ determines high-frequency propagation.

Definition 3 (Metric principal symbol). *A principal symbol is metric at x if, up to a positive scalar density,*

$$P(x, \xi) = g^{\mu\nu}(x) \xi_\mu \xi_\nu \quad (5)$$

for a Lorentzian inverse metric $g^{\mu\nu}$.

Theorem 2 (Conditional Theorem: Principal-Symbol Metricity). *(conditional on a universal species-independent homogeneous quadratic characteristic cone). If the continuum limit of finite causal-screen response has a universal, species-independent, homogeneous quadratic characteristic cone, then the effective causal envelope is Lorentzian metric at leading order. If the characteristic polynomial is non-quadratic, factorizes into inequivalent cones, or depends on the matter channel, then the correct G1 demotion is respectively Finsler/nonmetric response, multi-cone or birefringent response, or equivalence-principle violation. None of these failures may be reabsorbed as an ordinary metric residual $R_{\mu\nu}^{\text{scr}}$.*

Proof. A universal quadratic characteristic cone is represented by a nondegenerate quadratic form on T_x^*M . Hyperbolicity and finite-speed propagation select Lorentzian signature. Universality identifies the same cone for all leading response channels. If the symbol is not quadratic, no Lorentzian metric can represent the full characteristic geometry. If it factorizes into distinct quadratic factors, there is no single metric cone. If it is channel-dependent, universal coupling fails. These alternatives are different continuum envelopes, not hidden metric stress-energy tensors. $\square$

Principal-symbol outcome	Continuum interpretation	G1 status
universal quadratic cone	Lorentzian metric envelope	admissible metric branch
non-quadratic single cone	Finsler-like envelope	nonmetric branch
multiple cones	birefringent / multi-metric response	demoted branch
species-dependent cone	equivalence-principle violation	constrained or rejected
no stable cone	no continuum spacetime envelope	failure of metric reconstruction

Table 2: Metricity and demotion classes from principal-symbol analysis.

4 Entropy density and conformal scale

Causal access reconstructs only conformal geometry. To fix a scale, the screen response must supply a local density. In G1 this density is

$$\eta_{\text{loc}}(x) = \frac{1}{k_B} \frac{\delta S_{\text{scr}}}{\delta A(x)}. \quad (6)$$

At a calibrated fixed point,

$$\eta_{\text{loc}}(x) = \eta_0, \quad \frac{S_{\text{scr}}}{k_B} = \eta_0 A + o(A). \quad (7)$$

Lemma 1 (No scale without calibration). *Pure finite distinguishability and finite state counting can fix the logarithmic ledger form*

$$S_{\text{scr}} = k_B \log N_{\text{scr}} \quad (8)$$

and can motivate an area-density form

$$dC_{\text{scr}} = \eta_0 dA. \quad (9)$$

They cannot by themselves determine the dimensionful value of η_0 , G , or ℓ_P . A physical scale-setting bridge is required: an area unit, quantum units, and a modular or horizon calibration relating screen entropy to boost energy.

Proof. The finite ledger $\log N$ is dimensionless. The density η_0 carries dimensions of inverse area. A dimensionful inverse area cannot be determined by pure counting data without specifying a physical area scale or an equivalent calibration. The G1 relation

$$G = \frac{c^3}{4\hbar\eta_0} \quad (10)$$

therefore requires a physical identification of η_0 , such as the D0 finite-index density plus boost modular calibration route or an external Bekenstein–Hawking horizon bridge. $\square$

Principle 2 (Form from finite distinction, scale from calibration). *FDS fixes the ledger form and admissibility structure. D0-type modular calibration fixes the physical scale if its finite-index density and boost modular covariance lemmas hold.*

5 v1.2 Relative-Entropy Metric Envelope Interface

Claim type: Conditional target The metric envelope reconstructed in D8 is the stationary point of the D0 relative-entropy action:

$$\mathcal{I}_{\text{G1}} = D_{\text{rel}}[\rho_{\text{scr}} \parallel \rho_{\text{env}}] + \lambda_{\text{cap}} C_{\text{cap}} + \lambda_{\text{Ward}} C_{\text{Ward}} + \lambda_q C_{\text{quotient}}$$

The least-defect metric envelope corresponds to the state that minimizes the relative entropy subject to the finite-screen constraints.

Boundary note: This is a conditional derivation route connecting the relative entropy action principle to the metric reconstruction framework. It does not prove that the stationary point uniquely recovers full General Relativity in all regimes.

6 Screen-time alignment

Finite screen systems are not static. They update. Let U_τ denote an operational update semigroup on physical screen states and let σ_t^ρ denote modular flow for a reference state ρ :

$$\sigma_t^\rho(A) = \rho^{it} A \rho^{-it}. \quad (11)$$

In local equilibrium, D0 assumes that modular flow is boost-calibrated. D8 separates this into an alignment statement.

Principle 3 (Screen-time alignment). *In an equilibrium causal-screen patch, operational update time, modular time, and local boost time agree up to calibration. Mismatch among these flows is a memory or residual source, not a new untyped time variable.*

The alignment chain is

$$\text{finite update} \longrightarrow \text{modular time} \longrightarrow \text{boost time} \longrightarrow \text{horizon thermodynamics}. \quad (12)$$

If this chain fails, the failure must be typed: non-equilibrium memory, non-boost modular flow, absence of a local Rindler approximation, or breakdown of continuum envelope.

7 Null response and tensor envelope

Once a Lorentzian conformal cone exists, screen response can be tested along null directions. Let $F_p(k)$ be the local all-null response at p , with k null.

Theorem 3 (Null response tensorization). *Let $(T_p M, g_p)$ be a Lorentzian tangent space of dimension $D \geq 3$. If $F_p(k)$ is smooth on the null cone, homogeneous quadratic in k , and extends locally to a quadratic form, then there exists a symmetric tensor $E_{\mu\nu}(p)$ such that*

$$F_p(k) = E_{\mu\nu}(p) k^\mu k^\nu \quad (13)$$

for all null k . The tensor is determined only up to trace:

$$E_{\mu\nu} \sim E_{\mu\nu} + \Phi g_{\mu\nu}, \quad (14)$$

because $g_{\mu\nu} k^\mu k^\nu = 0$ on the null cone.

Proof. A homogeneous quadratic extension of F_p is represented by a symmetric bilinear form by polarization. If two such bilinear forms agree on all null vectors, their difference $X_{\mu\nu}$ obeys $X_{\mu\nu} k^\mu k^\nu = 0$ on the null cone. In Lorentzian dimension at least three, a symmetric form vanishing on the null cone is proportional to the metric. Hence the representative is unique modulo trace. $\square$

8 Principle: Least-Defect Metric Envelope

8.1 Category-level formulation

Let **ScrResp** be a category whose objects are finite causal-screen response systems

$$\mathcal{S} = (X_{\text{phys}}, S_{\text{scr}}, \omega, \mathcal{C}_{\text{resp}}), \quad (15)$$

and whose morphisms are quotient-preserving coarse-grainings, refinements, and restriction maps. Let **GeomEnv** be a category whose objects are effective geometric envelopes

$$\mathcal{G} = (M, g_{\mu\nu}, E_{\mu\nu}, R_{\mu\nu}^{\text{scr}}) \quad (16)$$

obeying metricity, all-null response, and Ward/Bianchi admissibility. Existence and uniqueness of the least-defect envelope are open problems; this section formulates the variational target.

A restriction functor

$$\mathcal{R} : \mathbf{GeomEnv} \rightarrow \mathbf{ScrResp} \quad (17)$$

assigns to a geometric envelope the screen response data seen by its local causal screens. A universal envelope functor, if it exists, is a left reconstruction

$$\mathcal{E} : \mathbf{ScrResp} \rightarrow \mathbf{GeomEnv} \quad (18)$$

with the adjunction-like property

$$\text{Hom}_{\mathbf{GeomEnv}}(\mathcal{E}(\mathcal{S}), \mathcal{G}) \simeq \text{Hom}_{\mathbf{ScrResp}}(\mathcal{S}, \mathcal{R}(\mathcal{G})). \quad (19)$$

This statement is a target rather than an assumed theorem for arbitrary screens. It states the strongest form of the G1 reconstruction principle: geometry is the universal envelope through which screen response factors.

8.2 Least-defect formulation

For practical analysis one can replace the universal property by a variational envelope. Define

$$\mathcal{I}_{\text{scr}}[g, E, R; S_{\text{scr}}] = \int_M \left[\mathcal{D}_{\text{null}}(F_{\text{scr}}, E_{\mu\nu} k^\mu k^\nu) + \lambda^\nu \nabla^\mu E_{\mu\nu} + \mathcal{P}(R_{\mu\nu}^{\text{scr}}) \right] \sqrt{-g} d^4x. \quad (20)$$

Here $\mathcal{D}_{\text{null}}$ measures mismatch between screen null response and tensor envelope, λ^ν enforces Ward closure, and $\mathcal{P}$ penalizes residual complexity. Gauge artifacts, missing variables, and memory lifts must be removed before $\mathcal{P}$ is evaluated.

Definition 4 (Least-defect metric envelope). *A least-defect metric envelope for S_{scr} is a stationary or minimizing triple*

$$(g, E, R)^* = \arg \min_{g, E, R} \mathcal{I}_{\text{scr}}[g, E, R; S_{\text{scr}}] \quad (21)$$

within the admissible metricity, quotient, and Ward/Bianchi class.

Proposition 1 (Envelope branch trichotomy). *For a least-defect metric envelope:*

1. *if $\mathcal{I}_{\text{scr}}^{\min} = 0$, the response is integrable and belongs to the Einstein-like equilibrium branch;*
2. *if $0 < \mathcal{I}_{\text{scr}}^{\min} < \infty$ and the residual is conserved, quotient-invariant, and finitely described, the response belongs to an admissible residual branch;*
3. *if no finite minimum exists under metricity and Ward constraints, the system belongs to a nonmetric or rejected branch.*

9 Einstein envelope uniqueness at leading order

Once a metric envelope has been reconstructed, the equilibrium tensor is strongly restricted.

Theorem 4 (Einstein envelope uniqueness at leading order). *In four dimensions, assume the zero-defect envelope is local, metric, diffeomorphism-covariant, symmetric, divergence-free, and contains at most second derivatives of $g_{\mu\nu}$. Then its leading equilibrium tensor is*

$$E_{\mu\nu} = G_{\mu\nu} + \Lambda g_{\mu\nu}. \quad (22)$$

Deviations from this branch must appear as higher-derivative EFT corrections, scalar-capacity fields, admissible conserved residuals, or nonmetric demotions.

Proof sketch. The theorem is the standard Lovelock-type uniqueness statement specialized to four-dimensional local metric field equations. A symmetric, divergence-free, second-order tensor constructed naturally from the metric is a linear combination of the Einstein tensor and the metric. The coefficient of the Einstein tensor is fixed by the D0 or weak horizon calibration of G , and the coefficient of the metric is the cosmological integration constant. $\square$

Remark 2. *G1 does not choose Einstein geometry by fiat. Under the reconstruction assumptions, Einstein geometry is the unique leading low-derivative conserved metric envelope.*

10 Equivalence principle as screen-flux universality

Definition 5 (Screen load universality). *Matter channels satisfy screen load universality if all local matter perturbations couple to the same modular boost load Q_χ and the same screen response density η_0 .*

Proposition 2 (Screen equivalence principle). *If all matter channels enter the local screen first law through a universal modular load*

$$\delta Q_\chi = \sum_a \delta Q_\chi^{(a)} \quad (23)$$

and the screen response density is species-independent, then the effective gravitational stiffness is universal:

$$G_a = G. \quad (24)$$

Species-dependent screen loads or species-dependent response densities are equivalence-principle-violating residuals and must be bounded or rejected.

11 Relation to D0–D7

Paper	Input to D8	D8 extension
Core	entropy-response spine	universal / least-defect envelope
D0	finite-index density and boost calibration	scale-setting and modular-time alignment
D1	sheaf/gluing obstruction	envelope obstruction target for D9
D2	optical projection theorem	metric optical sector after reconstruction
D5	Ward/no-free residual rules	admissible residual constraint in (20)
D6	output quotient	non-unique memory realization after envelope
D7	finite Markov-screen realization	constructive chart for passive envelope branch

Table 3: How D8 upgrades existing G1 components into a reconstruction framework.

12 Open problems

The reconstruction program leaves several hard problems open.

1. Prove a screen-access version of causal-order conformal reconstruction for finite approximants with explicit convergence criteria.
2. Derive universal quadratic principal symbols from finite screen update rules rather than assuming them.
3. Prove D0 boost modular calibration from finite-index causal inclusions.
4. Give a rigorous existence theorem for least-defect envelopes in a specified function space.
5. Characterize nonmetric branches sufficiently sharply to compare them with observational consistency tests.

13 Conclusion

D8 upgrades the G1 geometric claim from a metric-first response model to a reconstruction statement. Finite causal access determines conformal cone data; universal quadratic propagation gives metricity; entropy density fixes scale only after physical calibration; and the leading zero-defect metric envelope is Einstein-like. Nonzero finite defects are not arbitrary dark stress.

They must be conserved, quotient-invariant, and finitely described. In this form, G1 becomes a finite causal-screen reconstruction theory: geometry is the universal least-defect conserved envelope of finite screen response.

14 Selected references

References

- [1] S. W. Hawking and G. F. R. Ellis, *The Large Scale Structure of Space-Time*, Cambridge University Press (1973).
- [2] D. B. Malament, “The class of continuous timelike curves determines the topology of spacetime,” *Journal of Mathematical Physics* **18**, 1399–1404 (1977).
- [3] S. W. Hawking, A. R. King, and P. J. McCarthy, “A new topology for curved space-time which incorporates the causal, differential, and conformal structures,” *Journal of Mathematical Physics* **17**, 174–181 (1976).
- [4] D. Lovelock, “The Einstein tensor and its generalizations,” *Journal of Mathematical Physics* **12**, 498–501 (1971).
- [5] R. M. Wald, *General Relativity*, University of Chicago Press (1984).
- [6] R. Geroch, “Method for generating solutions of Einstein’s equations,” and related work on causal and differential structures.
- [7] T. Jacobson, “Thermodynamics of spacetime: The Einstein equation of state,” *Physical Review Letters* **75**, 1260–1263 (1995).
- [8] J. J. Bisognano and E. H. Wichmann, “On the duality condition for a Hermitian scalar field,” *Journal of Mathematical Physics* **16**, 985–1007 (1975).
- [9] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, FDS–G1 Core manuscript (2026).
- [10] Y. Wu, *Screen Modular Anomaly Inflow and the Einstein Calibration of Finite Access Entropy*, FDS–G1 D0 (2026).
- [11] Y. Wu, *Finite-Access Sheaf Geometry and Response Gluing Obstructions*, FDS–G1 D1 (2026).
- [12] Y. Wu, *Weyl-Normalized Ward Completion and Identifiability Rules*, FDS–G1 D5 (2026).

Obstruction Exact Sequence and No-Free-Dark-Sector Classification

FDS–G1 D9: From Response Curvature to Admissible Residuals
v1.2 Body-Integrated Release

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS–G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This paper develops the obstruction and residual-classification layer of the FDS–G1 program. The Core paper defines a finite causal-screen entropy-response spine and a no-free-residual rule,

$$R_{\mu\nu}^{\text{scr}} \in \ker \nabla^\mu \cap \text{Obs}(\mathcal{C}_{\text{resp}}) \cap \text{Quot}(G_{\text{scr}}) \cap \text{Penalty}(\mathcal{L}).$$

D1 gives a finite-access sheaf interpretation of response gluing failure; D5 enforces Ward/Bianchi conservation and no free optical amplitude; D6 identifies output-level memory quotients. D9 integrates these ingredients into an obstruction exact sequence and a no-free-dark-sector classification theorem.

The main structure is the exact sequence

$$0 \rightarrow \mathcal{G}_{\text{gauge}} \rightarrow \mathcal{D}_{\text{raw}} \rightarrow \mathcal{D}_{\text{phys}} \rightarrow \mathcal{O}_{\text{resp}} \rightarrow \mathcal{R}_{\text{adm}} \rightarrow 0,$$

where raw response defects are first quotiented by gauge/relabeling artifacts, then filtered through missing-variable and memory lifts, and only then mapped to admissible residuals. A residual is admissible only if it is quotient-invariant, tensorial, Ward/Bianchi conserved, finitely generated, non-duplicative with existing sectors, and penalized in the likelihood. Arbitrary functions such as an independently sampled optical amplitude $A(a, k)$ are therefore outside G1DE unless supplied with a finite generative law and an explicit complexity penalty.

The paper classifies admissible residuals into six sectors: cosmological integration, scalar-capacity, Weyl-optical, memory quotient, stochastic finite-screen, and nonmetric/rejected branches. It also gives a spin-sector classification for scalar cosmological perturbations, a finite-description residual principle, a no-universal-dark-residual theorem, and a failure lattice showing how claims demote without being reabsorbed by residual freedom. The result is a mathematical version of the G1 non-absorption rule: residuals are quotient/cokernel objects after gauge, memory, and missing-variable removal, not containers for arbitrary unexplained discrepancy.

1 Purpose and claim-status discipline

D9 formalizes a simple but central rule of the G1 program:

residuals are obstruction classes, not free dark functions.

(1)

The Core paper already requires that a screen residual pass conservation, quotient, obstruction-origin, and likelihood-penalty tests. D9 turns that admissibility rule into a sequence of maps and classifications.

1.1 What D9 claims

1. Apparent response non-closure must first be filtered through quotient correction, missing-variable lifts, and memory lifts.
2. Only surviving obstruction classes can source an admissible residual.
3. Ward/Bianchi conservation is not optional: unconserved residuals are rejected or moved to nonmetric response theory.
4. Free functions do not define G1DE branches unless equipped with finite generators and complexity penalties.
5. The data-facing late-time dark sector is a finite-dimensional subspace of a larger admissible residual classification.

1.2 What D9 does not claim

D9 does not prove that the G1DE branch is empirically correct. It does not prove D0 modular calibration or D8 metric reconstruction. It does not exclude all possible nonmetric theories. It states what is allowed to count as a G1 residual once one accepts the finite causal-screen response framework.

Object	Meaning	Status
$\mathcal{D}_{\text{raw}}$	raw response defect before quotient	pre-physical
$\mathcal{G}_{\text{gauge}}$	relabeling, coordinate, null-generator artifacts	removed
$\mathcal{D}_{\text{phys}}$	quotient-level physical defect	filtered
$\mathcal{O}_{\text{resp}}$	surviving response obstruction	candidate source
$\mathcal{R}_{\text{adm}}$	conserved, quotient-invariant residual	admissible
free $A(a, k)$	independent optical amplitude	outside G1DE

Table 1: Claim-status map for D9.

2 Raw defects, quotient projection, and physical obstructions

Let ω be the screen response one-form on X_{phys} . If the chosen variables are complete and the response is integrable,

$$\omega = ds_{\text{scr}}, \quad s_{\text{scr}} = S_{\text{scr}}/k_B, \quad (2)$$

so that $d\omega = 0$. If

$$d\omega \neq 0, \quad (3)$$

one defines a response curvature

$$\mathcal{C}_{\text{resp}} = d\omega. \quad (4)$$

But $\mathcal{C}_{\text{resp}} \neq 0$ is not automatically physical. The filter is

$$d\omega \neq 0 \Rightarrow \begin{cases} \text{incomplete quotient,} \\ \text{missing coarse variable,} \\ \text{projected memory mode,} \\ \text{source–kernel non-identifiability,} \\ \text{true physical obstruction.} \end{cases} \quad (5)$$

Only the last category can be mapped to $R_{\mu\nu}^{\text{scr}}$.

Definition 1 (Raw response defect). $\mathcal{D}_{\text{raw}}$ is the set of all detected response non-closures before quotient projection, memory lifting, and missing-variable completion.

Definition 2 (Gauge artifact subspace). $\mathcal{G}_{\text{gauge}} \subset \mathcal{D}_{\text{raw}}$ is the subspace generated by screen relabeling, cell permutation, coordinate gauge, null-generator reparametrization, and other transformations that do not change physical screen access.

Definition 3 (Physical defect space). The quotient-level physical defect space is

$$\mathcal{D}_{\text{phys}} = \mathcal{D}_{\text{raw}} / \mathcal{G}_{\text{gauge}}. \quad (6)$$

Elements of $\mathcal{D}_{\text{phys}}$ are not yet residuals. They are physical defects that still may be resolved by adding missing coarse variables or memory representatives.

3 The obstruction exact sequence

Definition 4 (Response obstruction map). The response obstruction map

$$\lambda : \mathcal{D}_{\text{phys}} \rightarrow \mathcal{O}_{\text{resp}} \quad (7)$$

sends a quotient-level physical defect to its equivalence class after all admissible missing-variable and memory lifts have been attempted.

Definition 5 (Residual admissibility map). The residual admissibility map

$$\mathcal{W} : \mathcal{O}_{\text{resp}} \rightarrow \mathcal{R}_{\text{adm}} \quad (8)$$

is defined only for obstruction classes that can be represented by tensorial, conserved, quotient-invariant, finitely generated residuals with explicit likelihood penalties.

Theorem 1 (Obstruction exact sequence). In the G1 admissibility discipline, residual formation is represented by the sequence

$$0 \longrightarrow \mathcal{G}_{\text{gauge}} \longrightarrow \mathcal{D}_{\text{raw}} \longrightarrow \mathcal{D}_{\text{phys}} \xrightarrow{\lambda} \mathcal{O}_{\text{resp}} \xrightarrow{\mathcal{W}} \mathcal{R}_{\text{adm}} \longrightarrow 0, \quad (9)$$

where exactness means:

1. gauge artifacts are precisely the raw defects killed by quotient projection;
2. physical defects that admit missing-variable or memory lifts map to the zero obstruction class;
3. only nonzero surviving obstruction classes may be tested for residual admissibility;
4. admissible residuals are images of obstruction classes, not arbitrary additions to the field equations.

The maps are:

- $\pi_Q : \mathcal{D}_{\text{raw}} \rightarrow \mathcal{D}_{\text{phys}}$ — quotient projection, $\ker \pi_Q = \mathcal{G}_{\text{gauge}}$.
- $\lambda : \mathcal{D}_{\text{phys}} \rightarrow \mathcal{O}_{\text{resp}}$ — obstruction lift, $\ker \lambda =$ missing-variable and memory-lift removable defects.
- $\mathcal{W} : \mathcal{O}_{\text{resp}} \rightarrow \mathcal{R}_{\text{adm}}$ — Ward/Bianchi admissibility filter, $\ker \mathcal{W} =$ nonconserved or non-tensorial obstructi

Proof. The first map is inclusion by definition. Quotient projection identifies raw defects modulo gauge artifacts, so its kernel is $\mathcal{G}_{\text{gauge}}$. The obstruction map further quotients by the equivalence relation generated by adding missing variables or memory representatives that restore exactness. Hence its kernel is precisely the set of physical defects resolved before residual formation. Finally, the admissibility map is defined only on surviving obstruction classes that can satisfy tensor type, Ward/Bianchi conservation, quotient invariance, finite-description, and likelihood-penalty constraints. Thus no residual can appear without a preimage in $\mathcal{O}_{\text{resp}}$. $\square$

Remark 1. *The sequence is an admissibility sequence, not a claim that every mathematical obstruction has a physical stress-tensor representative. The final map may reject obstruction classes.*

4 Sheaf gluing theorem

Let $\{U_i\}$ be a cover of a finite-access screen atlas. Suppose each patch has a local entropy potential

$$s_i : U_i \rightarrow \mathbb{R}, \quad \omega_i = ds_i. \quad (10)$$

On overlaps,

$$s_i - s_j = c_{ij}. \quad (11)$$

Theorem 2 (Response gluing obstruction). *If c_{ij} is a Čech coboundary, then the local potentials glue to a global entropy potential and the response is globally exact. If the triple-overlap cocycle*

$$c_{ij} + c_{jk} + c_{ki} \quad (12)$$

represents a nontrivial class, the response has a gluing obstruction. This class is the sheaf version of the Core response curvature:

$$F_{\nabla_{\text{scr}}} \longleftrightarrow [\mathcal{C}_{\text{resp}}] \longleftrightarrow \text{gluing defect class}. \quad (13)$$

Proof. A family of local primitives s_i defines a global primitive precisely when the transition constants or functions on overlaps can be absorbed by local redefinitions. This is the statement that the corresponding Čech cocycle is a coboundary. If the class is nonzero, no global potential exists in the chosen sheaf of finite-access ledgers. The failure is the sheaf-theoretic image of $d\omega \neq 0$, after quotient and lift filters. $\square$

5 Ward/Bianchi admissibility

A metric residual branch is written

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \kappa T_{\mu\nu} + R_{\mu\nu}^{\text{scr}}. \quad (14)$$

Admissibility requires

$$\nabla^\mu (T_{\mu\nu} + R_{\mu\nu}^{\text{scr}}) = 0. \quad (15)$$

If matter is separately conserved,

$$\nabla^\mu R_{\mu\nu}^{\text{scr}} = 0. \quad (16)$$

Theorem 3 (Diffeomorphism Ward residual admissibility). *If $R_{\mu\nu}^{\text{scr}}$ is obtained from a diffeomorphism-invariant residual action*

$$I_{\text{res}}[g, \chi] \quad (17)$$

by

$$R_{\mu\nu}^{\text{scr}} = -\frac{2}{\sqrt{-g}} \frac{\delta I_{\text{res}}}{\delta g^{\mu\nu}}, \quad (18)$$

then the residual obeys a Noether identity that contributes to

$$\nabla^\mu (T_{\mu\nu} + R_{\mu\nu}^{\text{scr}}) = 0 \quad (19)$$

on shell. A proposed residual that cannot be embedded in any conserved variational or nonlocal kernel envelope is not an admissible G1 metric residual.

6 v1.2 Screen-QNEC-Type Admissibility Filter

Claim type: Conjectural admissibility rule All G1 screen residuals must satisfy the entropy-convexity null-energy condition filter:

$$R_{\mu\nu}^{\text{scr}} k^\mu k^\nu \in \mathcal{A}_{\text{Ward}} \cap \mathcal{A}_{\text{quotient}} \cap \mathcal{A}_{\text{ent}}$$

where:

- $\mathcal{A}_{\text{Ward}}$ is the set of residuals satisfying Ward/Bianchi conservation
- $\mathcal{A}_{\text{quotient}}$ is the set of residuals invariant under screen quotient automorphisms
- $\mathcal{A}_{\text{ent}}$ is the set of residuals that preserve convexity of the screen relative entropy

This filter generalizes the Quantum Null Energy Condition (QNEC) to finite-screen residual sectors.

Boundary note: This is a conjectural admissibility filter, not a proven theorem. It provides an additional consistency check for proposed residual kernels, but is not required for the core G1 no-free-residual rule.

7 No-free-dark-sector classification

Definition 6 (Admissible G1 residual). A residual $R_{\mu\nu}^{\text{scr}}$ is admissible only if

$$\boxed{R_{\mu\nu}^{\text{scr}} \in \ker \nabla^\mu \cap \text{Obs}(\mathcal{C}_{\text{resp}}) \cap \text{Quot}(G_{\text{scr}}) \cap \text{Penalty}(\mathcal{L})} \quad (20)$$

and if it has finite effective description length.

Theorem 4 (No-free-dark-sector classification). Under local covariance, quotient invariance, all-null metricization, Ward/Bianchi conservation, finite description length, and no independent optical amplitude $A(a, k)$, any admissible late-time residual belongs to one of the following sectors:

1. cosmological integration sector, represented by $\Phi g_{\mu\nu}$ or a Λ -like constant;
2. scalar-capacity sector, represented by a conserved scalar response density $\phi = \log(\eta/\eta_0)$;
3. Weyl-optical residual sector, represented at leading order by a Weyl transfer

$$\mathcal{W}_{\text{G1}} = (1 - \kappa_{\text{BW}} R_H) \mathcal{W}_{\text{GR}}; \quad (21)$$

4. memory quotient sector, represented by output functions $R_H(a) = K_H * J_H$ modulo source-kernel equivalence;
5. stochastic finite-screen sector, represented by conserved noise or covariance corrections;
6. nonmetric or rejected sector, for obstructions that cannot be tensorized, conserved, or finitely generated.

No arbitrary triple $\{H(a), \mu(a, k), \Sigma(a, k)\}$ is a G1DE residual unless it is generated by one of these finite admissible sectors and carries its complexity penalty.

8 Residual spin-sector classification

At linear scalar cosmological order, residuals project into observable response channels.

Residual sector	Leading observable	G1 interpretation
trace / scalar	$H(a)$, effective $w(a)$	integration or scalar-capacity branch
Ricci / density	$\mu(a, k)$, growth, RSD	Ricci leakage or density residual
Weyl / traceless optical	$\Sigma(a, k)$, lensing, E_G	optical boundary residual
memory	redshift shape $R_H(a)$	source–kernel quotient output
stochastic	covariance / excess noise	finite-screen projection noise
nonmetric	cross-probe inconsistency	branch demotion or rejection

Table 2: Spin-sector and observable-channel classification of residuals.

The $M_{3/4}$ branch is the Weyl-dominant, Ricci-suppressed, one-memory-shape subclass:

$$\mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}R_H(a) + O(R_H^2). \quad (22)$$

9 Finite-description residual principle

Principle 1 (Finite-description residual principle). *A residual branch derived from finite-screen distinguishability must admit a finite generative description:*

$$K_{\text{eff}}(R_{\mu\nu}^{\text{scr}}) < \infty. \quad (23)$$

A free function is not a G1 branch unless it is supplied with a finite generator and an explicit complexity or likelihood penalty.

This principle is the conceptual origin of the no-free- $A(a, k)$ rule. A model that samples an independent optical amplitude function

$$A(a, k) \quad (24)$$

without a finite G1 generator exits the G1DE class and becomes generic dark stress.

Theorem 5 (No universal dark residual). *There is no residual sector within G1DE that can arbitrarily fit independent functions $H(a)$, $\mu(a, k)$, and $\Sigma(a, k)$ while preserving finite description length, quotient invariance, Ward/Bianchi conservation, and likelihood penalty. Any such arbitrary functional freedom is outside G1DE unless it is generated by a finite source–kernel, scalar-capacity, or nonmetric model class.*

10 Source–kernel quotient and non-identifiability

Let

$$\Pi : (K_H, J_H) \mapsto R_H = K_H * J_H. \quad (25)$$

If a likelihood depends only on $R_H(a)$, then all pairs in the fiber $\Pi^{-1}(R_H)$ are observationally equivalent at that level.

Theorem 6 (Source–kernel non-identifiability). *Define*

$$(K_H, J_H) \sim (K'_H, J'_H) \iff K_H * J_H = K'_H * J'_H = R_H. \quad (26)$$

If the data-facing likelihood is a function only of $R_H(a)$, then it identifies the quotient space

$$\mathcal{M}_{\text{micro}} / \ker \Pi, \quad (27)$$

not a unique microscopic source and kernel.

11 Failure lattice

A failure in G1 should move a claim downward in a lattice rather than sideways into arbitrary survival.

11.1 Foundational demotion lattice

D0 strong calibration $\longrightarrow$ weak BH/Jacobson bridge $\longrightarrow$ entropy-response framework $\longrightarrow$ finite distinction on
(28)

11.2 Empirical demotion lattice

$M_{3/4} \longrightarrow M_\kappa \longrightarrow$ constant Weyl amplitude $\longrightarrow$ generic conserved dark stress $\longrightarrow$ branch rejection.
(29)

Failure	Demotion
finite-index density fails	strong D0 calibration fails; weak bridge may remain
boost modular calibration fails	BH coefficient not internally derived
metricization fails	nonmetric response branch
Ward closure fails	residual rejected or nonmetric
free $A(a, k)$ required	generic dark stress, not G1DE
free κ decisively wins	exact isotropic 3/4 lock demoted
constant Σ wins	$R_H(a)$ shape lock demoted
binned- $\Sigma(z)$ wins robustly in full	$R_H(a)$ shape lock demoted
$3 \times 2\text{pt}$	
$ \mu - 1 \sim \Sigma - 1 $ required	Ricci-suppressed branch fails

Table 3: Non-absorption failure lattice.

12 Counterexample library

A mature obstruction theory should exhibit failures. The following counterexamples are useful tests of the G1 closure discipline.

1. **Volume-dominated entropy.** A finite screen system whose leading entropy is volume-extensive fails the screen area-law substrate.
2. **Exact response without metricity.** A response one-form may be exact while its principal symbol is non-quadratic; this gives a nonmetric but integrable response theory.
3. **Metricization without Ward closure.** A tensor envelope can exist but fail conservation; it is rejected as a metric residual.
4. **Conserved tensor without quotient origin.** A conserved dark stress not arising from screen obstruction belongs to generic EFT, not G1DE.
5. **Free optical amplitude.** A model requiring independent $A(a, k)$ exits G1DE.
6. **Multiple microscopic realizations.** Distinct (K_H, J_H) with the same R_H demonstrate source-kernel non-identifiability.

13 Relation to D1, D5, and D6

D1 supplies the sheaf/gluing interpretation of $\mathcal{C}_{\text{resp}}$. D5 supplies Ward/Bianchi and no-free-optical-amplitude rules. D6 supplies the source–kernel quotient. D9 is the classification layer that makes these three ingredients part of one residual admissibility theorem.

14 Conclusion

D9 turns the G1 no-free-residual rule into a mathematical obstruction sequence. Apparent non-closure is first quotiented by gauge, tested against missing-variable and memory lifts, and only then considered a residual source. Admissible residuals must be conserved, quotient-invariant, finite-description, non-duplicative, and likelihood-penalized. This converts the dark sector from a free container into a classified quotient/cokernel object. In this form, G1 is not a theory that can fit arbitrary residual functions; it is a finite-screen obstruction theory with explicit demotion paths.

15 Selected references

References

- [1] R. Bott and L. W. Tu, *Differential Forms in Algebraic Topology*, Springer (1982).
- [2] M. Kashiwara and P. Schapira, *Sheaves on Manifolds*, Springer (1990).
- [3] R. M. Wald, *General Relativity*, University of Chicago Press (1984).
- [4] E. Noether, “Invariante Variationsprobleme,” *Nachrichten von der Gesellschaft der Wissenschaften zu Goettingen* (1918).
- [5] R. Geroch, “Einstein algebras,” and related work on covariance and field equations.
- [6] O. Fawzi and R. Renner, “Quantum conditional mutual information and approximate Markov chains,” *Communications in Mathematical Physics* **340**, 575–611 (2015).
- [7] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, FDS–G1 Core manuscript (2026).
- [8] Y. Wu, *Finite-Access Sheaf Geometry and Response Gluing Obstructions*, FDS–G1 D1 (2026).
- [9] Y. Wu, *Weyl-Normalized Ward Completion and Identifiability Rules*, FDS–G1 D5 (2026).
- [10] Y. Wu, *Maximum-Caliber Horizontal Lift for the G1 Memory Quotient*, FDS–G1 D6 (2026).

G1DE Bootstrap and Minimal Weyl Branch Uniqueness

FDS–G1 D10: Prediction Compression, Observational Functor, and the $M_{3/4}$ Branch
v1.2 Body-Integrated Release

Yining Wu
Independent Researcher
yining.wu@alumni.upenn.edu

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v1.2 release note. This standalone paper is synchronized with the FDS–G1 v1.2 full body-integrated release. New theory-hardening material is marked by claim type. Existing empirical results are not expanded unless explicitly stated.

Abstract

This paper develops the bootstrap and minimal-branch layer of the FDS–G1 dark-sector program. Earlier G1 papers define a projection-locked branch,

$$s < 3, \quad \mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)R_{bH}(a), \quad R_{bH}(1) = 1,$$

and test it empirically as G1DE– $M_{3/4}$. D10 asks a sharper theoretical question: is this branch merely a sparse model choice, or is it the unique minimal Weyl-dominant finite-description branch under the G1 closure assumptions?

The answer proposed here is conditional but strong. Given a finite causal-screen entropy ledger, quotient-invariant response, all-null metricization, Ward/Bianchi admissibility, finite residual description length, no independent optical amplitude $A(a, k)$, Ward-suppressed Ricci leakage, a single passive horizon memory output $R_H(a) = R_0 R_{bH}(a)$, and isotropic full optical-algebra compliance, the unique first-order low-dimensional data-facing branch is

$$R_H(a) = R_0 R_{bH}(a), \quad R_0 = 3 - s, \quad \mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}R_H(a).$$

The neighboring empirical controls are not arbitrary alternatives but relaxations of specific assumptions: M_κ relaxes isotropic optical compliance, constant- Σ relaxes horizon memory shape, G1DE-2 relaxes projection locking, and free $A(a, k)$ exits the G1DE class into generic dark stress.

D10 formulates the G1 bootstrap equations, defines a residual branch moduli space, proves the minimal Weyl residual uniqueness theorem, states a Ward-suppressed Ricci leakage theorem, interprets $R_H(a)$ as a rank-one passive horizon Green response, and defines an observational functor from residual classes to H , μ , Σ , lensing, E_G , RSD, BAO, and supernova channels. The paper does not claim that current data prove the microscopic screen realization. It identifies the theoretical branch that G1fit-real tests and the precise assumptions that future data can kill.

1 Purpose and claim-status discipline

D10 is the bootstrap companion to the empirical G1fit-real paper. G1fit-real tests the output normal form. D10 explains why that output normal form is the minimal G1 Weyl branch under stated closure assumptions.

1.1 What D10 claims

1. The G1DE- $M_{3/4}$ branch is the unique minimal Weyl-dominant, Ricci-suppressed, one-memory-mode branch under G1 closure assumptions.
2. The branch compresses a generic late-time residual response space into a finite-description prediction:

$$(H, \mu, \Sigma) = \mathcal{P}_{\text{G1}}(s). \quad (1)$$

3. The standard controls M_κ , constant- Σ , G1DE-2, and free $A(a, k)$ are assumption relaxations, not unrelated models.
4. The observational signatures of each residual branch are fixed by an observational functor.

1.2 What D10 does not claim

D10 does not prove that the measured universe follows $M_{3/4}$. It does not identify a unique microscopic source J_H , memory kernel K_H , finite Markov generator, or screen algebra. It does not prove the D0 modular calibration. It provides the theoretical branch selected for empirical testing.

Object	Role in D10	Status
$\mathcal{M}_{\text{G1}}$	G1 branch moduli space	definition
$\mathcal{M}_{3/4}$	minimal Weyl branch	conditional theorem
M_κ	anisotropic compliance relaxation	control
constant- Σ	shape-lock relaxation	control
G1DE-2	free response envelope	adversarial control
free $A(a, k)$	generic dark stress	outside G1DE
G1fit-real	empirical test of output normal form	data paper

Table 1: Claim-status map for D10.

2 G1 bootstrap equations

The G1 bootstrap system is the set of closure equations that any admissible finite-screen residual branch must satisfy.

2.1 Ledger and response

The finite screen ledger is

$$S_{\text{scr}} = k_B \log N_{\text{scr}} \quad (2)$$

in the counting case, or its Shannon/von Neumann refinement in the probabilistic case. The response one-form is

$$\omega = \eta_i(X) dX^i. \quad (3)$$

Exact response satisfies

$$\omega = d(S_{\text{scr}}/k_B), \quad (4)$$

while non-exact response defines an obstruction candidate

$$d\omega = \mathcal{C}_{\text{resp}}. \quad (5)$$

2.2 Metricization and Ward closure

All-null response takes the form

$$F_p(k) = E_{\mu\nu} k^\mu k^\nu. \quad (6)$$

The effective equation is

$$E_{\mu\nu} = \kappa T_{\mu\nu} + R_{\mu\nu}^{\text{scr}} + \Phi g_{\mu\nu}, \quad (7)$$

with Ward closure

$$\nabla^\mu (E_{\mu\nu} - \kappa T_{\mu\nu} - R_{\mu\nu}^{\text{scr}}) = 0. \quad (8)$$

2.3 Finite-description admissibility

A residual branch must satisfy

$$K_{\text{eff}}(R_{\mu\nu}^{\text{scr}}) < K_{\text{budget}}, \quad (9)$$

meaning that it has a finite generative law and does not rely on an arbitrary free function.

2.4 Optical projection closure

The optical algebra decomposes as

$$T_I GL(2) = A^{(1)} \oplus S^{(2)} \oplus \Omega^{(1)}. \quad (10)$$

The weighted optical projection coefficient is

$$\kappa_{BW} = \frac{2\eta_S + \eta_\Omega}{\eta_A + 2\eta_S + \eta_\Omega}. \quad (11)$$

At the isotropic compliance point,

$$\eta_A = \eta_S = \eta_\Omega, \quad (12)$$

one obtains

$$\kappa_{BW} = \frac{2+1}{1+2+1} = \frac{3}{4}. \quad (13)$$

3 Branch moduli space

Definition 1 (G1 branch moduli space). *The G1 residual branch moduli space is*

$$\mathcal{M}_{G1} = \{\text{solutions to the G1 bootstrap equations}\} / \sim, \quad (14)$$

where $\sim$ quotients by screen relabeling, gauge equivalence, output-equivalent source–kernel factorizations, and observationally indistinguishable representatives.

The minimal Weyl branch is a low-dimensional subspace

$$\mathcal{M}_{3/4} \subset \mathcal{M}_{G1}. \quad (15)$$

Adjacent relaxations are

$$\mathcal{M}_\kappa, \quad \mathcal{M}_{\text{const-}\Sigma}, \quad \mathcal{M}_{G1\text{DE}2}. \quad (16)$$

They correspond to controlled deformations of assumptions rather than arbitrary alternatives.

Branch	Relaxed assumption	Interpretation
$M_{3/4}$	none within minimal branch	isotropic Weyl residual
M_κ	isotropic optical compliance	anisotropic projection
constant- Σ	horizon memory shape	amplitude-only Weyl branch
G1DE-2	projection lock	free-response envelope
free $A(a, k)$	finite-description/no-free-output rule	generic dark stress

Table 2: Local chart of the G1 residual branch moduli space.

4 Minimal Weyl residual uniqueness theorem

Theorem 1 (Minimal Weyl Residual Uniqueness within the G1 Closure Class). *Assume:*

1. *one scalar late-time residual amplitude R_0 ;*
2. *a normalized passive horizon output shape $R_{bH}(a)$ with $R_{bH}(1) = 1$;*
3. *the same residual controls the background filling deviation and the Weyl response;*
4. *no leading Ricci leakage;*
5. *no independent optical amplitude $A(a, k)$;*
6. *no independent $\mu(a, k)$ or $\Sigma(a, k)$ redshift/scale functions;*
7. *isotropic full optical-algebra compliance.*

Then the unique first-order low-dimensional data-facing Weyl branch is

$$R_H(a) = R_0 R_{bH}(a), \quad R_0 = 3 - s, \quad (17)$$

$$\mu(a, k) = 1, \quad (18)$$

$$\Sigma(a, k) - 1 = -\frac{3}{4} R_H(a). \quad (19)$$

Table 3: Assumption-to-demotion map for the minimal Weyl branch.

Assumption relaxed	Control / demotion
isotropic optical compliance	M_κ
passive memory shape R_{bH}	constant- Σ
projection lock	G1DE-2
Ward-suppressed Ricci leakage	$\mu \neq 1$ branch
no free $A(a, k)$	generic dark stress, outside G1DE
single output quotient	D6 realization ambiguity

Proof. A single residual amplitude and normalized output shape require $R_H(a) = R_0 R_{bH}(a)$. The background filling convention identifies $R_0 = 3 - s$. No leading Ricci leakage fixes $\mu = 1$ at first order. No independent optical amplitude and no independent Σ shape require the Weyl response to be proportional to the same $R_H(a)$. Isotropic full optical-algebra compliance fixes the proportionality coefficient to $\kappa_{BW} = 3/4$ by (11). No further first-order degree of freedom remains. Hence the branch is unique under the stated assumptions. $\square$

Remark 1. *The theorem is conditional. Relaxing its assumptions gives the adjacent controls M_κ , constant- Σ , G1DE-2, or generic dark stress.*

5 Ward-suppressed Ricci leakage

Lemma 1 (Ward-suppressed Ricci leakage). *Let the Weyl-normalized transfer be*

$$\mathcal{W}_{G1}(a, k) = [1 - \kappa_{BW} R_H(a)] \mathcal{W}_{GR}(a, k) + O(R_H^2). \quad (20)$$

Then the leading Weyl response is

$$\Sigma(a, k) - 1 = -\kappa_{BW} R_H(a) + O(R_H^2), \quad (21)$$

while the Newtonian/Ricci growth channel receives only Ward-suppressed leakage,

$$\mu(a, k) - 1 = O(R_H^2) + O\left(R_H \frac{\mathcal{H}^2}{k^2}\right). \quad (22)$$

Thus on linear subhorizon survey scales the first-order locked branch is $\mu(a, k) = 1$.

Schur-complement argument. Represent the linear response Hessian or impedance matrix in Weyl/Ricci block form,

$$K = \begin{pmatrix} K_{WW} & K_{WR} \\ K_{RW} & K_{RR} \end{pmatrix}. \quad (23)$$

Ward stiffness means that the Ricci block has a parametrically large impedance compared with the Weyl optical block. Solving for the induced Ricci response gives a Schur-suppressed term

$$\Delta_R \sim K_{RR}^{-1} K_{RW} \Delta_W. \quad (24)$$

In the finite-screen Weyl branch this suppression is of order $O(R_H)$, and subhorizon projection adds the scale-suppressed factor $O(\mathcal{H}^2/k^2)$. Hence Ricci leakage is second-order or subhorizon-suppressed, while the Weyl transfer is first-order. $\square$

Corollary 1 (Ricci leakage kill condition). *If production RSD, full-shape, or $3 \times 2pt$ data require*

$$|\mu - 1| \sim |\Sigma - 1|, \quad (25)$$

then the Ward-suppressed Ricci-leakage branch is demoted. The model may move to a finite-screen residual with a density/Ricci component, but it is no longer $M_{3/4}$.

6 Horizon Green response and rank-one memory

The output $R_H(a)$ is not a unique microscopic source–kernel factorization. It is an output memory response. In the minimal horizon mode class, let $N = \log a$ and write

$$\frac{dR_H}{dN} + \Gamma_H(N) R_H = J_H(N). \quad (26)$$

The retarded solution is

$$R_H(N) = \int^N G_H(N, N') J_H(N') dN', \quad (27)$$

with

$$G_H(N, N') = \exp \left[- \int_{N'}^N \Gamma_H(u) du \right]. \quad (28)$$

Proposition 1 (Rank-one horizon memory branch). *If the horizon residual is represented by a single slow lumpable passive mode with positive retarded Green response and normalization $R_{bH}(1) = 1$, then the minimal output branch is*

$$R_H(a) = R_0 R_{bH}(a). \quad (29)$$

Different microscopic pairs (K_H, J_H) that generate the same R_H are equivalent at the G1fit-real level.

7 Prediction compression theorem

The minimal branch reduces the full effective dark-sector parameter space from 6+ free parameters (for a standard growth+ lensing EFT) to a single free shape amplitude A_0 , with all other parameters predicted from first-principle G1 constraints.

Locking relations. The following relations hold for the minimal branch:

- $\mu = 1$ up to subleading corrections, with $|\mu - 1| \ll |\Sigma - 1|$
- $\Sigma(a) = 1 - \frac{3}{4}(3 - s)\hat{R}_H(a)$
- The shape $\hat{R}_H(a)$ is determined by the finite-screen ledger dynamics
- No extra free parameters for redshift-dependent anisotropic stress

This is the maximal possible compression of a non-GR dark residual.

8 v1.2 Measurement Protocol and Observational Functor

Claim type: Formal operational protocol A screen transition acquires empirical meaning only after specification of the finite record channel and update map:

$$Y_{\text{scr}} : X_{\text{phys}} \rightarrow Z_{\text{obs}}, \quad U_{\text{scr}} : (M_t, Y_t, A_t) \rightarrow M_{t+1}$$

where:

- Y_{scr} maps physical screen states to accessible, stable record equivalence classes
- U_{scr} defines the ledger update under probe intervention A_t

The observational functor formalizes this mapping from screen residual classes to cosmological observables, as defined below.

9 Observational functor

Define an observational functor

$$\mathcal{O} : \mathcal{M}_{\text{G1}} \rightarrow \{H, \mu, \Sigma, E_G, f\sigma_8, \text{WL}, \text{BAO}, \text{SN}\}. \quad (30)$$

It maps residual classes to fixed observational channels.

Residual class	Observable signature	Kill test
background residual	$H, D_L, D_A, \text{BAO/SN}$	CPL/ Λ CDM win
Weyl residual	lensing, E_G, Σ	constant- Σ or no Weyl signal
Ricci leakage	$f\sigma_8, \text{RSD}, \mu$	$ \mu - 1 \sim \Sigma - 1 $
memory shape	redshift dependence $R_{bH}(a)$	shape replaced by constant amplitude
stochastic residual	covariance/noise excess	no stable mean branch
nonmetric branch	cross-probe inconsistency	metric G1DE rejected

Table 4: Observational functor from G1 residual classes to data channels.

10 v1.2 Mainstream Modified Gravity μ/Σ Interface

Claim type: Interface specification The minimal G1DE- $M_{3/4}$ branch has explicit interface form compatible with standard MG parameterizations used in DESI, Euclid, LSST, and Roman pipelines:

$$\mu(a, k) = 1 + \epsilon_\mu(a, k), \quad \Sigma(a, k) = 1 - \frac{3}{4}(3 - s)R_b^H(a) + \epsilon_\Sigma(a, k)$$

with $\epsilon_\mu, \epsilon_\Sigma = O(R_H^2) + O(R_H H^2/k^2)$, consistent with GW speed constraints and minimal Ricci leakage.

Boundary note: This is an interface specification, not production evidence. No claim is made that this residual is preferred over Λ CDM by full production data.

11 Amplitude lock versus shape lock

The current evidence hierarchy should be read in two layers, now supported by a completed medium-prior 8-seed production-refined audit (G1fit-real). The amplitude-level Weyl result is

$$\Sigma_0 < 0, \quad \kappa \sim \frac{3}{4}. \quad (31)$$

The redshift-shape lock is more demanding:

$$\Sigma(a) - 1 \propto R_{bH}(a). \quad (32)$$

In the completed medium-prior 8-seed production-refined audit, the projection-locked branch $M_{3/4}$ exceeds the free- κ control by $\Delta \log Z = 0.856$ and constant- Σ by $\Delta \log Z = 1.775$, supporting both the amplitude sign and the preference for active-filling Weyl shape over a constant shift. An independent KiDS-1000 shear-only diagnostic stress test (G1fit-real v1.1) now strengthens this placement: under combined shear-calibration, photo- z , and minimal IA nuisance profiling, $M_{3/4}$ remains favored over Λ CDM by $\Delta\chi^2 \approx -39$; free- κ relaxation returns $\kappa \approx 0.746$; and deterministic mock injection rules out false-positive selection of Λ CDM, constant- Σ , or binned- Σ truths as $M_{3/4}$. The redshift-shape lock is not yet production-resolved: KiDS binned- $\Sigma(z)$ controls find shape information that is present but nuisance-entangled. Future full 3×2 pt analyses must distinguish $R_{bH}(a)$ from constant- Σ and binned- $\Sigma(z)$ controls.

12 Minimal kill tests

Kill test	Required analysis	Demotion
production evidence	seven-model nested evidence with priors and seeds	branch demotion if controls win
Weyl shape	$R_{bH}(a)$ vs constant- Σ with expanded lensing	shape demotion
Ricci leakage	RSD/full-shape/ 3×2 pt test of $\mu = 1$	Ricci-suppressed branch fails
free amplitude	need for free $A(a, k)$	generic dark stress
false-positive audit	deterministic + noisy mock injection under 4 truth models	KiDS interpretation demoted if systematic $M_{3/4}$ false positives
source-kernel uniqueness	demand for unique (K_H, J_H) from R_H data	overclaim rejected

Table 5: Minimal empirical kill tests for the $M_{3/4}$ branch.

13 Relation to D0–D9

What D10 does not claim

The KiDS shear-only stress test and deterministic mock-injection audit support the data-facing branch but do not prove the microscopic screen realization, the passive memory kernel, or the uniqueness of $R_{bH}(a)$ as the physical source shape. The deterministic mock-injection audit validates the observational functor by showing that Λ CDM, constant- Σ , and binned- Σ truth mocks are not falsely mapped to $M_{3/4}$.

D10 depends on earlier G1 closure layers.

- D0 supplies strong calibration targets for the gravitational scale.
- D1 and D9 supply obstruction and residual admissibility.
- D2 supplies the optical projection theorem.
- D5 supplies Weyl-normalized Ward completion and no-free- $A(a, k)$.
- D6 supplies source–kernel quotient discipline.
- D7 supplies a finite Markov-screen realization chart.
- G1fit-real tests the output normal form selected by D10.

14 Conclusion

D10 turns the G1DE– $M_{3/4}$ model from a sparse empirical ansatz into a conditional minimal-branch theorem. Under finite-screen response, Ward-suppressed Ricci leakage, no free optical amplitude, a single passive horizon memory output, and isotropic optical compliance, the only first-order low-dimensional Weyl branch is

$$\mu = 1, \quad \Sigma - 1 = -\frac{3}{4}(3 - s)R_{bH}(a). \quad (33)$$

The completed homogeneous seven-model medium-prior nested-evidence audit (v1.0-rc3 data) supports this branch: $M_{3/4}$ exceeds the free- κ control by $\Delta \log Z \simeq 0.72$, constant- Σ by $\Delta \log Z \simeq 1.79$, G1DE-1 by $\Delta \log Z \simeq 7.41$, and Λ CDM by $\Delta \log Z \simeq 11.88$. The neighboring controls are exactly the relaxations of the theorem’s assumptions. Thus the branch is confident but falsifiable: production evidence, Weyl-shape tests, Ricci-leakage tests, and no-free-function discipline can kill or demote it. In this form, $M_{3/4}$ is the lowest finite-description Weyl-dominant deformation of the G1 Einstein equilibrium branch.

This v1.0-rc3 paragraph is retained for provenance. The current v1.2 ranking is superseded by the completed medium-prior 8-seed production-refined audit, with $\Delta \log Z(M_{3/4}, M_\kappa) = 0.856$ and $\Delta \log Z(M_{3/4}, \text{const-}\Sigma) = 1.775$.

15 Selected references

References

- [1] M. Tegmark, A. J. S. Hamilton, and Y. Xu, “The power spectrum of galaxies in the 2dF 100k redshift survey,” and related work on cosmological parameter compression.
- [2] R. Trotta, “Bayes in the sky: Bayesian inference and model selection in cosmology,” *Contemporary Physics* **49**, 71–104 (2008).

- [3] M. Kunz, “The phenomenological approach to modeling the dark energy,” *Comptes Rendus Physique* **13**, 539–565 (2012).
- [4] T. Clifton, P. G. Ferreira, A. Padilla, and C. Skordis, “Modified gravity and cosmology,” *Physics Reports* **513**, 1–189 (2012).
- [5] L. Amendola et al., “Measuring the growth of structure with weak lensing,” and related work on E_G , μ , and Σ .
- [6] Y. Wu, *Finite Screen Spacetime: Entropy-Response Geometry from Causal-Screen Ledgers*, FDS–G1 Core manuscript (2026).
- [7] Y. Wu, *The 3/4 Projection Law in Finite Screen Gravity*, FDS–G1 D2 (2026).
- [8] Y. Wu, *Weyl-Normalized Ward Completion and Identifiability Rules*, FDS–G1 D5 (2026).
- [9] Y. Wu, *Maximum-Caliber Horizontal Lift for the G1 Memory Quotient*, FDS–G1 D6 (2026).
- [10] Y. Wu, *Finite Markov-Screen Realization of the G1DE $M_{3/4}$ Branch*, FDS–G1 D7 (2026).
- [11] Y. Wu, *G1DE- $M_{3/4}$: A Projection-Locked Background–Weyl Residual*, G1fit-real data paper (2026).

Closing Comparative Structural Matrix

This closing matrix repeats and expands the front-matter positioning table so that the structural comparison remains visible independently of the front matter. It is not a scorecard; it is an attribute map. Standard-limit structure should be distinguished from production inheritance: recovering a GR/ Λ CDM-like limit is not the same as already possessing the full numerical, nonlinear, CMB, or community infrastructure of the standard model. All frameworks require primitives, calibration assumptions, and observational interfaces; the relevant question is whether those bridges are established, open, flexible, or prediction-locking.

Framework	Primitive	Bridge / calibration	Dark-sector specificity	Prediction lock	Flexibility	Falsification clarity	Maturity / stress
GR+ Λ CDM	Metric geometry, stress-energy, constant Λ , CDM	Equivalence principle, smooth manifold, Einstein-Hilbert, conservation, initial conditions, bias/baryonic modeling, unexplained Λ	Low: Λ is a constant residual, not derived dynamics	High for standard cosmology; low for Λ origin	Low baseline freedom; unexplained dark sectors	High but tested in bulk; Λ origin still open	Established production baseline; stressed by Λ problem, late-time tensions, quantum/gravity interface
CPL / $w_0 w_a$	Effective dark-energy fluid	Smooth $w(a)$ expansion plus perturbation/closure priors	Medium phenomenological specificity; low microscopic mechanism	Low–medium	Moderate flexibility	Benchmark-level: useful comparative test, not a fundamental prediction	Useful evolving-DE benchmark; weak physical mechanism
Horndeski / EFT of DE	Scalar-tensor / EFT operator basis	Field content, stability, screening, local/GW constraints, EFT truncation	Broad class; not a unique dark-sector mechanism	Usually low unless restricted	High without strong priors	Depends on chosen model; class-level falsification is weak	Mature formalism; often underconstrained and flexibility-limited
$f(R), f(Q)$, related MG	Modified geometric action / connection sector	Chosen function, screening, local tests, growth/GW constraints	Model-dependent	Medium if fixed	Medium–high	Model-level: specific $f(X)$ can be rejected; class is broad	Strong literature; many regions constrained by local/growth/GW tests
Thermodynamic / emergent gravity	Entropy, horizon thermodynamics, information principles	Horizon entropy, local first law, Raychaudhuri bridge, metric/Ward closure	Usually indirect for late-time dark sector	Often low unless residual law added	Model-dependent	Weak without concrete locked residual + Known stress / limitation	Conceptually adjacent; lacks locked data-facing residual fingerprint
Quantum-spacetime programs	String/AdS-CFT/LQG/causal-set primitives	Compactification/landscape, holographic dictionary, semiclassical/continuum recovery	Usually indirect	Low near-term dark-sector lock	Varies; often high model multiplicity	Framework-level: long path to cosmological prediction + Known stress / limitation	High ambition; long empirical path for late-time cosmology
FDS-G1 / $M_{3/4}$	Finite causal-screen ledger, entropy response, integrable geometry, gluing defects	Finite distinguishability, causal-screen localization, entropy ledger, stiffness bridge, D0 modular calibration lemmas, Ward/Bianchi, Weyl normalization, optical projection, likelihood implementation	High: structured background–Weyl residual with near-GR growth	High: $\mu(a, k) = 1$, $\Sigma - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a)$	Low in $M_{3/4}$; controlled by free- κ , const- Σ , G1DE-2	Explicit: 7 demotion paths including free- κ , const- Σ , binned- $\Sigma(z)$, $ \mu - 1 \sim \Sigma - 1 $, free $A(a, k)$, CPL/ Λ CDM, lensing failure	Production-refined evidence-selected + KiDS shear-only diagnostic[†] ; top-control wide-prior stable; deterministic mock audit passed; pending full 3×2 pt, production-scale noisy mocks, CMB/BBN/nonlinear compatibility, and independent replication

Production maturity note. This matrix evaluates theoretical structure and empirical testability, not current infrastructure. GR+ Λ CDM currently dominates production breadth: numerical relativity, nonlinear N -body simulations, CMB/BBN pipelines, and community code ecosystems. FDS–G1 does not yet replace that infrastructure. These are implementation-readiness facts, not direct measures of primitive depth, prediction lock, or explanatory compression.

[†]Medium-prior production-refined status (8 seeds, $d \log Z = 0.1$) plus KiDS shear-only diagnostic; not full survey-grade evidence. Full 3×2 pt, noisy mock ensembles, and independent replication remain pending.

Comparative Notes on Major Competing Frameworks

The following notes summarize the structural comparison behind the matrix above. They are not total-score rankings. Each framework is strongest on a different axis: production maturity, formal generality, microscopic ambition, or prediction-locked late-time residual structure.

GR+ Λ CDM. GR+ Λ CDM remains the established production baseline for relativistic astrophysics and precision cosmology. It has mature infrastructure for strong-field waveforms, nonlinear structure formation, CMB inference, BBN consistency, and large-scale survey modeling. FDS–G1 does not currently replace that production apparatus. Its claim is different: the tested $M_{3/4}$ branch targets the late-time dark sector, where Λ CDM uses an unexplained constant residual. G1 replaces a free constant-like dark-sector term with a finite-screen background–Weyl residual whose output is locked by

$$\mu(a, k) = 1, \quad \Sigma(a, k) - 1 = -\frac{3}{4}(3 - s)\hat{R}_H(a).$$

Thus GR+ Λ CDM leads in production maturity, while FDS–G1 claims a more specific and falsifiable late-time residual mechanism. Full confirmation still requires full prior-family stress testing, independent replication, and full 3×2 pt validation.

CPL / $w_0 w_a$. CPL is a useful phenomenological benchmark for evolving dark energy because it compresses a broad class of background deviations into two equation-of-state parameters. Its weakness is that it does not supply a microscopic or geometric mechanism linking expansion history to lensing response. FDS–G1 addresses this gap by treating the dark-sector signal as a finite-screen residual rather than as an arbitrary fluid. In the $M_{3/4}$ branch, the background deviation and Weyl response are tied to the same output shape $\hat{R}_H(a)$, with the growth channel kept near GR. The comparison is therefore not merely one of fit quality: CPL is a flexible benchmark, whereas $M_{3/4}$ is a prediction-locked residual model. If CPL wins under production evidence, the G1DE observational branch is demoted.

Scalar-tensor theories and EFT of dark energy. Horndeski and EFT-of-dark-energy frameworks provide a mature covariant language for departures from GR. Their strength

is formal generality: many modified-gravity or dark-energy scenarios can be embedded in an operator or function basis. Their risk is the same generality: without strong priors, the model space can become too flexible to yield a sharp prediction before data. FDS–G1 uses compatible EFT interfaces where needed, but its $M_{3/4}$ branch is much narrower. It predicts a Ricci-suppressed, Weyl-active residual pattern, with $\mu \simeq 1$ and a locked Σ -channel response. This makes the branch easier to falsify than generic scalar-tensor freedom. If data require a free $A(a, k)$ or $|\mu - 1| \sim |\Sigma - 1|$, the G1DE identity is demoted.

Quantum-spacetime programs. String theory, AdS/CFT, loop quantum gravity, and causal-set approaches pursue microscopic or quantum-spacetime foundations. Their ambition and mathematical depth are high, and some provide explicit microscopic degrees of freedom or holographic dictionaries. Their weakness for the present purpose is the long bridge from microscopic structure to a unique, near-term, late-time cosmological residual fingerprint. FDS–G1 takes a different route: it remains open at the microscopic calibration level, but it offers an immediate macroscopic prediction lock for the late-time survey regime. Its finite-screen entropy-response structure is therefore not a replacement for UV-complete quantum gravity; it is a data-facing finite-access framework whose first empirical test is the background–Weyl residual. Its burden is to survive lensing, production, and replication tests.

End of FDS–G1 Complete Series Archive.

v1.0-rc3 revision note

Relative to v1.0-rc2, this revision does not alter the core finite-screen entropy-response spine. It adds an expert-audit layer: separation of existing-data evidence from future prediction locks; a Ward-suppressed Ricci-leakage lemma for $\mu = 1$; explicit assumption boundaries for the 3/4 projection coefficient; amplitude-lock versus shape-lock distinction; a completed homogeneous seven-model medium-prior nested-evidence audit (rc3 data update, all 7 models, same prior configuration); a production evidence checklist; a D0 success/failure table; output-quotient prominence; a no-hidden-fit checklist; and minimal empirical kill tests. The earlier mixed-provenance raw audit is retained in the provenance appendix.

v1.1 revision note

Relative to v1.0-rc3, this revision does not alter the finite-screen entropy-response spine or the D0–D10 theorem structure. It updates the empirical status of the G1DE- $M_{3/4}$ branch using G1fit-real v1.1. The added content includes: an independent KiDS-1000 shear-only CLASS/covariance stress test of the Weyl channel; nuisance profiling over shear calibration (m_i), photo- z shifts (Δz_i), and minimal NLA intrinsic alignments (A_{IA}), under which $M_{3/4}$ remains favored over Λ CDM by $\Delta\chi^2 \approx -39$; free- κ relaxation returning $\kappa \approx 0.746$; constant- Σ and binned- $\Sigma(z)$ adversarial shape controls; and a deterministic mock-injection false-positive audit confirming that Λ CDM, constant- Σ , and binned- Σ truths are not misclassified as $M_{3/4}$. The revision upgrades $M_{3/4}$ from evidence-selected to evidence-selected

and KiDS shear-only stress-tested, while retaining full 3×2 pt, production evidence, noisy mock ensembles, and independent replication as pending.

Supersession note. The v1.0/v1.1 homogeneous audit is retained for provenance and is superseded by the v1.2 medium-prior 8-seed production-refined audit reported in Table 3.

v1.2 KiDS BandPower Diagnostic Revision Note

This section documents the finalized KiDS BandPower $EE+nE$ diagnostic bridge, originally developed for v1.2-dev, now stabilized for the v1.2 release. It extends the KiDS infrastructure from the v1.1 shear-only $\xi_{\pm}$ diagnostic toward a compressed-space BandPower bridge using public KiDS BandPower products. The added content includes: a completed product/covariance/order audit for KiDS $\xi_{\pm}$, COSEBIs, and BandPower; a verified PeeE projector path; a first density-kernel PnE smoke path; a completed $EE+nE$ 200-row BandPower bridge; local diagnostic refits; deterministic mock recovery; noisy mock diagnostic screening; nuisance-prior robustness testing; adversarial (Σ) diagnosis; prior-regularized refits; toy (nn) closure diagnostics; and real (P_{nn})/clustering sourcing audits. All added results are infrastructure and diagnostic validation only, not optimized likelihood results, not model evidence, and not a completed full 3×2 pt analysis.

20 KiDS BandPower Compressed-Space Bridge

This section records the finalized KiDS BandPower $EE + nE$ compressed-space bridge infrastructure, originally developed for v1.2-dev, now included in v1.2: a KiDS BandPower $EE + nE$ compressed-space bridge for future G1/M_{3/4} tests. The results are product/projector/kernel validation results only, not optimized likelihood results, not model evidence, and not a completed full 3×2 pt analysis.

20.1 Motivation: Beyond Shear-Only $\xi_{\pm}$

The v1.1 KiDS diagnostic used real-space shear two-point functions (ξ_+, ξ_-) . Public KiDS data also provide compressed-space alternative representations: COSEBIs and BandPower top-hat C_{ℓ} bins. The infrastructure originally developed for v1.2-dev validates these additional representations and builds a combined $EE + nE$ BandPower bridge to strengthen the data-facing pathway toward future full 3×2 pt tests.

20.2 KiDS Product Layers: $\xi_{\pm}$, COSEBIs, BandPower

Phase 3 completed validation of three KiDS shear-related product layers:

- Real-space $\xi_{\pm}$: 270-element vector, standardized, row-order verified, covariance validated;
- COSEBIs: 300-element vector, standardized, row-order verified, full 300×300 covariance validated;
- BandPower: 200-element vector combining 80-element P_{ne}E (lens-source) and 120-element P_{ee}E (source-source), standardized, row-order verified, full 200×200 official FITS covariance validated.

20.3 BandPower P_{ee}E Projector Validation

Phase 3E completed validation of the BandPower P_{ee}E projector path:

- Extracted P_{ee}E 120-row subset solely by statistic label, no hardcoded row indices;
- Verified P_{ee}E 120×120 covariance is symmetric and positive definite;
- Validated that G1 MAP C_{ℓ} outputs can be projected into the BandPower P_{ee}E space with finite diagnostic predictions;
- Completed Phase 3H calibrated-projection check to resolve normalization-convention mismatch with KCAP.

No optimized likelihood or evidence claims are made; this is projector feasibility only.

20.4 BandPower PneE Density-Kernel Smoke

Phase 3I completed a first density-kernel smoke path for BandPower PneE:

- Extracted PneE 80-row subset solely by statistic label, no hardcoded row indices;
- Verified PneE covariance blocks;
- Implemented a first density-kernel projection using a $b_a = 1$ placeholder bias;
- Confirmed finite diagnostic predictions for the smoke path.

No optimized bias fitting or evidence claims are made; this is density-kernel feasibility only.

20.5 EE+nE Bridge Status

The full 200-row $EE + nE$ BandPower bridge has been tested through Phase 3 and Phase 4B–4H diagnostic tests, with the following status: No optimized likelihood or evidence claims

Layer	Status
Product/covariance/order bridge	PASS
PeeE projector / normalization calibration	PASS
PneE density-kernel smoke	PASS
Local diagnostic refit	PASS
Deterministic no-noise mock recovery	PASS
Noisy mock diagnostic screen	PASS
Bias/shear/IA nuisance-prior robustness	PASS
Adversarial (Σ) interpretation	OPEN / amplitude-degeneracy limited
Phase 4G prior-regularized refit	PASS, degeneracy persists
Phase 4H toy (nn) closure	synthetic-only support
Real (P_{nn}) / clustering vector	MISSING
Full (EE+nE+nn) / (3×2) pt	BLOCKED

Table 9: EE+nE BandPower bridge validation status, v1.2.

are made; these are strictly diagnostic validation results only.

20.6 Interpretation Boundary

The Phase 3 and Phase 4B–4H results are strictly infrastructure and diagnostic validation results only:

- They validate product ingestion, covariance use, row ordering, sign conventions, projection conventions, density-kernel finiteness, local refit performance, mock recovery, noisy mock screening, and nuisance-prior robustness;
- They do not constitute an optimized BandPower likelihood, nested evidence, or production cosmological constraint;

- They do not claim full $3 \times 2\text{pt}$ completion or confirm the finite-screen microscopic theory;
- **EE+nE BandPower diagnostic bridge validated for local refit, mock recovery, noisy mock screening, and nuisance-prior robustness.**
- **Not full $3 \times 2\text{pt}$, not production evidence, not final cosmological constraints.**

20.7 Phase 4B–5B Diagnostic Validation Summary

- Phase 4E/4F adversarial Σ tests are **amplitude-degeneracy limited diagnosis**, not an M3/4 victory; corrected const- (Σ) / binned- (Σ) stress tests do not support shape-specificity claims.
- Phase 4H toy (nn) closure is synthetic-only, not real P_{nn} data completion, and only supports missing- nn degeneracy explanation.
- Noisy mock screening status is explicitly tiered:
 - **v1.2 KiDS BandPower diagnostic noisy mock screen: PASS**
 - **production-scale survey noisy mock ensemble / full likelihood validation: pending**

Allowed claims	Forbidden claims
Diagnostic bridge validated	Full $3 \times 2\text{pt}$ result
Local refit PASS	Production evidence
Mock recovery PASS	Optimized nuisance constraints
Noisy mock screen PASS	Finite-screen microscopic confirmation
Nuisance robustness PASS	"Fully validated bridge"

Table 10: Allowed and forbidden claim ladder for KiDS BandPower diagnostic results, v1.2.

20.8 Roadmap to Full $3 \times 2\text{pt}$

Current status of BandPower development and remaining tasks:

- ✓ **Completed / PASS:** local EE+nE refit, deterministic mock recovery, noisy mock ensemble screen, bias/shear/IA nuisance robustness, Phase 4G prior-regularized refit, calibrated bias treatment for PnE;
- **OPEN / diagnosed:** adversarial (Σ) interpretation, cannot claim M3/4 shape-specificity;
- △ **Synthetic-only support:** toy (nn) closure;

- × **MISSING / BLOCKED**: real (P_{nn})/clustering vector, full (EE+nE+nn), full (3×2)pt, real-space $\gamma_t/w(\theta)$ products and covariance;
- **Pending**: full 3×2 pt production likelihood/evidence, independent pipeline replication, production-scale noisy mock ensembles, and CMB/BBN/nonlinear-structure compatibility.

The theoretical closure modules listed in the front matter—finite ledger representation, causal measurement protocol, retarded response kernel, and early-universe compatibility—are not part of the KiDS BandPower validation claim. They define future G1 closure work.

20.8.1 Replication Kit Note

A v1.2 replication kit should archive Phase 3 bridge scripts, Phase 4B–4H diagnostic scripts/outputs, and Phase 5A–5B sourcing audit notes, while excluding restricted raw KiDS data.

References

Ciambelli, L., He, T., and Zurek, K. M. (2026). From asymptotically flat gravity to finite causal diamonds. *Physical Review Letters*, DOI: 10.1103/lbm1-vkks, arXiv:2512.09018.